

MEGN 301

Mechanical Integration & Design

Master Reference Document

Systems Engineering Process & Technical Reference

Colorado School of Mines

Department of Mechanical Engineering

A comprehensive guide to the design-build process from problem definition through operations and maintenance, with technical reference material for electromechanical system design.

Contents

How to Use This Document	xi
I The Design-Build Process	1
1 Problem Definition & Stakeholder Engagement	2
1.1 Defining the Problem	2
1.1.1 Market Analysis and Competitive Positioning	2
1.1.2 Customer Discovery and the Entrepreneurial Mindset	2
1.1.3 Market Analysis and Competitive Positioning	3
1.1.4 Customer Discovery and the Entrepreneurial Mindset	4
1.1.5 Problem vs. Solution Thinking	4
1.1.6 The Five-Element Framework	4
1.1.7 Scope and Boundaries	5
1.2 Stakeholder Engagement	5
1.2.1 Identifying Stakeholders	5
1.2.2 The Power/Interest Grid	5
1.2.3 Needs Elicitation Techniques	6
1.2.4 Managing Conflicting Needs	7
1.2.5 Needs vs. Wants	7
1.3 From Problem to ConOps	7
1.3.1 Concept of Operations	7
1.3.2 Use Cases and Operational Scenarios	7
1.4 Artifact Handoff: What Chapter 1 Produces	8
1.5 Artifact Handoff: What Chapter 1 Produces	9
1.6 Practice Questions	9
1.6.1 Answers	10
2 Writing Requirements & Defining Architecture	12
2.1 Writing Requirements	12
2.1.1 The System Design Requirements Document (SDRD)	12
2.1.2 The System Design Requirements Document (SDRD)	12
2.1.3 What Is a Requirement?	12
2.1.4 Requirement Types	13
2.1.5 Requirements Hierarchy and Traceability	13
2.1.6 The Requirements Quality Checklist	13
2.1.7 Writing Good Requirements: Five Rules	14
2.2 Defining Architecture	14
2.2.1 Functional Decomposition	14
2.2.2 From Functions to Architecture Blocks	15
2.2.3 From Functions to Architecture Blocks	15
2.2.4 Interface Definition	15

2.2.5	Architecture Patterns	15
2.2.6	Requirements–Architecture Iteration	15
2.3	Environmental and Regulatory Constraints	16
2.3.1	Requirements–Architecture Iteration	16
2.4	Environmental and Regulatory Constraints	16
2.5	Trade Studies and Decision Making	16
2.5.1	The Trade Study Process	16
2.5.2	Pugh Matrix vs. Weighted Decision Matrix	16
2.5.3	Worked Example: MCU Trade Study	17
2.6	Requirements-to-Architecture Flow	17
2.7	Artifact Handoff: What Chapter 2 Produces	19
2.8	Artifact Handoff: What Chapter 2 Produces	19
2.9	Practice Questions	19
2.9.1	Answers	20
3	Concept Generation & Early Design	21
3.1	Reverse Engineering	23
3.1.1	The Reverse Engineering Process	23
3.1.2	What Reverse Engineering Reveals	23
3.1.3	Reverse Engineering Outputs	23
3.1.4	From Reverse Engineering to Concept Generation	24
3.2	Reverse Engineering	24
3.2.1	The Reverse Engineering Process	24
3.2.2	What Reverse Engineering Reveals	24
3.3	Generating Concepts	24
3.3.1	Diverge Then Converge	24
3.3.2	Brainstorming Rules	25
3.3.3	SCAMPER	25
3.3.4	Morphological Charts	25
3.3.5	Bio-Inspired Design	25
3.3.6	Stage 1: Pugh Screening Matrix	25
3.3.7	Stage 2: Weighted Decision Matrix	25
3.4	Risk Assessment	25
3.4.1	The Risk Matrix	26
3.4.2	Risk Mitigation Strategies	26
3.5	Failure Modes and Effects Analysis (FMEA)	26
3.5.1	Conducting a Design-Phase FMEA	26
3.5.2	Prototype Types in Detail	27
3.6	Prototyping and Feasibility	27
3.6.1	Feasibility Assessment	27
3.7	Design Documentation	28
3.7.1	Bill of Materials (BOM)	28
3.7.2	Wiring Diagrams and Schematics	28
3.7.3	Software Architecture and Flowcharts	28
3.7.4	CAD Models and Engineering Drawings	28
3.8	Artifact Handoff: What Chapter 3 Produces	29
3.9	Practice Questions	30
3.9.1	Answers	30

4	Verification, Validation & Test Plans	32
4.1	Verification vs. Validation	32
4.2	Four Verification Methods: TAID	32
4.3	Writing Test Plans	33
4.3.1	What Makes a Good Test Plan	33
4.3.2	Test Procedure Structure	33
4.3.3	Writing Good Procedure Steps	34
4.4	Handling Results	34
4.4.1	PASS	34
4.4.2	FAIL	35
4.4.3	PARTIAL PASS	35
4.4.4	Root Cause Analysis	35
4.5	The VCRM	35
4.5.1	VCRM Evolution Over Time	35
4.5.2	End-to-End Traceability Example	35
4.6	V&V and the V-Model	36
4.7	Statistical Confidence and Test Rigor	36
4.7.1	Developmental Testing vs. Production Qualification	36
4.7.2	Practical Implications for Student Projects	36
4.7.3	The Test Environment Matters	36
4.7.4	Measurement Uncertainty	36
4.8	Edge Case and Boundary Testing	36
4.9	Minimum Viable V&V for MEGN 301	37
4.10	External Team Testing	37
4.11	Artifact Handoff: What Chapter 4 Produces	37
4.12	Statistical Confidence and Test Rigor	38
4.13	Edge Case and Boundary Testing	38
4.14	Minimum Viable V&V for MEGN 301	38
4.15	External Team Testing	38
4.16	Artifact Handoff: What Chapter 4 Produces	39
4.17	Practice Questions	39
4.17.1	Answers	39
4.17.2	Answers (continued)	40
5	Subsystem Integration	41
5.1	Integration Strategy	43
5.1.1	Bottom-Up Integration	43
5.1.2	Integration Stages and the V-Model	43
5.1.3	Integration Sequence Planning	43
5.1.4	Test Gates	43
5.1.5	Critical Subsystem Demonstrations	43
5.1.6	The N ² (N-Squared) Diagram	43
5.2	Artifact Handoff: What Chapter 5 Produces	44
5.3	Interface Management	44
5.3.1	Interface Control Documents (ICDs)	44
5.4	Debugging	45
5.5	Configuration Management	45
5.5.1	Critical Subsystem Demonstrations	45

5.5.2	The N ² (N-Squared) Diagram	45
5.6	Artifact Handoff: What Chapter 5 Produces	46
5.7	Practice Questions	46
5.7.1	Answers	47
6	Operations & Maintenance	48
6.1	Lifecycle Thinking	50
6.2	Design for Serviceability	50
6.2.1	Reliability Metrics	50
6.2.2	The Bathtub Curve	50
6.2.3	FMECA: Failure Modes, Effects, and Criticality Analysis	50
6.2.4	Writing Maintainability Requirements	50
6.3	Artifact Handoff: What Chapter 6 Produces	51
6.4	Maintenance Planning	51
6.5	Documentation, Training, and End-of-Life	51
6.5.1	Design for Disassembly (DfD)	51
6.5.2	The Bathtub Curve	52
6.5.3	FMECA: Failure Modes, Effects, and Criticality Analysis	52
6.5.4	Writing Maintainability Requirements	52
6.6	Artifact Handoff: What Chapter 6 Produces	53
6.7	Practice Questions	53
6.7.1	Answers	53
II	Technical Reference	55
7	Electrical & Fluid Connectors	56
7.1	Electrical Connectors	56
7.1.1	Connector Anatomy	56
7.1.2	Gender Convention	56
7.1.3	Keying and Polarization	56
7.1.4	Connection Types	57
7.1.5	Harsh-Environment Connectors	57
7.1.6	Termination Methods	58
7.1.7	Key Electrical Specifications	59
7.1.8	Electrical Connector Failure Modes	61
7.2	Fluid System Connectors	63
7.2.1	Threaded Pipe Fitting Standards	63
7.2.2	Tube Fittings and Quick-Connect Couplings	65
7.2.3	Key Fluid Connector Specifications	66
7.2.4	Fluid Connector Failure Modes	68
7.3	Selection and Design Practice	69
7.3.1	Connector Selection Logic	69
7.3.2	Design Rules of Thumb	70
7.3.3	Key Standards Reference	70
7.4	Applying Connectors in Your Design Projects	71
7.5	Quick-Reference Summary	71
7.6	Artifact Handoff: What This Chapter Supports	72

7.7	Practice Questions for Quiz Preparation	72
7.7.1	Conceptual Questions	72
7.7.2	Application / Scenario Questions	73
7.7.3	Answers	74
7.7.4	Application Scenario Answers	75
8	Power Supplies, Motors & Pumps	77
8.1	Power Supply Selection	77
8.1.1	What a Power Supply Does	77
8.1.2	Linear Regulators (LDO)	77
8.1.3	Switching Regulators (DC-DC Converters)	77
8.1.4	AC-DC Power Supplies	78
8.1.5	Key Power Supply Specifications	79
8.1.6	Power Supply Failure Modes	79
8.2	DC Motors	79
8.2.1	Motor Fundamentals	79
8.2.2	Brushed DC Motors	80
8.2.3	Brushless DC (BLDC) Motors	80
8.2.4	Stepper Motors	80
8.2.5	Servo Motors	81
8.2.6	Key Motor Specifications	81
8.2.7	Motor Drive Electronics	81
8.3	Pumps	82
8.3.1	Positive Displacement vs. Dynamic	82
8.3.2	Pump Types	82
8.3.3	Key Pump Specifications	83
8.3.4	Worked Example: Motor Selection for a Belt Conveyor	83
8.4	Selection and Design Practice	84
8.4.1	Unified Selection Framework	84
8.4.2	Design Rules	84
8.5	Artifact Handoff: What This Chapter Supports	85
8.6	Practice Questions	85
8.6.1	Conceptual Questions	85
8.6.2	Application / Scenario Questions	86
8.6.3	Answers	86
9	Power Architecture & Circuit Safety	88
9.1	Power Architecture	88
9.1.1	System-Level Design	88
9.1.2	Grounding Strategy	88
9.1.3	Battery Integration	89
9.1.4	Power Budget Spreadsheet: How to Build One	90
9.2	Protection Circuits	90
9.2.1	Overvoltage and Undervoltage Protection (OVP / UVP)	91
9.2.2	Overcurrent Protection (OCP)	91
9.2.3	Reverse Polarity and Back-Current Protection	92
9.2.4	Transient Voltage Suppression	93
9.3	Electromechanical Integration	94

9.3.1	Gear Reduction	94
9.3.2	Output Mechanisms	94
9.4	Safety Design Checklist	95
9.5	Practice Questions	95
9.5.1	Conceptual Questions	95
9.5.2	Application / Scenario Questions	96
9.5.3	Answers	96
10	Mechanical Fasteners & Bearings	99
10.1	Threaded Fasteners	99
10.1.1	Bolt and Screw Anatomy	99
10.1.2	Thread Standards	99
10.1.3	Bolt Grades and Strength	99
10.1.4	Torque and Preload	100
10.1.5	Locking Methods	100
10.1.6	Non-Threaded Fasteners	101
10.2	Bearings	101
10.2.1	What Bearings Do	101
10.2.2	Bearing Types	101
10.2.3	Bearing Selection	102
10.2.4	Mounting Bearings	102
10.2.5	Failure Modes	102
10.2.6	Thread Engagement Table	103
10.2.7	Threaded Inserts for 3D-Printed Parts	103
10.2.8	Bolt Selection Checklist	103
10.3	Design Rules of Thumb	103
10.4	Practice Questions	104
10.4.1	Answers	104
11	Mechanical Power Transmission	105
11.1	Fundamental Relationships	105
11.2	Belt and Pulley Drives	105
11.2.1	V-Belts	105
11.2.2	Timing Belts (Synchronous)	105
11.2.3	Flat and Round Belts	105
11.3	Chain Drives	106
11.4	Gear Trains	106
11.4.1	Gear Train Design Process	106
11.4.2	Backlash and Precision	107
11.5	Shafts and Couplings	107
11.5.1	Shaft Design	107
11.5.2	Shaft-Hub Connections	107
11.5.3	Couplings	107
11.6	Linear Motion	107
11.6.1	Lead Screws	107
11.6.2	Ball Screws	107
11.6.3	Rack and Pinion	107
11.6.4	Belt-Driven Linear Stages	107

11.7 Linkages	108
11.8 Selection Framework	108
11.9 Practice Questions	109
11.9.1 Answers	109
12 Designing for Injection Molding	110
12.1 Injection Molding Fundamentals	110
12.1.1 Process Overview	110
12.1.2 The Four-Phase Cycle	110
12.1.3 Material Selection	111
12.2 Design for Manufacturing (DFM) Rules	112
12.2.1 Rule 1: Uniform Wall Thickness	112
12.2.2 Rule 2: Draft Angles	112
12.2.3 Rule 3: Ribs and Bosses	113
12.2.4 Rule 4: Radii and Fillets	113
12.2.5 Rule 5: Undercuts	113
12.2.6 Rule 6: Gate Location	114
12.3 Common Defects	114
12.3.1 Worked DFM Example: Sensor Enclosure	115
12.4 Tooling and Economics	115
12.4.1 Mold Construction	116
12.4.2 Mold Classifications (SPI)	116
12.4.3 Part Cost Economics	116
12.5 Warpage	116
12.5.1 Causes of Warpage	116
12.5.2 Designing to Minimize Warpage	117
12.6 Surface Finish	117
12.6.1 SPI (PIA) Finish Standards	117
12.6.2 Textured Surfaces and Draft	117
12.7 Insert Molding and Overmolding	117
12.7.1 Insert Molding	117
12.7.2 Overmolding	118
12.8 Living Hinges and Snap-Fit Clips	118
12.8.1 Living Hinges	118
12.8.2 Snap-Fit Clips	118
12.9 Artifact Handoff: What This Chapter Supports	118
12.10 DFM Checklist	118
12.11 Practice Questions	119
12.11.1 Conceptual Questions	119
12.11.2 Application / Scenario Questions	120
12.11.3 Answers	120
12.11.4 Application / Scenario Questions	121
12.11.5 Answers to Application Questions	122

13 Printed Circuit Board Design Fundamentals	123
13.1 PCB Structure and Materials	123
13.1.1 Layer Stack-Ups	123
13.1.2 Key Specifications	123
13.1.3 Component Packages	124
13.2 Design Workflow	124
13.3 Schematic Capture Best Practices	124
13.3.1 Schematic Organization	125
13.3.2 Component Selection and Footprint Verification	125
13.3.3 Design Rule Configuration	126
13.4 Critical Layout Rules	126
13.4.1 Decoupling Capacitors	126
13.4.2 Ground Plane	126
13.4.3 Component Placement and Routing	126
13.5 Fabrication, Assembly, and Bring-Up	127
13.5.1 Fabrication Files	127
13.5.2 Fab Houses	127
13.5.3 Board Bring-Up Protocol	127
13.5.4 Design Review Checklist	128
13.5.5 Development Board Integration	128
13.6 EMI, Noise, and Signal Integrity	128
13.6.1 The Return Current Problem	128
13.6.2 Analog-Digital Separation	128
13.6.3 Motor Noise Mitigation on PCB	129
13.7 Thermal Management on PCBs	129
13.8 Common Mistakes	129
13.8.1 Development Boards vs Custom PCBs	130
13.9 Artifact Handoff: What This Chapter Supports	130
13.10 Practice Questions	130
13.10.1 Answers	131
14 Microcontrollers & the ESP32	133
14.1 MCU Fundamentals	133
14.1.1 What Is a Microcontroller?	133
14.1.2 MCU Platform Comparison	133
14.2 ESP32 Deep Dive	134
14.2.1 Safe GPIO Selection	135
14.2.2 Wi-Fi and Bluetooth	135
14.2.3 Power Modes	135
14.3 Communication Protocols	135
14.4 Firmware and Integration	136
14.4.1 Firmware Best Practices	136
14.4.2 System Integration	136
14.5 Artifact Handoff: What This Chapter Supports	136
14.5.1 Development Environment Setup	136
14.5.2 Common Integration Patterns	136
14.5.3 Battery Sizing Worked Example	137
14.6 Sensor Interfacing Patterns	137

14.6.1	Analog Sensors	137
14.6.2	Digital Sensors (I2C)	137
14.6.3	Motor and Actuator Control	137
14.7	Debugging and Troubleshooting	138
14.7.1	Serial Monitor Debugging	138
14.7.2	Common ESP32 Failure Modes	138
14.8	Practice Questions	139
14.8.1	Answers	139
III	Major Project Milestones	141
15	Preliminary Design Review (PDR)	143
15.1	Purpose and Scope	143
15.2	Deliverables	143
15.2.1	Problem Definition and Background Research	143
15.2.2	System Requirements and Traceability	143
15.2.3	Concept Development and Selection	143
15.2.4	Risk and Feasibility	143
15.2.5	Initial Test Plan	144
15.2.6	Concept Development and Selection: Detail	144
15.2.7	Budget and Schedule	144
15.2.8	Team Organization	144
15.3	Evaluation Criteria	144
15.4	Practice Questions	145
15.4.1	Answers	145
16	Critical Design Review (CDR)	147
16.1	Purpose and Scope	147
16.2	Deliverables	147
16.2.1	Finalized Technical Design	147
16.2.2	Design Analysis and Verification	147
16.2.3	Manufacturing and Procurement	147
16.2.4	Updated Test Plan	148
16.2.5	Software and Firmware	148
16.2.6	Thermal and Structural Analysis	148
16.2.7	Integration and Test Readiness	148
16.2.8	Risk Register Update	148
16.2.9	Procurement Status	148
16.3	Evaluation Criteria	148
16.4	Practice Questions	149
16.4.1	Answers	149
17	Final Design Review (FDR)	151
17.1	Purpose and Scope	151
17.2	Deliverables	151
17.2.1	System Verification and Validation Results	151
17.2.2	Final Design Documentation	151

17.2.3 As-Designed vs. As-Built Comparison	151
17.2.4 Design Process Documentation	152
17.2.5 Operations and Maintenance Materials	152
17.2.6 External Team Testing Results	152
17.2.7 Demonstration Video	152
17.2.8 As-Designed vs. As-Built Comparison: Template	152
17.2.9 Lessons Learned: Beyond “Start Earlier”	152
17.2.10 Reflection and Future Work	153
17.3 Evaluation Criteria	153
17.3.1 Budget Final Accounting	153
17.3.2 Schedule Performance	153
17.4 Evaluation Criteria	153
17.5 The Complete Review Progression	154
17.6 Practice Questions	154
17.6.1 Answers	154
Glossary	156
Exam Quick-Reference	158
Why Projects Fail	159
ABET Outcome Mapping	161
GreenShield: Complete Worked Example	162
Quick Reference Tables	164
References	166

How to Use This Document

This reference document accompanies the MEGN 301 lecture modules and is organized to follow the complete design-build lifecycle of an engineering project.

Part I: The Design-Build Process covers systems engineering from start to finish. The six chapters follow the natural project timeline, from defining the problem through delivering and supporting the final system. Use these chapters in order as your project progresses through design reviews:

- **Chapter 1** (Problem Definition) and **Chapter 2** (Requirements) support your work before and at **PDR**.
- **Chapter 3** (Concept Generation) supports the transition from PDR to **CDR**.
- **Chapter 4** (V&V) and **Chapter 5** (Integration) support your work between CDR and **FDR**.
- **Chapter 6** (Operations & Maintenance) supports system handoff and your FDR deliverables.

Part II: Technical Reference provides detailed design guidance for common electromechanical subsystems. These chapters are reference material; consult them when your project involves connectors, power supplies, motors, pumps, protection circuits, gear trains, injection-molded parts, or printed circuit boards.

Every chapter includes practice questions with worked answers. Use them for self-study and exam preparation.

Part III: Major Project Milestones provides detailed guidance for the three formal design reviews, PDR, CDR, and FDR, with deliverable checklists, evaluation criteria, and common failure patterns. These chapters explain what each review evaluates, what evidence you must present, and how the three milestones form a coherent narrative arc across the project.

The Running Case Study: GreenShield Frost Protection System

Throughout Part I, teal-bordered **GS** boxes follow a single design, the GreenShield automated greenhouse frost protection system, from problem definition through operations. Watch for these boxes to see how each artifact builds on the last.

GreenShield is an Example, Not a Template

The GreenShield system illustrates *how* to move from problem statement to requirements, how to conduct a trade study, and how to write a test plan. It is **not** a prescription for your project. Your problem, stakeholders, requirements, and architecture will be different. Use GreenShield to understand the process, not to copy components or design decisions.

What Should I Be Reading Right Now?

- **Just started your project?** → Chapter 1 (Problem Definition), Chapter 2 (Requirements)
- **Preparing for PDR?** → Chapter 3 (Concept Generation), Chapter 13 (PDR Deliverables)
- **Designing hardware?** → Chapters 7–12 (Technical Reference for your subsystem types)
- **Preparing for CDR?** → Chapter 14 (CDR Deliverables), verify BOM and schematics
- **Integration is failing?** → Chapter 5 (Integration), Chapter 9 (Power Architecture)
- **Motor won't spin?** → Chapter 8 §8.2 (Motors & Drivers), Chapter 9 (Brown-Out Blues)

- **Sensor readings are noisy?** → Chapter 9 §9.3 (Grounding), Chapter 12 (PCB Layout Rules)
- **Preparing for FDR?** → Chapter 4 (V&V), Chapter 6 (O&M), Chapter 15 (FDR Deliverables)
- **Studying for the exam?** → Practice questions at the end of every chapter

Sprint-to-Chapter Mapping

Sprint	Weeks	Primary Activities	Chapters	Key Deliverables
1	1–2	Team formation, problem definition, stakeholder analysis	Ch. 1	Problem statement, stakeholder map, ConOps
2	3–4	Requirements, architecture, competitive analysis	Ch. 2, 3	SDRD, functional decomposition
3	5–6	Concept generation, trade studies, preliminary design	Ch. 3	Morph. chart, decision matrix, PDR
4	7–8	Detailed design, procurement, prototyping	Ch. 7–12	BOM, schematics, CAD, orders
5	9–10	Fabrication, subsystem testing, integration begins	Ch. 4, 5	Subsystem test results, CDR
6	11–13	System integration, V&V execution, debugging	Ch. 4, 5, 9	VCRM, test reports, system demos
7	14–15	Final V&V, documentation, O&M package	Ch. 6	O&M manual, as-built docs
Final	16	Final design review, reflection, knowledge transfer	Part III	FDR presentation & package

Table 1: Sprint-to-chapter mapping for a 16-week MEGN 301 project.

Part I

The Design-Build Process

Chapter 1

Problem Definition & Stakeholder Engagement

1.1 Defining the Problem

1.1.1 Market Analysis and Competitive Positioning

Before defining your problem, understand what already exists. Conduct a **competitive search** appropriate to your project type:

- **Commercial product:** Search Amazon, patent databases (Google Patents), and review sites. Document 3–5 competing products with price, features, and limitations. Your problem statement must articulate what gap remains.
- **Research/instrumentation:** Search Google Scholar and IEEE Xplore for prior solutions. Document the state of the art and identify the specific capability gap.
- **Process improvement:** Benchmark current performance metrics against industry standards or competitor processes.

The competitive analysis feeds directly into your problem statement’s “WHY IT MATTERS” element; you cannot justify a project without knowing what already exists.

1.1.2 Customer Discovery and the Entrepreneurial Mindset

Engineering value creation begins with deep customer understanding. The KEEN framework emphasizes three C’s: **Curiosity** (investigate the problem space broadly), **Connections** (link technical knowledge to stakeholder needs and market realities), and **Creating Value** (design solutions that deliver measurable benefit).

Customer discovery is the practice of testing assumptions about the problem *before* committing to a solution. Talk to at least 5 potential users. Ask open-ended questions: “Walk me through the last time this problem occurred.” “What did you try? Why didn’t it work?” Document answers; they become the foundation of your ConOps.

Synthesis: After interviews, group raw needs into clusters: (1) critical needs (mentioned by most personas, high impact) become SHALL requirements; (2) important needs (mentioned by some, moderate impact) become SHOULD goals; (3) nice-to-have needs (mentioned rarely) become COULD features. This maps directly to MoSCoW prioritization.

Patent Landscape Search

Search Google Patents and USPTO for 2–3 relevant utility patents. Document: patent number, title, assignee, key claims, and a freedom-to-operate assessment (does your concept infringe any active claims?). This is not a legal opinion; it is an engineering awareness exercise that reveals prior art and protected design approaches.

Standards and Regulatory Landscape

Identify applicable standards early: UL/CSA for electrical safety, FCC Part 15 for radiated emissions, IP ratings for environmental protection, food-contact regulations if applicable. Standards become non-functional requirements. Discovering a regulatory requirement at CDR (instead of PDR) causes expensive redesign.

AI-Assisted Customer Persona Development

When direct stakeholder access is limited, use AI tools to generate customer personas. Describe the problem domain and ask the AI to role-play as different stakeholder types: the primary user, the maintainer, the regulator, the adjacent-system owner. Conduct the same interview questions you would use with real stakeholders. This is not a substitute for real customer discovery; it is a complement that helps you prepare better questions for real interviews and identify needs you might otherwise miss.

Document each persona interview with: persona description, key quotes, identified needs (stated and unstated), and implications for requirements.

The Mom Test

Never ask “Would you use this?”; people say yes to be polite. Instead ask about *past behavior*: “When did you last face this problem? What did you do?” Past actions predict future behavior far better than hypothetical opinions.

1.1.3 Market Analysis and Competitive Positioning

Before defining your problem, understand what already exists. A market analysis answers three questions: Who else has tried to solve this problem? What did they build? And where are the opportunities to do better?

Part A: Competitive Product Survey. Identify at least three competing products or existing solutions. For each, document features, strengths, weaknesses, price point, and user reviews. Synthesize: What features are universally present (table stakes)? What features are missing or poorly executed (opportunities)? What do customers complain about most (unmet needs)?

Part B: Patent Landscape. Search patents.google.com for relevant utility patents. Record: (1) filing date (patents expire after 20 years), (2) abstract, and (3) claims. The purpose is freedom to operate: can your design avoid infringing active claims?

Reading Patents Efficiently

Read the abstract first. Then skip to the claims at the end. Claims define the legal boundaries. If your design does not practice every element of an independent claim, you do not infringe. Expired patents contain detailed engineering you can freely use.

The competitive analysis feeds directly into the “WHY IT MATTERS” element of your problem statement; you cannot justify a project without knowing what already exists.

1.1.4 Customer Discovery and the Entrepreneurial Mindset

Engineering teams that design in isolation build systems nobody wants. Customer discovery is the practice of systematically interviewing potential users to validate your assumptions about the problem before committing to a solution.

Three components of the entrepreneurial mindset (KEEN 3 C’s):

Curiosity: Ask open-ended questions that reveal unstated needs. “Tell me about the last time you had this problem” yields richer insight than “Would you use a product that does X?”

Connections: Link what you learn across interviews. If three different user types independently mention the same pain point, that is a strong signal of a real need.

Creating Value: Translate insights into requirements that deliver measurable value. A feature is only valuable if it solves a problem the customer actually has.

From Customer Discovery to Requirements

The output of customer discovery is a set of validated needs, organized by priority:

1. **Must-have needs** (mentioned by most users, significant pain): become **SHALL** requirements.
2. **Important needs** (mentioned by some users, moderate impact): become **SHOULD** goals.
3. **Nice-to-have needs** (mentioned rarely, low impact): become **COULD** features, pursued if budget allows.

This maps directly to the MoSCoW framework discussed later in this chapter.

1.1.5 Problem vs. Solution Thinking

Engineers instinctively jump to solutions. The discipline of problem definition requires staying in the **problem space** (what is wrong, who is affected, what the consequences are) before entering the **solution space** (technologies, components, architectures).

Critical test: Can someone read your statement and propose a fundamentally different solution? If yes, you are in the problem space. If the statement implies only one approach, you have prematurely entered the solution space.

Solution Masquerading as a Problem

Bad: “We need to build an Arduino-based temperature monitoring system with a solar panel.” This specifies technology before the problem is understood. **Good:** “Greenhouse operators cannot maintain optimal growing temperatures in unheated hoop houses during Colorado Front Range winters, resulting in \$12,000/season crop losses.” This is solution-agnostic.

1.1.6 The Five-Element Framework

1. **WHO:** Who experiences the problem? Be specific about roles and context.
2. **WHAT:** What is the measurable gap between current and desired states?
3. **WHERE:** In what physical, operational, or environmental context?
4. **WHY IT MATTERS:** What are the consequences of not solving it? (Cost, safety, health.)

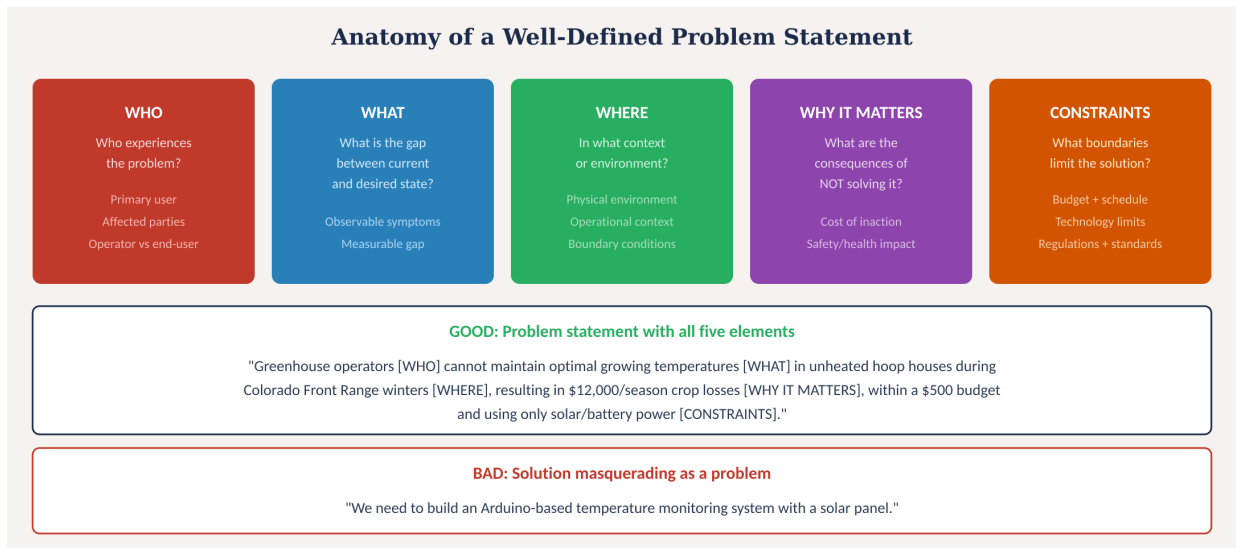


Figure 1.1: Five-element problem statement framework with good and bad examples.

5. **CONSTRAINTS:** What boundaries limit the solution? (Budget, schedule, technology, regulations.)

Constraints directly become non-functional requirements. The “why it matters” element justifies the project’s existence.

1.1.1.7 Scope and Boundaries

Scope defines what is inside and outside the project. Without it, scope creep expands the project indefinitely. The **system boundary** distinguishes what you build from what the environment provides. Capture this in a **context diagram**: your system as a single block with all external interfaces (power, data, user, mechanical, environmental) labeled.

1.2 Stakeholder Engagement

1.2.1 Identifying Stakeholders

A stakeholder is anyone who affects, is affected by, or has an interest in the project. **Missing a stakeholder = missing requirements = integration surprises.**

Nine categories to check: primary users, beneficiaries, sponsor/funder, maintainers, regulators, integrators (adjacent systems), adversaries, trainers, and disposal handlers. Students consistently miss maintainers, regulators, and adjacent-system owners.

1.2.2 The Power/Interest Grid

Classify each stakeholder by their power (ability to influence the project) and interest (how much they care). Four quadrants with distinct strategies: **Manage closely** (high power, high interest, key decision makers), **Keep satisfied** (high power, low interest), **Keep informed** (low power, high interest, valuable input), **Monitor** (low power, low interest).

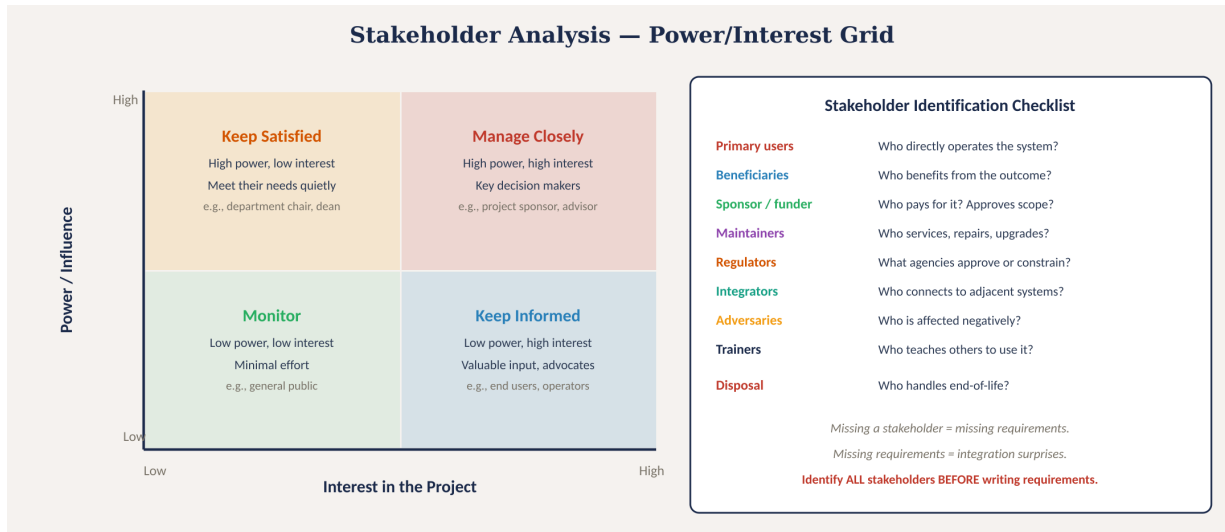


Figure 1.2: Power/interest grid with engagement strategies and nine-category identification checklist.

1.2.3 Needs Elicitation Techniques

Needs Elicitation Methods — Choosing the Right Technique				
Method	How It Works	Best For	Strengths	Watch Out
Interviews	One-on-one structured conversation with prepared questions	Deep understanding of individual needs, motivations, context	Rich qualitative data; uncovers hidden needs; builds rapport	Time-intensive; interviewer bias; small sample
Surveys	Structured questionnaire distributed to many stakeholders	Quantifying preferences, priorities, and demographics at scale	Large sample; statistical analysis; low cost per respondent	Low response rate; cannot probe deeply; question bias
Observation	Watch users perform tasks in their real environment	Understanding actual behavior vs stated behavior	Reveals workarounds and pain points users don't mention	Time-intensive; Hawthorne effect; requires access
Workshops	Facilitated group session with multiple stakeholders together	Building consensus; resolving conflicts; brainstorming	Surfaces disagreements early; group creativity; efficient	Dominant voices suppress others; groupthink risk
Prototyping	Build quick models (paper, 3D print, mockup) for user feedback	Validating assumptions early; testing form/fit/ergonomics	Tangible; surfaces misunderstandings; iterations fast	Stakeholders may fixate on prototypes as final design
Document Analysis	Review existing specs, manuals, incident reports, standards	Understanding legacy systems, regulations, known issues	No stakeholder time required; objective; comprehensive	Documents may be outdated; missing tacit knowledge

Figure 1.3: Six elicitation methods with best-use cases, strengths, and watch-outs.

Use **2–3 methods**; single-method elicitation misses needs. Strongest combination for engineering projects: **interviews** (capture stated needs) + **observation** (capture unstated needs) + **prototyping** (validate assumptions).

The Best Question in Engineering

End every stakeholder interview with: “What should I have asked that I didn’t?” This single question catches more missing requirements than any structured technique.

1.2.4 Managing Conflicting Needs

Conflicts are normal. Prioritize using **MoSCoW**: Must (becomes SHALL), Should (becomes SHOULD), Could (if budget allows), Won't (explicitly excluded). When conflicts are irreconcilable, the sponsor or primary user has final authority, but the decision must be documented.

1.2.5 Needs vs. Wants

A **need** is something without which the system fails its mission, it becomes a SHALL requirement. A **want** enhances value but the system works without it, it becomes a SHOULD goal pursued only if resources allow. Separate them early.

1.3 From Problem to ConOps

1.3.1 Concept of Operations

The ConOps describes how the system will be used from the stakeholder's perspective, written in **plain language** (no engineering jargon). Contents: system mission, current situation, operational scenarios (normal, degraded, emergency, maintenance), user roles, environmental conditions, external interfaces, and performance expectations.

The ConOps is the document the stakeholder reviews and approves **before requirements are written**. Misunderstandings caught here cost minutes; caught in requirements, they cost weeks.

1.3.2 Use Cases and Operational Scenarios

A **use case** describes a specific user-system interaction to achieve a goal. An **operational scenario** extends the use case into a realistic narrative with timing, weather, user state, and system state. Scenarios reveal requirements that abstract analysis misses.

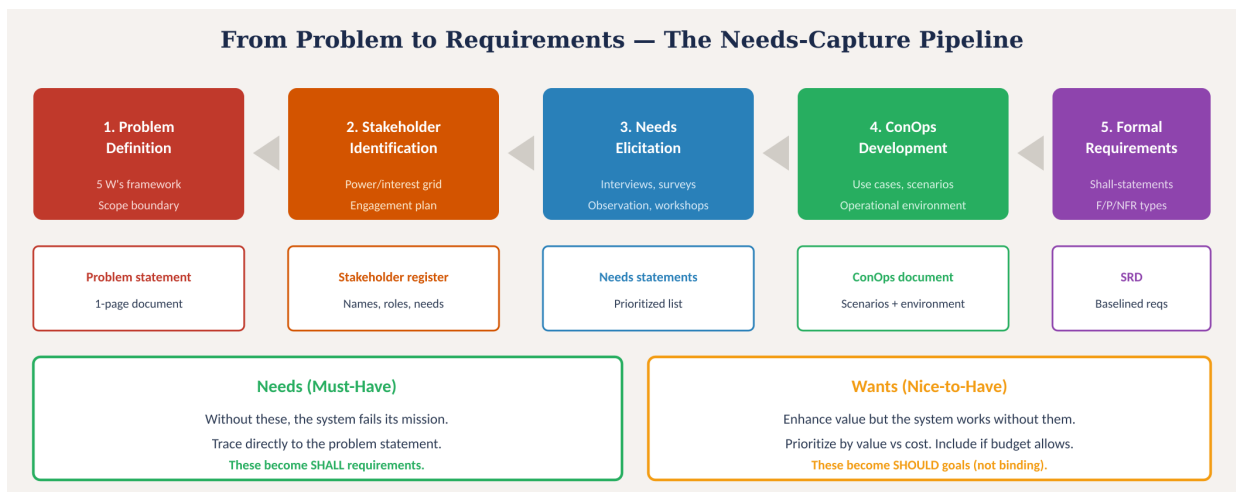


Figure 1.4: Five-stage needs-capture pipeline from problem to formal requirements with deliverables.

GS GreenShield: ConOps to Requirements Mapping

System: GreenShield. Automated Greenhouse Frost Protection System.

ConOps excerpt: “When nighttime temperature drops below 3 °C, the system activates supplemental heating and alerts the operator via SMS. In degraded mode (cellular failure), the system continues autonomous heating control using local logic.”

Requirements derived:

- **SYS-REQ-001:** The system shall monitor air temperature continuously at ≥ 1 Hz.
- **SYS-REQ-002:** The system shall activate the heater when temperature $\leq$ user-defined setpoint (default 3 °C).
- **SYS-REQ-003:** The system shall send an SMS alert within 60 s of heater activation.
- **SYS-REQ-004:** The system shall continue autonomous heating control during communication failure.

Every ConOps sentence should map to at least one requirement. If it does not, either the ConOps is too vague or requirements are missing.

Industry SE Parallels

- **INCOSE SEH 4th ed.:** Sec. 4.1 (Stakeholder Needs and Requirements Definition), Sec. 4.2 (System Requirements Definition).
- **NASA SP-2016-6105 Rev 2:** Ch. 4 (System Design Processes), Sec. 4.1 (Stakeholder Expectations Definition).
- **IEEE 29148:** Standard for requirements engineering processes.

In industry, the ConOps is often a formal deliverable reviewed at a System Requirements Review (SRR), which precedes the Preliminary Design Review. The MoSCoW prioritization method maps to what INCOSE calls “needs prioritization” within stakeholder requirements definition.

1.4 Artifact Handoff: What Chapter 1 Produces

Artifact	Description	Forward Reference
Problem Statement	Five-element statement (who/what/where/why/constraints)	Ch. 2 (requirements derivation)
Stakeholder Map	Power/interest grid with engagement strategies	Ch. 2, Ch. 4 (validation)
Concept of Operations	Plain-language usage scenarios (normal, degraded, emergency)	Ch. 2, Ch. 6 (O&M)
Competitive Analysis	3–5 competing solutions with gap identification	Ch. 3 (concept generation)
Context Diagram	System boundary with external interfaces labeled	Ch. 2 (interface requirements)

GS GreenShield: ConOps to Requirements Mapping

System: GreenShield. Automated Greenhouse Frost Protection System.

ConOps excerpt: “When nighttime temperature drops below 3 °C, the system activates supplemental heating and alerts the operator via SMS. In degraded mode (cellular failure),

the system continues autonomous control using local logic.”

Requirements derived:

- SYS-REQ-001: The system shall monitor air temperature continuously at ≥ 1 Hz.
- SYS-REQ-002: The system shall activate the heater when temperature $\leq$ setpoint (default 3 °C).
- SYS-REQ-003: The system shall send an SMS alert within 60 s of heater activation.
- SYS-REQ-004: The system shall continue autonomous heating control during communication failure.

Every ConOps sentence should map to at least one requirement. Unmapped sentences indicate vague ConOps or missing requirements.

Industry SE Parallels

- **INCOSE SEH 4th ed.:** §4.1 Stakeholder Needs Definition, §4.2 System Requirements Definition.
- **NASA SP-2016-6105 Rev 2:** Ch. 4.1 Stakeholder Expectations Definition.
- **IEEE 29148:** Requirements engineering processes.

1.5 Artifact Handoff: What Chapter 1 Produces

Artifact	Description	Used In
Problem Statement	Five-element statement (who/what/where/why/constraints)	Ch. 2
Stakeholder Map	Power/interest grid with engagement strategies	Ch. 2, 4
Concept of Operations	Plain-language usage scenarios with operational modes	Ch. 2, 6
Competitive Analysis	3–5 existing solutions with gap identification	Ch. 3

1.6 Practice Questions

1. Rewrite this solution-as-problem into a proper five-element problem statement: “We need to design a solar-powered weather station using a Raspberry Pi.”
2. List all stakeholders for a MEGN 301 design project that designs an automated pet feeder for elderly users. Use the nine-category checklist.
3. You interview a greenhouse operator who says “I need a temperature display.” Through observation, you notice she checks the display, then walks to the heater and turns it on manually. What is her *actual* need? How does this change the requirements?
4. Classify these stakeholders on a power/interest grid for a campus parking management system: (a) university president, (b) daily commuter student, (c) campus police, (d) city traffic department, (e) disability services office.
5. Write a 1-paragraph ConOps for an automated frost protection system for a small greenhouse, including at least two operational scenarios (normal and degraded mode).
6. **Market analysis:** existing pet feeders on Amazon range \$30–\$150. Strengths: app connectivity, portion control. Weaknesses: require Wi-Fi (elderly users may lack technical skill), small

hoppers (2–3 day capacity), no medication dispensing. Gap: a system designed specifically for elderly users with large-print controls, cellular connectivity (no Wi-Fi setup), and caregiver alerting.

7. **Customer discovery approach:** interview 3–5 elderly pet owners and 2–3 caregivers. Key questions: “Show me how you feed your pet today.” “What happens when you travel or are hospitalized?” “How comfortable are you with smartphone apps?” Expected insights: manual feeding is a daily routine that provides companionship (do not eliminate it entirely); missed feedings happen during health crises, not routine days; caregivers need alerts, not control.
8. A sponsor wants the system under \$500, the user wants a touchscreen display, and the maintainer wants modular components. These conflict. How do you resolve this using the MoSCoW framework?

1.6.1 Answers

1. “Small-farm operators [WHO] lack access to real-time, site-specific weather data [WHAT] at remote agricultural locations without grid power [WHERE], leading to crop management decisions based on regional forecasts that miss micro-climate variations by up to 5 ° C [WHY], within a \$300 budget using only renewable power sources [CONSTRAINTS].”
2. Primary users: elderly pet owner (operator). Beneficiaries: the pet; family members who worry remotely. Sponsor: the pet owner or their family. Maintainers: family member or home-care aide who refills food and troubleshoots. Regulators: food-safety standards for pet food storage; electrical safety (UL). Integrators: home Wi-Fi network; smartphone (if app-connected). Adversaries: the pet (may try to defeat the mechanism). Trainers: whoever teaches the elderly user. Disposal: recycling or trash service.
3. Her actual need is automated frost protection, not a temperature display. The display is her current workaround for a lack of automation. The real requirement is: “The system shall activate the supplemental heater when temperature drops below a user-defined setpoint.” The display becomes a secondary want, not the primary need. This changes the architecture from a passive monitor to an active control system.
4. (a) University president: high power, low interest → Keep Satisfied. (b) Commuter student: low power, high interest → Keep Informed. (c) Campus police: high power, high interest → Manage Closely (they enforce parking rules). (d) City traffic department: moderate power, low interest → Keep Satisfied (jurisdiction over surrounding roads). (e) Disability services: moderate power, high interest → Manage Closely (ADA compliance is non-negotiable).
5. “The frost protection system monitors greenhouse air temperature continuously. During normal operation, when temperature drops below the user-defined threshold (default 3 ° C), the system automatically activates the supplemental heater and sends an alert to the operator’s phone. The operator can acknowledge the alert, adjust the setpoint, or override the system remotely. In degraded mode (communication failure), the system continues autonomous control using local logic, activating the heater based on its last-known setpoint and logging events locally until communication is restored. The operator discovers the communication failure on their next manual check or when they notice the absence of the daily status report.”
6. MoSCoW: **Must:** system cost under \$500 (sponsor constraint; non-negotiable). Modular components for field replacement (maintainer need; essential for long-term usability). **Should:**

visual status display (user need; but a low-cost LCD or LED indicators at \$5 may satisfy the need without a \$50 touchscreen). **Could:** touchscreen interface (user want; only if budget allows after must-haves). **Won't:** full-color graphical touchscreen in v1.0. Resolution: the display is a legitimate need but the *touchscreen* is a want. Propose a 2-line LCD (\$3) for v1.0 with a touchscreen upgrade path for v2.0 if the sponsor increases budget.

Chapter 2

Writing Requirements & Defining Architecture

2.1 Writing Requirements

2.1.1 The System Design Requirements Document (SDRD)

The SDRD is the single authoritative document that captures all system-level requirements, their rationale, traceability, and verification methods. Standard structure:

1. **Introduction:** system overview, scope, applicable documents.
2. **System Description:** ConOps summary, system boundary, external interfaces.
3. **Requirements:** organized by functional, performance, and non-functional categories. Each includes: unique ID, shall-statement, rationale, parent traceability, verification method (T/A/I/D).
4. **Traceability Matrix:** forward (need → requirement) and backward (requirement → need).
5. **Verification Plan Summary:** method assignments and planned test phases.

The SDRD is a living document; baselined at PDR and updated under change control through CDR and FDR.

2.1.2 The System Design Requirements Document (SDRD)

The SDRD is the single authoritative document capturing all system-level requirements, their rationale, traceability, and verification methods. Structure:

1. **Introduction:** system overview, scope, applicable documents.
2. **System Description:** ConOps summary, system boundary, external interfaces.
3. **Requirements:** organized by functional, performance, and non-functional categories. Each entry: unique ID, shall-statement, rationale, parent traceability, verification method (T/A/I/D).
4. **Traceability Matrix:** forward (need → requirement) and backward (requirement → need).
5. **Verification Plan Summary:** method assignments and planned test phases.

The SDRD is baselined at PDR and updated under change control through CDR and FDR.

2.1.3 What Is a Requirement?

A requirement is a formal statement of a capability or constraint the system must satisfy. It uses the word **shall** to indicate mandatory compliance. “The system *shall* measure ambient temperature” is a requirement. “Should” and “will” are non-binding.

A well-formed requirement has: (1) a unique ID (SYS-REQ-042) for traceability, (2) a single shall-statement with one testable condition, (3) measurable acceptance criteria with units and tolerance, (4) a rationale explaining why it exists, and (5) a verification method (Test, Analysis, Inspection, or Demonstration).

Common Failures

Vague: “shall be fast.” **Compound:** “shall measure AND display.” **Untestable:** “shall be user-friendly.” **Gold-plated:** “shall use an Arduino” (specifies implementation, not need). Each failure produces downstream rework.

2.1.4 Requirement Types

- **Functional:** What the system shall *do*. Actions: sense, compute, transmit, actuate, display. “The system shall measure ambient temperature.”
- **Performance:** How *well* it does it. Quantitative criteria. “... with accuracy of $\pm 0.5^\circ \text{C}$ over -20 to $+60^\circ \text{C}$ at $\geq 1 \text{ Hz}$.” A functional requirement without a paired performance requirement is incomplete.
- **Non-Functional (NFR):** Constraints on *how* it is built. Environmental (“operate at -20 to $+60^\circ \text{C}$ ”), regulatory (“comply with FCC Part 15”), material (“RoHS-compliant”), reliability (“ ≥ 1000 hrs MTBF”), interface (“CAN 2.0B”).

2.1.5 Requirements Hierarchy and Traceability

Requirements decompose top-down: stakeholder needs → system requirements → subsystem requirements → component specifications. Each level adds detail. Every child requirement traces to a parent (backward traceability). Every parent is decomposed into children (forward traceability). An orphan requirement has no justification; an undecomposed requirement has no implementation.

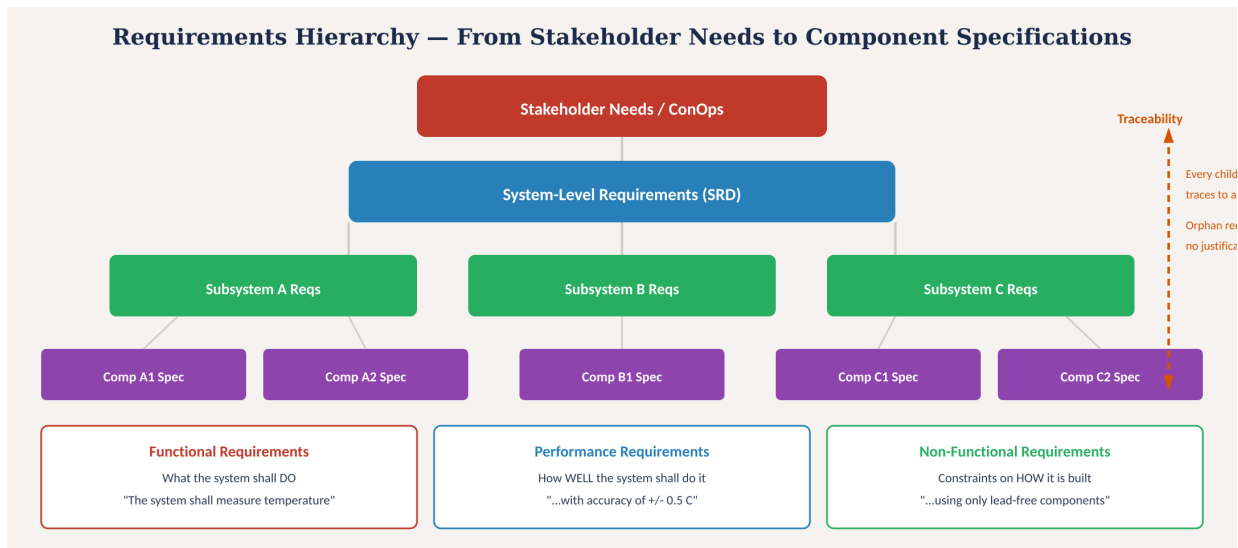


Figure 2.1: Requirements hierarchy from stakeholder needs through system, subsystem, and component levels with traceability and three requirement types.

2.1.6 The Requirements Quality Checklist

Before baselining any requirement, verify it passes all seven tests:

1. **Necessary:** does it trace to a stakeholder need? (No gold-plating.)

2. **Unambiguous:** can it be interpreted only one way?
3. **Complete:** does it include all conditions (range, tolerance, units)?
4. **Consistent:** does it conflict with any other requirement?
5. **Verifiable:** can you write a test procedure for it right now?
6. **Traceable:** does it have a unique ID and parent link?
7. **Implementation-free:** does it specify WHAT, not HOW?

A requirement that fails any test must be rewritten before PDR.

2.1.7 Writing Good Requirements: Five Rules

1. **One shall per requirement:** compound requirements are untestable.
2. **Quantitative criteria:** numbers, units, tolerance. “Fast” is not a requirement.
3. **Testable as written:** if you cannot envision a test procedure, rewrite.
4. **Implementation-free:** state WHAT, not HOW. “ ≥ 12 analog inputs” not “use Arduino Mega.”
5. **Include rationale:** WHY does this requirement exist? Prevents zombie requirements.

2.2 Defining Architecture

2.2.1 Functional Decomposition

Define WHAT the system must do (functions) before deciding HOW (physical components). Break the mission into primary functions, sub-functions, and leaf functions. Each leaf maps to a physical component. Use verb-noun naming: Sense Temperature, Process Data, Provide Power.

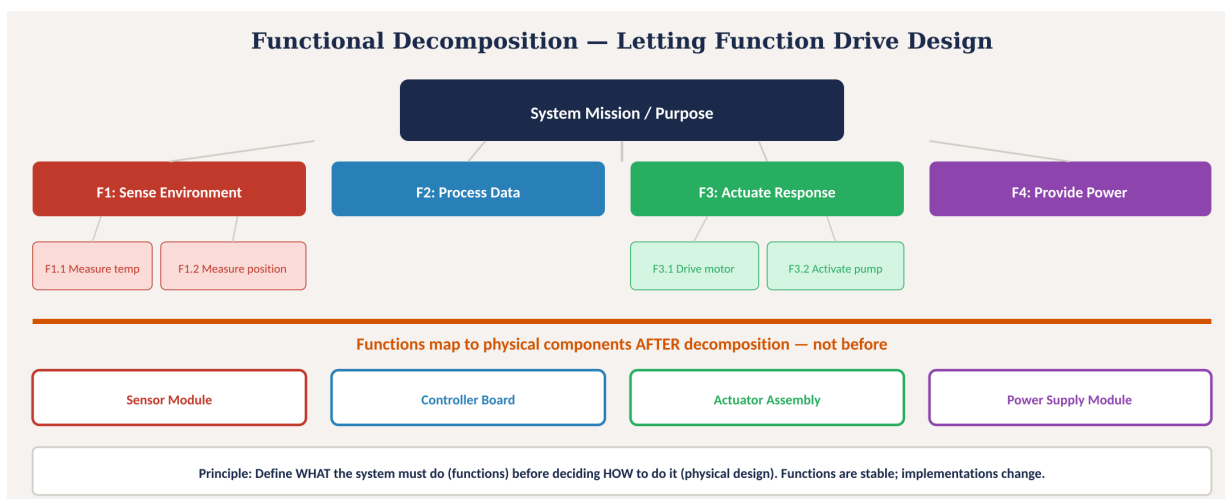


Figure 2.2: Functional decomposition: functions defined before physical allocation. Functions are stable; implementations change.

2.2.2 From Functions to Architecture Blocks

Functional decomposition produces a tree of functions (verb-noun). Architecture allocation maps each leaf function to a physical module. This mapping is one-to-many (one function may require multiple modules) and many-to-one (one module may implement multiple functions). Document this in an **allocation matrix**: rows = functions, columns = physical modules, cells = primary/supporting role.

The allocation matrix reveals integration complexity: modules that implement many functions are high-risk single points of failure.

2.2.3 From Functions to Architecture Blocks

Functional decomposition produces a tree of functions (verb-noun pairs). Architecture allocation maps each leaf function to a physical module. This mapping is one-to-many (one function may require multiple modules) and many-to-one (one module may implement multiple functions). Document this in an **allocation matrix**: rows = functions, columns = physical modules, cells = primary/supporting role.

The allocation matrix reveals integration complexity: modules implementing many functions are high-risk. Modules sharing functions require careful interface definition.

2.2.4 Interface Definition

Once functions map to modules, define interfaces between them with Interface Control Documents (ICDs): what crosses the boundary (signals, power, data), in what format (protocols, voltage levels), with what constraints (bandwidth, latency).

Start Your FMEA During Architecture: Not at FDR

Failure Mode and Effects Analysis (FMEA) is most valuable *before* hardware is ordered. As you allocate functions to physical modules (this chapter), immediately ask: “What happens if this module fails?” Chapter 3 provides the full FMEA framework and a worked GreenShield example. Performing FMEA during architecture (Chapters 2–3) enables you to design out critical failure modes. Waiting until O&M (Chapter 6) turns FMEA into an autopsy on a failed project.

2.2.5 Architecture Patterns

Centralized: one controller. Simple but single point of failure. **Distributed**: multiple controllers on a bus (CAN, I²C). More robust, enables parallel development. **Hierarchical**: layered control. Choose based on reliability, complexity, and team structure.

2.2.6 Requirements–Architecture Iteration

Requirements and architecture co-evolve. Initial requirements drive a candidate architecture. That architecture reveals implementation constraints (weight, power, thermal) that feed back as new requirements or modified performance targets. This iteration is normal and healthy, but must be **documented**. Each iteration produces a revised SDRD version with change rationale.

Convergence test: when two consecutive iterations change fewer than 5% of requirements, the design is stable enough to baseline at PDR.

2.3 Environmental and Regulatory Constraints

Every system operates in an environment that imposes constraints. Systematically address:

- **Temperature:** operating range, storage range, thermal shock.
- **Moisture/IP rating:** splash (IPX4), rain (IPX5), submersion (IPX7/8).
- **Vibration/shock:** transport, operational, handling drop.
- **EMC:** FCC Part 15 Class B (residential), IEC 61000 (EU).
- **Safety:** UL/CSA/CE marking, OSHA regulations, university lab safety policies.
- **Materials:** RoHS compliance, California Prop 65, food-contact (FDA 21 CFR).

2.3.1 Requirements–Architecture Iteration

Requirements and architecture co-evolve. Initial requirements drive a candidate architecture; that architecture reveals implementation constraints (weight, power, thermal) that feed back as new requirements or modified targets. This iteration is normal, but must be **documented**. Each iteration produces a revised SDRD version with change rationale.

Convergence test: when two consecutive iterations change fewer than 5% of requirements, the design is stable enough to baseline at PDR.

2.4 Environmental and Regulatory Constraints

Every system operates in an environment that imposes constraints. Systematically address:

- **Temperature:** operating range, storage range, thermal shock.
- **Moisture/IP:** splash (IPX4), rain (IPX5), submersion (IPX7/8).
- **Vibration/shock:** transport, operational, handling drop height.
- **EMC:** FCC Part 15 Class B (residential), IEC 61000 (EU).
- **Safety:** UL/CSA/CE marking, OSHA, university lab safety policies.
- **Materials:** RoHS compliance, California Prop 65, food-contact (FDA 21 CFR).

2.5 Trade Studies and Decision Making

2.5.1 The Trade Study Process

1. Define the decision and evaluation criteria (derived from requirements). Weight criteria (sum to 1.0).
2. Identify 3–5 viable candidates. No straw-men.
3. Rate each option against each criterion using data, not preference.
4. Score (raw $\times$ weight), sum, select. Check sensitivity: if ± 0.05 weight change flips the winner, investigate further.
5. Document everything: matrix, data sources, weight rationale, dissenting opinions.

2.5.2 Pugh Matrix vs. Weighted Decision Matrix

Pugh (screening): rate options as +, S, – against a datum. Fast. No weights. Eliminates clearly inferior options. **Weighted matrix (selection):** numerical ratings $\times$ weights. Rigorous, quantitative, traceable. Best practice: Pugh to screen $\rightarrow$ weighted matrix to select.

Weighted Decision Matrix — Systematic Trade Study Example								
Criteria	Weight	Option A Brushed DC		Option B Stepper		Option C BLDC + ESC		Notes
		Raw	Wtd	Raw	Wtd	Raw	Wtd	
Torque at speed	0.25	3	0.75	4	1.00	5	1.25	Must meet 0.5 Nm @ 200 RPM
Position accuracy	0.20	2	0.40	5	1.00	4	0.80	Open-loop vs closed-loop
Cost (motor+driver)	0.20	5	1.00	3	0.60	2	0.40	Budget constrained
Efficiency	0.15	3	0.45	2	0.30	5	0.75	Battery life critical
Control complexity	0.10	5	0.50	4	0.40	2	0.20	Team skill level
Size / weight	0.10	4	0.40	3	0.30	4	0.40	Envelope limited
TOTAL	1.00		3.50		3.60		3.80	WINNER
Rating scale: 1 (poor) to 5 (excellent). Weighted score = raw × weight. Decision is traceable, documented, and defensible at design review.								

Figure 2.3: Weighted decision matrix example: motor selection with six criteria, three options.

2.5.3 Worked Example: MCU Trade Study

Decision: Select an MCU platform for a battery-powered environmental sensor node.

Criterion (Weight)	ESP32	STM32L4	ATmega328P
Power consumption (0.25)	3 (0.75)	5 (1.25)	4 (1.00)
Analog channels (0.20)	5 (1.00)	4 (0.80)	4 (0.80)
Cost (0.20)	5 (1.00)	3 (0.60)	5 (1.00)
Flash/RAM (0.15)	5 (0.75)	5 (0.75)	2 (0.30)
Ecosystem (0.10)	5 (0.50)	3 (0.30)	5 (0.50)
Wi-Fi/BLE (0.10)	5 (0.50)	1 (0.10)	1 (0.10)
Total	4.50	3.80	3.70

Table 2.1: MCU trade study for battery-powered sensor node.

Sensitivity check: If “Power consumption” weight increases from 0.25 to 0.30 (taking 0.05 from Wi-Fi), STM32L4 total rises to 4.05 and ESP32 drops to 4.40; winner unchanged. Selection is robust.

Decision: ESP32. Despite higher power draw, the built-in Wi-Fi eliminates a separate module (\$8 saved, one fewer interface, one fewer failure mode). For battery life, use deep sleep aggressively (Section 12.4).

2.6 Requirements-to-Architecture Flow

The five stages produce specific deliverables: ConOps, requirements document, functional tree, trade study reports, and architecture document (block diagrams + BOM + ICDs). V&V results feed backward through the chain when failures occur.

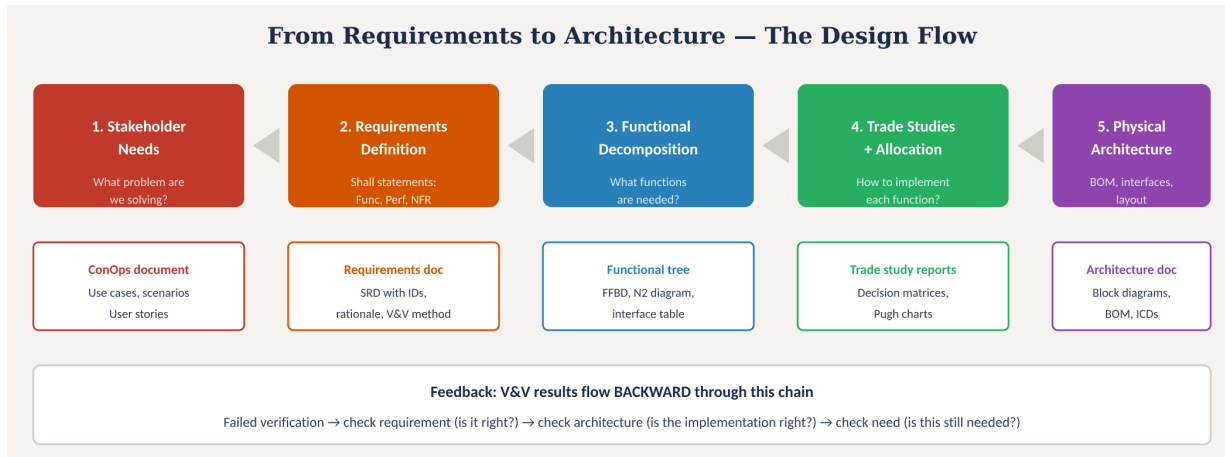


Figure 2.4: Five-stage design flow: stakeholder needs → requirements → functional decomposition → trade studies → architecture.

GS GreenShield: Functional Decomposition

Top-level function: Protect greenhouse crops from frost damage.

Decomposition:

- **Sense Environment** → Measure Air Temp, Measure Soil Temp, Detect Heater Status
- **Process Data** → Compare to Setpoint, Apply Hysteresis, Log Data
- **Control Actuator** → Activate/Deactivate Heater Relay
- **Communicate Status** → Send SMS Alert, Display Local Status, Receive Remote Commands
- **Provide Power** → Regulate Supply Voltage, Manage Battery, Protect Circuits

Each leaf function maps to a physical component. “Measure Air Temp” → DS18B20 sensor.

“Regulate Supply Voltage” → 5 V buck converter.

Worked Example. GreenShield Architecture: Grouping functions into subsystems yields: Sensor subsystem (DS18B20 + soil probe), Controller subsystem (ESP32), Actuator subsystem (relay + heater), Communications subsystem (SIM800L GSM module), and Power subsystem (12 V adapter + 5 V buck + 3.3 V LDO).

Industry SE Parallels

- **INCOSE SEH 4th ed.:** Sec. 4.2 (System Requirements Definition), Sec. 4.3 (Architecture Definition).
- **NASA SP-2016-6105 Rev 2:** Ch. 4.2 (Technical Requirements Definition), Ch. 4.3 (Logical Decomposition).
- **IEEE 1233:** Guide for developing system requirements specifications.

Artifact	Description	Forward Reference
SDRD	System requirements with traceability and V&V assignments	Ch. 3–6, PDR baseline
Functional Decomposition	Function tree from mission to leaf functions	Ch. 3 (concept generation)
Allocation Matrix	Functions mapped to physical modules	Ch. 5 (integration)
Trade Study Reports	Weighted decision matrices with sensitivity analysis	Ch. 3, PDR

2.7 Artifact Handoff: What Chapter 2 Produces

GS GreenShield: Functional Decomposition

Top-level function: Protect greenhouse crops from frost damage.

Decomposition:

- **Sense Environment** → Measure Air Temp, Measure Soil Temp, Detect Heater Status
- **Process Data** → Compare to Setpoint, Apply Hysteresis, Log Data
- **Control Actuator** → Activate Heater Relay, Deactivate Heater Relay
- **Communicate Status** → Send SMS Alert, Display Local Status, Receive Remote Commands
- **Provide Power** → Regulate Supply Voltage, Manage Battery, Protect Circuits

Each leaf maps to a physical component. “Measure Air Temp” → DS18B20 sensor. “Regulate Supply Voltage” → 5 V buck converter.

Industry SE Parallels

- **INCOSE SEH 4th ed.:** §4.2 System Requirements Definition, §4.3 Architecture Definition.
- **NASA SP-2016-6105 Rev 2:** Ch. 4.2 Technical Requirements, Ch. 4.3 Logical Decomposition.
- **IEEE 1233:** Guide for developing system requirements specifications.

2.8 Artifact Handoff: What Chapter 2 Produces

Artifact	Description	Used In
SDRD	System requirements with traceability and V&V assignments	Ch. 3–6, PDR
Functional Decomposition	Function tree from mission to leaf functions	Ch. 3
Allocation Matrix	Functions mapped to physical modules	Ch. 5
Trade Study Reports	Weighted decision matrices with sensitivity analysis	Ch. 3, PDR

2.9 Practice Questions

1. Rewrite this bad requirement as a well-formed shall-statement: “The system should be fast and accurate.”

2. A system requirement states “shall measure temperature to $\pm 0.5^\circ \text{C}$.” Decompose this into three subsystem requirements for the sensor, ADC, and display subsystems.
3. Your project has a requirement: “shall use an Arduino Mega 2560.” Is this a valid requirement? Why or why not? Rewrite it properly.
4. Explain the difference between forward and backward traceability. Why do you need both?
5. Create a weighted decision matrix comparing three MCU options for a battery-powered sensor node. Define at least five criteria with weights.
6. Why should you perform functional decomposition BEFORE selecting physical components?
7. A trade study produces scores of 3.80 vs. 3.75. What should you do before accepting the winner?

2.9.1 Answers

1. Two requirements: “SYS-REQ-001: The system shall respond to user input within 200 ms of activation.” “SYS-REQ-002: The system shall measure ambient temperature with accuracy of $\pm 0.5^\circ \text{C}$.”
2. (a) “The sensor shall output a signal proportional to temperature with linearity $\leq \pm 0.2^\circ \text{C}$ over -20 to $+60^\circ \text{C}$.” (b) “The ADC shall digitize the sensor signal with resolution $\leq 0.1^\circ \text{C}$ and total error $\leq \pm 0.15^\circ \text{C}$.” (c) “The display shall present the temperature value with update rate $\geq 1 \text{ Hz}$ and display resolution of 0.1°C .” Note: $\sqrt{0.2^2 + 0.15^2} + \text{display} < 0.5^\circ \text{C RSS}$.
3. Not valid; it specifies implementation (HOW) rather than need (WHAT). Rewrite: “SYS-REQ-XXX: The controller shall provide ≥ 12 analog input channels, ≥ 4 serial interfaces, and $\geq 256 \text{ KB}$ flash memory. Rationale: sensor count and communication needs.”
4. Forward: parent $\rightarrow$ children. Checks completeness, every requirement is decomposed. Backward: child $\rightarrow$ parent. Checks justification, every spec traces to a need. Without forward, requirements go unimplemented. Without backward, unnecessary work gets done.
5. Criteria might include: power consumption (0.25), analog channels (0.20), cost (0.20), flash/RAM (0.15), ecosystem/library support (0.10), package size (0.10). Options: STM32L4, ESP32, AT-mega328P.
6. Functions are stable across implementations. Decomposing functionally separates the problem space from the solution space, enabling objective trade studies. Jumping to hardware locks the design before alternatives are explored.
7. Sensitivity analysis: vary each weight by ± 0.05 and check if the winner changes. A 0.05 margin is very tight. Investigate the criteria where the options diverge most. Consider prototyping both finalists.

Chapter 3

Concept Generation & Early Design

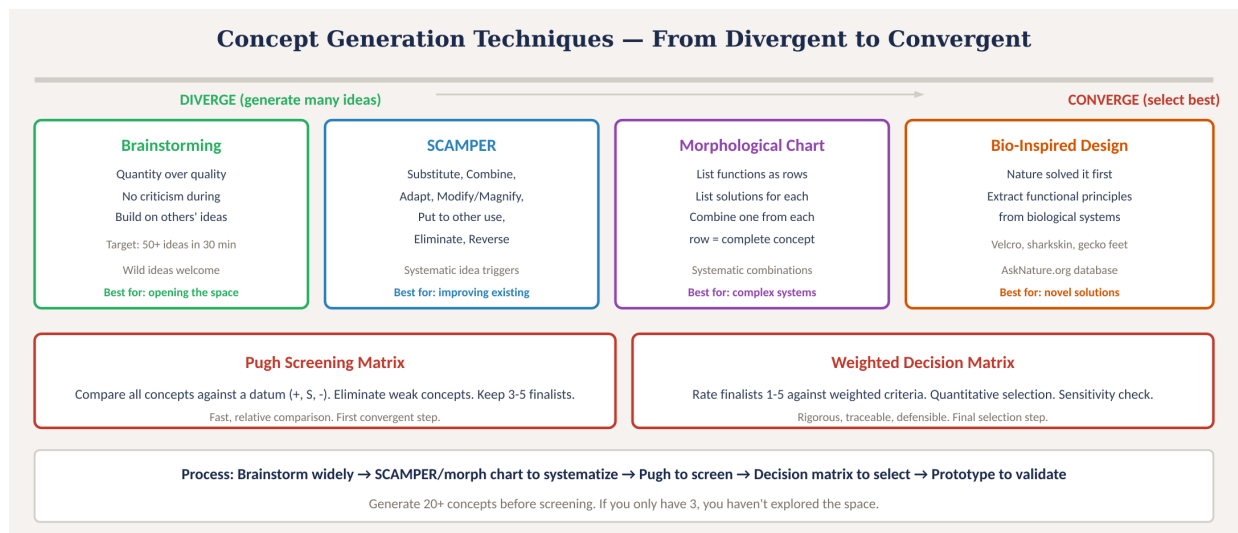


Figure 3.1: Concept generation techniques — divergent to convergent.

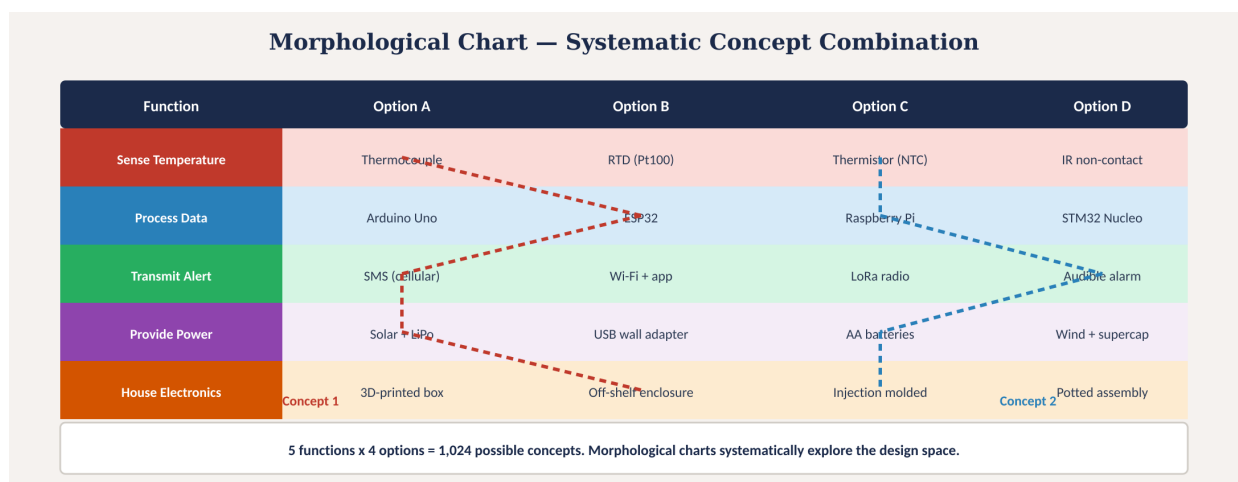


Figure 3.2: Morphological chart with systematic concept combinations.

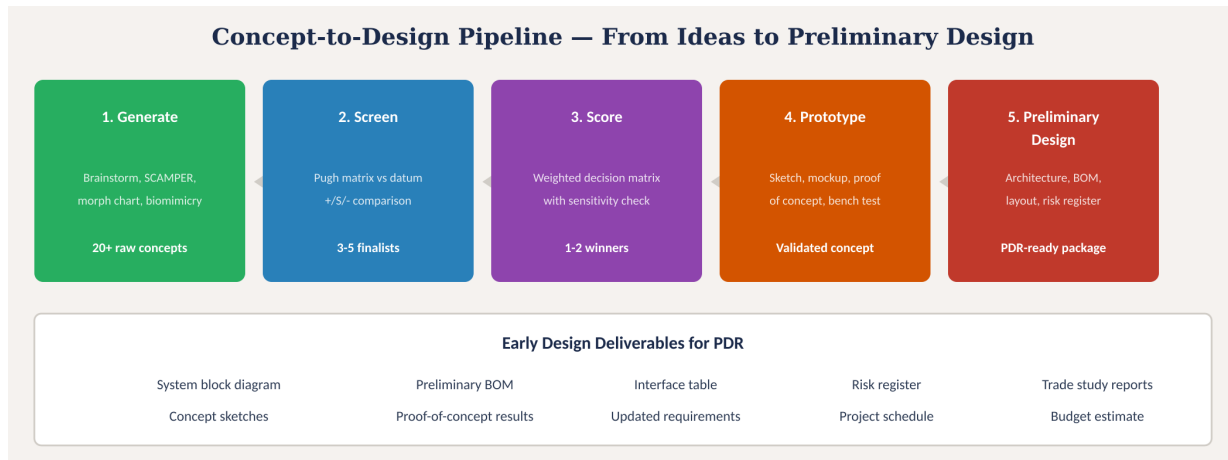


Figure 3.3: Concept-to-design pipeline with PDR deliverables.

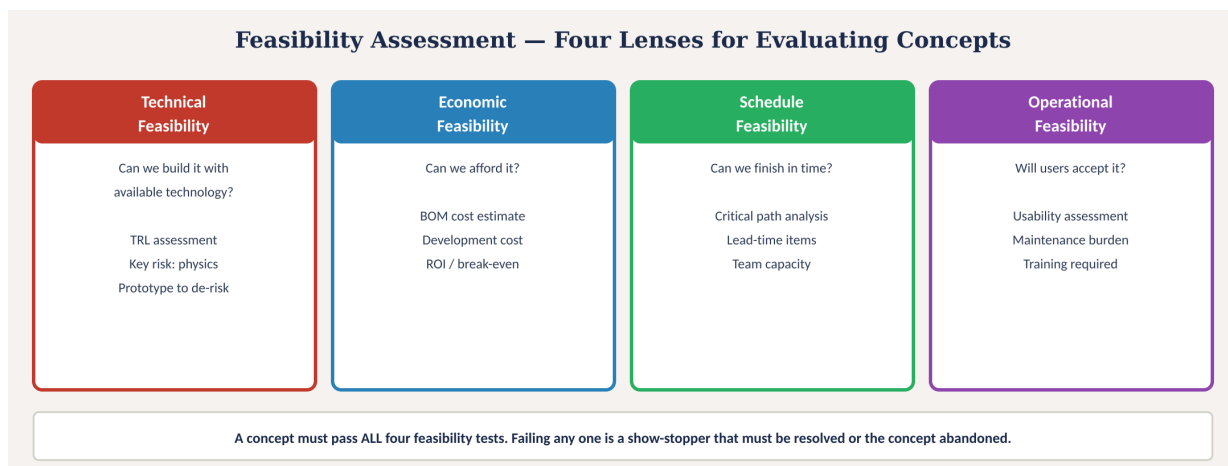


Figure 3.4: Four-lens feasibility assessment.

3.1 Reverse Engineering

Reverse engineering is a structured process of disassembling an existing product to understand how it works. It builds engineering intuition and directly informs concept generation.

3.1.1 The Reverse Engineering Process

1. **Document the intact product:** photograph from all angles, measure overall dimensions, weigh it, record all labels and markings.
2. **Disassemble systematically:** remove fasteners in order, photograph each step, bag and label hardware. Use the right tools; forcing components causes damage that obscures design intent.
3. **Measure and catalog:** calipers for every component. Record materials (magnet test for ferrous metals, density test for plastics). Identify standard parts (bearings, fasteners, seals) by markings.
4. **Create the BOM:** part number, description, quantity, material, source, estimated cost.
5. **Create an exploded view:** CAD or carefully arranged photograph with callout labels.
6. **Functional analysis:** map each component to a function. Which are structural? Functional? Cosmetic? Note any clever design solutions.

3.1.2 What Reverse Engineering Reveals

Assembly sequence (what goes in first must come out last. DFA insight). Material choices and why they were made. Manufacturing processes (injection molding witness lines, machining marks, weld joints). Standard vs. custom parts ratio. Design-for-cost decisions. Failure modes on used products (wear patterns).

3.1.3 Reverse Engineering Outputs

The analysis produces artifacts that feed directly into design:

- **Component inventory** with identified materials and manufacturing methods; informs DFM decisions (Injection Molding chapter).
- **Interface inventory** of all accessible mounting points, electrical connections, and mechanical features; basis for ICDs (Chapter 2).
- **Constraint list** documenting what your design cannot change; feeds requirements as non-functional constraints.
- **Design insights** about proven approaches, failure modes, and opportunities; informs concept generation and FMEA.

Non-Destructive Constraint

In MEGN 301, you must return the COTS product undamaged at the end of each class period. No drilling, cutting, gluing, or permanent modifications. Your add-on system must interface through existing features: existing screw holes, connectors, friction fits, clamps, or custom brackets. This mirrors real-world accessory development where your product must work with

an existing platform without voiding its warranty.

3.1.4 From Reverse Engineering to Concept Generation

The reverse engineering analysis directly feeds concept generation:

1. The interface inventory defines the physical envelope and connection options for your morphological chart.
2. The constraint list eliminates infeasible concepts before time is wasted developing them.
3. Design insights from the existing product provide proven approaches to build upon.
4. The component inventory reveals standard parts that you can reuse or specify.

3.2 Reverse Engineering

Before generating new concepts, understand how existing solutions work. Reverse engineering builds engineering intuition and directly informs concept generation.

3.2.1 The Reverse Engineering Process

1. **Document the as-received state:** photograph from all angles before disassembly. Record weight, dimensions, labels, model numbers.
2. **Systematic disassembly:** remove fasteners in order, photograph each step, bag and label all hardware. Use the right tools; forcing components obscures design intent.
3. **Identify components and materials:** measure with calipers (not rulers). Magnet test for ferrous metals. Identify standard parts (bearings, fasteners, seals) by markings.
4. **Analyze the architecture:** create a block diagram showing subsystem connections. Identify functional groups, signal flows, and power distribution.
5. **Assess design decisions:** Why did the original designers make each choice? What manufacturing processes were used (injection molding witness lines, machining marks, weld joints)?
6. **Create deliverables:** BOM, exploded view photograph or CAD, functional analysis report.

3.2.2 What Reverse Engineering Reveals

Assembly sequence (what goes in first comes out last; this drives DFA decisions). Material choices and manufacturing methods. Standard vs. custom parts ratio. Design-for-cost decisions. Wear patterns on used products that reveal failure modes.

3.3 Generating Concepts

3.3.1 Diverge Then Converge

The concept generation process has two distinct phases. **Divergent** thinking generates as many ideas as possible without evaluation — brainstorming, SCAMPER, morphological charts, and bio-inspired design. **Convergent** thinking systematically evaluates and selects — Pugh screening matrices and weighted decision matrices. Generate at least 20 concepts before screening any of them.

3.3.2 Brainstorming Rules

Defer judgment (no criticism during ideation). Go for quantity (target 50+ ideas in 30 minutes). Build on others' ideas ("yes, and..."). Encourage wild ideas (they contain seeds of insight). Sketch, don't just talk.

3.3.3 SCAMPER

Seven systematic triggers: Substitute, Combine, Adapt, Modify/Magnify, Put to other use, Eliminate, Reverse. Apply each to existing concepts or competitor products to generate variations.

3.3.4 Morphological Charts

List functions as rows (from functional decomposition), brainstorm 3–5 implementation options per function as columns. A complete concept = one option from each row. A 5×4 chart produces 1,024 combinations — systematically exploring the design space.

3.3.5 Bio-Inspired Design

Extract functional principles from biological systems. Velcro from burdock burrs. Sharkskin for drag reduction. Use AskNature.org to search by function. TRIZ provides 40 inventive principles for resolving design contradictions.

3.3.6 Stage 1: Pugh Screening Matrix

Rate each concept against a datum (baseline concept) as better (+), same (*S*), or worse (−). No weights, no numbers; just qualitative comparison. Sum +’s and −’s. Eliminate concepts with many −’s. The survivors proceed to the weighted matrix.

Rules: at least 3 concepts plus the datum. Criteria from requirements only. If two concepts score similarly, do not eliminate either.

3.3.7 Stage 2: Weighted Decision Matrix

Rate surviving concepts 1–5 against weighted criteria (weights sum to 1.0). Calculate weighted score per option. Check sensitivity: vary each weight ± 0.05 ; if the winner changes, the selection is fragile.

Common traps: Straw-man alternatives (including clearly inferior options to make the favorite “win”; every candidate must be legitimate). Anchoring on a favorite (assign ratings independently before group discussion). Criteria not from requirements (if a criterion does not trace to a requirement, remove it). Post-hoc weight adjustment (lock weights before scoring).

Stage 1: Pugh Screening — rate concepts as +/S/- against a datum. Fast, no weights. Eliminates clearly inferior options. **Stage 2: Weighted Decision Matrix** — rate finalists 1–5 against weighted criteria. Quantitative, traceable. Sensitivity check: if ± 0.05 weight change flips the winner, investigate further.

3.4 Risk Assessment

Risk management begins during concept selection, not after design. A **risk** is any event or condition that could prevent the system from meeting requirements. Every concept carries risks; the question is which are acceptable and which require mitigation.

3.4.1 The Risk Matrix

Risks are characterized by **likelihood** (1–5) and **consequence** (1–5). The product $L \times C$ gives the Risk Priority Number (RPN).

Score	Likelihood	Consequence
1	Very unlikely (<5%)	Negligible (cosmetic, workaround exists)
2	Unlikely (5–20%)	Minor (schedule slip <1 week, minor rework)
3	Possible (20–50%)	Moderate (schedule slip 1–2 weeks, partial re-design)
4	Likely (50–80%)	Major (missed milestone, significant redesign)
5	Almost certain (>80%)	Catastrophic (project failure, safety hazard)

Table 3.1: Risk likelihood and consequence scales.

Action thresholds: RPN 1–4 = monitor, 5–9 = mitigate, 10–15 = mitigate aggressively, 16–25 = redesign or eliminate. Maintain a risk register from PDR through FDR, updating status at each review.

3.4.2 Risk Mitigation Strategies

Four options (in order of preference): **Avoid** (change the design to eliminate the risk), **Mitigate** (reduce likelihood or consequence), **Transfer** (shift risk to a vendor, insurer, or partner), **Accept** (document and monitor; only for low-RPN items).

GS GreenShield: Risk Register at PDR

Risk	L	C	RPN	Mitigation
Solar panel insufficient in winter	4	4	16	Battery margin + Dec test
LoRa blocked by terrain	3	3	9	Site survey + repeater
Relay fails closed (heater on)	2	5	10	Thermal fuse backup
Thermocouple accuracy inadequate	3	3	9	In-situ calibration
MCU brown-out during Wi-Fi TX	4	3	12	470 μ F decoupling
Red-zone: solar panel (16) and MCU brown-out (12) require active mitigation before CDR.				

3.5 Failure Modes and Effects Analysis (FMEA)

FMEA systematically identifies potential failure modes, their effects, and their causes *during design*; before hardware exists.

3.5.1 Conducting a Design-Phase FMEA

For each component or function:

1. List every way it can fail (**failure modes**).
2. Determine the effect of each failure on the system (**effects**).
3. Identify root causes.
4. Rate Severity (S), Occurrence (O), and Detection (D) on 1–10 scales.
5. Calculate $RPN = S \times O \times D$. Higher RPN = higher priority for corrective action.
6. Design corrective actions for the top-5 RPN items.

Component	Failure Mode	Effect	S	O	D	RPN
Temp sensor	Reads high	Heater never activates; crop loss	8	3	4	96
Relay	Stuck closed	Heater runs continuously; fire risk	9	2	6	108
MCU	Brown-out reset	System offline during freeze event	7	4	5	140
GSM module	No signal	Alert not sent; operator unaware	5	3	3	45
Power supply	Fails open	Complete system loss	8	2	7	112

Table 3.2: Example design-phase FMEA for GreenShield frost protection system.

The MCU brown-out (RPN 140) is the highest-priority item; address it with decoupling capacitors, a voltage supervisor, and separate power routing (see Chapter 9).

3.5.2 Prototype Types in Detail

Proof of Concept (PoC): validates the riskiest technical assumption. Build the minimum hardware/software to answer one question: “Does the physics/algorithm/mechanism work?” Example: a breadboard circuit with just the sensor and MCU to verify temperature accuracy before designing the full system. Build time: 1–3 days. Quality: ugly is fine.

Form/Fit prototype: verifies physical compatibility. 3D-print the enclosure, mount real components, check cable routing and assembly sequence. No electronics need to function; this is a geometry check. Build time: 2–4 days. **Always print at 1:1 scale and test-fit real components.**

Looks-Like/Works-Like: a near-complete system that demonstrates both form and function. This is your CDR deliverable. It should be close to the final design but may use breadboard wiring or temporary mounting. Build time: 1–3 weeks.

Minimum Viable Product (MVP): the simplest version that delivers value to the stakeholder. It meets the MUST requirements but may defer SHOULD and COULD features. The MVP is what you demonstrate at FDR.

Build the Highest-Risk PoC First

Identify the single technical element most likely to fail. Build a PoC for that element before anything else. If it fails, you learn early and can pivot. If it works, confidence increases and the rest of the design proceeds on solid ground.

3.6 Prototyping and Feasibility

Four prototype types: Proof of Concept (does the physics work?), Form/Fit (does it fit the space?), Looks-Like/Works-Like (can the team build the full system?), and Minimum Viable Product (does the stakeholder accept it?). Build the PoC for the highest-risk element FIRST.

3.6.1 Feasibility Assessment

Every concept must pass four tests: **Technical** (can we build it?), **Economic** (can we afford it?), **Schedule** (can we finish in time?), **Operational** (will users accept it?). Failing any one is a show-stopper.

3.7 Design Documentation

Complete design documentation enables someone else to build, test, and maintain your system without your presence.

3.7.1 Bill of Materials (BOM)

Required columns: item number, part number, description, quantity, material, vendor, vendor part number, unit cost, lead time, status (ordered/received/installed). Organize hierarchically: Level 0 = system, Level 1 = subsystems, Level 2 = assemblies, Level 3 = parts. **Every part must appear in the BOM.**

3.7.2 Wiring Diagrams and Schematics

Schematics show electrical connections using standard symbols; the “source code” of hardware. **Wiring diagrams** show physical routing with connector pin assignments and wire colors/gauges. Both are required: schematics for design intent, wiring diagrams for assembly.

3.7.3 Software Architecture and Flowcharts

Document firmware with: (1) state-machine diagram showing all states and transitions, (2) data flow diagram (sensor → processing → actuator), (3) pin mapping table (GPIO → function → component), (4) communication protocol definitions (message formats, baud rates, addresses).

3.7.4 CAD Models and Engineering Drawings

CAD models define geometry. Engineering drawings communicate that geometry with dimensions, tolerances, surface finishes, and notes. Every custom part requires a drawing. Standard parts reference vendor part numbers; no drawing needed.

Plan B Components

For every critical component, identify a Plan B; an alternative from a different vendor meeting the same requirements. Lead times slip, parts go out of stock, and vendors make mistakes. A Plan B list saves weeks of panic.

Free Electronics Supplies

Before ordering, check the MEGN 301 electronics inventory in the ME shop. Common components (resistors, capacitors, LEDs, wire, headers, breadboards) are available at no cost. Check the shared inventory spreadsheet first.

The Spring Break Shipping Trap

Orders placed the week before spring break often arrive *during* break when no one is on campus. Plan so critical parts arrive at least one week before any closure. International suppliers (AliExpress, LCSC): allow 3–4 weeks. McMaster-Carr ships next-day to Golden.

GS GreenShield: Morphological Chart

Function	Option A	Option B	Option C
Sense Temp	DS18B20 digital	BME280 (T/H/P)	Thermocouple + amp
Control Logic	ESP32 (Wi-Fi)	Arduino Nano	RPi Zero
Heat Actuator	Ceramic heater + relay	Heat lamp + relay	Heated water loop
Communication	SMS via SIM800L	Wi-Fi + cloud	LoRa to base station
Power Source	120 VAC wall adapter	12V solar + battery	PoE

Concept 1 (Grid-Connected): DS18B20 + ESP32 + Ceramic heater + Wi-Fi + Wall adapter. Lowest cost, simplest, requires AC power.

Concept 2 (Off-Grid Solar): BME280 + ESP32 + Ceramic heater + LoRa + Solar+battery. Grid-independent, higher cost, battery sizing critical.

Concept 3 (Industrial): Thermocouple + RPi + Heated water loop + SMS + PoE. Highest capability and cost.

GS GreenShield: Risk Assessment for Concept 2 (Off-Grid)

Technical risks: (1) Solar panel sizing for Colorado winters, December insolation is 3.5 peak sun hours vs. 6.5 in July. Battery must sustain 3 consecutive cloudy days. (2) LoRa range through polycarbonate glazing, 3–6 dB attenuation at 915 MHz.

Schedule risk: Solar charge controller + BMS integration adds 2 weeks. If the team is behind at CDR, this is a schedule-buster.

Economic risk: Solar (\$25) + charge controller (\$15) + LiPo (\$20) + LoRa (\$12 ea) = \$84 for power/comms vs. \$8 for Concept 1.

Verdict: High-risk for 16-week schedule. Recommend Concept 1 with upgrade path.

Industry SE Parallels

- **INCOSSE SEH 4th ed.:** Sec. 4.3 (Architecture Definition), Sec. 4.4 (Design Definition).
- **NASA SP-2016-6105 Rev 2:** Ch. 4.4 (Design Solution Definition), Ch. 6.6 (Decision Analysis).
- **IEEE 1220:** Standard for systems engineering process management.

3.8 Artifact Handoff: What Chapter 3 Produces

Artifact	Description	Forward Reference
Morphological Chart	Functions × implementation options	PDR documentation
Trade Study / Decision Matrix	Weighted evaluation of top concepts	PDR deliverable
Selected Concept Package	Block diagram, preliminary BOM, risk assessment	Ch. 4, CDR
Design Documentation	BOM, schematics, CAD, software architecture	Ch. 5 (integration), FDR

3.9 Practice Questions

1. Create a morphological chart for a campus recycling station with at least 4 functions and 3 options each. Trace two complete concept paths.
2. Your team brainstormed only 5 concepts in 20 minutes. What went wrong? What facilitation techniques would you use to increase output?
3. A trade study produces scores of 3.82 vs 3.79 between two motor options. What do you do?
4. Your proof-of-concept test shows the sensor accuracy is 1.2°C against a 0.5°C requirement. What are your options?
5. You are reverse-engineering a consumer drone. During disassembly you find a small metal weight glued inside one motor arm. What is its likely purpose, and what does this tell you about the manufacturer's design process?
6. Your team's BOM lists "motor" with no part number, no vendor, and no specifications. Why is this a problem, and what minimum information must every BOM entry contain?

3.9.1 Answers

1. Example morphological chart for campus recycling station:

Function	Option A	Option B	Option C
Identify material	RFID tag	Camera + ML	Weight sensor
Sort mechanism	Rotating drum	Conveyor + diverter	Gravity chute
User interface	Touchscreen	Button panel	Voice commands
Compaction	Hydraulic press	Screw auger	Gravity (none)

Concept 1: RFID + Conveyor + Touchscreen + Hydraulic = high accuracy, high cost, complex.

Concept 2: Camera + Gravity chute + Button + None = moderate accuracy, low cost, simple.

2. Too few concepts: (a) Criticism crept into brainstorming (evaluation during ideation kills creativity). (b) Team anchored on a favorite concept. (c) No systematic technique used. Fixes: enforce "no criticism" rule with a timer, use SCAMPER to force variations, require sketches not just words, set a minimum of 20 ideas before stopping.
3. A 0.03 margin is within noise. Sensitivity analysis: vary each weight by ± 0.05 . If the winner changes, the selection is not robust. Options: prototype both finalists, gather better data on the most differentiating criterion, or choose based on a secondary factor (availability, team familiarity).
4. Options: (a) Improve the sensor (better calibration, thermal isolation, averaging algorithm). (b) Change the sensor (switch to RTD from thermocouple, higher accuracy, higher cost). (c) Relax the requirement (stakeholder may accept $\pm 1.0^{\circ}\text{C}$, verify with stakeholder before changing). (d) Add a correction factor (calibrate against a reference and apply polynomial correction in firmware). Always investigate the requirement validity before redesigning hardware.

5. The weight is likely for vibration balancing. The motor arm, with motors and propellers, has a center of gravity that is not perfectly symmetric. The manufacturer added a counterweight to tune the balance and reduce vibration. This tells you the manufacturer performs dynamic balancing as a post-assembly step; a quality control measure that indicates vibration was a problem in earlier designs.
6. “Motor” without specifications is useless for procurement, testing, or replacement. A vendor cannot fill this order. Minimum BOM entry: manufacturer (e.g., Pololu), part number (e.g., 4843), description (“12V 100RPM DC gearmotor”), key specs (voltage, no-load speed, stall torque, rated current), quantity, unit cost, vendor, and lead time.

Chapter 4

Verification, Validation & Test Plans

4.1 Verification vs. Validation

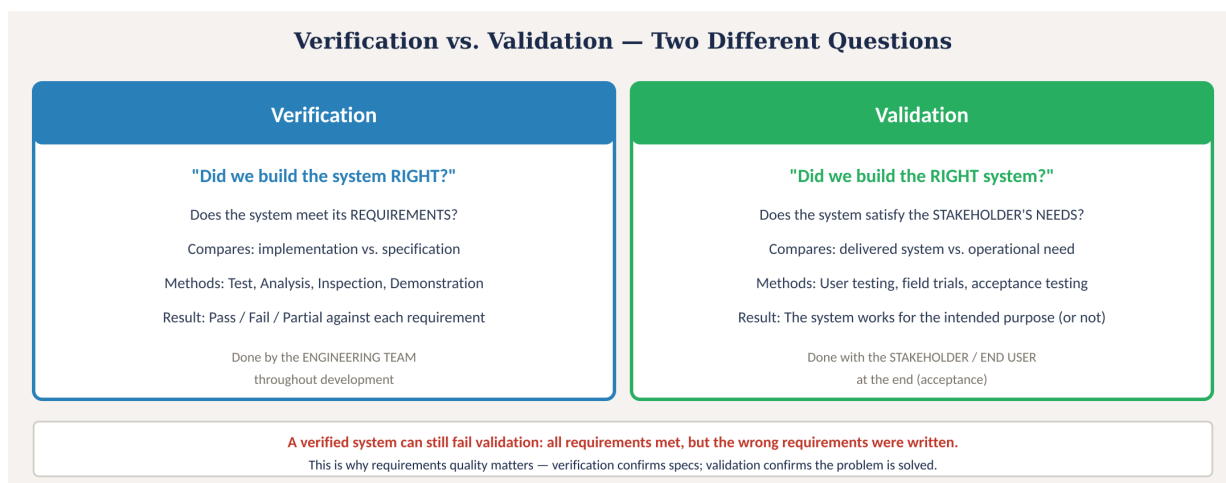


Figure 4.1: Verification (meets requirements) vs. validation (meets needs); two different questions.

Verification asks: "Did we build the system RIGHT?" It compares the implemented system against the written specification. Done by the engineering team using Test, Analysis, Inspection, or Demonstration (TAID). Each requirement receives an individual PASS/FAIL verdict.

Validation asks: "Did we build the RIGHT system?" It compares the delivered system against the stakeholder's actual operational need. Done with the stakeholder through user testing, field trials, or acceptance testing.

A Verified System Can Fail Validation

All requirements met perfectly, but the wrong requirements were written. This is why requirements quality matters; verification confirms specs; validation confirms the problem is actually solved. You need both.

4.2 Four Verification Methods: TAID

- **Test (T, \$ ~ \$60% of reqs):** Stimulate the system and measure the response against quantitative criteria. Most rigorous. Preferred whenever feasible.

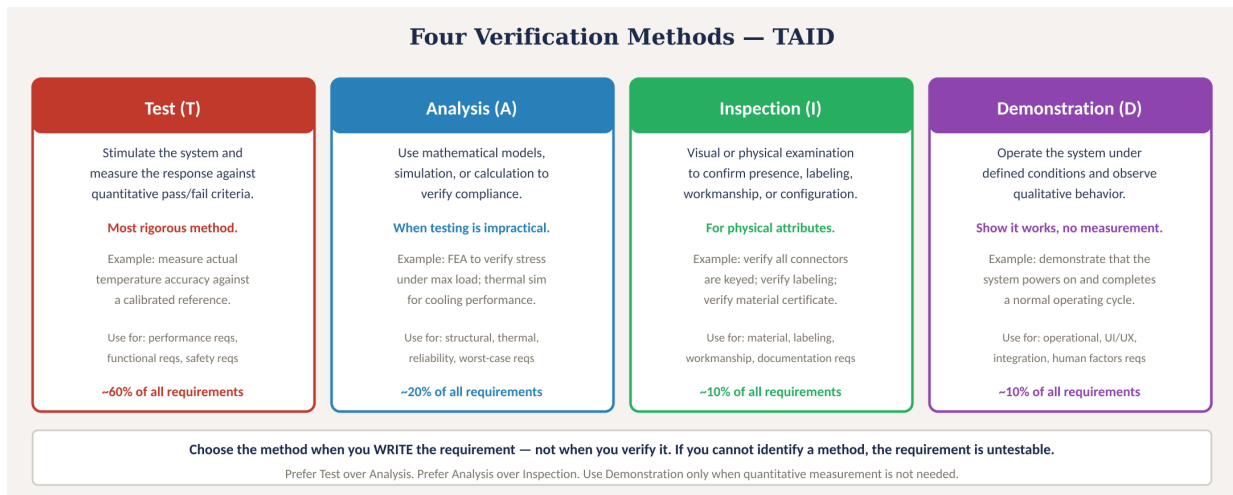


Figure 4.2: Test, Analysis, Inspection, and Demonstration with usage percentages and selection guidance.

- **Analysis (A, \$ ~ \$20%)**: Mathematical models, simulation, or calculation. Used when testing is impractical (FEA, thermal sim, worst-case analysis). Must document assumptions.
- **Inspection (I, \$ ~ \$10%)**: Visual/physical examination; presence, labeling, workmanship, material certificates.
- **Demonstration (D, \$ ~ \$10%)**: Operate under defined conditions and observe qualitative behavior. No measurement. Used when quantitative testing is not needed.

Assign the Method When You Write the Requirement

If you cannot identify a verification method, the requirement is untestable and must be rewritten. The method check is the single best quality gate for requirements.

4.3 Writing Test Plans

4.3.1 What Makes a Good Test Plan

The gold standard: another engineer with no project knowledge can reproduce your results exactly. Five qualities: **Traceable** (maps to requirement IDs), **Reproducible** (specific enough for independent replication), **Binary** (quantitative pass/fail, no judgment calls), **Complete** (all sections filled), **Executed as written** (no ad-hoc deviations).

4.3.2 Test Procedure Structure

Every test procedure contains: (1) Purpose; requirement IDs verified; (2) Test setup; equipment with model numbers and calibration; (3) Preconditions; system state before starting; (4) Procedure steps; numbered, single-action, imperative voice, with expected results; (5) Pass/fail criteria; quantitative thresholds from requirements; (6) Data recording; format, resolution, blank table; (7) Safety precautions; (8) Results and verdict; (9) Signatures; (10) Attachments.

Anatomy of a Test Plan — What Every Test Procedure Must Contain	
Test Procedure Template — TP-XXX: [Test Name]	
1. Purpose	Which requirement(s) does this test verify? List requirement IDs. Example: "Verifies SYS-REQ-003, SYS-REQ-004."
2. Test Setup	Equipment, instruments, fixtures, software, environmental conditions, and calibration status. Another engineer must reproduce your setup.
3. Preconditions	What must be true BEFORE the test starts? System powered, warmed up, calibrated, firmware version, ambient conditions verified.
4. Procedure Steps	Numbered, unambiguous steps. Each step is a single action + expected result. "Step 3: Apply 25.0 C reference. Record DUT reading. Expected: 24.5-25.5 C."
5. Pass/Fail Criteria	Quantitative thresholds with units and tolerances. Derived directly from the requirement. No judgment calls — the criteria are binary.
6. Data Recording	What data to capture, in what format, at what resolution. Include a data table template. Raw data must be retained for traceability.
7. Safety Precautions	Hazards, PPE requirements, emergency procedures. Required for any test involving high voltage, pressure, heat, or moving parts.
8. Results and Verdict	PASS: all criteria met. FAIL: any criterion not met (document failure and root cause). PARTIAL PASS: some criteria met, action required.
9. Signatures	Tester, witness, date, equipment serial numbers, environmental conditions at time of test. Makes the record legally defensible.
10. Attachments	Photos, oscilloscope captures, data files, calibration certificates. Raw evidence supporting the verdict.

Figure 4.3: Complete test procedure template with all 10 required sections.

4.3.3 Writing Good Procedure Steps

Each step is a single action with an expected result. Use specific values ("Set reference bath to $25.0 \pm 0.1^\circ \text{C}$ "), not vague instructions ("check temperature"). Include wait times for stabilization. Imperative voice: "Set..." "Record..." "Verify..."

4.4 Handling Results

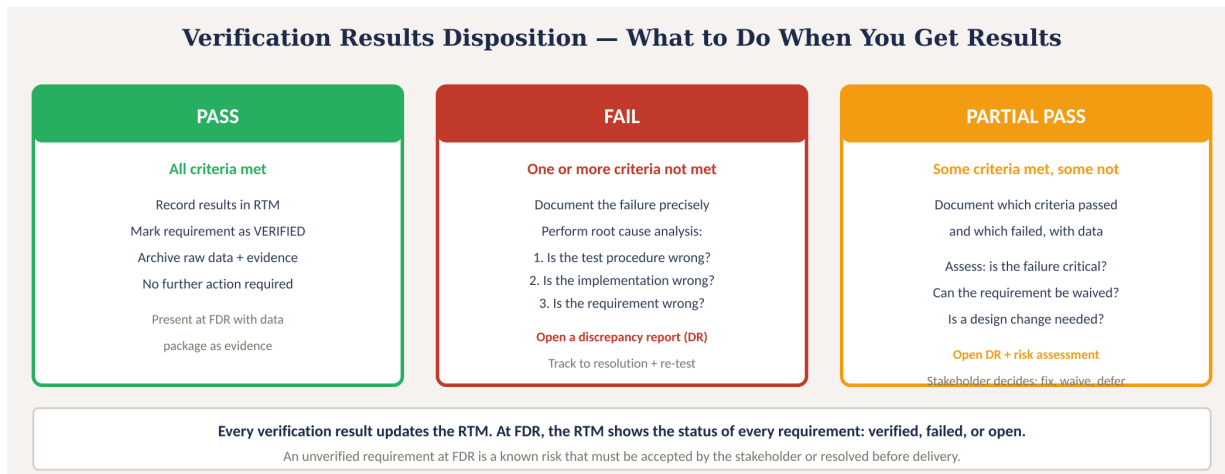


Figure 4.4: Three verification outcomes: PASS, FAIL, PARTIAL PASS with disposition workflows.

4.4.1 PASS

All criteria met. Record results in RTM, mark requirement VERIFIED, archive raw data. Include actual measurement values, not just "PASS."

4.4.2 FAIL

One or more criteria not met. Root cause analysis in order: (1) Is the test procedure wrong? (2) Is the implementation wrong? (3) Is the requirement wrong? Open a Discrepancy Report (DR), assign corrective action, track to resolution, re-test after correction.

4.4.3 PARTIAL PASS

Some criteria met, some not. Document which passed and which failed with data. Three resolution paths: **Fix** (redesign + re-test), **Waive** (stakeholder formally accepts the deviation with documented risk), or **Defer** (postpone with a plan and deadline). A waiver transfers risk to the stakeholder.

4.4.4 Root Cause Analysis

Three possible root causes, investigated in order: (1) the test is wrong (procedure error, calibration issue); (2) the implementation is wrong (most common); (3) the requirement is wrong (overly conservative or based on incorrect assumptions). Changing a requirement requires formal change control. Never silently.

4.5 The VCRM

The Verification Cross-Reference Matrix links every requirement to its method, procedure, and status. Maintain from day one. At FDR, it tells the complete story. Example columns: Req ID, Text, Type, Method, Test Proc ID, Status, Notes.

4.5.1 VCRM Evolution Over Time

The VCRM is not a static document; it evolves at each project milestone:

- **At PDR:** all requirements listed, methods assigned, status = “Planned” for all.
- **At CDR:** some subsystem-level tests executed, status updated. Analysis-method requirements may show results from design calculations.
- **At FDR:** all requirements have a final status. The VCRM is the single-page executive summary of your entire V&V campaign.

4.5.2 End-to-End Traceability Example

Stakeholder need: “I need to know when the greenhouse is too cold.” → **System requirement:** SYS-REQ-003: “The system shall send an SMS alert within 60 s of heater activation.” → **Verification method:** Test. → **Test procedure:** TP-003: Trigger heater activation, start stopwatch, verify SMS received within 60 s, record actual time for 3 trials. → **Result:** Trial 1: 12 s, Trial 2: 14 s, Trial 3: 11 s. Mean = 12.3 s. Verdict: PASS (12.3 < 60). → **VCRM entry:** SYS-REQ-003 | Test | TP-003 | PASS | 12.3 s mean.

This chain, need → requirement → method → procedure → result → VCRM, is what “requirements-driven design” means in practice.

4.6 V&V and the V-Model

The V-Model maps development phases (left side) to verification phases (right side): stakeholder needs $\leftrightarrow$ validation, system requirements $\leftrightarrow$ system testing, subsystem requirements $\leftrightarrow$ integration testing, component specs $\leftrightarrow$ unit testing. Implementation is at the bottom. Test planning happens on the LEFT side; design tests while writing requirements.

4.7 Statistical Confidence and Test Rigor

4.7.1 Developmental Testing vs. Production Qualification

How many times must you run a test? For **developmental testing** (MEGN 301): 3 repetitions minimum per test point. Report mean and range. For **production qualification** (industry): sample sizes from statistical requirements; typically $n \geq 30$ for parametric statistics, or MIL-STD-1916 sampling plans.

4.7.2 Practical Implications for Student Projects

Run each test 3 times, report all three values plus the mean, flag any result where the spread exceeds 20% of the mean. Report measurement uncertainty: if your instrument resolution is 0.1°C and your requirement is $\pm 0.5^\circ\text{C}$, you have adequate measurement capability. If your instrument resolution is 0.5°C , you cannot meaningfully verify a $\pm 0.5^\circ\text{C}$ requirement.

4.7.3 The Test Environment Matters

Test conditions that differ from operational conditions produce invalid results. Document and control: ambient temperature, humidity, power supply source (bench supply vs. battery vs. wall adapter; each behaves differently), electromagnetic environment (lab bench vs. field), and mechanical mounting (loose on bench vs. installed in enclosure). Any deviation from the intended operating environment must be noted in the test report with an assessment of impact.

4.7.4 Measurement Uncertainty

Every measurement has uncertainty. Your instrument's resolution, accuracy, and calibration status all contribute. Rule of thumb: the measurement system should be $\geq 4\times$ more accurate than the requirement tolerance. If your requirement is $\pm 0.5^\circ\text{C}$, your measurement system must be accurate to $\pm 0.125^\circ\text{C}$ or better. Document measurement uncertainty in every test report.

4.8 Edge Case and Boundary Testing

Requirements specify ranges ("0 to 50°C ," "0 to 100% RH"). Test at: the **nominal** operating point, **both boundary values**, and at least one point **outside** the boundary to verify graceful degradation (not catastrophic failure).

The Room-Temperature Trap

Testing only at room temperature ($\sim 22^\circ\text{C}$) is the most common V&V shortcut. If your requirement says $0\text{--}50^\circ\text{C}$, you *must* test at 0°C and 50°C . Use the ME department environ-

mental chamber or an ice bath + heat gun. A system that works at 22 °C but fails at 0 °C does not meet the requirement.

4.9 Minimum Viable V&V for MEGN 301

The minimum V&V package for student projects:

1. A VCRM with every requirement assigned a method and status.
2. A written test procedure (all 10 sections) for every T-method requirement.
3. Test results with actual measured data (not just “pass”).
4. For FAIL or PARTIAL PASS: root cause analysis and corrective action.
5. Peer review: another team executes at least one procedure. If they cannot reproduce results, the procedure is inadequate.

4.10 External Team Testing

Having another team execute your test procedures verifies reproducibility and uncovers ambiguities. **Preparing:** provide the procedure, equipment list, and safety briefing. Do NOT explain verbally; the document must stand alone. Observe without intervening (unless safety). Collect written feedback on every unclear step.

GS GreenShield: VCRM Fragment at Three Milestones

Req ID	Requirement	Method	At PDR	At CDR
SYS-REQ-001	Temp meas. at ≥ 1 Hz	T	Planned	In Progress
SYS-REQ-002	Heater at $\leq$ setpoint	T	Planned	PASS
SYS-REQ-003	SMS within 60 s	T	Planned	Planned
SYS-REQ-004	Autonomous mode on failure	D	Planned	Planned
SYS-REQ-005	Accuracy ± 0.5 °C	T	Planned	PARTIAL

At **FDR:** all rows should show PASS, FAIL with Discrepancy Report, or waived with stakeholder approval. SYS-REQ-005 PARTIAL PASS (0.6 °C error at 0 °C boundary) requires resolution.

Industry SE Parallels

- **INCOSE SEH 4th ed.:** Sec. 4.10 (Verification), Sec. 4.11 (Validation).
- **NASA SP-2016-6105 Rev 2:** Ch. 5.3 (Verification), Ch. 5.4 (Validation).
- **IEEE 1012:** Standard for System, Software, and Hardware V&V.

4.11 Artifact Handoff: What Chapter 4 Produces

Artifact	Description	Forward Reference
VCRM	Cross-reference matrix linking reqs to test status	CDR, FDR deliverables
Test Procedures	Reproducible procedures for every T-method requirement	Ch. 5 (integration test)
Test Reports	Results with data, verdict, and any DRs	FDR documentation

4.12 Statistical Confidence and Test Rigor

How many times must you run a test? For **developmental testing** (MEGN 301): 3 repetitions minimum per test point. Report mean and range. For **production qualification**: sample sizes from statistical requirements ($n \geq 30$ for parametric statistics, or MIL-STD-1916 sampling plans).

For MEGN 301: run each test 3 times, report all three values plus the mean, flag any result where spread exceeds 20% of mean.

4.13 Edge Case and Boundary Testing

Requirements specify ranges (“0 to 50 °C”, “0 to 100% RH”). Test at: the **nominal** operating point, **both boundary values**, and at least one point **outside** the boundary to verify graceful degradation.

The Room-Temperature Trap

Testing only at 22 °C is the most common V&V shortcut in student projects. If your requirement says 0–50 °C, you must test at 0 °C and 50 °C. Use the ME department environmental chamber or an ice bath + heat gun.

4.14 Minimum Viable V&V for MEGN 301

The minimum V&V package:

1. A VCRM with every requirement assigned a method and status.
2. Written test procedures (all 10 sections) for every T-method requirement.
3. Test results with actual measured data (not just “pass”).
4. For FAIL or PARTIAL PASS: root cause analysis and corrective action plan.
5. Peer review: another team executes at least one of your procedures.

4.15 External Team Testing

Having another team execute your test procedures verifies reproducibility and uncovers ambiguities the authoring team is blind to.

Preparation: (1) Provide the procedure, equipment list, and safety briefing. (2) Do NOT explain verbally; the document must stand alone. (3) Observe without intervening (unless safety). (4) Collect written feedback on unclear steps.

GS GreenShield: VCRM at Three Milestones

Req ID	Requirement	Method	CDR	FDR
SYS-001	Temp at ≥ 1 Hz	T	In Progress	PASS
SYS-002	Heater at $\leq$ setpoint	T	PASS	PASS
SYS-003	SMS within 60 s	T	Planned	PASS
SYS-004	Autonomous on comm fail	D	Planned	PASS
SYS-005	Accuracy ± 0.5 °C	T	PARTIAL	PASS

Industry SE Parallels

- **INCOSE SEH 4th ed.:** §4.10 Verification, §4.11 Validation.
- **NASA SP-2016-6105 Rev 2:** Ch. 5.3 Verification, Ch. 5.4 Validation.
- **IEEE 1012:** System, Software, and Hardware V&V.

4.16 Artifact Handoff: What Chapter 4 Produces

Artifact	Description	Used In
VCRM	Reqs linked to test status	CDR, FDR
Test Procedures	Reproducible procedures for T-method reqs	Ch. 5
Test Reports	Results with data, verdict, and DRs	FDR

4.17 Practice Questions

1. Explain how a system can pass all verification tests but fail validation. Give a concrete example.
2. Classify each requirement by verification method (T/A/I/D): (a) “Shall withstand 50 N load without permanent deformation.” (b) “Shall display the company logo on the front panel.” (c) “Shall complete a power-on sequence within 10 seconds.” (d) “Shall have a thermal margin of $\geq 20^\circ \text{C}$ at max power under worst-case ambient.”
3. Write a complete test procedure (all 9 sections) for verifying: “SYS-REQ-007: The system shall measure ambient temperature with accuracy of $\pm 0.5^\circ \text{C}$ over 0 to 50°C .”
4. A test result shows 0.6°C error against a requirement of $\pm 0.5^\circ \text{C}$. What is the verdict? Describe the three possible root causes and the investigation sequence.
5. Your VCRM shows 18 PASS, 2 FAIL, and 1 PARTIAL PASS out of 21 requirements at FDR. What do you present to the stakeholder?

4.17.1 Answers

1. A greenhouse monitoring system meets all temperature accuracy, data rate, and communication requirements (verified). But the stakeholder discovers the display is unreadable in direct sunlight; a need that was never captured as a requirement. The system was built right (verification), but it was not the right system (validation failed). This is a requirements-capture failure.
2. (a) Test; apply 50 N load and measure deformation. (b) Inspection; visually confirm logo presence and placement. (c) Demonstration; power on and time the sequence. (d) Analysis; thermal simulation under worst-case conditions.
3. Purpose: verifies SYS-REQ-007. Setup: NIST-traceable temperature bath ($\pm 0.05^\circ \text{C}$), calibrated reference thermometer, DUT powered and warmed 30 min. Preconditions: DUT firmware v2.1, ambient $23 \pm 3^\circ \text{C}$. Steps: (1) Set bath to 0.0°C , wait 10 min, record DUT and reference. (2) Repeat at 10, 20, 30, 40, 50°C . Pass/fail: $|\text{DUT} - \text{reference}| \leq 0.5^\circ \text{C}$ at

every point. Data: table with columns for setpoint, reference, DUT, error. Safety: bath uses water, standard lab PPE. Signatures: tester, witness, date.

4. Verdict: FAIL ($0.6 > 0.5$). Investigation: (1) Check test: was the reference bath stable? Was the reference thermometer calibrated? Repeat with verified equipment. (2) Check implementation: sensor calibration coefficients, ADC linearity, thermal coupling. (3) Check requirement: was $\pm 0.5^\circ\text{C}$ justified? If the application can tolerate $\pm 1.0^\circ\text{C}$, a requirements change (with stakeholder approval) may be appropriate. Open a DR regardless.
5. Present the VCRM showing all 21 requirements with status. For the 18 PASS items, show evidence summaries. For the 2 FAIL items, present root cause analysis, corrective actions, and re-test plan with timeline. For the PARTIAL PASS, present which criteria passed, which failed, risk assessment, and recommendation (fix, waive, or defer). The stakeholder decides on the partial pass and accepts or rejects the open items.

4.17.2 Answers (continued)

6. **External testing value:** When your team writes and executes its own test procedures, you unconsciously fill in gaps with tacit knowledge (“oh, you need to wait 30 seconds after power-on before reading the sensor”, but the procedure does not say that). An external team has no tacit knowledge. Every ambiguity, missing step, and unclear criterion is exposed. The external team’s feedback is a direct measure of your documentation quality.
7. **Boundary testing strategy:** For a requirement “ $0\text{--}50^\circ\text{C} \pm 0.5^\circ\text{C}$ ”: test at 0°C (boundary), 25°C (nominal), and 50°C (boundary). Also test at -5°C and 55°C to verify graceful degradation (system should report out-of-range, not crash or give false readings). For humidity requirement “ $0\text{--}100\%$ RH”: test at 0% (desiccant chamber), 50% (ambient), and 95% (humidity chamber; 100% causes condensation that is a separate failure mode). Report all data points, not just pass/fail.

Chapter 5

Subsystem Integration

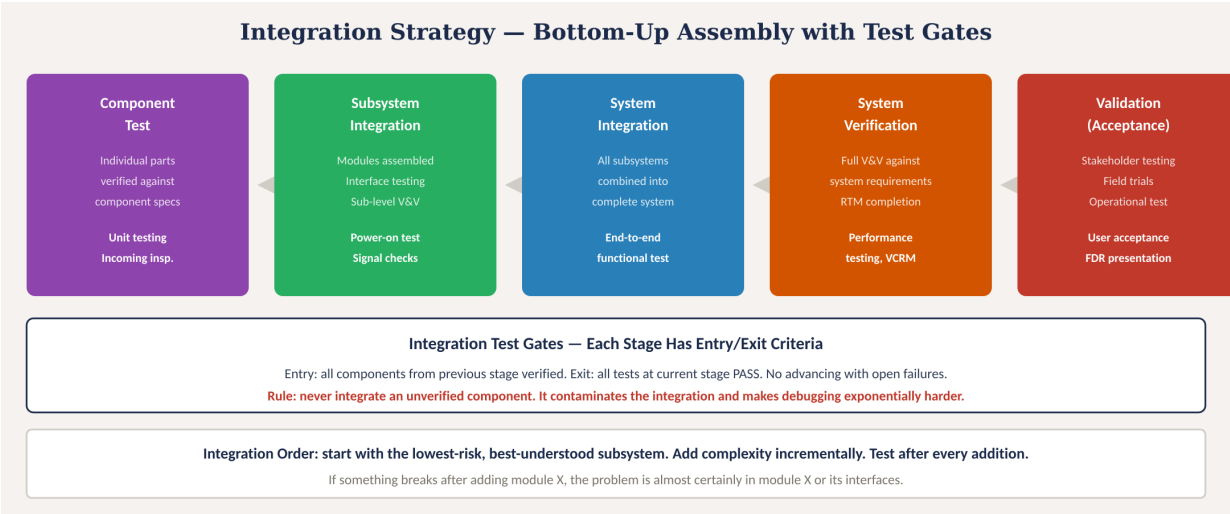


Figure 5.1: Bottom-up integration with test gates.

Interface Management — Where Integration Problems Live

Interface Type	What Crosses	What to Specify	Common Failures
Electrical	Power, signals, data	Voltage levels, protocols, pin assignments, grounding	Wrong voltage, missing ground, protocol mismatch
Mechanical	Force, torque, vibration	Mounting holes, tolerances, alignment, fasteners	Misaligned holes, interference fits, wrong thread
Thermal	Heat flow	Thermal path, contact resistance, airflow, max temp	Hot spots, insufficient cooling, thermal runaway
Software	Data, commands, status	API, message format, timing, error handling	Mismatched data types, timing race conditions
Fluid	Liquid, gas, vacuum	Fitting type, pressure rating, flow direction, seals	Wrong fitting standard, leak at connection
Human	Information, control	Display layout, control placement, labels, ergonomics	Confusing UI, unlabeled controls, poor visibility

80% of integration failures occur at interfaces. Define every interface in an ICD BEFORE building. Test interfaces BEFORE system integration.

Figure 5.2: Interface types and common failure modes.

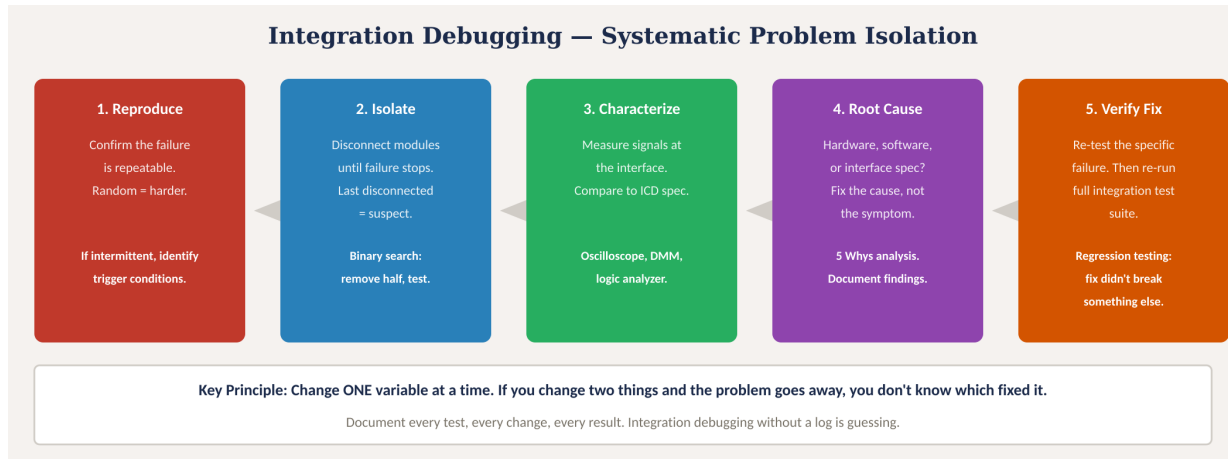


Figure 5.3: Five-step systematic debugging process.

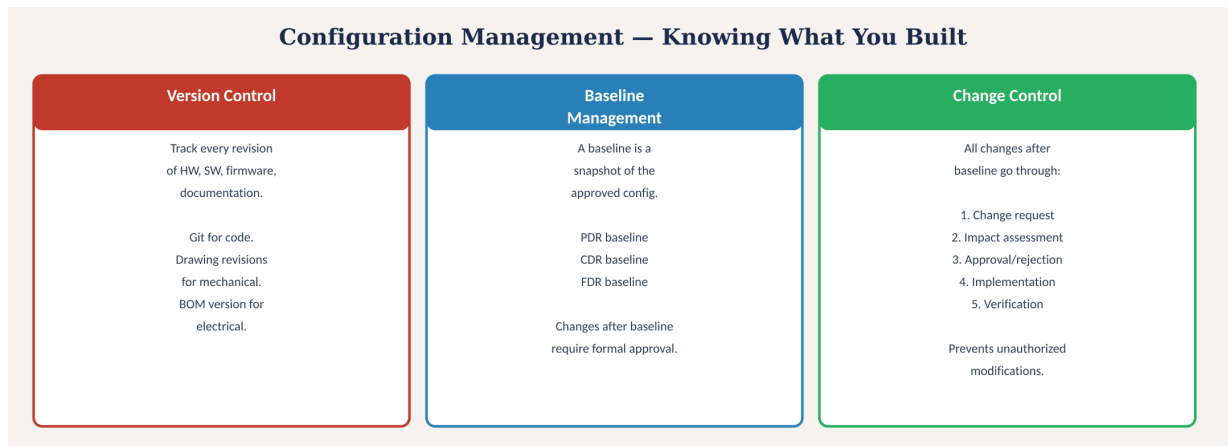


Figure 5.4: Configuration management pillars.

5.1 Integration Strategy

5.1.1 Bottom-Up Integration

Verify components individually → integrate into subsystems → test each subsystem → combine into system → system test. At each stage, add ONE new element and test. If something breaks, the new element or its interface is the suspect. **Never big-bang** (assemble everything at once).

5.1.2 Integration Stages and the V-Model

The V-Model maps integration from bottom-up: **Component verification** (each part works alone), **Subsystem integration** (components within a subsystem work together), **System integration** (subsystems work together), **System verification** (system meets all requirements), **Validation** (system meets stakeholder needs). Each stage has specific entry and exit criteria. Never advance with open failures.

5.1.3 Integration Sequence Planning

The order of integration matters. Rules: (1) Integrate power first; nothing works without power. (2) Add the lowest-risk subsystem next. (3) Add the highest-risk subsystem last (so problems are isolated to one new element). (4) Never add two new elements simultaneously (if something breaks, you cannot isolate the cause).

5.1.4 Test Gates

Each integration stage has entry and exit criteria. Entry: all components from the previous stage verified. Exit: all tests at the current stage PASS. Never advance with open failures.

5.1.5 Critical Subsystem Demonstrations

Before full system integration, each subsystem must pass a standalone demonstration. Three steps: (1) **Plan**; define pass/fail criteria, (2) **Execute**; run with data recording (video for design review), (3) **Document**; 1-page summary with data, photos, verdict. This is a CDR deliverable.

5.1.6 The N^2 (N-Squared) Diagram

The N^2 diagram maps interfaces between subsystems. Modules on the diagonal; cell (i, j) = what module i sends to module j . Dense rows/columns = high integration risk.

GS GreenShield: N^2 Diagram (5×5)

	Sensor	MCU	Relay	Comms	PSU
Sensor	—	Temp (I2C)			
MCU	Config	—	GPIO	UART TX	
Relay		Status	—		
Comms		UART RX		—	
PSU	3.3V	3.3V	12V	3.3V	;

PSU interfaces with everything. MCU receives from everything. Both are high-risk integration points. The I2C bus requires defined pull-up resistor ownership.

GS GreenShield: I²C Pull-Up Failure

The sensor team assumed the MCU board had pull-ups. The MCU team assumed the sensor breakout had them. Neither did. I2C floated, communication was intermittent, 8 hours of debugging. **Fix:** every ICD must state which side provides pull-ups, termination resistors, and bias voltages.

Case Study: Diagnosing Brown-Out Blues

A greenhouse controller reset randomly on relay activation. Scope showed 800 mV dip on 3.3 V rail (below 2.7 V brown-out threshold). Root cause: relay coil inrush through shared power trace. Fix: 470 μ F cap on MCU rail + separate relay power trace. Regression: 100 relay cycles, zero resets.

Industry SE Parallels

- **INCOSE SEH 4th ed.:** §4.6 Implementation, §4.7 Integration.
- **NASA SP-2016-6105 Rev 2:** Ch. 5.1 Implementation, Ch. 5.2 Integration.

5.2 Artifact Handoff: What Chapter 5 Produces

Artifact	Description	Used In
Integration Plan	Build sequence with test gates	CDR
N ² Diagram	Interface matrix	ICD verification
Integration Test Results	Subsystem and system-level data	FDR

5.3 Interface Management

80% of integration failures occur at interfaces. Six types: electrical (voltage, protocols, grounding), mechanical (mounting, tolerances, alignment), thermal (heat flow, contact resistance), software (APIs, data formats, timing), fluid (fittings, pressure, seals), and human (displays, controls, ergonomics).

Write Interface Control Documents (ICDs) DURING architecture — not during integration. The ICD defines what crosses each boundary, in what format, with what constraints. Every interface needs a single owner.

5.3.1 Interface Control Documents (ICDs)

An ICD defines what crosses each boundary, in what format, with what constraints. Minimum ICD content for each interface:

1. **Interface name and number** (e.g., ICD-003: MCU ↔ Relay Driver).
2. **Physical connection:** connector type, pin assignments, wire gauge/color.
3. **Electrical characteristics:** voltage levels, current draw, impedance, timing.
4. **Protocol:** I2C address, baud rate, message format, error handling.
5. **Ownership:** which side provides power, pull-ups, termination.
6. **Test method:** how you verify the interface works (oscilloscope measurement, loopback test).

Every ICD must be signed by the owners of both sides of the interface. Unsigned ICDs are the #1 root cause of integration failures.

5.4 Debugging

Five-step process: (1) Reproduce the failure reliably. (2) Isolate by disconnecting modules until the failure stops. (3) Characterize by measuring signals at the interface against the ICD. (4) Root-cause: hardware, software, or interface spec? (5) Verify the fix, then regression-test. **Change ONE variable at a time.**

5.5 Configuration Management

Three pillars: **Version control** (Git for code, revision letters for drawings, versioned BOMs), **Baseline management** (PDR/CDR/FDR snapshots with formal change control after baseline), **Change control** (change request → impact assessment → approval → implementation → verification).

5.5.1 Critical Subsystem Demonstrations

Before full integration, each subsystem should pass a standalone demonstration. Three steps: (1) **Plan**: define what the subsystem demonstrates, test equipment needed, pass/fail criteria. (2) **Execute**: run the demonstration with data recording; video for design review. (3) **Document**: 1-page summary with data, photos, verdict; a CDR deliverable.

5.5.2 The N^2 (N-Squared) Diagram

The N^2 diagram maps interfaces between subsystems. Modules on the diagonal; cell (i, j) describes what module i sends to module j . Empty cells = no interface. Dense rows = high-integration-risk modules.

Building one: list modules on the diagonal; for each pair ask “does module i send anything to j ?”; fill in signal type, protocol, voltage, data rate.

GS GreenShield: N^2 Diagram

	Sensor	MCU	Relay	Comms	PSU
Sensor	—	Temp (I2C)			
MCU	Config	—	GPIO high/low	UART TX	
Relay		Status (GPIO)	—		
Comms		UART RX		—	
PSU	3.3V	3.3V	12V	3.3V	;

PSU row is dense (interfaces with everything). MCU column is dense (receives from everything). Both are high-risk integration points. The I2C interface between Sensor and MCU requires defined pull-up ownership.

GS GreenShield: Integration Failure: I²C Pull-Up Error

The sensor team assumed the MCU board provided I2C pull-ups. The MCU team assumed the breakout board had them. Neither did. Result: bus floated, intermittent communication, 8 hours of debugging.

Root cause: ICD did not specify pull-up ownership. **Fix:** every ICD must explicitly state which side provides pull-ups, termination resistors, and bias voltages.

Case Study: Diagnosing the Brown-Out Blues

A greenhouse controller worked on the bench but reset randomly in the field. Debugging: (1) Reproduced only when relay switched on. (2) Isolated: relay disconnected = stable. (3) Characterized: oscilloscope showed 800 mV dip on 3.3 V rail at relay switch-on. (4) Root cause: relay coil inrush drew 150 mA through shared trace, drooping MCU supply below brown-out threshold. (5) Fix: 470 μ F cap on MCU rail, separate relay power trace. Regression: 100 relay cycles, zero resets.

Industry SE Parallels

- **INCOSE SEH 4th ed.:** Sec. 4.6 (Implementation), Sec. 4.7 (Integration).
- **NASA SP-2016-6105 Rev 2:** Ch. 5.1 (Product Implementation), Ch. 5.2 (Product Integration).

5.6 Artifact Handoff: What Chapter 5 Produces

Artifact	Description	Forward Reference
Integration Plan	Build sequence, test gates, interface test list	CDR deliverable
N ² Diagram	Interface matrix with signal/power definitions	ICD verification
Integration Test Results	Subsystem and system-level test data	Ch. 4 (VCRM), FDR

5.7 Practice Questions

1. Your system works with subsystems A and B connected, but fails when you add subsystem C. Describe your debugging approach step by step.
2. List all interfaces for a system with a sensor module, controller board, motor driver, and power supply. Which interfaces are highest risk?
3. Your teammate changes the firmware without telling anyone and the system stops working. What configuration management failure occurred?
4. Write entry and exit criteria for the subsystem integration stage of a greenhouse monitoring system.
5. Draw an N² diagram for a system with four modules: temperature sensor, ESP32 MCU, relay driver, and OLED display. Identify the highest-risk interface and explain why.
6. Your team skipped critical subsystem demonstrations at CDR. During integration, the motor driver outputs 5 V logic but the MCU expects 3.3 V. What process failure caused this, and how would a proper ICD have prevented it?

5.7.1 Answers

1. Debugging approach: (1) Reproduce: confirm failure occurs with A+B+C but not A+B. (2) Isolate: is it C itself or the C-to-A/C-to-B interface? Disconnect C from B, connect only C-to-A. Does it work? Then C-to-B interface is the suspect. (3) Characterize: measure signals at the C-to-B interface against the ICD. Check voltage levels, timing, protocol. (4) Root cause: is the interface spec wrong, or is C's output wrong? (5) Fix the specific issue, re-test, then add B back.
2. Interfaces: Sensor↔Controller (I2C data + power), Controller↔Motor Driver (PWM signal + power), Motor Driver↔PSU (high-current power), Controller↔PSU (logic power), Sensor↔PSU (sensor power). Highest risk: Controller↔Motor Driver (PWM noise coupling, ground bounce from motor current spikes) and Motor Driver↔PSU (inrush current, ground loops).
3. Configuration management failure: no change control process. The teammate made an uncontrolled change. Fix: establish version control (Git for firmware), require peer review before merge, tag releases, and never deploy untested code to the integration hardware.
4. Entry criteria: all subsystem tests passed, all ICDs signed, all subsystem firmware at tagged release, test equipment available. Exit criteria: all subsystem-to-subsystem interface tests pass, system powers up and executes basic sequence, no brown-out or reset events in 30-minute soak test.
5. N² diagram: MCU column is densest (receives from all three others). MCU↔Relay Driver interface is highest risk because it crosses voltage domains (3.3 V logic to 12 V relay coil) and the relay switching causes noise on shared power lines.
6. The 5 V logic on a 3.3 V MCU input would have been caught by an ICD that specifies: "Motor driver PWM input: 3.3 V logic, active high, 1 kHz." The ICD would have flagged the voltage incompatibility during the CDR documentation review, before any hardware was built.

Chapter 6

Operations & Maintenance

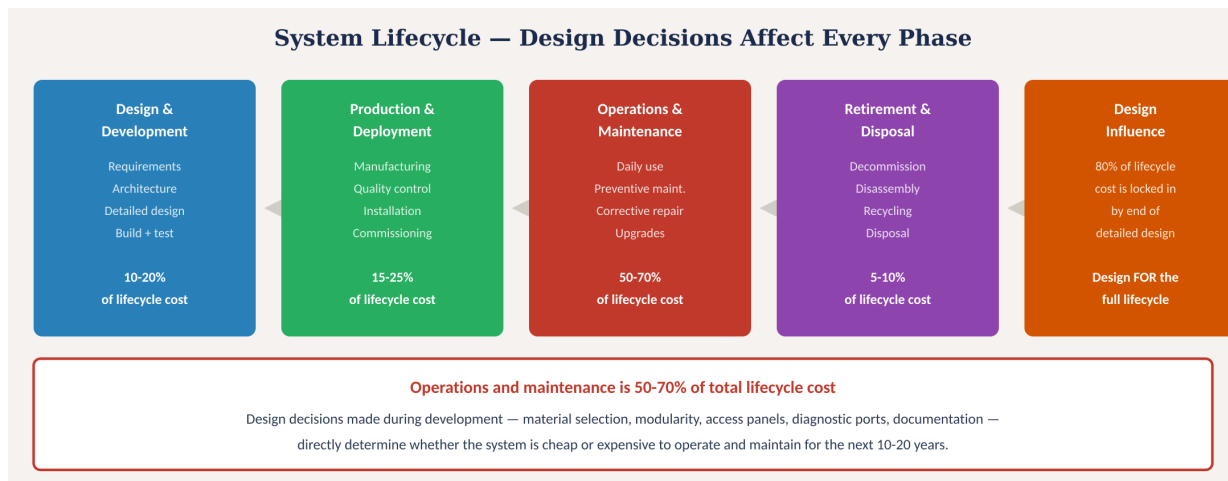


Figure 6.1: System lifecycle with cost distribution.

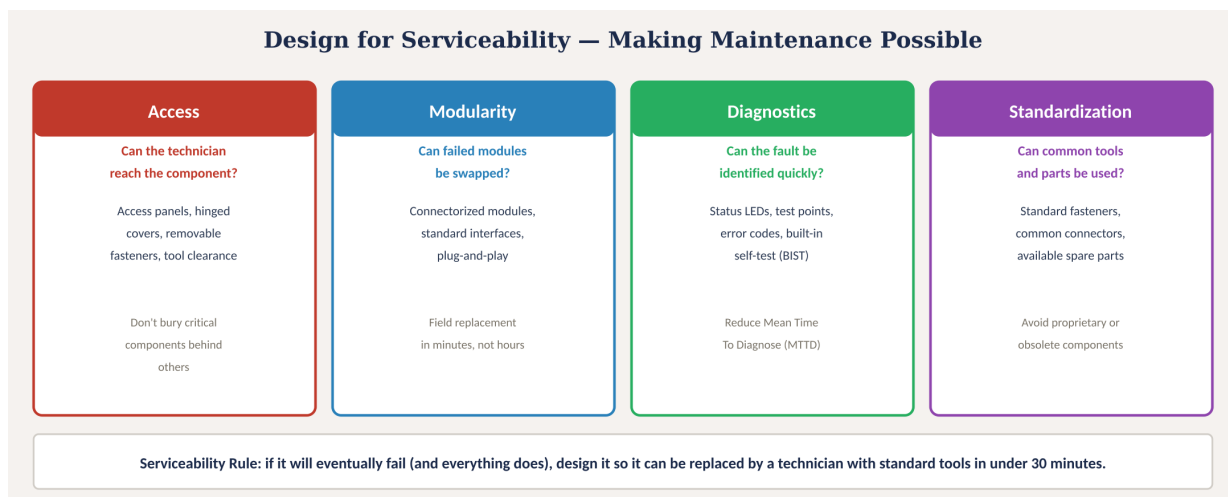


Figure 6.2: Four serviceability principles.

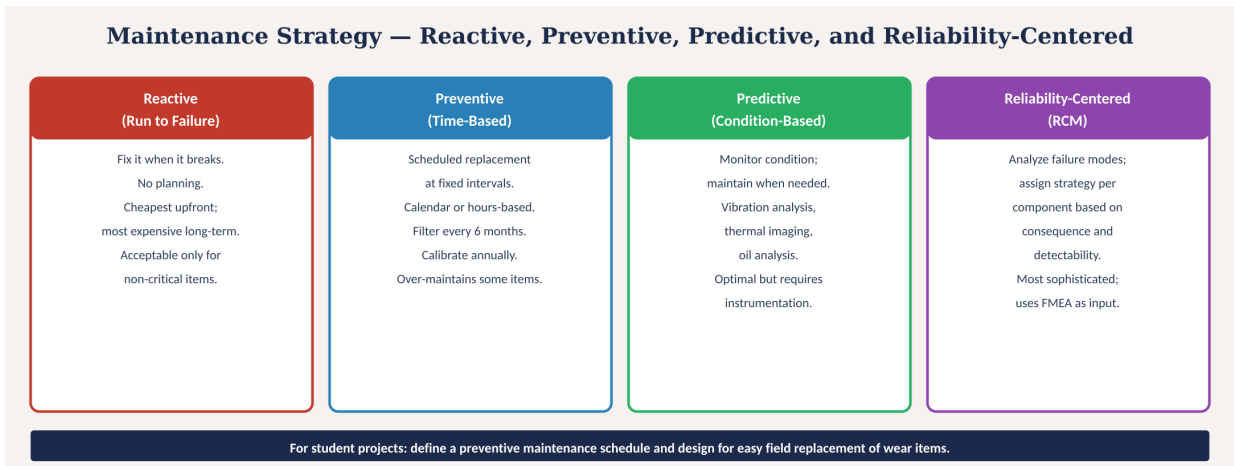


Figure 6.3: Maintenance strategies from reactive to RCM.

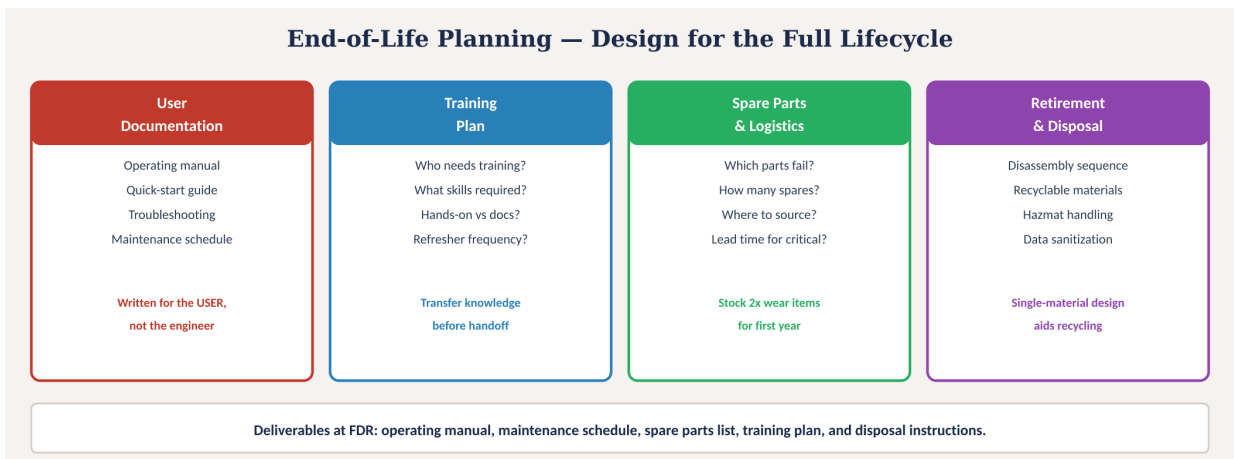


Figure 6.4: End-of-life planning deliverables.

6.1 Lifecycle Thinking

Operations and maintenance accounts for 50–70% of total lifecycle cost — far more than development. 80% of that cost is locked in by decisions made during concept and preliminary design. This means design-for-serviceability, maintainability, and disposability must be considered DURING design.

6.2 Design for Serviceability

Four principles: **Access** (can the technician reach the component? Use access panels, hinged covers, quick-release fasteners), **Modularity** (can failed modules be swapped? Connectorized, independently testable), **Diagnostics** (can the fault be identified? Status LEDs, test points, error codes, BIST), **Standardization** (can common tools and parts be used? Standard fasteners, multi-source components).

6.2.1 Reliability Metrics

MTBF (Mean Time Between Failures) and MTTR (Mean Time To Repair) combine to give Availability:

$$A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

Improving MTTR through better serviceability design is often cheaper than improving MTBF through better components. FMEA identifies failure modes and prioritizes design improvements.

6.2.2 The Bathtub Curve

Component failure rate follows three phases: (1) **Infant mortality** (decreasing rate): manufacturing defects fail early. Mitigate with burn-in testing. (2) **Useful life** (constant rate): random failures at steady rate; the MTBF region. (3) **Wear-out** (increasing rate): fatigue, corrosion, seal degradation. Mitigate with preventive replacement.

For student projects, most failures occur in Phase 1 (build quality). Burn in your system for 48+ hours before final demo.

6.2.3 FMECA: Failure Modes, Effects, and Criticality Analysis

FMECA extends FMEA by adding **Criticality Analysis**. For MEGN 301, use simplified FMEA with RPN (Risk Priority Number = Severity × Occurrence × Detection, each 1–10). Focus design effort on top-5 RPN items.

6.2.4 Writing Maintainability Requirements

Examples of maintainability requirements:

- “The system shall allow sensor replacement without disassembling the enclosure.”
- “The system shall provide visual indication (LED) of power, communication, and fault status.”
- “The system shall allow firmware update via USB without opening the enclosure.”
- “All field-replaceable modules shall use tool-free connectors (no soldering).”

GS GreenShield: O&M Requirement Trace

Req ID	O&M Requirement	Parent Need
OM-001	Sensor replacement w/o enclosure disassembly	Operator cannot solder
OM-002	Status LEDs for power, comms, fault	Remote monitoring need
OM-003	Battery replacement <5 min	Off-grid solar concept

Industry SE Parallels

- **INCOSE SEH 4th ed.:** §4.12–4.15 (Transition, Operation, Maintenance, Disposal).
- **NASA SP-2016-6105 Rev 2:** Ch. 6.3 Reliability, Maintainability, and Availability.
- **MIL-STD-1629A:** FMECA procedures.

6.3 Artifact Handoff: What Chapter 6 Produces

Artifact	Description	Used In
O&M Manual	Quick-start, operating, maintenance, troubleshooting guides	FDR
FMEA/FMECA	Failure mode analysis with RPN ranking	Design iteration
Spare Parts List	Wear items with vendors and intervals	Handoff package

6.4 Maintenance Planning

Four strategies: **Reactive** (run to failure — cheapest upfront, most expensive long-term), **Preventive** (scheduled replacement at fixed intervals), **Predictive** (condition-based monitoring), **Reliability-Centered** (FMEA-driven strategy per component). For student projects: define a preventive schedule and design for easy field replacement.

Build a maintenance schedule: identify wear items, define replacement intervals, create a spare parts list with vendors and lead times, and establish a calibration schedule.

6.5 Documentation, Training, and End-of-Life

Four documentation types: Quick-Start Guide (1 page, 10-minute setup), Operating Manual (complete reference by task), Maintenance Manual (step-by-step procedures with photos), Troubleshooting Guide (symptom-based flowcharts).

Training plan: identify audiences (operators, maintainers, trainers), methods (hands-on, written, video), and competency assessment. Knowledge transfer: document design rationale, known limitations, and workarounds.

6.5.1 Design for Disassembly (DfD)

Sustainability begins with designing systems that can be taken apart. Write this as a formal requirement:

DfD-REQ-001: The system shall be fully disassembled into its base material categories (metal, polymer, electronics, hazardous) using only standard hand tools (Phillips screwdriver, hex key set, pliers) in ≤ 5 minutes by a single technician.

This is a **testable requirement**; verify it by demonstration (TAID method “D”) at FDR. Time the disassembly; if it exceeds 5 minutes, the design fails the requirement. Practical rules: use fasteners instead of adhesives, avoid mixed-material bonding (epoxy on sensors is fast but makes recycling impossible), label material type on each component or enclosure panel, and design snap-fits to be releasable (not permanently locked).

End-of-life: design for disassembly (fasteners not adhesives), single-material design for recyclability, hazardous material handling (batteries, solvents), data sanitization.

6.5.2 The Bathtub Curve

Component failure rate follows the “bathtub curve”:

1. **Infant mortality** (decreasing rate): manufacturing defects and weak components fail early. Mitigate with burn-in testing and quality inspection.
2. **Useful life** (constant rate): random failures at a steady rate. This is the MTBF region.
3. **Wear-out** (increasing rate): fatigue, corrosion, seal degradation, bearing wear. Mitigate with preventive replacement before onset.

For student projects, most failures occur in Phase 1 (build quality). Burn-in your system for 48+ hours before the final demonstration.

6.5.3 FMECA: Failure Modes, Effects, and Criticality Analysis

Important: Your FMEA should have *started* during concept generation and architecture (Chapters 2–3), not here. By the O&M phase, you are updating the FMEA with field-observed failure data and refining RPN scores, not creating it from scratch. If your team’s first FMEA appears at FDR, it is too late to design out any failure mode it reveals.

FMECA extends FMEA by adding **Criticality Analysis**. For MEGN 301, use simplified FMEA with RPN (Risk Priority Number = Severity $\times$ Occurrence $\times$ Detection, each 1–10). Focus design effort on the top 5 RPN items.

6.5.4 Writing Maintainability Requirements

Examples of often-neglected maintainability requirements:

- “The system shall allow sensor replacement without disassembling the enclosure.”
- “The system shall provide LED indication of power, communication, and fault status.”
- “The system shall allow firmware update via USB without opening the enclosure.”
- “All field-replaceable modules shall use tool-free connectors.”

GS GreenShield: O&M Requirement Trace

Req ID	O&M Requirement	Parent Need
OM-001	Sensor replacement without enclosure disassembly	Operator cannot solder
OM-002	Status LEDs for power, comms, fault	Remote monitoring
OM-003	Battery replacement < 5 min	Off-grid concept
OM-004	All wear items documented with intervals	Long-term operation

Industry SE Parallels

- **INCOSE SEH 4th ed.:** Sec. 4.12–4.15 (Transition, Operation, Maintenance, Disposal).
- **NASA SP-2016-6105 Rev 2:** Ch. 6.3 (Reliability, Maintainability, Availability).
- **IEC 60300:** Dependability management standards.
- **MIL-STD-1629A:** Procedures for performing FMECA.

6.6 Artifact Handoff: What Chapter 6 Produces

Artifact	Description	Forward Reference
O&M Manual	Quick-start, operating, maintenance, troubleshooting guides	FDR deliverable
FMEA/FMECA	Failure mode analysis with RPN ranking	Design iteration
Spare Parts List	Wear items with vendors, lead times, replacement intervals	Handoff package

6.7 Practice Questions

1. A system has $MTBF = 2000$ hrs and $MTTR = 24$ hrs. Calculate availability. Propose two design changes to improve availability, one targeting MTBF and one targeting MTTR.
2. Write a quick-start guide (1 page, numbered steps) for a greenhouse temperature monitoring system.
3. Your project sponsor asks why you budgeted \$200 for spare parts on a \$2000 project. Justify the expenditure.
4. Identify all wear items in an electromechanical system with a DC motor, belt drive, pressure sensor, and lithium battery. Define replacement intervals for each.
5. Sketch the bathtub curve and label all three regions. Your system fails twice in the first week then runs flawlessly for 6 months. What region were the initial failures in, and how should you have prevented them?
6. Write a simplified FMEA (Severity/Occurrence/Detection, 1–10 each) for a greenhouse frost protection system with at least 5 failure modes. Calculate RPN for each.
7. Your sponsor asks why you need an O&M manual for a one-semester student project. Argue why it is still necessary.

6.7.1 Answers

1. $Availability = MTBF / (MTBF + MTTR) = 2000 / (2000 + 24) = 98.8\%$. Improving MTBF: use industrial-grade components with higher rated life; add redundancy (dual sensors). Improving MTTR: add diagnostic LEDs and test points for faster fault isolation; use modular, connectorized design so a failed module can be swapped in <5 minutes instead of 24 hours.
2. Quick-start guide: (1) Unbox system. (2) Mount sensor probe inside greenhouse at plant height using provided bracket. (3) Connect sensor cable to controller box (keyed connector). (4) Plug wall adapter into 120 VAC outlet. (5) Verify green POWER LED is on. (6) Set temperature

threshold using the dial on the front panel (default: 3 °C). (7) Verify amber MONITORING LED blinks once per second. (8) Test: hold ice cube near sensor. Within 30s, red HEATER LED should illuminate. Setup complete.

3. The \$200 spare parts budget is insurance against project failure. Motors wear out, sensors break, PCBs develop faults during testing, and connectors get damaged during repeated assembly/disassembly. Without spares, a single component failure can halt the project for 1–3 weeks (ordering + shipping). At $\$200/\$2000 = 10\%$ of project cost, this is standard engineering practice. Industry typically budgets 5–15% of project cost for spares and consumables.
4. Wear items: DC motor brushes (replace at 2000–5000 hrs), belt (inspect at 500 hrs, replace at 1000 hrs or on visible cracking), pressure sensor diaphragm (inspect annually, replace at first sign of drift), lithium battery (capacity degrades; replace when capacity <80% of rated, typically 300–500 charge cycles). Additionally: all O-rings and seals in fluid connections (annual replacement), and any connectors subject to repeated mating cycles.
5. Bathtub curve: decreasing (infant mortality), constant (useful life), increasing (wear-out). Two failures in week one = infant mortality phase. Prevention: (a) burn-in test for 48+ hours before deployment, (b) inspect all solder joints and connections during assembly, (c) verify all power rails under load before system-level testing. The flawless 6-month period = useful life phase with constant (low) failure rate.
6. FMEA example (abbreviated): Sensor reads high (S=8, O=3, D=4, RPN=96), Relay stuck closed (S=9, O=2, D=6, RPN=108), MCU brown-out (S=7, O=4, D=5, RPN=140), GSM no signal (S=5, O=3, D=3, RPN=45), Power supply fails open (S=8, O=2, D=7, RPN=112). Action priority: MCU brown-out first (decoupling + supervisor), then power supply (add status LED + fuse), then relay (add thermal fuse), then sensor (add second sensor for comparison).
7. The O&M manual is necessary because: (a) it is a primary FDR deliverable, the project is not complete without it; (b) the instructor and future teams need to operate, maintain, and build upon your work; (c) writing the manual forces you to think about failure modes, maintenance procedures, and spare parts, improving the design; (d) in industry, a product without O&M documentation is undeliverable.

Part II

Technical Reference

Chapter 7

Electrical & Fluid Connectors

7.1 Electrical Connectors

7.1.1 Connector Anatomy

Every electrical connector consists of three fundamental elements:

1. **Contacts.** The conductive pins (male) and sockets (female) that carry electrical current. Typical base metals are phosphor bronze and beryllium copper (BeCu), plated with either gold (best corrosion resistance, lowest contact resistance) or tin (lower cost, adequate for benign environments).
2. **Housing.** The insulating body that mechanically positions and electrically isolates the contacts. Common housing materials include:
 - Nylon (PA66); general purpose, good toughness, low cost.
 - PBT (polybutylene terephthalate); higher stiffness, better chemical resistance.
 - LCP (liquid crystal polymer); highest temperature rating, used in reflow-soldered connectors.
3. **Termination.** The mechanical and electrical joint between the wire conductor and the contact. The five major termination methods are detailed in Section 7.1.6.

7.1.2 Gender Convention

- **Male** (plug/header): pins protrude from the connector face.
- **Female** (receptacle/socket): pins are received into sockets.
- In wire-to-wire assemblies, one cable carries the plug, the other the receptacle.
- In wire-to-board applications, the board-mounted header is typically male.

7.1.3 Keying and Polarization

Connectors use physical features to ensure correct mating orientation and prevent damage from incorrect insertion. Common keying methods include key slots, asymmetric housing profiles, color coding, and differentiated contact arrangements.

Design Rule

Always specify keyed connectors when your design has multiple similar-looking connections. Incorrect mating can destroy components, and in student projects this is one of the most common causes of electrical failure on first power-up.

7.1.4 Connection Types

Electrical connectors are classified by *what* they connect:

Type	Description	Typical Examples
Wire-to-Wire	Joins two discrete wires or cable harnesses. Both sides are free-hanging connectors.	Molex Mini-Fit Jr., JST SM, Deutsch DT, automotive weatherpacks
Wire-to-Board	Connects a wire harness to a printed circuit board (PCB). One side is a cable connector; the other is soldered to the board.	JST XH/PH, Molex KK 254, pin headers with crimp housings
Board-to-Board	Connects two PCBs directly, often in stacked or mezzanine configurations. No wires involved.	Board-to-board headers, mezzanine connectors, card-edge connectors

Table 7.1: Electrical connector classification by connection type.

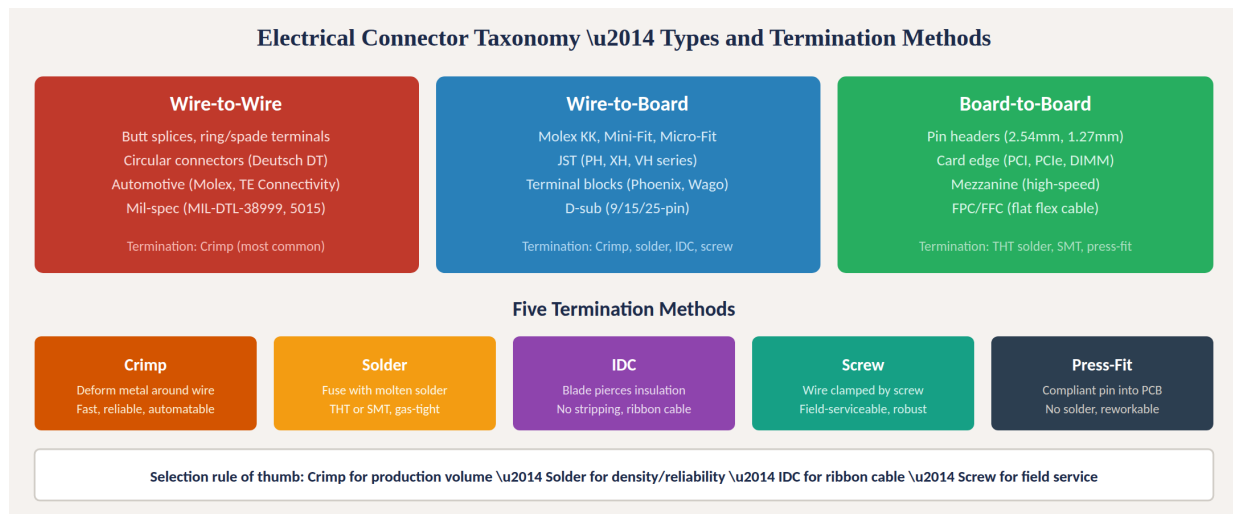


Figure 7.1: Electrical connector taxonomy by connection type with five termination methods [1], [2].

7.1.5 Harsh-Environment Connectors

Students designing vehicle harnesses, outdoor systems, or launch hardware will encounter connectors engineered for abuse. Understanding the physical components of these systems is essential for specifying them correctly.

Deutsch DT / DTM Series (automotive/off-highway): These use a wedgelock that mechanically retains all contacts simultaneously, independent of the wire seal. Each cavity receives

a silicone wire seal that grips the insulation OD, providing IP67/IP69K protection. Contacts are stamped-and-formed with a barrel crimp; the crimp must capture both the conductor and the insulation for strain relief. Deutsch connectors are the de facto standard for SAE Baja, Formula SAE, and agricultural equipment harnesses.

MIL-DTL-38999 Series III (aerospace/defense): A bayonet-coupled circular connector with a self-locking coupling ring that resists vibration. Uses machined contacts (not stamped) with interference-fit rear grommets for environmental sealing. Rated for extreme vibration (per MIL-STD-810), wide temperature ranges (-65 to $+200^{\circ}\text{C}$), and EMI shielding via conductive shell-to-shell contact. If your design involves launch hardware, flight systems, or classified defense work, 38999 is the starting point.

Common to both: Proper barrel crimps are critical; the crimp must form a gas-tight cold weld in the conductor barrel *and* a secure grip in the insulation barrel. Using generic pliers instead of a calibrated ratchet tool produces crimps that look acceptable but fail under vibration or thermal cycling.

7.1.6 Termination Methods

The termination method is how the wire conductor is mechanically and electrically bonded to the connector contact. Each method involves different tools, skill levels, and performance characteristics.

Crimp

A precision die compresses a metal barrel around a stripped wire conductor, creating a gas-tight joint through cold-welding at the metal interface.

- **Strengths:** Fast, repeatable, no heat required, excellent vibration resistance, gas-tight joint resists corrosion.
- **Limitations:** Requires calibrated crimp tools (hand ratchet or pneumatic); die must match the contact exactly. Poor crimps are invisible failures.
- **Best for:** Production wiring, automotive harnesses, harsh-environment connectors.
- **Quality verification:** Pull-test a sample of crimps per IPC/WHMA-A-620. Cross-section crimps periodically to verify cold-weld zone.

Pin Extraction: Use the Right Tool

In prototype environments, students inevitably insert a contact into the wrong cavity. **Never** yank the wire or use a paperclip to extract a crimped pin; this damages the housing retention lance and the contact, creating an unreliable connection that will fail in service. Every connector family has a specific extraction tool (typically a thin-walled sleeve that depresses the retention lance). Keep the correct extraction tools on hand alongside your crimp tools; they cost a few dollars and save hours of rework and scrapped housings.

Solder

Molten solder alloy (typically SAC305 lead-free or Sn63/Pb37) forms a metallurgical bond between wire and contact.

- **Strengths:** Low contact resistance, well-understood process, easy visual inspection.
- **Limitations:** Requires heat (risk of insulation damage), not reworkable without desoldering tools, solder joints are brittle under vibration and thermal cycling.

- **Best for:** Board-level connections (through-hole and SMT), prototype wiring, low-vibration environments.

Insulation Displacement Contact (IDC)

A sharp, slotted contact blade pierces through wire insulation and makes contact with the conductor, all in a single operation.

- **Strengths:** Extremely fast, no stripping required, good for mass-termination of ribbon cables.
- **Limitations:** Limited to small wire gauges (typically 22–28 AWG), not suitable for high-vibration environments, contact can relax over time.
- **Best for:** Ribbon cable connectors, low-current signal wiring, internal PCB connections.

Screw Terminal

A screw clamp compresses the wire conductor against a metal plate or barrel inside the terminal block.

- **Strengths:** Tool-free or simple screwdriver, field-serviceable, wide wire gauge range, easy to inspect.
- **Limitations:** Bulky, screws can loosen under vibration, relatively high contact resistance compared to crimp.
- **Best for:** Panel-mount terminal blocks, DIN-rail wiring, power distribution, industrial controls.

Press-Fit (Compliant Pin)

A specially shaped contact pin (often an “eye-of-needle” or “action pin” geometry) is pressed into a plated through-hole on a PCB, creating a gas-tight interference fit.

- **Strengths:** No solder required, excellent reliability, eliminates solder joint fatigue, good for backplane connectors.
- **Limitations:** Requires controlled insertion force and tooling, board must have proper hole tolerances, not field-reworkable.
- **Best for:** Backplane connectors, automotive ECU headers, high-reliability board-level connections.

Termination Method Selection Summary

7.1.7 Key Electrical Specifications

When specifying an electrical connector in your design documentation, you must define each of the following parameters.

Current Rating

The maximum continuous current per contact, in amperes. This depends on contact size, material, plating type, and number of loaded contacts.

Termination Method Selection Tradeoffs and Applications				
Method	How It Works	Strengths	Limitations	Best For
Crimp	Tool compresses metal contact around stripped wire	Fast, consistent, gas-tight Highly automatable	Requires specific tool per contact; pull-test needed	Production wire harnesses, automotive, mil/aero
Solder (THT/SMT)	Molten solder fuses wire to contact or pad on PCB	Strong mechanical bond, excellent for high density	Labor-intensive (hand); heat can damage components	PCB assembly, prototyping, mil/aero high-reliability
IDC	Sharp blade pierces wire insulation; cold-weld joint	No stripping; mass-terminate ribbon cables instantly	Limited wire gauge range; not for harsh environments	Ribbon cable, telecom, internal PC connections
Screw / Clamp	Wire clamped by screw or spring-cage mechanism	No special tools; field reworkable; vibration-resistant	Larger footprint; slower assembly than crimp	Industrial panels, DIN rail, power distribution, HVAC
Press-Fit	Compliant pin pressed into plated through-hole on PCB	No solder; no heat; gas-tight; reworkable; high-reliability	Requires precise hole sizing; press equipment needed	Automotive ECUs, backplanes, high-pin-count connectors

Selection hierarchy: Define current/signal → Choose connector family → Select termination based on volume, environment, and serviceability [1]

Figure 7.2: Termination tradeoff matrix; mechanism, strengths, limitations, and best applications [1], [2].

Thermal Derating

Always derate for multi-pin loaded connectors. When adjacent contacts all carry current, thermal stacking raises the temperature inside the housing, reducing each contact's effective capacity. A connector rated at 10 A per contact at 25 °C may only safely handle 7 A per contact at 85 °C ambient. Manufacturers provide derating curves; never ignore them.

Bridging to Wire Selection (AWG)

The connector's current rating sets the *minimum* wire gauge you can use. Standard practice: select wire whose ampacity (per NEC Table 310.16 or MIL-W-22759 for aerospace) meets or exceeds the derated connector contact rating. Common reference points for chassis wiring in free air at 30 °C: 22 AWG ≈ 7 A, 18 AWG ≈ 10 A, 16 AWG ≈ 13 A, 14 AWG ≈ 17 A, 12 AWG ≈ 23 A, 10 AWG ≈ 33 A. Always verify that the connector contact accepts your chosen wire gauge; crimp contacts are rated for a specific AWG range (e.g., 18–22 AWG), and using the wrong gauge produces unreliable crimps.

Voltage Rating

The maximum working voltage (VAC or VDC). This is determined by creepage (surface distance) and clearance (air gap) distances between adjacent contacts, as specified by IEC 61984 [2].

Important: Always specify the *rated working voltage*, not the dielectric withstand (breakdown) voltage. Working voltage includes safety margin; breakdown voltage does not.

Contact Resistance

The electrical resistance measured across a mated contact pair, typically in milliohms (mΩ). Gold-plated contacts achieve <20 mΩ; tin-plated contacts may range from 20–50 mΩ when new.

Why it matters: At 10 A, even 50 mΩ of contact resistance produces:

$$P = I^2 R = (10)^2 \times 0.050 = 0.5 \text{ W of heat per contact} \quad (7.1)$$

Contact resistance increases with mating cycles and in corrosive environments. Always verify with a milliohm meter during commissioning.

IP Rating (Ingress Protection)

Defined by IEC 60529 [3], the IP code uses two digits:

- **First digit (0–6)**: Protection against solid objects, from “no protection” (0) to “dust-tight” (6).
- **Second digit (0–9K)**: Protection against liquids, from “no protection” (0) to “high-pressure, high-temperature washdown” (9K).

IP Rating	Meaning	Typical Application
IP20	Finger-safe; no liquid protection	Indoor panel wiring
IP54	Dust-protected; splash-proof	General industrial
IP67	Dust-tight; 1 m immersion for 30 min	Outdoor, vehicle, marine
IP68	Dust-tight; continuous submersion	Submersible equipment
IP69K	Dust-tight; high-pressure washdown	Food/pharma equipment

Table 7.2: Common IP ratings and their applications.

Mating Cycles

The rated number of connect/disconnect operations before mechanical or electrical degradation. Ranges vary enormously:

- 25–50 cycles: permanent/semi-permanent installations.
- 500–1,000 cycles: periodic maintenance access.
- 5,000–10,000+ cycles: test fixtures, instrumentation.

Gold plating significantly outlasts tin plating. Contact wear directly increases resistance over mating life.

7.1.8 Electrical Connector Failure Modes

Understanding failure modes is essential for designing robust connections. These are listed roughly in order of how commonly they cause problems in practice.

Fretting Corrosion

Mechanism: Micro-motion from vibration (amplitudes as small as $10\mu\text{m}$) wears through the contact plating, exposing base metal. Oxides form on the exposed metal, creating high-resistance debris particles that accumulate at the contact interface.

Why it’s dangerous: Fretting corrosion is the most insidious failure mode because it develops slowly, manifests as intermittent contact, and is difficult to diagnose. Resistance may fluctuate between normal and open-circuit depending on vibration state.

Mitigations:

- Gold plating (resists oxide formation).
- Higher contact normal force (resists micro-motion).
- Strain-relief on cables (reduces transmitted vibration).
- Secondary locking features on connectors.

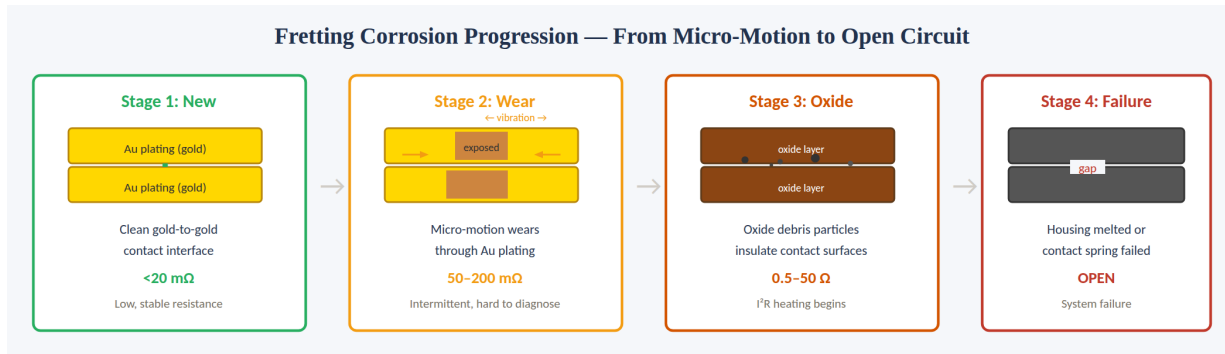


Figure 7.3: Fretting corrosion progression, from clean gold-plated contacts through oxide accumulation to open-circuit failure. Note resistance escalation across stages.

Figure 7.3 illustrates the four-stage progression of fretting corrosion. At Stage 1, clean gold-plated contacts exhibit stable resistance below 20 mΩ. Vibration-induced micro-motion gradually wears through the plating at Stage 2, intermittently exposing base metal and raising resistance to 50–200 mΩ, often misdiagnosed as a “loose connection.” By Stage 3, accumulated oxide debris insulates the contact surfaces, resistance climbs to the ohm range, and I^2R heating accelerates further degradation. Stage 4 is outright failure: melted housing, collapsed contact springs, or open circuit. Because Stage 2 symptoms are intermittent and position-dependent, fretting often passes bench-top continuity checks only to fail under dynamic load, a pattern students must learn to recognize.

Overheating (I^2R Heating)

Mechanism: Exceeding the current rating causes resistive heating at the contact interface. Heat accumulates in the housing, softening thermoplastic materials and potentially igniting adjacent insulation or components.

Mitigations:

- Never exceed rated current; derate for ambient temperature.
- Select contacts with lower resistance (gold plating, larger cross-section).
- Distribute high currents across multiple contacts.

Creep and Stress Relaxation

Mechanism: Contact springs gradually lose their clamping force at elevated temperatures. The contact normal force decreases over time, increasing resistance and accelerating other failure modes.

Mitigations:

- Specify beryllium copper (BeCu) contacts for any application above 60 °C continuous. BeCu resists creep far better than brass or phosphor bronze.
- Overspecify contact normal force for high-temperature applications.

Moisture Ingress

Mechanism: Water bridges between contacts, causing electrochemical corrosion, increased resistance, or direct short circuits.

Mitigations:

- Specify appropriate IP rating per IEC 60529 [3].

- Use sealed connectors with O-ring face seals and wire seals for any outdoor, vehicle, or wash-down application.
- Apply conformal coating on exposed PCB connections.

7.2 Fluid System Connectors

7.2.1 Threaded Pipe Fitting Standards

Two incompatible tapered thread standards dominate global pipe fitting. Geography determines which you encounter.

NPT: National Pipe Taper (ASME B1.20.1)

- **Thread angle:** 60° with pointed crests and roots.
- **Taper:** 1:16 ($\frac{1}{4}$ inch per inch of thread length, equivalent to $\frac{3}{4}$ inch per foot).
- **Sealing mechanism:** Thread interference creates a metal-to-metal wedge seal. Because the thread form has clearance at crests and roots, NPT joints *always* require supplemental sealing; either PTFE (Teflon) tape or anaerobic pipe dope.
- **NPTF variant:** “Dryseal” NPT uses a modified thread form with tighter root/crest tolerances that crushes metal at the interface, theoretically sealing without sealant. In practice, sealant is still recommended.
- **Size range:** 1/8 in. to 24 in. nominal pipe size (NPS).
- **Pressure rating:** Up to 15000 psi depending on material and size.
- **Geography:** Dominant in the United States and Canada.

BSP: British Standard Pipe (ISO 228 / ISO 7-1)

- **Thread angle:** 55° Whitworth form with rounded crests and roots.
- **Two sub-types:**
 - **BSPT** (designation “R”): Tapered threads, seals like NPT via thread interference plus sealant.
 - **BSPP** (designation “G”): Parallel (straight) threads that *do not* seal on the threads. Sealing is achieved with a bonded washer or O-ring compressed against a flat face.
- **Size range:** 1/8 in. to 6 in. (inch-denominated but used globally outside North America).
- **Pressure rating:** Up to 6000 psi typical.
- **Geography:** Dominant in Europe, Asia, and most of the world outside North America.

Thread Galling: The Point of No Return

In stainless steel or aluminum, forcing a mismatched or cross-threaded fitting with a wrench causes **galling**; a cold-weld that permanently destroys both components. Once galling begins, reaching for a bigger wrench turns a \$50 component into a \$100 mistake (two destroyed parts

plus the replacement).

The Finger Test: if you cannot thread a fitting at least three full turns by hand, **STOP**. You are either cross-threaded or mixing thread standards. No amount of force or sealant can fix galled threads.

Prevention for stainless-on-stainless connections: apply nickel-based anti-seize compound (e.g., Permatex Nickel Anti-Seize or Loctite LB 8023) to the male threads before assembly. Anti-seize reduces friction and prevents the metal-to-metal adhesion that initiates galling. This is mandatory for any stainless steel fluid fitting in your design.

Critical Incompatibility

NPT (60° pointed threads) and BSP (55° rounded threads) are **NOT** interchangeable. They may appear to engage for a few turns but will not seal properly. Cross-threading causes galling, thread damage, and leaks that may develop catastrophically under pressure. **Always** verify thread standard with a thread pitch gauge before assembly.

Feature	NPT (ASME B1.20.1)	BSP (ISO 228 / ISO 7-1)
Thread angle	60°, pointed crests/roots	55° Whitworth, rounded crests/roots
Taper	1:16 (all NPT tapered)	BSPT tapered; BSPP parallel
Sealing	Thread interference + sealant	BSPT: thread + sealant; BSPP: washer/O-ring
Geography	U.S., Canada	Europe, Asia, rest of world
Max pressure	15000 psi	6000 psi typical

Table 7.3: NPT vs. BSP thread comparison.

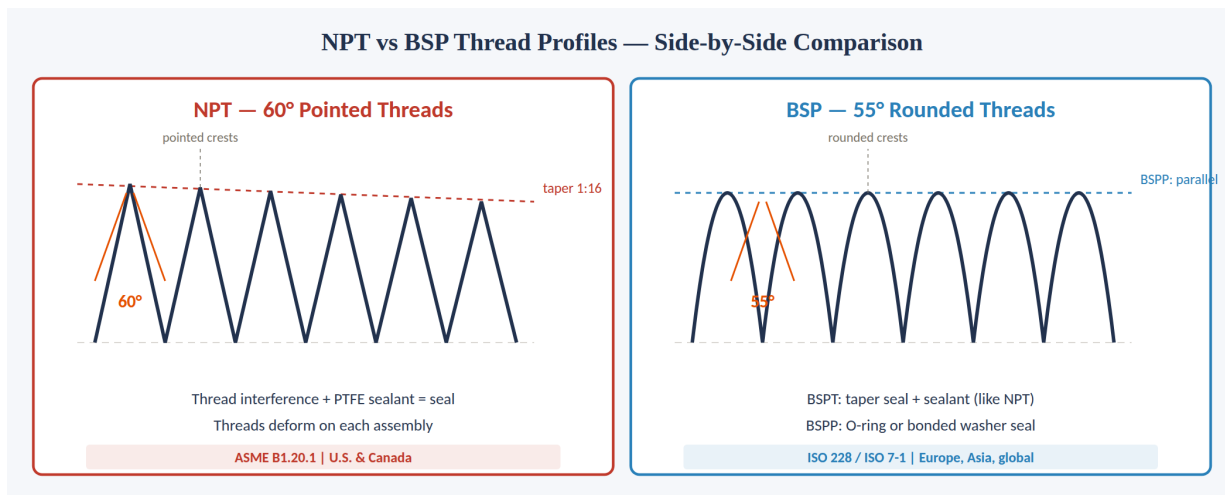


Figure 7.4: NPT vs. BSP thread profiles; note the pointed 60° crests on NPT versus the rounded 55° Whitworth crests on BSP. These subtle geometric differences make the two standards completely incompatible.

Figure 7.4 places the two thread forms side by side so the geometric distinction is immediately visible. NPT's 60° pointed crests create a wedging action as the taper engages; metal deforms

at each assembly, which is why PTFE sealant is required and why each reassembly cycle slightly degrades the thread. BSP's 55° Whitworth form uses rounded crests that are less prone to galling but follow an entirely different pitch schedule. The BSPP (parallel) variant does not seal on the thread at all; the thread merely provides clamping force while an O-ring or bonded washer creates the pressure boundary against a flat face. This is a critical distinction: applying PTFE tape to a BSPP fitting is a common student mistake that serves no purpose and can actually prevent proper washer compression.

7.2.2 Tube Fittings and Quick-Connect Couplings

Unlike threaded pipe fittings (which connect pipes), tube fittings connect *tubing*; a product specified by outside diameter (OD) and wall thickness, with tighter dimensional tolerances than pipe.

JIC 37° Flare (SAE J514)

The tube end is flared to a 37° cone that mates with a matching seat in the fitting body. The joint seals metal-to-metal; no sealant, tape, or elastomeric seal is required. The fitting uses a straight (non-tapered) thread; the thread provides clamping force only and does not contribute to sealing.

- Reassemblable without degradation (unlike taper threads which deform with each assembly).
- The hydraulic industry standard worldwide.
- Pressure ratings to 10000 psi.
- AN (Army–Navy) fittings are the military equivalent of JIC, with identical 37° geometry.

Compression Fittings (Swagelok-type)

One or two ferrules (metal rings) are swaged (compressed) onto the tube OD when the nut is tightened. The ferrule bites into the tube, creating both a grip and a seal.

- No welding, brazing, flaring, or special tools required.
- Tube OD is the sizing basis (1/4 in., 3/8 in., 1/2 in., etc.).
- **Gaugeable:** Many designs include a visual indicator (gap gauge) so you can verify proper assembly by inspection.
- Pressure ratings up to 20000 psi for medium-pressure configurations.
- Brands: Swagelok, Parker CPI/A-LOK, Ham-Let.

Push-to-Connect

The tube pushes into the fitting body. An internal collet (toothed ring) grips the tube OD, and an O-ring provides the seal. Assembly is one-handed and tool-free.

- Common in pneumatics (air systems), typically up to $\sim$ \$250 psi.
- Common brands: SMC, Festo, John Guest.
- Fast assembly but limited to lower pressures and non-critical sealing applications.

Quick-Disconnect (QD) Couplings

Two halves, a plug and a socket, connect and disconnect by hand, often with a locking sleeve or collar. Self-sealing valves in one or both halves prevent fluid loss on disconnect.

- **Flat-face:** Minimal fluid spillage on disconnect, used in clean or environmentally sensitive applications.
- **Poppet-valve:** Spring-loaded valve seals on disconnect, less expensive but more spillage.

- Common applications: hydraulic tool circuits, coolant loops, test equipment.

Fitting Family	Pressure	Seal Type	Key Characteristic
JIC 37° Flare	10000 psi	Metal-to-metal	Reassemblable; hydraulic standard
Compression	20000 psi	Ferrule swage	No special tools; gaugeable
Push-to-Connect	250 psi	O-ring	Tool-free; pneumatics
Quick-Disconnect	Varies	Valve + O-ring	Hand-connect; self-sealing

Table 7.4: Tube fitting and coupling comparison.

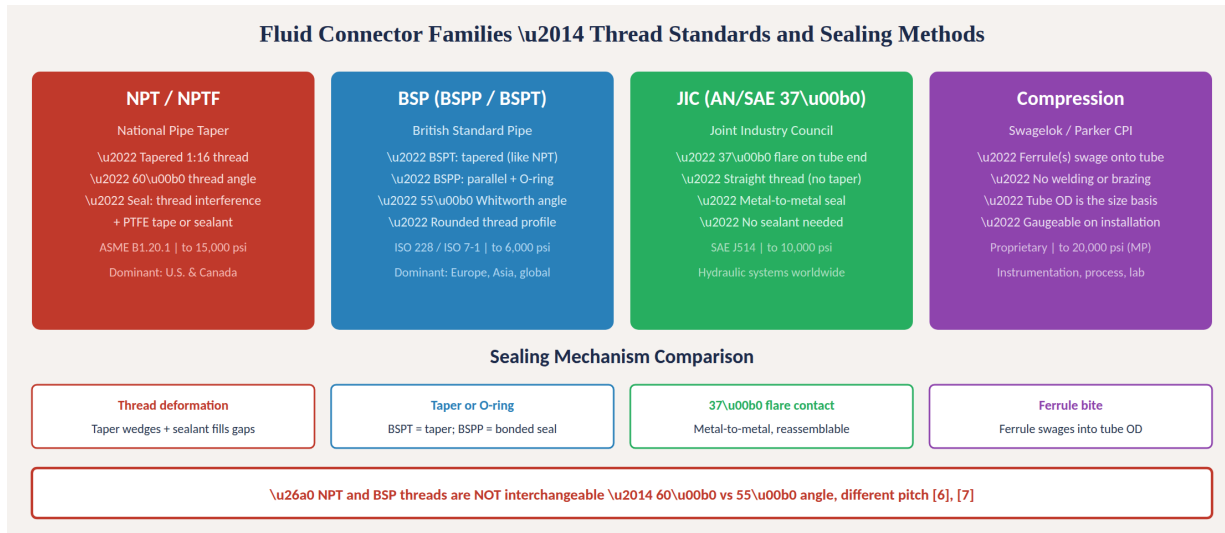


Figure 7.5: Four major fluid connector families with thread standards, pressure ratings, and sealing mechanisms.

Figure 7.6 shows the internal geometry of each sealing method so students can visualize *why* proper torque and assembly procedures matter. In the NPT taper seal, the thread flanks themselves deform to create a pressure boundary, which is why sealant is mandatory to fill the remaining micro-gaps and why repeated assembly progressively damages the seal surfaces. In the JIC 37\u00b0 flare, the cone-shaped tube end presses against a matching seat in the fitting body; the straight thread provides clamping force only. This metal-to-metal geometry is fully reassemblable because the sealing surfaces do not deform. In compression fittings, the front ferrule bites radially into the tube OD while the back ferrule provides axial grip; the nut drives both ferrules during initial makeup. Once swaged, the ferrule is permanently deformed, which limits reassembly. In the BSPP face seal, the parallel thread clamps the fitting against the port face while an O-ring in a groove creates the pressure boundary, the thread contributes zero sealing.

7.2.3 Key Fluid Connector Specifications

When specifying a fluid connector in your design documentation, define each of the following.

Working Pressure

The maximum continuous operating pressure (psi or bar). Always specify *working* pressure, not burst pressure. Standard safety factors:

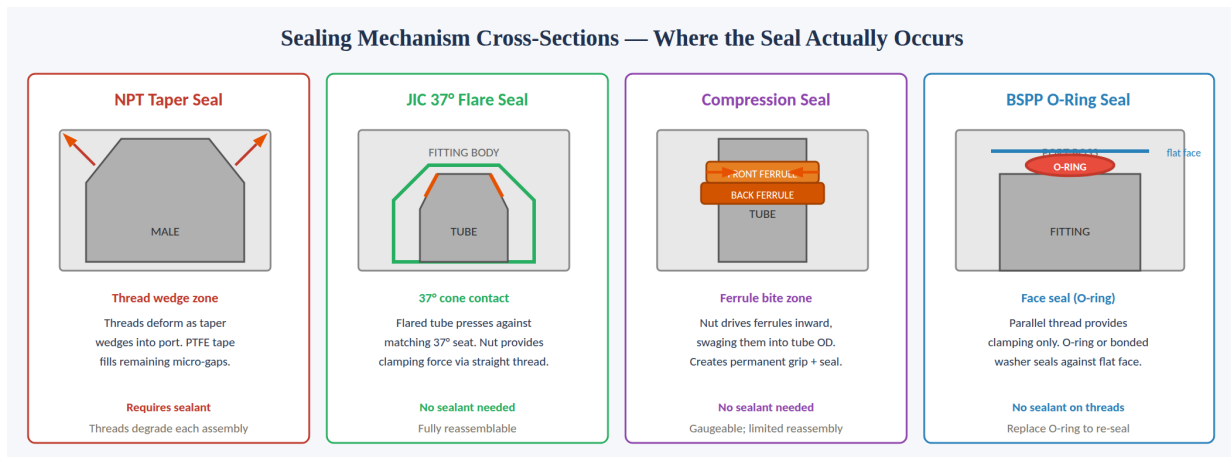


Figure 7.6: Sealing mechanism cross-sections showing exactly where each seal forms: NPT thread wedge, JIC 37° flare cone contact, compression ferrule bite zone, and BSPP face-seal O-ring.

- Hydraulic: burst $\geq 4\times$ working.
- Pneumatic: burst $\geq 3\times$ working.
- **Design rule:** Never operate above 25% of burst pressure.

Temperature Range

Temperature determines seal material selection:

Seal Material	Range	Notes
Buna-N (Nitrile, NBR)	−40 to 250 °F	General purpose; petroleum-compatible
Viton (FKM)	−20 to 400 °F	Chemical resistance; fuel systems
PTFE (Teflon)	−100 to 500 °F	Widest range; chemically inert; no elasticity
EPDM	−65 to 300 °F	Steam/water service; not oil-compatible

Table 7.5: Seal material temperature ranges and applications.

Always verify seal compatibility with both the fluid *and* the operating temperature.

Fluid Compatibility

Verify fitting body material and seal material against the working fluid using manufacturer chemical compatibility charts:

- **Stainless steel:** Resists most chemicals; highest cost.
- **Brass:** Common for water, air, and general service; not for ammonia or acetylene.
- **Carbon steel:** Standard for hydraulic oil systems; requires corrosion protection.

Thread / Tube Size

- **Threaded fittings:** Specified by Nominal Pipe Size (NPS). Note that NPS does *not* equal the actual outside diameter. For example, 1/2 in. NPS pipe has an OD of 0.840 in.
- **Tube fittings:** Specified by tube outside diameter (OD). Common sizes: 1/4 in., 3/8 in., 1/2 in.

- **Always measure with calipers and thread gauges.** Never assume a size by visual inspection.

End Configuration

Select the geometry that connects the fitting to your system: straight, elbow (45° or 90°), tee, cross, union, reducer, or bulkhead.

Minimize Joints

Every fitting joint is a potential leak point. Design your routing to minimize the total number of fittings in a line. Avoid stacking street elbows in series; use a single 90° elbow or a pre-bent tube run instead.

7.2.4 Fluid Connector Failure Modes

Cross-Threading

Mechanism: Attempting to mate NPT into BSP (or vice versa), or even same-standard threads started at an angle, destroys thread profiles and causes immediate or delayed leaks.

Mitigations:

- Always verify thread standard with a thread pitch gauge before assembly.
- Start threads by hand for at least two full turns before applying a wrench.
- NPT: 60° pointed threads. BSP: 55° rounded threads. They *feel* similar but are not compatible.

Over-Tightening

Mechanism: Excessive torque cracks port bosses (especially aluminum and cast iron), deforms compression ferrules beyond recovery, and splits flare cones. Over-tightening destroys more fittings than under-tightening.

Mitigations:

- Always use manufacturer-specified torque values.
- Use a calibrated torque wrench, not a “feel” wrench.
- For Swagelok-type fittings, follow the turns-past-finger-tight specification exactly.

Seal Degradation

Mechanism: O-rings and gaskets deteriorate from chemical attack (incompatible fluid), UV exposure, thermal cycling, and simple age. Seals swell, harden, crack, or take a permanent compression set.

Mitigations:

- Establish replacement intervals based on seal material and service conditions.
- Keep spare seal kits on hand.
- Verify seal material compatibility with the fluid and temperature range before installation.

Vibration Fatigue

Mechanism: Cyclic stress at tube-to-fitting junctions nucleates fatigue cracks. Under pressure, these cracks propagate to failure; potentially a sudden, high-pressure fluid release.

Mitigations:

- Use proper tube support clamps spaced per manufacturer recommendations.

- Route tubing to avoid sharp bends and unsupported cantilever lengths.
- Include flexible hose segments at vibration sources (pumps, engines).

High-Pressure Fluid Injection Injury

A hydraulic line leak may be invisible; a pinhole stream at 2000 psi can penetrate skin and inject fluid deep into tissue, causing catastrophic damage that requires emergency surgery within hours. **Never** check for hydraulic leaks by running a bare hand along a pressurized line. Use a piece of cardboard or a thermal camera instead. This is one of the most serious and least-understood hazards in hydraulic system work, and students designing or testing any pressurized fluid system must be explicitly aware of it before commissioning.

7.3 Selection and Design Practice

7.3.1 Connector Selection Logic

Whether electrical or fluid, connector selection follows a universal logic sequence:

1. **Rating:** What current, voltage, pressure, or flow must it handle?
2. **Environment:** Temperature range, vibration, moisture/IP, chemical exposure?
3. **Serviceability:** How often must it be connected/disconnected? What tools are available?
4. **Cost:** What is the budget per connection, including tooling?
5. **Standards:** Which industry/regulatory standards apply (IEC, SAE, ASME, MIL)?
6. **Space:** What are the physical envelope constraints?

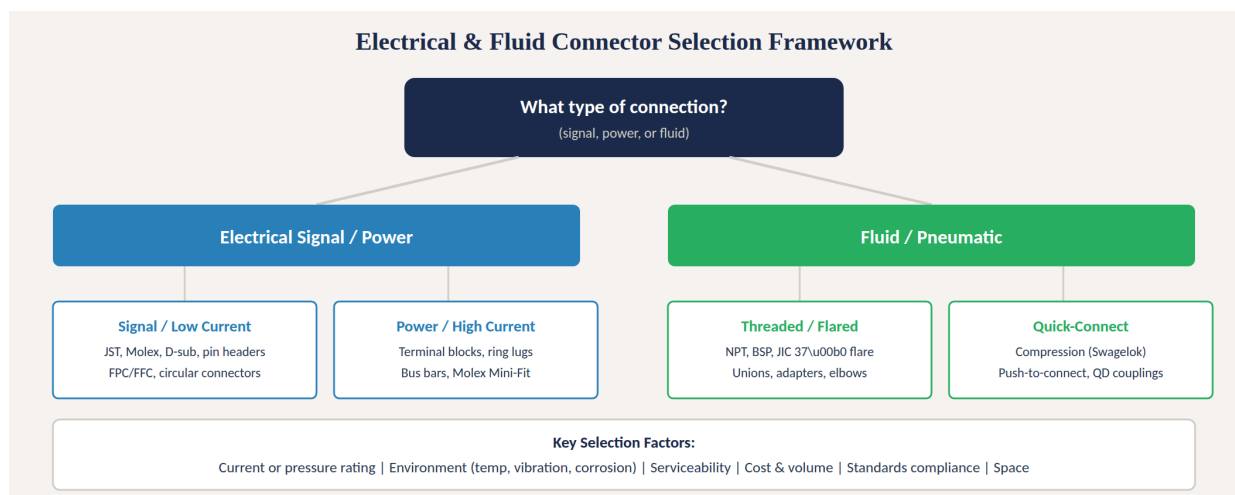


Figure 7.7: Decision tree for electrical and fluid connector selection.

Spec the Connector First

The connector determines wire gauge, tube OD, termination tooling, and housing envelope. Select the connector family first (based on rating, environment, and serviceability), then design the wiring or tubing around it, not the other way around.

7.3.2 Design Rules of Thumb

1. **Derate everything.** Never operate at 100% of rated current, pressure, or temperature. A 25% safety margin is minimum engineering practice; 50% is professional practice for critical systems.
2. **Design for Assembly (DFA).** Provide wrench clearance around fittings. Group fluid connections for leak-test access. Use consistent connector families within a subsystem to reduce tool count and spare parts inventory.
3. **For fluid systems:** Verify thread standard with a gauge before assembly. Use manufacturer torque specifications. Pressure-test every joint before commissioning; hydrostatic test at $1.5\times$ working pressure per ASME B31.3 [8].
4. **For electrical systems:** Pull-test a sample of crimps per IPC/WHMA-A-620. Verify contact resistance with a milliohm meter. Document every connector in your wire harness schedule.
5. **Document every choice.** Record part number, manufacturer, specifications, and selection rationale in your Bill of Materials (BOM). Your future self and your teammates will need this information for procurement, assembly, and troubleshooting.

7.3.3 Key Standards Reference

Standard	Org	Scope
IEC 61984 [2]	IEC	Electrical connector safety requirements, testing, and performance criteria
IEC 60529 [3]	IEC	IP ingress protection rating system (IP67, IP68, IP69K)
MIL-DTL-38999 [4]	U.S. DoD	Military circular connectors for high-performance, harsh-environment applications
UL 486A-486B	UL	Safety certification for wire connectors: crimp, solder, and IDC terminations
SAE J514 [5]	SAE	JIC 37° flare hydraulic tube fittings; dimensions and performance requirements
ASME B1.20.1 [6]	ASME	NPT tapered pipe thread form, dimensions, and tolerances
ISO 228 / ISO 7-1 [7]	ISO	BSP parallel (BSPP) and tapered (BSPT) pipe thread standards
ASME B31.3 [8]	ASME	Process piping code; materials, fabrication, inspection, and testing

7.4 Applying Connectors in Your Design Projects

Incorrect connector selection is one of the most common failure modes in student design projects. The following workflow ensures systematic coverage.

1. **Create a connection schedule early.** List every electrical and fluid joint in your design. Classify each by type (wire-to-wire, wire-to-board, NPT, JIC, etc.). This becomes a living document through your design reviews.
2. **For every electrical connection, specify:** connector family, pin count, current rating (derated), voltage rating, IP rating, termination method, wire gauge, and mating cycles.
3. **For every fluid connection, specify:** fitting standard (NPT, BSP, JIC, or compression), nominal size, working pressure, seal material, fluid compatibility, and assembly torque.
4. **Build a complete BOM** with manufacturer part numbers and procurement sources. Recommended suppliers:
 - Electrical: Digi-Key, Mouser, Newark, Allied.
 - Fluid: Swagelok, Parker, McMaster-Carr, Grainger.
5. **Design review milestones:**
 - **PDR:** Present your connection schedule.
 - **CDR:** Present datasheets and torque specifications.
 - **FDR:** Demonstrate assembly procedures and test results.
6. **Test every connection before commissioning:** Pull-test electrical crimps, measure contact resistance with a milliohm meter, and pressure-test all fluid joints at $1.5\times$ working pressure.

7.5 Quick-Reference Summary

Five Key Takeaways

1. Electrical connectors are classified by *what they connect* (wire-to-wire, wire-to-board, board-to-board) and *how they terminate* (crimp, solder, IDC, screw, press-fit).
2. For every electrical connection, specify: current rating, voltage rating, IP rating, contact resistance, and mating cycles, and derate for temperature.
3. Fluid connectors require matching thread standards. $NPT \neq BSP$. JIC 37° flare and compression fittings avoid the pitfalls of tapered-thread sealing.
4. Failure modes differ by domain: electrical joints fail from fretting corrosion and I^2R overheating; fluid joints fail from cross-threading and seal degradation.
5. Every connector specification follows the same logic: **rating** $\rightarrow$ **environment** $\rightarrow$ **serviceability** $\rightarrow$ **cost** $\rightarrow$ **standards** $\rightarrow$ **space**. Document everything in your BOM.

Artifact	Description	Forward Reference
Connector Specifications	Selected connectors with ratings in BOM	Ch. 5 (ICDs)
Wire/Tube Routing	Harness design and fluid line layout	Assembly drawings

7.6 Artifact Handoff: What This Chapter Supports

Requirements Mapping: From Connector Specs to Shall-Statements

Connector specifications generate system requirements:

- “Outdoor deployment in rain” → SYS-REQ: All external connectors shall be rated IP67 or higher.
- “Field-replaceable sensor” → SYS-REQ: Sensor connections shall use keyed, tool-free connectors rated for ≥ 100 mating cycles.
- “Motor draws 8 A continuous” → CONN-REQ: Motor power connectors shall be rated for ≥ 10 A continuous (20% margin) with 12 AWG wire termination.
- “Hydraulic test pressure 200 psi” → CONN-REQ: All fluid fittings shall be rated for ≥ 800 psi working pressure ($4\times$ safety factor per hydraulic standard).

Every connector in your BOM must trace to a requirement. If you cannot write a shall-statement for it, you have not justified the selection.

7.7 Practice Questions for Quiz Preparation

Use these questions to test your understanding. Answers follow on the next page.

7.7.1 Conceptual Questions

1. What are the three fundamental elements of every electrical connector?
2. A connector is rated at 15 A per contact at 25 °C. Your design has an ambient temperature of 75 °C and all 12 contacts are loaded. Why should you *not* assume 15 A per contact is safe?
3. Explain why fretting corrosion is considered the “most insidious” electrical connector failure mode.
4. You need to connect a hydraulic line rated at 5000 psi that will be disconnected for maintenance monthly. Which fitting family is most appropriate? Why?
5. A colleague suggests using PTFE tape on a BSPP fitting. Is this correct? Explain.
6. What is the difference between rated working voltage and dielectric withstand voltage? Which should appear in your design documentation?
7. Why does NPS (Nominal Pipe Size) create confusion when specifying fittings?
8. Describe two mitigations for vibration fatigue in fluid tubing systems.
9. You measure 50 m Ω contact resistance on a connector carrying 8 A. How much power is dissipated as heat at that contact? Is this a concern?
10. Compare the sealing mechanisms of NPT, JIC 37° flare, and compression fittings. Which requires sealant?

7.7.2 Application / Scenario Questions

11. Your design team is designing an outdoor weather station. You need a 6-pin connector for sensor signals (all under 500 mA) that will be installed once and never disconnected. What IP rating, plating, and mating-cycle specification would you select? Justify your choices.
12. A pneumatic system in your project uses 1/4 in. OD nylon tubing at 80 psi. Recommend a fitting type and explain why.
13. Your design uses both NPT and BSP components (sourced from U.S. and European manufacturers). What steps must you take to ensure safe assembly?
14. You are specifying connectors for a cooling loop carrying a 50/50 ethylene glycol–water mix at 200 °F and 50 psi. What seal material would you select? What body material?
15. At your CDR, a reviewer asks: “How did you verify your crimp terminations are reliable?” Describe the test procedure you would reference.
16. **Diagnostic scenario:** Your team’s prototype passes a low-voltage continuity check on the bench (all connections show $<1\ \Omega$), but the system experiences intermittent sensor dropouts and a motor controller that occasionally faults when the prototype is running on a vibration table or driving over rough terrain. Where do you start your investigation, and what failure modes should you suspect? Describe the diagnostic steps you would take, referencing at least three concepts from this module.

7.7.3 Answers

1. Contacts (conductive pins/sockets), housing (insulating body), and termination (wire-to-contact joint).
2. Two derating effects compound: (a) elevated ambient temperature reduces the thermal headroom of the housing, and (b) thermal stacking from all 12 contacts being loaded raises the internal housing temperature further. The manufacturer's derating curve must be consulted; actual safe current per contact could be as low as 10 A or less.
3. Fretting corrosion develops slowly from micro-vibration, manifests as intermittent (not permanent) high resistance, is invisible without disassembly, and is difficult to reproduce during troubleshooting. By the time symptoms are consistent, oxide debris has accumulated extensively.
4. JIC 37° flare (SAE J514). It handles 10000 psi (well above 5000 psi), provides a metal-to-metal seal that doesn't degrade with reassembly, and is designed for repeated disconnect/reconnect without special tools beyond standard wrenches. Compression would also work but ferrules deform on initial assembly, making reassembly less reliable.
5. Incorrect. BSPP (G-type) uses parallel threads that do *not* seal on the threads. Sealing is achieved with a bonded washer or O-ring against a flat face. PTFE tape is used on *tapered* threads (NPT, BSPT) where the thread interference provides the primary seal. Applying tape to BSPP serves no useful purpose and may prevent proper washer compression.
6. Working voltage is the maximum voltage at which the connector is rated for continuous, safe operation, including creepage and clearance margins per IEC 61984. Dielectric withstand (breakdown) voltage is the voltage at which insulation fails; it is a destructive test limit, not an operating specification. Always specify working voltage in design documentation.
7. NPS is a nominal designation, not a physical dimension. For example, 1/2 in. NPS has an OD of 0.840 in. Engineers unfamiliar with this convention may attempt to match fittings by measuring OD, leading to incorrect specifications. Always reference the NPS table and verify with thread gauges.
8. (a) Install tube support clamps at manufacturer-recommended intervals to eliminate unsupported spans. (b) Include flexible hose segments at vibration sources (pumps, engines) to decouple cyclic motion from rigid tubing.
9. $P = I^2 R = (8)^2 \times 0.050 = 3.2 \text{ W}$. Yes, this is a concern; 3.2 W of localized heat will raise the contact and housing temperature significantly. This level of contact resistance suggests degraded contacts that should be inspected and potentially replaced.
10. NPT: Thread interference creates a wedge seal, but clearance at crests/roots requires supplemental sealant (PTFE tape or pipe dope). JIC 37°: Metal-to-metal cone seal at the flared tube end; no sealant required. Compression: Ferrule swaged onto tube OD creates the seal mechanically; no sealant required. Only NPT (and BSPT) requires sealant.
11. IP67 (dust-tight, survives 1 m immersion for rain/flooding protection). Gold plating (resists corrosion in outdoor environment with humidity and temperature cycling). 25–50 mating cycles (permanent installation, only disconnected for rare maintenance). Lower mating cycle connectors are less expensive and can be more robust.

12. Push-to-connect fittings. At 80 psi with nylon tubing, this is well within the 250 psi range of push-to-connect. Benefits: tool-free assembly, compatible with nylon tubing, easy reconfiguration, and widely available in 1/4 in. OD size from SMC, Festo, or John Guest.
13. (a) Create a thread identification chart for the project listing which ports are NPT and which are BSP. (b) Verify every connection with a thread pitch gauge ($60^\circ = \text{NPT}$, $55^\circ = \text{BSP}$) before assembly. (c) If adaptation is needed, use NPT-to-BSP adapter fittings from a reputable manufacturer. Never force-thread. (d) Label all ports on your assembly drawings with thread standard.
14. Seal: Viton (FKM); compatible with ethylene glycol, rated to 400°F (above the 200°F operating temperature). Body: stainless steel or brass (both compatible with glycol-water). Avoid carbon steel unless corrosion-protected, as glycol solutions can be mildly corrosive to unprotected ferrous metals.
15. Reference IPC/WHMA-A-620 standard. Procedure: (a) Pull-test a statistical sample of crimps at the specified force for the wire gauge and contact size. (b) Cross-section a sample crimp to verify the cold-weld zone under magnification. (c) Measure contact resistance across each mated pair with a milliohm meter; verify $<20\text{ m}\Omega$ for gold-plated contacts. (d) Document results with crimp force values, pass/fail criteria, and inspector sign-off.
16. This scenario synthesizes three interconnected failure modes. **Primary suspect: fretting corrosion.** The hallmark is a system that passes static bench testing but fails under dynamic load, exactly the intermittent, vibration-dependent behavior described in the fretting progression (Section 7.1.8, Figure 7.3). Diagnostic steps: (a) Measure contact resistance across every mated connector pair with a milliohm meter *while the system is subject to vibration* (tap on connectors, flex cables). Resistance that fluctuates from milliohms to ohms on disturbance confirms fretting. (b) Check for I^2R heating: use a thermal camera to identify hot connectors under load. A connector running 10 A through a fretting contact at $500\text{ m}\Omega$ dissipates 5 W, enough to soften a nylon housing. (c) Inspect strain relief: if cables transmit vehicle vibration directly to the contact zone, fretting is accelerated regardless of plating. (d) Check connector specifications: are tin-plated contacts being used in a high-vibration application where gold plating is warranted? Is the contact normal force sufficient? (e) For the motor controller faults specifically, consider whether the power connector is derated for the actual ambient temperature inside the enclosure under operation. Resolution: replace affected contacts with gold-plated equivalents, add proper strain-relief clamps, install secondary locking features (CPA clips), and retest under dynamic load with milliohm monitoring.

7.7.4 Application Scenario Answers

11. Redesigning a student project from soldered connections to connectors:

Step 1: Identify every wire-to-wire and wire-to-board connection in the system. Create a connection table: source, destination, signal type (power/signal/data), current, voltage, mating cycles expected.

Step 2: Group connections by subsystem boundary. Each subsystem-to-subsystem boundary gets one multi-pin connector (e.g., sensor module connects to controller via one 4-pin JST-XH: VCC, GND, SDA, SCL).

Step 3: Select connector families. For low-current signals ($<1\text{ A}$): JST-XH (wire-to-board, 2.5 mm pitch, rated 3 A, keyed). For power ($>1\text{ A}$): XT30 (rated 30 A, gendered, compact)

or Anderson Powerpole (genderless, stackable). For board-to-board: standard 2.54 mm pin headers with locking housings.

Step 4: Create a wiring diagram showing every connector with pin assignments, wire colors, and wire gauges. This diagram is a CDR deliverable.

Step 5: Label every connector pair with matching alphanumeric codes (J1/P1, J2/P2, etc.) on both the connector and the wiring diagram.

Benefits: any subsystem can be disconnected, tested independently, and reconnected. Failed modules can be swapped without soldering. The system can be disassembled for transport and reassembled on-site.

Chapter 8

Power Supplies, Motors & Pumps

8.1 Power Supply Selection

8.1.1 What a Power Supply Does

A power supply converts electrical energy from a source, AC mains, battery, solar panel, or USB, into one or more regulated DC voltage rails that your circuits and actuators require. Every component in your design has a specific voltage and current requirement. A typical student project might need 3.3 V for the microcontroller, 5 V for sensors and logic, and 12 V or 24 V for motors and solenoids. The power supply must deliver all of these simultaneously with adequate regulation and current capacity.

8.1.2 Linear Regulators (LDO)

Linear regulators are the simplest power supply topology. They are essentially a controllable series resistance that drops the input voltage to the desired output, dissipating the excess as heat.

$$P_{\text{waste}} = (V_{\text{in}} - V_{\text{out}}) \times I_{\text{load}}, \quad \eta = \frac{V_{\text{out}}}{V_{\text{in}}} \times 100\% \quad (8.1)$$

An LM7805 producing 5 V from 12 V at 1 A wastes 7 W as heat with only 42% efficiency. A low-dropout (LDO) regulator reduces the minimum input-to-output voltage difference to as little as 200 mV, improving efficiency when the differential is small.

When to Use an LDO

Use LDOs when: (1) the voltage drop is small ($< 2\text{ V}$), (2) current is under $\sim 1\text{ A}$, (3) noise matters (analog sensors, audio, ADCs), or (4) simplicity matters (only 2 capacitors, no inductor needed). Common devices: LM7805 (5 V/1 A), AMS1117-3.3 (3.3 V/1 A), MCP1700 (3.3 V/250 mA, ultra-low quiescent current).

8.1.3 Switching Regulators (DC-DC Converters)

Switching regulators chop the input voltage at high frequency (100 kHz–2 MHz) and use an inductor to transfer energy efficiently, achieving 85–95% efficiency.

Buck (Step-Down): $V_{\text{out}} < V_{\text{in}}$

The workhorse of DC-DC conversion. A switching FET, inductor, and capacitor step voltage down. Typical efficiency 85–95%. Common devices: LM2596 (3 A), MP1584 (3 A), TPS54302 (3 A). Pre-

built module boards are available for under \$2; recommended for student projects because the critical PCB layout is already done.

Example: 12 V battery $\rightarrow$ 5 V at 2 A. At 90% efficiency, waste heat is only 1.1 W compared to 14 W for a linear regulator.

Boost (Step-Up): $V_{\text{out}} > V_{\text{in}}$

Stores energy in an inductor during the switch-on phase, then releases it at higher voltage during switch-off. Typical efficiency 80–93%. Common devices: MT3608 (2 A), TPS61088 (10 A). Important limitation: most boost converters cannot fully disconnect the load from the input.

Buck-Boost: $V_{\text{out}} \geq V_{\text{in}}$

Handles input voltage that swings above and below the output; essential for battery applications. A LiPo cell ranges from 4.2 V (full) to 3.0 V (depleted). If your circuit needs 3.3 V, the input crosses the output during discharge. Only buck-boost works through this transition. Common: TPS63020 (3.6 A), LTC3780.

EMI from Switching Regulators

All switching regulators generate electromagnetic interference at their switching frequency. This can corrupt ADC readings, cause radio interference, and inject noise into audio circuits. Always follow the datasheet layout recommendations: keep switching traces short, use a continuous ground plane, and place input/output capacitors as close to the regulator as physically possible. When powering noise-sensitive circuits, add an LDO downstream of the switcher as a noise filter.

8.1.4 AC-DC Power Supplies

For systems that plug into AC mains (120/240 VAC), you need an AC-to-DC conversion stage with galvanic isolation to separate lethal mains voltage from safe low-voltage outputs.

Student Project Rule

Never design your own AC-DC converter. Always use a pre-certified wall adapter (UL/CE listed) that provides a regulated DC output. These are fully enclosed, safety-certified, and eliminate all mains-voltage hazards from your design. If a wall adapter cannot meet your requirements, use an enclosed industrial supply (Mean Well, TDK-Lambda) with proper enclosure design.

LiPo Battery Safety: This Is Not Optional

A lithium-polymer battery that is overcharged, over-discharged, punctured, or short-circuited can undergo thermal runaway: **fire and toxic gas release** within seconds. A LiPo fire in a campus lab is a safety emergency that can end careers and close facilities. **Mandatory rules for MEGN 301:**

- **Never** charge a LiPo without a functional Battery Management System (BMS) with cell-level overvoltage, undervoltage, and overcurrent protection.
- **Never** leave a charging LiPo unattended.
- **Always** charge and store LiPo batteries in a fireproof LiPo safety bag.

- **Never** use a battery with a puffy or swollen pouch; it is generating gas internally.
- Violation of these rules is grounds for immediate lab access revocation.

8.1.5 Key Power Supply Specifications

Specification	Description
Output Voltage	Nominal output with tolerance (e.g., 5.0 V $\pm 2\%$). Must match each load rail.
Output Current	Max continuous delivery. Must exceed total load current with $\geq 20\%$ margin.
Regulation	Line regulation: output change when input varies. Load regulation: output change when load varies. Expressed as % or mV.
Ripple & Noise	Peak-to-peak AC on the DC output. Switchers: 20–100 mV. LDOs: <1 mV.
Efficiency	$P_{\text{out}}/P_{\text{in}} \times 100\%$. LDO: 40–80%. Buck: 85–95%. Determines waste heat and battery life.
Isolation & Safety	Galvanic isolation for AC-DC. Look for UL/CSA/CE marks.

Table 8.1: Key power supply specifications.

8.1.6 Power Supply Failure Modes

- **Overcurrent:** Motor inrush can be 6–10 $\times$ running current. Size for peak, not average.
- **Thermal failure:** Every supply has a derating curve; output current decreases with ambient temperature.
- **Input transients:** Voltage spikes from motor switching. Mitigate with input capacitors and TVS diodes.
- **Capacitor aging:** Electrolytic caps dry out, increasing ripple. Use 105 °C rated caps.

8.2 DC Motors

8.2.1 Motor Fundamentals

Every DC motor operates on the Lorentz force: a current-carrying conductor in a magnetic field experiences a force. Two equations govern motor behavior:

$$\tau = K_t \times I \quad (\text{torque is proportional to current}) \quad (8.2)$$

$$V_{\text{back-EMF}} = K_e \times \omega \quad (\text{back-EMF is proportional to speed}) \quad (8.3)$$

At stall ($\omega = 0$), back-EMF is zero and current is limited only by winding resistance: $I_{\text{stall}} = V/R_{\text{winding}}$. This is typically 6–10 $\times$ the rated running current and causes rapid overheating.

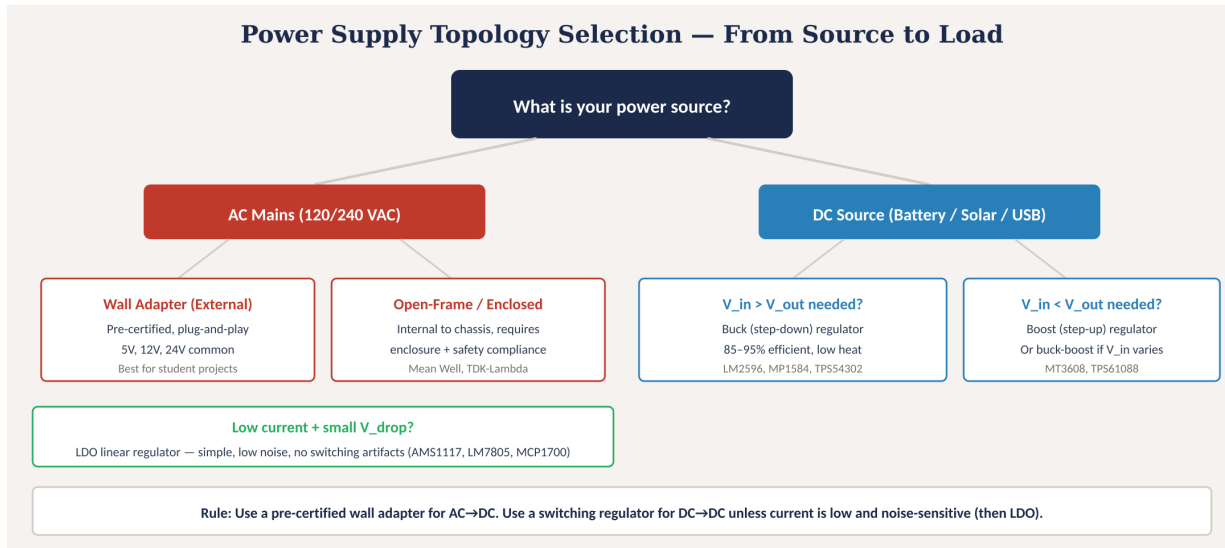


Figure 8.1: Power supply topology decision tree, from source type through voltage conversion to specific regulator selection.

8.2.2 Brushed DC Motors

Simplest motor type. Mechanical commutation via carbon brushes on a copper commutator. Apply voltage, it spins; reverse polarity, it reverses. Speed control via PWM. The torque-speed curve is linear from stall torque at zero speed to no-load speed at zero torque.

Advantages: Cheap, simple to drive (just PWM + H-bridge), widely available.

Limitations: Brush wear limits lifetime (1,000–5,000 hours), EMI from brush arcing, moderate efficiency (70–85%).

8.2.3 Brushless DC (BLDC) Motors

Permanent magnets on the rotor, three-phase windings on the stator. An Electronic Speed Controller (ESC) sequentially energizes the stator phases. The motor *cannot function without its ESC*; they are an inseparable system.

Advantages: 85–95% efficiency, 10,000+ hour life, higher power density, flat torque curve in the mid-range, lower EMI.

Limitations: Requires ESC (adds cost and complexity), cannot be driven directly from DC, commutation tuning matters.

8.2.4 Stepper Motors

Rotate in discrete angular steps (typically $1.8^\circ = 200$ steps/rev). Open-loop positioning: count steps to know position without an encoder. NEMA 17 is the standard frame size for student projects. Driver boards (A4988, TMC2209) accept step/direction signals from any microcontroller.

Advantages: Precise positioning, high holding torque, no encoder needed.

Limitations: Torque drops rapidly with speed, runs hot (always energized), skips steps silently under overload.

8.2.5 Servo Motors

A motor + encoder + controller in a closed-loop system. Hobby servos (SG90, MG996R) are self-contained with PWM control. Industrial servos use BLDC motors with high-resolution encoders and dedicated controllers running PID loops at kHz rates.

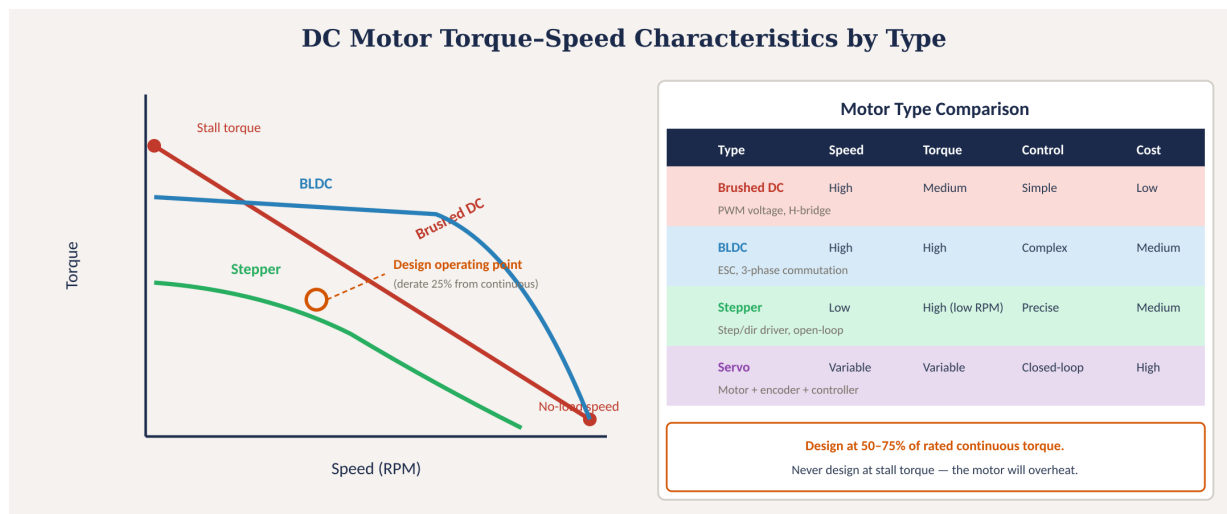


Figure 8.2: Torque-speed characteristics for brushed DC (linear), BLDC (flat mid-range), and stepper (high-torque at low RPM) with comparison table.

8.2.6 Key Motor Specifications

Specification	Description
Rated Voltage	Nominal operating voltage. Must match your supply rail.
No-Load Speed	Max speed with no load (RPM). Actual speed under load is lower.
Stall Torque	Max torque at zero speed. NOT your design point ; stall current overheats the motor.
Rated / Continuous Torque	Torque the motor delivers indefinitely. Your design ceiling. Derate to 50-75%.
Rated Current	Current at rated torque. Sizes your driver and power supply.
Torque Constant (K_t)	Torque per ampere (N·m/A). Calculate required current: $I = \tau_{\text{load}} / K_t$.

Table 8.2: Key motor specifications.

Never Design at Stall Torque

Stall current = $V_{\text{rated}} / R_{\text{winding}}$, typically 6-10× the rated running current. At stall, all electrical power converts to heat in the windings. A motor will overheat and fail within seconds to minutes. Always design at 50-75% of the *continuous* rated torque.

8.2.7 Motor Drive Electronics

Every motor type needs a matched driver circuit:

- **H-Bridge (Brushed DC):** Four transistors for bidirectional current. L298N (2A, legacy), BTS7960 (43A), DRV8871 (3.6A). **Must include flyback diodes**; back-EMF from inductive loads will destroy unprotected transistors.
- **ESC (BLDC):** Converts DC + throttle signal to three-phase AC. Drone ESCs are cheap and widely available. Use bidirectional or 4-quadrant ESC for reversing.
- **Stepper Driver:** Converts step + direction signals into motor phase currents. A4988 (2A), DRV8825 (2.5A), TMC2209 (2A, silent). **Set the current limit potentiometer before powering on.**
- **Servo Controller:** Hobby servos need only a PWM signal (1–2 ms pulse, 50 Hz). Industrial servos require dedicated controllers.

8.3 Pumps

8.3.1 Positive Displacement vs. Dynamic

Positive displacement (PD) pumps trap a fixed volume per cycle and force it into the discharge. Flow is proportional to speed and nearly independent of pressure. Self-priming. Pulsating flow. Types: gear, diaphragm, peristaltic, piston, vane.

Dynamic (centrifugal) pumps impart kinetic energy to the fluid via a spinning impeller. Flow varies with discharge pressure (follows a pump curve). Smooth, continuous flow. *Not* self-priming. Best for high-volume, lower-pressure applications.

8.3.2 Pump Types

Gear Pumps

Two meshing gears trap fluid between teeth and housing. Self-priming, handles viscous fluids (oil, glycol). Constant flow rate. Used in hydraulic systems and lubrication. Primary failure mode: wear from particle contamination. Always use inline filtration.

Diaphragm Pumps

A flexible diaphragm oscillates to create suction and discharge. Seal-less design isolates fluid from moving parts. Self-priming, handles corrosive and abrasive fluids. Pulsating flow requires dampeners.

Peristaltic Pumps

Rollers compress a flexible tube, pushing fluid forward. Zero contamination risk; fluid only contacts the tube. Easy maintenance: swap the tube. Gentle on shear-sensitive fluids. Limited to $\sim$ \$30 psi. Tube fatigue is the primary maintenance concern.

Centrifugal / Impeller Pumps

Spinning impeller accelerates fluid radially. Smooth, high-volume flow. Requires flooded suction (not self-priming). Poor with viscous fluids. Small 12V DC impeller pumps are widely available for student cooling and circulation projects.

Pump Selection Matrix — Type, Performance Envelope, and Best Application					
Pump Type	Category	Flow / Pressure	Strengths	Limitations	Best For
Gear	Positive displacement	Med flow High pressure	Self-priming, constant flow, handles viscous	Wear from particles, not for solids	Hydraulics, lubrication
Diaphragm	Positive displacement	Low-med flow Med pressure	Seal-less, handles corrosives, self-priming	Pulsating flow, diaphragm fatigue	Chemical, metering
Peristaltic	Positive displacement	Low flow Low pressure	No contamination, easy tube swap, reversible	Tube wear, pulsating, limited pressure	Biomedical, food/beverage
Centrifugal	Dynamic (rotodynamic)	High flow Low-med pressure	Smooth flow, high volume, low maintenance	Not self-priming, poor w/ viscous fluids	Cooling, water circulation
Vacuum / Air	Varies (diaphragm/vane)	Low flow Negative pressure	Creates suction for gripping, sampling	Noise, limited vacuum depth, heat	Pick-and-place, gas sampling
Selection priority: Flow rate → Pressure → Fluid compatibility → Self-priming need → Pulsation tolerance → Budget					

Figure 8.3: Pump selection matrix; type, performance envelope, strengths, limitations, and best applications.

8.3.3 Key Pump Specifications

Specification	Description
Flow Rate	Volume per unit time (L/min, mL/min). Must meet demand with $\geq 25\%$ margin.
Discharge Pressure / Head	Max pressure (psi/bar) or total dynamic head (ft/m). Must exceed system losses.
Fluid Compatibility	Wetted materials must be compatible with the pumped fluid. Check seal charts.
Self-Priming	PD pumps: yes. Centrifugal: no (must be flooded).
Motor Integration	Many small pumps include integrated 12V/24V DC motors; specify the assembly.

Table 8.3: Key pump specifications.

8.3.4 Worked Example: Motor Selection for a Belt Conveyor

Requirements: Move 1 kg payload at 0.1 m/s on a horizontal belt conveyor. Belt drive with 50 mm diameter driven roller.

Step 1: Calculate load torque. Conveyor force = payload weight + belt friction = $(1.0 \times 9.81) + (1.0 \times 9.81 \times 0.3) = 12.75 \text{ N}$ (assuming $\mu = 0.3$ for belt on roller).

Torque at roller = $F \times r = 12.75 \times 0.025 = 0.319 \text{ N}\cdot\text{m}$.

Step 2: Calculate required speed. Roller angular velocity = $v/r = 0.1/0.025 = 4.0 \text{ rad/s} = 38.2 \text{ RPM}$.

Step 3: Select motor with margin. Required motor torque (with $2\times$ margin): $0.319 \times 2 = 0.638 \text{ N}\cdot\text{m}$ at 38 RPM. A typical NEMA 17 stepper (0.44 N·m rated) is insufficient. Options: (a) 12 V DC gearmotor with 50:1 ratio (Pololu 37D: 0.8 N·m at 50 RPM, \$25), or (b) NEMA 17 stepper with 5:1 planetary gearbox ($0.44 \times 5 \times 0.95 = 2.09 \text{ N}\cdot\text{m}$ at 40 RPM, \$40).

Step 4: Verify power budget. $P = \tau \times \omega = 0.319 \times 4.0 = 1.28 \text{ W}$ mechanical. At 70% motor efficiency: $P_{\text{electrical}} = 1.28/0.70 = 1.83 \text{ W}$. At 12 V: $I = 1.83/12 = 0.15 \text{ A}$ running. Stall current (Pololu 37D): 5.5 A. Power supply must handle 5.5 A inrush.

Step 5: Select driver. DC gearmotor: L298N H-bridge (2 A continuous, inadequate for 5.5 A stall). Better: BTS7960 (43 A, \$8). Include flyback diodes (integrated in BTS7960).

BOM entries: Motor: Pololu 4756 (12V 50:1 37D gearmotor), \$24.95. Driver: BTS7960 module, \$7.50. Mounting: Pololu 1084 bracket, \$7.49.

8.4 Selection and Design Practice

8.4.1 Unified Selection Framework

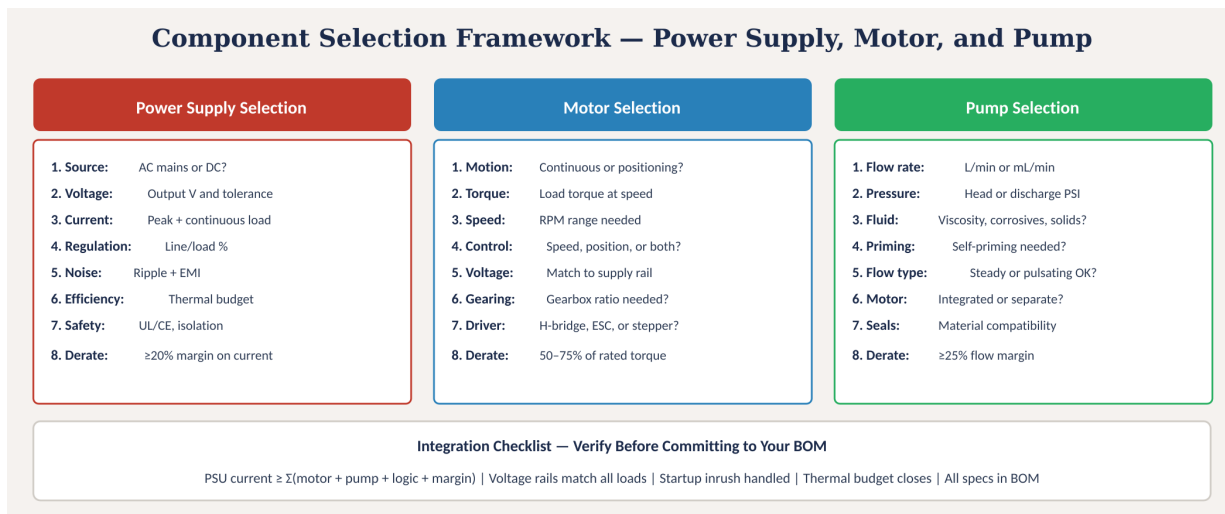


Figure 8.4: Parallel selection flows for power supply, motor, and pump with integration verification checklist.

8.4.2 Design Rules

- Build a power budget spreadsheet first.** List every component with its voltage rail, continuous current, and peak current. Sum the columns. Add 25% margin.
- Calculate mechanical load torque before selecting a motor.** $\tau = F \times r$ for linear-to-rotary. Include friction, inertia, and acceleration torque. Select a motor with rated torque $\geq 1.5 \times$ your load.
- For fluid systems, map your complete flow circuit,** pipe lengths, fittings, elevation, filters, to calculate total system head loss. Select a pump whose curve crosses your system curve with margin.
- Motor inrush is the #1 power supply problem.** A motor rated at 2 A running may draw 15 A at startup. Size your supply for inrush, or use soft-start circuits.
- Separate power and signal grounds.** Run motor power on dedicated wires back to the supply. Never share ground wires between motors and sensitive electronics.

6. **Document everything in your BOM:** part number, manufacturer, voltage, current, torque, flow rate, and procurement source.

Five Key Takeaways

1. Power supplies: use pre-certified wall adapters for AC-DC; switching regulators for efficient DC-DC; LDOs only for low-current, noise-sensitive loads.
2. Motors: match type to control need; brushed DC for simplicity, BLDC for efficiency, stepper for open-loop positioning, servo for closed-loop precision. Every motor needs a driver.
3. Pumps: positive displacement for precise flow and self-priming; centrifugal for high-volume smooth flow. Match wetted materials to your fluid.
4. Derating is non-negotiable: 20% current margin on power supplies, 25–50% torque margin on motors, 25% flow margin on pumps.
5. Integration: build a power budget first, size for inrush, separate power and signal grounds, document every selection in your BOM.

8.5 Artifact Handoff: What This Chapter Supports

Artifact	Description	Forward Reference
Power Budget	Voltage rails, current draws, margin analysis	Ch. 9 (protection), Ch. 5 (integration)
Motor Selection	Torque/speed calcs, driver selection, BOM entry	Ch. 11 (transmission), BOM
Pump Selection	Flow/pressure calcs, wetted material verification	BOM, test plan

8.6 Practice Questions

8.6.1 Conceptual Questions

1. What is the fundamental difference between how a linear regulator and a switching regulator convert voltage?

Requirements Mapping: From Motor Curves to Shall-Statements

A motor's torque–speed curve directly generates testable requirements:

- “System shall move payload at 0.5 m/s” → Calculate wheel RPM from diameter, verify the motor's torque at that RPM exceeds load torque with $\geq 50\%$ margin.
- “Motor runs continuously” → MOT-REQ: Motor current at operating point shall be $\leq 75\%$ of continuous rated current. Calculate: $I_{req} = \tau_{load}/K_t$.
- “Battery life 4 hours” → PSU-REQ: Battery capacity $\geq I_{avg} \times 4 \times 1.3$ (aging margin).
- “Pump delivers 2 L/min at 10 psi” → PUMP-REQ: Pump shall deliver ≥ 2.5 L/min at ≥ 12 psi system head loss (25% margin).

If your requirement says “0.5 m/s” but the motor cannot produce the required torque at the corresponding RPM, the requirement and hardware are in conflict. The torque–speed curve resolves this *before* you order parts.

2. A 12 V motor has a winding resistance of $0.5\ \Omega$ and a rated current of 4 A. What is the stall current? Why is this dangerous?
3. Explain why a BLDC motor cannot function without an ESC, while a brushed DC motor can operate with just a battery connection.
4. Your project needs to regulate a LiPo battery (3.0–4.2 V) down to 3.3 V. Why would a standard buck converter fail during battery discharge?
5. Compare the self-priming capability of gear pumps versus centrifugal pumps. How does this affect installation design?
6. Why must you include flyback diodes when driving a motor with an H-bridge? What happens without them?
7. A power supply is rated at 10 A at 25°C . If your enclosure reaches 50°C , can you still draw 10 A? Explain.
8. What is the primary advantage of a peristaltic pump over a gear pump for biomedical applications?

8.6.2 Application / Scenario Questions

9. Your design uses a 24 V wall adapter powering two DC motors (24 V, 3 A each running, 20 A stall each) and an Arduino (5 V, 200 mA). Design the power architecture: what regulators are needed, what is the minimum supply current rating, and how do you handle motor inrush?
10. You need to pump ethylene glycol coolant at 2 L/min through a closed cooling loop with 3 m of tubing, six fittings, and no elevation change. Which pump type do you recommend and why? What seal material?
11. Your team selects a NEMA 17 stepper for a linear stage. The required force is 20 N at 50 mm/s using a 10 mm lead screw. Calculate the required torque and verify whether a standard NEMA 17 (0.4 N·m rated) is adequate.
12. **Diagnostic scenario:** Your prototype's motor operates normally when powered from a bench supply but stutters and resets the microcontroller when powered from the project's 12 V battery pack. The battery is fully charged. What is the most likely cause, and how do you fix it?

8.6.3 Answers

1. A linear regulator dissipates excess voltage as heat (acts as a variable resistor). A switching regulator rapidly switches the input on/off and uses an inductor to transfer energy efficiently, achieving 85–95% efficiency versus 40–80% for linear.
2. $I_{\text{stall}} = V/R = 12/0.5 = 24\ \text{A}$; six times the rated current. At stall, all 288 W (24×12) converts to heat in the windings. The motor will overheat and fail within seconds.
3. A brushed DC motor has mechanical commutation (brushes and commutator) that automatically reverses current in the rotor windings. A BLDC motor's rotor has permanent magnets and the stator windings must be energized in a precise three-phase sequence by the ESC, which reads rotor position via Hall sensors or back-EMF sensing. Without the ESC, there is no commutation and no rotation.

4. A buck converter requires $V_{\text{in}} > V_{\text{out}}$ to regulate. When the LiPo discharges below 3.3 V (which happens as the cell depletes toward 3.0 V), the buck cannot maintain the output. A buck-boost converter is required because it handles the input crossing the output voltage.
5. Gear pumps are self-priming: they can draw fluid up from below the pump inlet by creating suction through the gear tooth spaces. Centrifugal pumps are not self-priming and require a flooded suction (pump inlet below fluid level). This means centrifugal pumps must be installed below the tank or use a foot valve and check valve to maintain prime.
6. A motor is an inductive load. When current is suddenly switched off, the collapsing magnetic field generates a voltage spike (back-EMF) that can be many times the supply voltage. Without flyback diodes, this spike destroys the H-bridge transistors. Flyback diodes clamp the spike by providing a current path for the decaying magnetic energy.
7. No. Every power supply has a thermal derating curve. At 50 °C, the supply has less thermal headroom to dissipate heat. A typical derating might reduce capacity to 70–80% of the 25 °C rating, meaning 7–8 A. Always check the datasheet derating curve.
8. The fluid in a peristaltic pump only contacts the inside of a disposable tube; no seals, rotors, or metal surfaces touch the fluid. This eliminates contamination risk and allows easy sterilization by swapping the tube. A gear pump has fluid contact with gears, housing, and shaft seals, making sterilization far more difficult.
9. Architecture: wall adapter provides 24 V rail directly to motors via H-bridges. A buck converter (e.g., LM2596) steps 24 V down to 5 V for the Arduino. Minimum supply current: continuous = $2 \times 3 + 0.2 = 6.2 \text{ A}$, but inrush = $2 \times 20 = 40 \text{ A}$ simultaneously. A 10 A supply with bulk capacitance ($\geq 4700 \mu\text{F}$) on the motor rail to absorb inrush, plus staggered motor startup in firmware (start one motor, wait 500 ms, start the second), is the practical solution.
10. A small DC centrifugal impeller pump is appropriate: the loop is closed (no priming issue if pump is at the bottom), flow is moderate, ethylene glycol is low viscosity, and smooth non-pulsating flow is preferred for cooling. Seal material: Viton (FKM) or EPDM; both compatible with ethylene glycol at moderate temperatures.
11. Torque = force $\times$ lead / ($2\pi \times$ efficiency). $\tau = 20 \times 0.010 / (2\pi \times 0.9) = 0.035 \text{ N}\cdot\text{m}$. The NEMA 17 at 0.4 N·m rated torque provides $0.4/0.035 = 11.4\times$ margin. However, at the required speed: $50 \text{ mm/s} / 10 \text{ mm/rev} = 5 \text{ rev/s} = 300 \text{ RPM}$. Check the stepper's torque-speed curve at 300 RPM; torque typically drops to 40–60% of holding torque, giving 0.16–0.24 N·m. Still adequate ($4.6\text{--}6.9\times$ margin), but verify on the actual datasheet curve.
12. Most likely cause: the battery cannot supply the motor inrush current. Unlike a bench supply (which can deliver virtually unlimited current), a battery has internal resistance. During motor startup, the inrush current (6–10 $\times$ running) causes a voltage drop across the battery's internal resistance, drooping the 12 V rail. If it drops below the buck converter's minimum input or the microcontroller's brown-out threshold ($\sim \$2.7 \text{ V}$ for most AVR/ARM), the MCU resets. Fix: add a large bulk capacitor ($\geq 4700 \mu\text{F}$) on the battery rail close to the motor driver, use a battery with lower internal resistance (higher C-rating), implement soft-start in firmware, and add a separate regulated supply for the MCU with adequate input capacitance.

Chapter 9

Power Architecture & Circuit Safety

9.1 Power Architecture

9.1.1 System-Level Design

A well-designed power system flows from source through three functional stages before reaching the loads. The **input stage** is where power enters through a single, well-defined connector. This is where you place first-line protections: a keyed connector (prevents miswiring), reverse polarity protection, an input fuse, and a TVS diode for transient suppression. The **regulation stage** converts raw input voltage into the specific rails your system needs, commonly cascading from the highest voltage down (e.g., 24 V input → 12 V motor rail → 5 V sensor rail → 3.3 V MCU rail). The **distribution stage** delivers regulated power to each load through individual protection, per-rail fuses or electronic current limiters that prevent a fault in one subsystem from cascading.

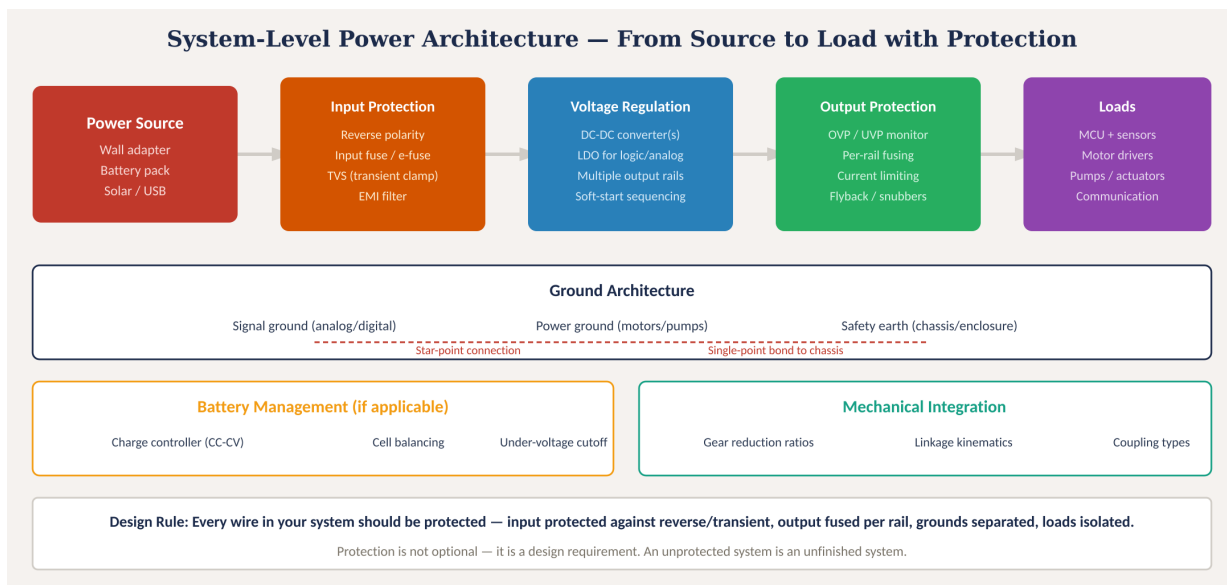


Figure 9.1: System-level power architecture from source to load with protection stages, ground architecture, battery management, and mechanical integration.

9.1.2 Grounding Strategy

Ground is not a magical zero-volt reference; it is a real conductor with real resistance and inductance. When a motor draws 5 A through a ground trace with 50 mΩ resistance, there is a 250 mV

drop along that trace. If your ADC's ground reference shares that path, your sensor readings are corrupted.

Star-Point Grounding

Run separate ground conductors from each subsystem back to a single point near the power supply. Motor ground, sensor ground, and MCU ground each get their own conductor. They meet at *one star point*. This prevents motor return current from flowing through sensor ground paths.

Three Ground Domains

- **Signal ground:** analog sensors, ADCs, digital logic. Low current, noise-sensitive.
- **Power ground:** motors, pumps, solenoids, heaters. High current, noisy.
- **Safety earth:** metal enclosure, chassis. Fault-current path for AC-powered systems.

The #1 Debugging Time Sink

Poor grounding is the single most common cause of difficult-to-diagnose intermittent failures in student projects. If your sensor readings are noisy, your MCU resets when a motor starts, or your serial communication drops characters; check your grounding first.

AC Mains Safety

Use a Class II (double-insulated) wall adapter whenever possible; no earth ground connection is needed. If you must use an internal open-frame supply, bond the metal enclosure to earth ground through a 3-prong inlet with proper strain relief. **Never defeat a ground prong.**

Signal Isolation

Use optocouplers or digital isolators (Si8620, ADUM1201) to pass signals between high-voltage and low-voltage domains. Essential when controlling AC loads (relays, triacs) from a microcontroller.

9.1.3 Battery Integration

Lithium cells (LiPo, Li-ion, LiFePO₄) store enormous energy density and can cause fire or explosion if mishandled. A Battery Management System (BMS) is **mandatory**:

- Overcharge cutoff: 4.2 V/cell maximum.
- Deep-discharge cutoff: 3.0 V/cell minimum (irreversible damage below this).
- Overcurrent / short-circuit shutoff.
- Cell balancing for multi-cell packs.

Common modules: TP4056 (single-cell charger, CC-CV profile), BQ25895 (buck-boost charger IC), BQ76940 (multi-cell BMS). During development, always store and charge lithium batteries in a fireproof LiPo-safe bag. **Never charge unattended without a BMS.**

DC Generation Sources (Solar, Alternator)

Solar panels produce variable voltage; use an MPPT charge controller to extract maximum power. Always include a blocking diode to prevent battery discharge through the panel at night. For alternators: rectify the AC output, regulate with a charge controller, and add TVS/MOV protection against load-dump voltage spikes if the battery disconnects while the

alternator is running.

9.1.4 Power Budget Spreadsheet: How to Build One

The power budget is a spreadsheet with one row per component and the following columns:

Component	Rail	I_{typ} (mA)	I_{peak} (mA)	Duty (%)	I_{avg} (mA)
ESP32 (active)	3.3V	80	240	10	8.0
ESP32 (deep sleep)	3.3V	0.01	—	90	0.009
DS18B20 sensor	3.3V	1.5	1.5	100	1.5
SIM800L (TX)	4.0V	200	2000	1	2.0
Relay coil	12V	80	80	5	4.0
3.3V rail total					9.5
3.3V rail peak					241.5
12V rail total					4.0

Table 9.1: GreenShield power budget example.

The peak column sizes your regulators and fuses. The average column sizes your battery. Add 25% margin to both. If the SIM800L's 2 A peak on the 4 V rail shares a regulator with the ESP32, that regulator must handle 2.24 A peak; a common sizing mistake.

9.2 Protection Circuits

The philosophy is **defense in depth**: layer multiple protections so that each catches what the previous one missed. No single device guards against all threats.

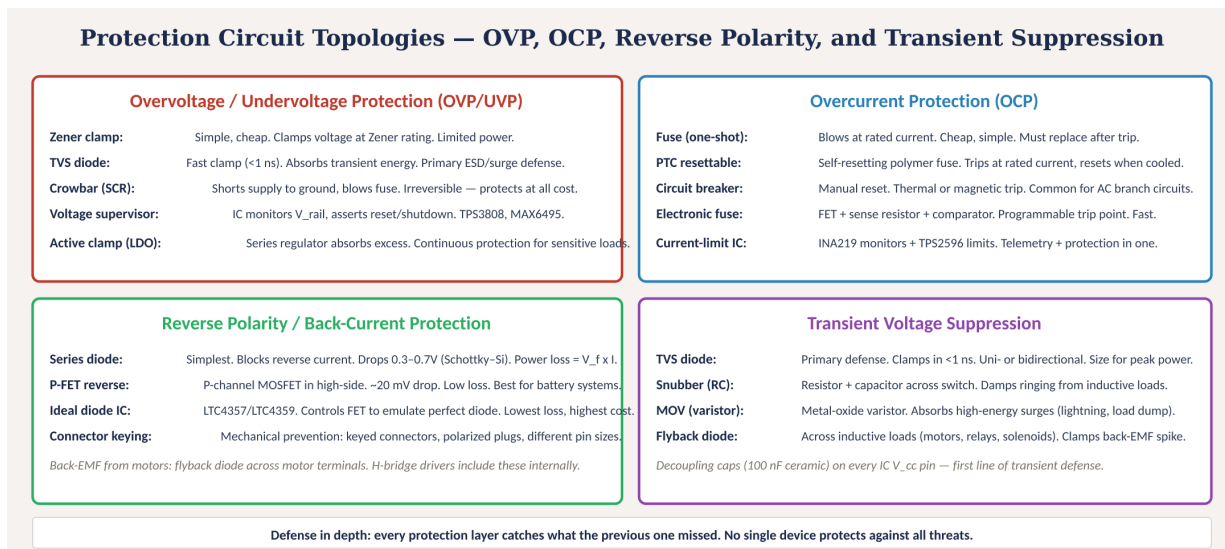


Figure 9.2: Four protection categories: OVP/UVLP, OCP, reverse polarity/back-current, and transient suppression with specific device options.

9.2.1 Overvoltage and Undervoltage Protection (OVP / UVP)

Overvoltage pushes components beyond absolute maximum ratings; gate oxide breakdown, latch-up, permanent damage. **Undervoltage** causes unpredictable behavior: brown-outs, corrupted flash memory, and erratic execution.

Zener Clamp

A Zener diode across the supply rail conducts when voltage exceeds its breakdown value, clamping the rail. Simple and cheap but limited by power dissipation ($P = (V_{\text{spike}} - V_Z) \times I$). Select V_Z slightly above your nominal rail (e.g., 5.6 V Zener for a 5 V rail).

TVS Diode

A Transient Voltage Suppressor; essentially a heavy-duty bidirectional Zener optimized for high peak power. Responds in <1 ns. The primary defense against ESD, inductive spikes, and power supply transients. **Every power input in a professional design has a TVS diode.**

Selection: standoff voltage $\geq$ operating voltage; clamping voltage $<$ component absolute maximum rating. Common series: SMAJ, SMBJ, 1.5KE.

Voltage Supervisor IC

Monitors the supply rail and asserts a reset or shutdown signal when voltage is out of range. Does not clamp; it shuts the system down gracefully. Essential for microcontroller brown-out protection. Common: TPS3808 (adjustable OVP/UVP), MAX6495 (OVP with FET control), TLV803 (simple reset).

Crowbar Circuit

An SCR (thyristor) fires when voltage exceeds a threshold, shorting the supply to ground and blowing the upstream fuse. **Irreversible**; the fuse must be replaced. Used as a last resort when the overvoltage source has enough sustained energy to overwhelm a clamp.

9.2.2 Overcurrent Protection (OCP)

Every power rail must have current limiting. A short circuit without protection delivers the full energy of the supply into the fault.

Fuses (One-Shot)

A fusible element melts at its rated current. **Fast-blow** for sensitive electronics; **slow-blow** (time-delay) for motor circuits where inrush exceeds steady-state. The fuse must blow *before* the wire or component is damaged. Must be accessible for replacement.

PTC Resettable Fuses

Polymer PTC (positive temperature coefficient) devices. Resistance increases dramatically at the trip point, limiting current. Self-resets when cooled. Slower than traditional fuses. Best for USB ports and sensor circuits where the fault is intermittent.

Electronic Fuses (eFuse)

A MOSFET in series with the load, controlled by a current-sense amplifier. Programmable trip point, fast response, automatic retry or latched shutdown. Provides current *limiting* (not just cutting); can hold current at a safe level during motor inrush. Common: TPS2596, LTC4368, MAX14690.

Circuit Breakers

Thermal or magnetic trip mechanism with manual reset. Common for AC branch circuits (15 A, 20 A per NEC). In DC systems, used for battery main disconnects. **Must be DC-rated**; an AC breaker may not safely interrupt DC arcs because DC does not cross zero.

Fuse + Wire Sizing Rule

The fuse must blow before the wire overheats. This means the fuse rating must be *below* the wire's ampacity. Size them together using NEC Table 310.16 or AWG ampacity charts: the fuse protects the wire, the wire carries the current, and the load defines the requirement.

9.2.3 Reverse Polarity and Back-Current Protection

A battery connected backwards forward-biases body diodes of every downstream MOSFET and IC, often destroying the entire board in milliseconds.

Series Diode

Simplest approach. A Schottky diode (\$ ~ \$0.3 V drop) or silicon diode (\$ ~ \$0.7 V drop) in the positive rail blocks reverse current. Power loss = $V_f \times I_{\text{load}}$. Best for low-current (<2 A) circuits where the voltage drop is acceptable.

P-Channel MOSFET

A P-FET in the high-side power path conducts when polarity is correct (body diode initially conducts, pulling gate low, fully enhancing the FET) and blocks when reversed. Voltage drop = $I \times R_{DS(\text{on})}$, typically 10–50 mΩ. At 5 A with 20 mΩ: only 0.1 V drop, 0.5 W loss. **Best for battery-powered systems**. Common: Si2301, IRLML6402, DMG2307L.

Ideal Diode IC

An IC controlling an external N-FET to emulate a perfect diode with \$ ~ \$20 mV forward drop. Lowest loss, fastest reverse blocking, highest cost. LTC4357, LTC4359, MAX40200. Also used for OR-ing redundant supplies and solar/battery combining.

Mechanical Prevention

Keyed connectors, polarized plugs, and different-sized pins physically prevent reverse connection. The first line of defense. Always use polarized connectors for DC power.

9.2.4 Transient Voltage Suppression

Every time you switch an inductive load, motor, relay, solenoid, you create a voltage transient. The physics: $V = -L \times dI/dt$. Current cannot change instantaneously in an inductor, so the collapsing magnetic field generates a spike that can reach hundreds of volts from a 12 V motor.

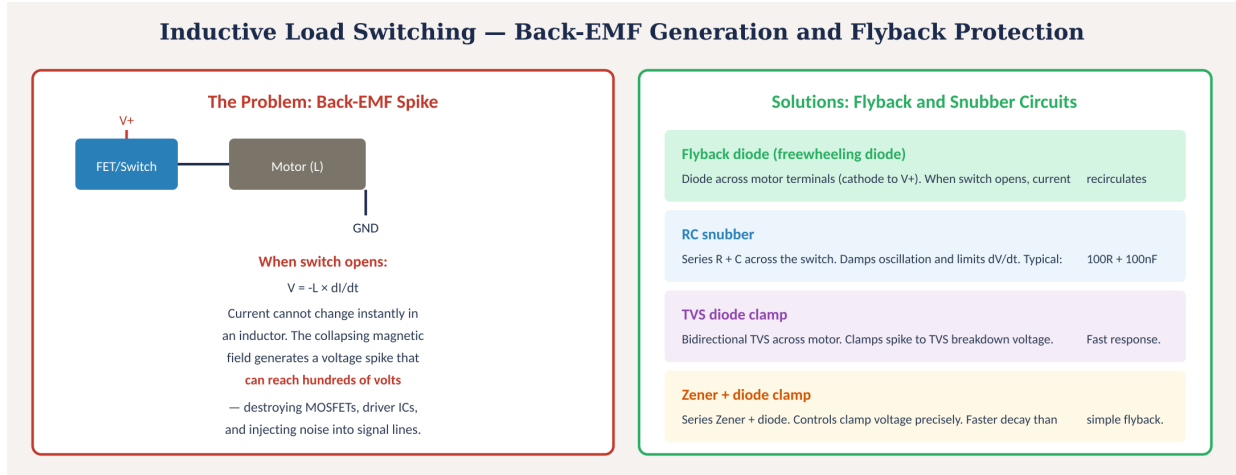


Figure 9.3: Back-EMF generation from inductive load switching and four protection solutions: flyback diode, RC snubber, TVS clamp, and Zener+diode.

Flyback Diode (Freewheeling Diode)

The single most important protection for inductive loads. A diode across the motor, relay, or solenoid (cathode to V+) provides a current path when the switch opens. Without it, the collapsing field generates a spike that destroys the switching device. Use a fast-recovery or Schottky diode. H-bridge driver ICs include these internally, but if you drive a relay or solenoid with a bare MOSFET, you **must** add one externally.

RC Snubber

A resistor-capacitor network across a switch or relay contact. The capacitor absorbs the initial spike; the resistor dissipates energy and damps oscillation. Typical starting values: 100 Ω + 100 nF for relay contacts. Tune by observing the transient on an oscilloscope.

TVS Diode (Board-Level)

Place a TVS across each power rail at the board edge where external cables connect. Absorbs energy from ESD, cable transients, and motor switching spikes propagating through the power bus.

Decoupling Capacitors

100 nF ceramic capacitors on **every** IC V_{CC} pin, as close to the pin as physically possible. These absorb high-frequency transients before they reach the IC. For power regulators, add bulk capacitance (10–100 μ F) per the datasheet. **Decoupling is not optional.**

9.3 Electromechanical Integration

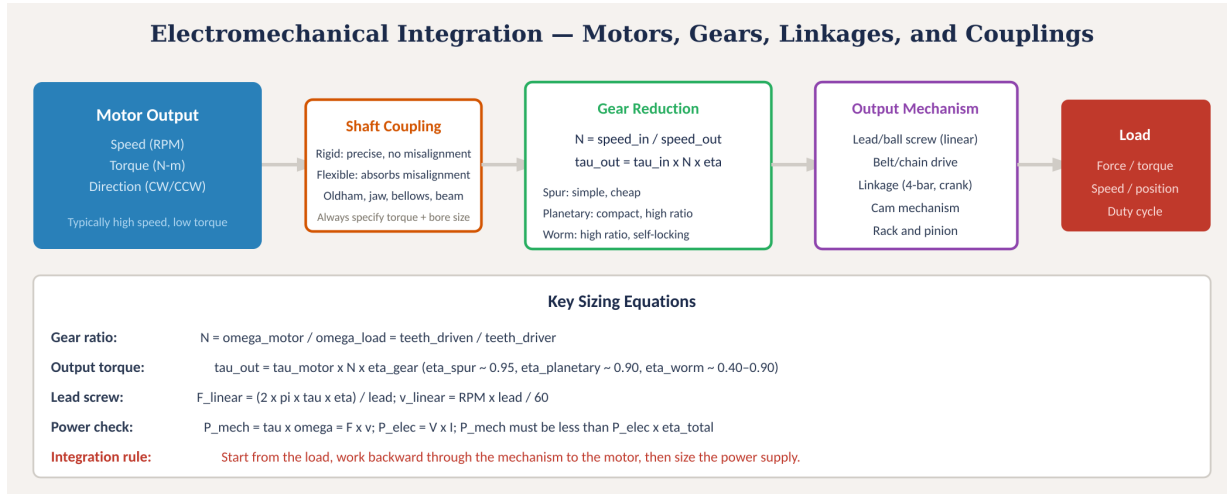


Figure 9.4: Motor-to-load power flow through coupling, gear reduction, and output mechanism with key sizing equations.

9.3.1 Gear Reduction

Motors run efficiently at high speed and low torque. Loads typically need the opposite. Gearboxes trade speed for torque:

$$\tau_{\text{out}} = \tau_{\text{motor}} \times N \times \eta_{\text{gear}} \quad (9.1)$$

where N is the gear ratio ($\omega_{\text{in}}/\omega_{\text{out}}$) and η is the gearbox efficiency.

Type	Efficiency	Characteristics
Spur	95–98%	Simplest, cheapest. External parallel-axis mesh. Available in stock gear-motors.
Planetary	85–95%/stage	Compact, coaxial, high torque density. 3:1 to 100:1 per stage.
Worm	40–90%	Very high ratios (20:1–300:1) in one stage. Self-locking at high ratios.

Table 9.2: Gear reduction types and efficiencies.

9.3.2 Output Mechanisms

- **Shaft couplings:** rigid (zero misalignment tolerance) or flexible (jaw, Oldham, bellows, beam). Beam couplings are the standard for stepper motors.
- **Lead/ball screws:** convert rotary to linear. $F = 2\pi\tau\eta/\text{lead}$. Lead screws are self-locking; ball screws are not.
- **Timing belt drives** (GT2, HTD): positive engagement, no slip. Standard for CNC and 3D printers.
- **Linkages** (4-bar, crank-slider, cam): complex output paths. Torque varies through the cycle; size the motor for peak, not average.

Integration Rule

Start from the load and work backward: calculate the force/torque and speed the load requires → select the mechanism and gear ratio → calculate the motor torque → select the motor → calculate the current draw → size the power supply. The power check: $P_{\text{mech}} = \tau\omega = Fv$ must be $\leq P_{\text{elec}} \times \eta_{\text{total}}$.

9.4 Safety Design Checklist

Every item must be addressed before commissioning:

1. Input has reverse polarity protection (P-FET or series diode).
2. Input has a fuse or electronic current limiter rated for expected peak current.
3. TVS diode on every power input where external cables connect.
4. Voltage supervisor on MCU power rail (brown-out reset).
5. Flyback diode on every inductive load not protected by the driver IC.
6. Per-rail fusing; each output rail has its own current protection.
7. 100 nF decoupling capacitor on every IC V_{CC} pin.
8. Signal and power grounds separated, joined at a single star point.
9. Enclosure bonded to earth ground (if using internal AC-DC supply).
10. All wire gauges sized for their maximum current with appropriate derating.

Requirements Mapping: From Protection Specs to Shall-Statements

Protection circuit design generates system requirements:

- “System survives reverse battery” → SYS-REQ: The power input shall include reverse-polarity protection limiting leakage to <1 mA.
- “Motor inrush must not reset MCU” → PWR-REQ: The MCU supply rail shall maintain ≥ 2.7 V during motor inrush transients up to 15 A for 100 ms.
- “ESD at user connectors” → SYS-REQ: User-accessible interfaces shall withstand ± 8 kV contact discharge per IEC 61000-4-2.

Build your power budget spreadsheet *first*, then derive protection requirements from worst-case currents and voltages.

9.5 Practice Questions

9.5.1 Conceptual Questions

1. Explain why star-point grounding prevents motor noise from corrupting ADC readings. What happens if motor current returns through a shared ground trace?
2. A TVS diode has a standoff voltage of 12 V and a clamping voltage of 19.9 V at 10 A. Is this suitable for protecting a 12 V circuit whose components have a 20 V absolute maximum rating?

3. Compare a series Schottky diode versus a P-channel MOSFET for reverse polarity protection at 5 A. Calculate the power loss for each (Schottky $V_f = 0.3\text{ V}$, P-FET $R_{DS(on)} = 20\text{ m}\Omega$).
4. Why must a DC circuit breaker be specifically rated for DC voltage? What is the hazard of using an AC-rated breaker on a DC circuit?
5. A relay coil has 50 mH inductance and draws 100 mA at 12 V. If the switch opens in $1\text{ }\mu\text{s}$, estimate the back-EMF spike voltage ($V = L \times dI/dt$). Why does this destroy a 60 V MOSFET?
6. What is the difference between a crowbar circuit and a TVS diode clamp? When would you use each?
7. Explain why decoupling capacitors must be placed as close to the IC power pin as possible, not just “somewhere on the same board.”

9.5.2 Application / Scenario Questions

8. Your robot uses a 4S LiPo battery (14.8 V nominal, 16.8 V max charge, 12.0 V cutoff) powering two DC motors (5 A each continuous, 30 A each stall), a Raspberry Pi (5 V, 2.5 A), and three servos (6 V, 1 A each peak). Design the complete power architecture: draw a block diagram showing every rail, regulator, and protection device. Specify part numbers where possible.
9. Your project team’s prototype works perfectly on the bench but fails intermittently in the field. Symptoms: the motor runs fine, but the microcontroller occasionally resets, and analog sensor readings drift by 200 mV when the motor switches direction. Diagnose the root cause and propose a fix using concepts from this module.
10. A classmate argues that a flyback diode is unnecessary because “the motor driver IC already handles back-EMF.” Under what conditions is the classmate correct? Under what conditions is the classmate dangerously wrong?
11. Design a protection scheme for a solar panel (18 V open-circuit, 5 A short-circuit) charging a 3S LiPo (11.1 V nominal) that also powers a 12 V water pump. Include reverse-current blocking, charge control, and load protection.

9.5.3 Answers

1. In a shared ground, motor current (I_M) flows through the same trace as sensor return current. The trace has resistance R , so the sensor sees a ground voltage offset of $V_{\text{noise}} = I_M \times R$. With star-point grounding, motor current returns through its own dedicated conductor, so the sensor ground conductor carries only the small sensor current. The noise voltage on the sensor ground drops from $I_M \times R$ to $I_{\text{sensor}} \times R$, which is typically orders of magnitude smaller.
2. Yes; the standoff voltage (12 V) is at or above the operating voltage, so the TVS does not conduct during normal operation. The clamping voltage (19.9 V) is below the 20 V absolute maximum, so the TVS clamps before the components are damaged. However, the margin is very tight (0.1 V). A TVS with a 16.7 V or lower clamping voltage would provide better margin. Always verify the clamping voltage at the expected peak current, not just the rated value.
3. Schottky: $P = V_f \times I = 0.3 \times 5 = 1.5\text{ W}$. P-FET: $P = I^2 \times R_{DS} = 25 \times 0.020 = 0.5\text{ W}$. The P-FET dissipates one-third the power. At 5 A, the Schottky also drops the voltage rail by 0.3 V, which may push the rail below the minimum operating voltage of downstream components. The

P-FET drops only 0.1 V. For battery-powered systems where efficiency and voltage headroom matter, the P-FET is strongly preferred.

4. AC current crosses zero 120 times per second (at 60 Hz), which naturally extinguishes any arc at the breaker contacts. DC current never crosses zero, so a DC arc can sustain indefinitely. An AC-rated breaker's contacts and arc chute may not be designed to extinguish a sustained DC arc, potentially causing the breaker to fail to interrupt the circuit, or worse, catch fire internally.
5. $V = L \times dI/dt = 0.050 \times (0.100/0.000001) = 5000 \text{ V}$. This 5 kV spike (in theory) far exceeds the 60 V MOSFET's drain-source breakdown voltage. In practice, the MOSFET avalanches and the spike is clamped at the breakdown voltage, but the energy dissipated during avalanche can destroy the device. A flyback diode clamps the spike to approximately $V_{\text{supply}} + V_f \approx 12.7 \text{ V}$, well within the MOSFET's rating.
6. A TVS diode *clamps* the voltage; it conducts just enough to limit the voltage and absorbs the transient energy, then stops conducting when the voltage returns to normal. Operation is non-destructive. A crowbar circuit *shorts* the supply to ground via an SCR, deliberately blowing the upstream fuse. It is destructive (fuse replacement required) but provides guaranteed protection when the overvoltage has enough sustained energy to overwhelm a TVS. Use TVS for transient spikes (ESD, inductive switching); use crowbar for sustained overvoltage from a failed regulator.
7. Decoupling caps work by providing a low-impedance path for high-frequency transients to bypass the IC rather than entering through the power pins. The trace between the capacitor and the IC pin has inductance proportional to its length. A long trace adds enough inductance to negate the capacitor's effectiveness at high frequencies. Placing the cap within a few millimeters of the pin minimizes this parasitic inductance and keeps the bypass impedance low at the frequencies where transients occur.
8. Architecture: Battery $\rightarrow$ main fuse (80 A, covers dual motor stall) $\rightarrow$ reverse polarity P-FET (IRLML6402 or similar) $\rightarrow$ TVS diode (18 V standoff) $\rightarrow$ power distribution bus. Motor rail: direct from battery through individual 40 A slow-blow fuses per motor, H-bridge drivers with integrated flyback diodes. 5 V rail: buck converter (e.g., LM2596 or MP1584, 3A rated) from battery bus, with 100 μF bulk cap + 100 nF ceramic decoupling. 6 V servo rail: buck converter (adjustable output set to 6 V, 5A rated, TPS54560) with per-servo PTC fuses. BMS: 4S BMS module with 12.0 V cutoff, 16.8 V overcharge protection, and cell balancing. Grounds: star-point near main fuse; separate motor ground bus and logic ground bus.
9. Root cause: ground-coupled motor noise. When the motor reverses direction via the H-bridge, the current transient creates a voltage spike on the shared ground that (a) momentarily shifts the MCU's ground reference, triggering a brown-out reset, and (b) offsets the ADC's reference voltage by 200 mV, corrupting readings. Fix: (a) separate motor power ground and MCU signal ground, joining at a star point near the battery; (b) add a voltage supervisor (TPS3808) on the MCU rail to hold the MCU in clean reset during transients rather than allowing corrupted execution; (c) add bulk capacitance (470 μF) on the MCU's supply rail to absorb the transient; (d) add a TVS diode on the motor power bus.
10. The classmate is correct when using a properly rated H-bridge driver IC (like L298N, DRV8871, BTS7960) that includes integrated flyback/freewheeling diodes across the output MOSFETs. These diodes handle the back-EMF from the motor. The classmate is **dangerously wrong**

when: (a) driving a relay, solenoid, or valve directly from a bare MOSFET or BJT; these loads have no integrated protection; (b) the motor's back-EMF exceeds the driver's clamping capability; or (c) using a driver module where the "protection diodes" are actually for the logic section, not the motor outputs. **Rule:** verify in the driver's datasheet that flyback diodes are explicitly rated for the motor's inductive energy.

11. Solar input: blocking diode (Schottky, e.g., MBR2045) prevents battery discharge through panel at night. MPPT charge controller (e.g., Genasun GV-5 or Victron 75/15) sized for 3S LiPo with 12.6 V charge termination. Reverse polarity: P-FET on both the solar input and the battery connector. TVS diode (18 V standoff) on the solar input for transient protection. Load: battery feeds pump through a slow-blow fuse (3 A for a 2 A pump). Flyback diode across pump motor terminals. Voltage supervisor monitors battery voltage and disables pump via a load switch when battery drops below 10.5 V to protect cells. The charge controller's MPPT algorithm ensures maximum solar harvest even as panel voltage varies with sun angle and temperature.

Chapter 10

Mechanical Fasteners & Bearings

Every mechanical system is held together by fasteners and supported by bearings. Selecting the wrong bolt grade, under-torquing a critical joint, or choosing a bearing that cannot handle the load will cause your system to fail; often catastrophically and without warning. This chapter covers the fastener and bearing knowledge you need to specify these components correctly in your BOM and engineering drawings.

10.1 Threaded Fasteners

10.1.1 Bolt and Screw Anatomy

A threaded fastener converts rotational torque into axial clamping force (preload) that holds a joint together. Key parameters: **nominal diameter** (major diameter, e.g., M6 = 6 mm), **pitch** (distance between threads; metric specifies pitch in mm; imperial uses TPI), **thread length**, and **head type** (hex, socket, flat, pan, button).

Bolts pass through clearance holes and clamp with a nut. **Screws** thread into a tapped hole. **Machine screws** use fine threads in tapped metal. **Self-tapping screws** form their own threads in plastic or sheet metal.

10.1.2 Thread Standards

Metric and Imperial Do Not Mix

M6×1.0 and 1/4-20 UNC look nearly identical (6.0 mm vs. 6.35 mm diameter). They will partially engage but strip immediately under load. Always verify with a thread gauge or package markings.

Unified (inch): UNC (Coarse) for general assembly, fewer turns, tolerant of damage. UNF (Fine) for vibration-sensitive and precision applications. **Metric (ISO):** Designated $Md \times p$ (e.g., M6×1.0). Coarse pitch is the default. Fine pitch for vibration resistance. Metric is the global standard, use for new designs unless interfacing with existing inch hardware.

10.1.3 Bolt Grades and Strength

SAE (inch): Grade 2 (no marks, 74ksi, general purpose), Grade 5 (3 radial lines, 120 ksi, most common structural), Grade 8 (6 radial lines, 150 ksi, high strength). **ISO (metric):** Class 4.6 ($\approx$ Gr. 2), 8.8 ($\approx$ Gr. 5), 10.9 ($\approx$ Gr. 8), 12.9 (socket heads, 1220 MPa). The first digit $\times 100$ gives the UTS in MPa; the product of digits gives yield.

Threaded Fastener Grade Markings & Strength Comparison		
SAE (Inch) Grades Grade 2 — No marks Low carbon · 74 ksi tensile · General purpose Grade 5 — 3 radial lines Med. carbon quenched · 120 ksi · Most common Grade 8 — 6 radial lines Alloy quenched · 150 ksi · High strength Grade 2 bolts look unmarked. Never assume grade from appearance. Always verify from package markings.	ISO (Metric) Classes Class 4.6 — Marked "4.6" Low carbon · 400 MPa · ≈ Grade 2 Class 8.8 — Marked "8.8" Med. carbon · 800 MPa · ≈ Grade 5 Class 10.9 — Marked "10.9" Alloy · 1040 MPa · ≈ Grade 8 Class 12.9 — Marked "12.9" Alloy · 1220 MPa · Socket heads First digit × 100 = UTS (MPa) Product = yield strength (MPa)	Torque-Preload $T = K \cdot F \cdot d$ T = torque (in-lb or N-m) K = nut factor (0.20 typical dry) F = desired preload (lb or N) d = nominal bolt diameter Locking Methods: • Nyloc nut — reusable 3–5× • Thread-locker (Loctite 242/271) • Nord-Lock — wedge locking • Safety wire — aerospace/motorsport Split lock washers are ineffective — avoid.

Figure 10.1: SAE and ISO bolt grade markings with strength data and torque-preload relationship.

Specify the Grade

Never list just “M6 hex bolt” in your BOM. Always specify: M6×1.0×25, Class 8.8, zinc plated. Missing the grade means the vendor ships 4.6 at half the strength.

10.1.4 Torque and Preload

The purpose of tightening a bolt is to generate **preload** (clamping force). The torque-preload relationship:

$$T = K \cdot F \cdot d \quad (10.1)$$

where T = applied torque, K = nut factor (0.20 dry steel, 0.15 lubricated, 0.12 anti-seize), F = desired preload (typically 75% of proof load), d = nominal diameter. **90% of applied torque is lost to friction**; only 10% generates clamping force.

10.1.5 Locking Methods

Vibration loosens fasteners. Four reliable methods:

- **Nyloc nut**: nylon insert deforms around threads. Reusable 3–5×. Max 120 °C.
- **Thread-locker**: Loctite 242 (blue, removable) for general use; 271 (red, permanent) for critical joints.
- **Nord-Lock washers**: wedge-locking. Positive mechanical lock independent of friction. Gold standard.
- **Safety wire**: drilled bolt heads wired in pairs. Aerospace and motorsport.

Split Lock Washers Are Ineffective

Despite ubiquity, split lock washers have been shown in NASA and Junker vibration testing to provide no meaningful locking action. They flatten at low preloads. Use nyloc nuts or thread-locker instead.

10.1.6 Non-Threaded Fasteners

Rivets: permanent fasteners for sheet materials. Blind (pop) rivets install from one side. **Pins:** dowel (precision alignment), roll/spring (self-retaining), clevis (quick connect/disconnect for linkages). **Retaining rings:** external E-clips on shafts, internal in bores; inexpensive axial retention for bearings and gears. **Snap-fits:** integral plastic features; cantilever design with strain $\leq 2\%$ for ABS, $\leq 4\%$ for PP.

MEGN 301 Sourcing

McMaster-Carr ships next-day to Golden and stocks every fastener here. Bolt Depot and Albany County Fasteners for bulk. For 3D-printed snap-fits, prototype in PLA, final in PETG or nylon.

10.2 Bearings

10.2.1 What Bearings Do

Bearings support a rotating or sliding shaft while minimizing friction. Every bearing balances: **load capacity, speed, precision, and life.**

10.2.2 Bearing Types

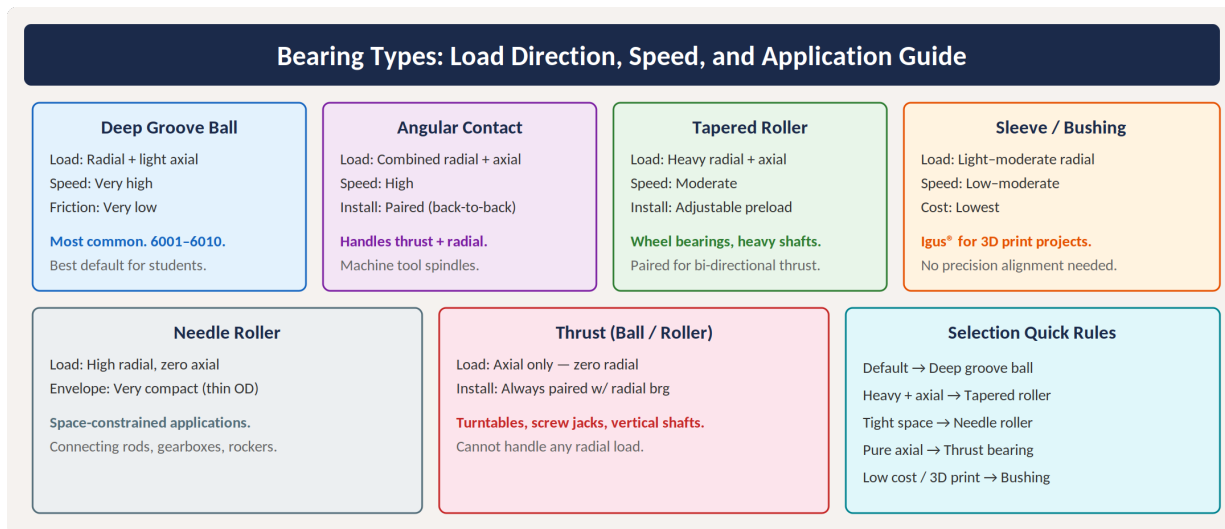


Figure 10.2: Six bearing types with load direction, speed capability, and application guidance.

Deep groove ball (6000 series): default for students. Radial + light axial, high speed, low cost. **Angular contact:** combined radial + axial; installed in pairs. **Tapered roller:** heavy combined loads (wheel bearings). **Needle roller:** compact, high radial, zero axial. **Thrust:** axial only; always pair with a radial bearing. **Sleeve/bushing:** no rolling elements, low cost, quiet. Igus iglide® polymer bushings are excellent for 3D-printed housings.

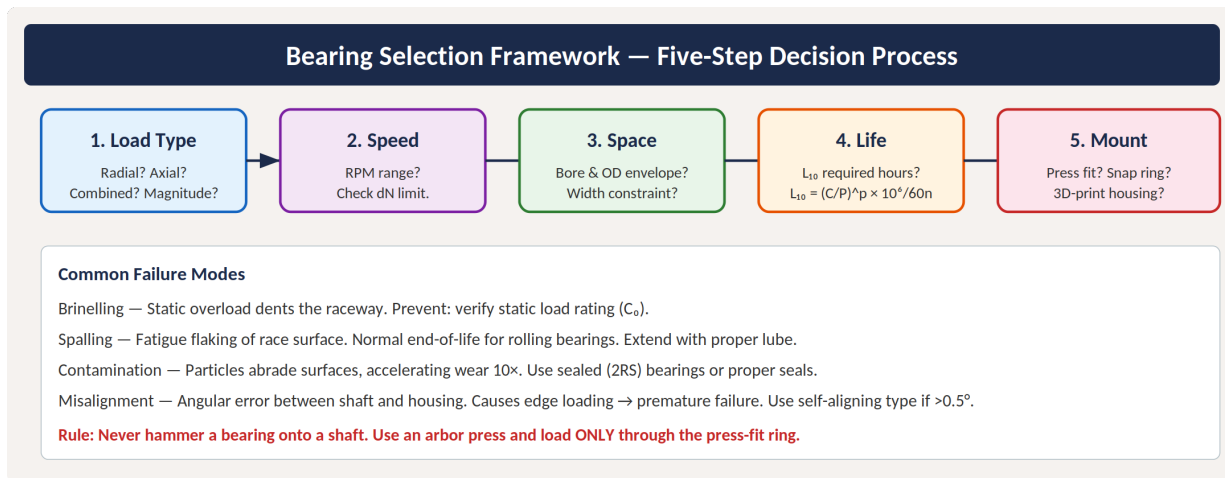


Figure 10.3: Five-step bearing selection framework with common failure modes.

10.2.3 Bearing Selection

Five steps: (1) **Load type and magnitude**; calculate equivalent dynamic load P . (2) **Speed**; check limiting RPM. (3) **Space**; bore and OD constraints. (4) **Life**; $L_{10} = (C/P)^p \times 10^6 / (60n)$ hours. (5) **Mounting**; press fit, retaining ring, adhesive.

Bearing Sizing for Student Projects

(1) Select shaft diameter. (2) Look up matching bearing (8 mm shaft → 608-2RS, \$2). (3) Verify $C \geq 3 \times \text{max load}$. (4) Order 2RS (sealed) variant. Done.

10.2.4 Mounting Bearings

Press fit: bore interference fit on shaft (0.005–0.015 mm). Use an arbor press; press only on the fitted ring.

Mounting Bearings in 3D Prints

FDM tolerance ± 0.2 –0.3 mm. For bearing pockets: design 0.1–0.2 mm undersize and ream to fit, or 0.05 mm oversize with retaining compound (Loctite 638). Always test-fit with a prototype print.

Never Hammer a Bearing

Striking a bearing transmits shock through rolling elements, creating brinelling dents. These cause vibration, noise, and premature failure. Always use an arbor press or soft-jaw vise.

10.2.5 Failure Modes

Brinelling: static overload dents raceway. **Spalling:** fatigue flaking (normal end-of-life). **Contamination:** particles accelerate wear 10×; use sealed (2RS) bearings. **Misalignment:** angular error causes edge loading.

10.2.6 Thread Engagement Table

Material	Min Engagement	Preferred	Notes
Steel	$1.0d$	$1.5d$	Standard for structural joints
Aluminum	$1.5d$	$2.0d$	Softer threads require more engagement
Cast iron	$1.5d$	$2.0d$	Brittle; threads can crack if overtorqued
Plastic (ABS)	$2.0d$	$2.5d$	Use coarse thread; consider threaded insert
Brass	$1.5d$	$2.0d$	Avoid for high-torque applications

Table 10.1: Minimum thread engagement by material.

10.2.7 Threaded Inserts for 3D-Printed Parts

Direct threading into FDM plastic is unreliable; the threads strip under moderate torque. Three better options: **Heat-set brass inserts** (pressed in with a soldering iron; strongest, most professional), **press-fit inserts** (pushed into undersized holes; adequate for light loads), and **captive nuts** (hex nut trapped in a designed pocket during printing). Heat-set inserts (McMaster #94180A series) are the recommended solution for MEGN 301 projects.

10.2.8 Bolt Selection Checklist

1. Determine joint loads (tension, shear, or combined).
2. Select thread standard (metric for new designs).
3. Select grade/class based on load with $\geq 2\times$ safety factor.
4. Size diameter from shear/tension calculations.
5. Determine length: grip length + nut + washer + 2–3 exposed threads.
6. Specify head type based on tool access and flush requirements.
7. Specify locking method based on vibration environment.
8. Calculate and specify torque value.
9. Add all details to BOM entry.

10.3 Design Rules of Thumb

1. Thread engagement: min $1.5d$ in steel, $2d$ in aluminum, $2.5d$ in plastic.
2. Bolt spacing: min $3d$ center-to-center, $2d$ from edge.
3. Inner ring press fit to shaft; outer ring sliding fit to housing (rotating-shaft case).
4. Always specify grade, material, finish, and torque in assembly documentation.

Requirements Mapping: From Fastener Specs to Shall-Statements

Mechanical joining generates system requirements:

- “Vibration environment (field deployment)” → MECH-REQ: All structural fasteners shall use positive locking (nyloc, thread-locker, or Nord-Lock). Split washers shall not be used.
- “Enclosure opens for maintenance” → MECH-REQ: Enclosure fasteners shall be captive (retained in the cover after removal) and require only a #2 Phillips driver.
- “Shaft supports 50 N radial load at 1500 RPM” → BRG-REQ: Shaft bearing shall have

$L_{10} \geq 5000$ hrs at the specified load and speed. Bearing type: deep-groove ball, sealed (2RS).

10.4 Practice Questions

1. Your design uses an M8×1.25 bolt in tapped aluminum. Minimum thread engagement depth? If limited to 10 mm, is this sufficient?
2. A joint experiences severe vibration. Compare nyloc, Loctite 242, and Nord-Lock. Recommend one.
3. Your BOM lists “1/4-20 bolt, qty 12” with no other info. What is wrong? Rewrite it properly.
4. Select a bearing for an 8 mm shaft at 3000 RPM with 50 N radial load. Specify designation, type, expected life.
5. A teammate suggests press-fitting bearings into PLA housings. Identify two problems and propose solutions.
6. Belt-driven stage idler pulley bearings grind after one week. Likely causes and diagnosis?
7. An M6 Class 8.8 bolt has proof load ≈ 16.3 kN. Calculate torque for 75% proof load preload ($K = 0.20$).

10.4.1 Answers

1. $\text{Min} = 2 \times 8 \text{ mm} = 16 \text{ mm}$ for aluminum. $10 \text{ mm} = 1.25d$; insufficient. Use through-bolt with nut, Heli-Coil insert, or thicker boss.
2. Nord-Lock for severe vibration with frequent disassembly. Loctite 271 (red) for permanent assembly. Nyloc for general use.
3. Missing: grade, head type, length, material/finish, torque. Correct: “1/4-20 UNC $\times$ 1.00 in hex cap, SAE Grade 5, zinc, qty 12. Torque: 8.0 ft-lb with Loctite 242.”
4. 608-2RS: 8 mm bore, 22 mm OD. $C = 3.45$ kN, $P = 0.05$ kN. $L_{10} = (3.45/0.05)^3 \times 10^6 / (60 \times 3000) = 918,000$ hrs.
5. PLA: low HDT ($\sim \$60^\circ\text{C}$), creeps under load $\rightarrow$ pocket enlarges. Tolerance ± 0.2 mm makes press fit unreliable. Fix: use PETG/nylon, ream pocket, or use retaining compound.
6. Contamination, misalignment, or open (unsealed) bearings. Diagnosis: spin by hand (gritty = contamination), check housing alignment, inspect seals. Replace with 2RS type.
7. $F = 0.75 \times 16,300 = 12,225$ N. $T = 0.20 \times 12,225 \times 0.006 = 14.7$ N·m.

Chapter 11

Mechanical Power Transmission

Power transmission converts motor output (torque and speed) into the motion your system needs. Every element, belt, chain, gear, shaft, linkage, lead screw, trades speed for torque (or vice versa), and every one introduces losses. Selecting the right transmission is the bridge between your electrical power budget and the mechanical work your system must perform.

11.1 Fundamental Relationships

Three equations govern all power transmission:

$$\text{Gear ratio: } GR = N_2/N_1 = \omega_1/\omega_2 = \tau_2/\tau_1 \quad (11.1)$$

$$\text{Power: } P = \tau \times \omega \quad (11.2)$$

$$\text{Efficiency: } \eta = P_{\text{out}}/P_{\text{in}} \quad (11.3)$$

Speed down $\rightarrow$ **torque up** by the gear ratio; power constant minus losses. Multi-stage efficiency compounds: $0.95^3 = 85.7\%$.

11.2 Belt and Pulley Drives

11.2.1 V-Belts

Wedge cross-section grips by friction. Allows slip (protects downstream). 90–95% efficient. Requires tension adjustment.

11.2.2 Timing Belts (Synchronous)

Toothed belts mesh with toothed pulleys; no slip, positive drive. 97–99% efficient.

GT2 Timing Belts for Student Projects

GT2 profile (2 mm pitch, 6 mm width) is the standard for student projects, 3D printers, and hobby CNC. \$3–8/meter on Amazon. Pulleys 16–60 tooth for \$2–5. Always use flanged pulleys to prevent belt tracking.

11.2.3 Flat and Round Belts

Flat belts on crowned pulleys for long center distances. Round belts (O-ring) for low-load, compact drives.

11.3 Chain Drives

Roller chain: positive engagement, 95–98% efficient, high loads and shock tolerance. #25 (1/4 in pitch) for student projects. Minimum 17 teeth on small sprocket.

Chain Requires Lubrication

Dry chain wears at 10× the rate of lubricated chain. Apply chain lubricant (not WD-40) every 20–50 operating hours. A worn chain skips teeth; sudden and dangerous.

11.4 Gear Trains

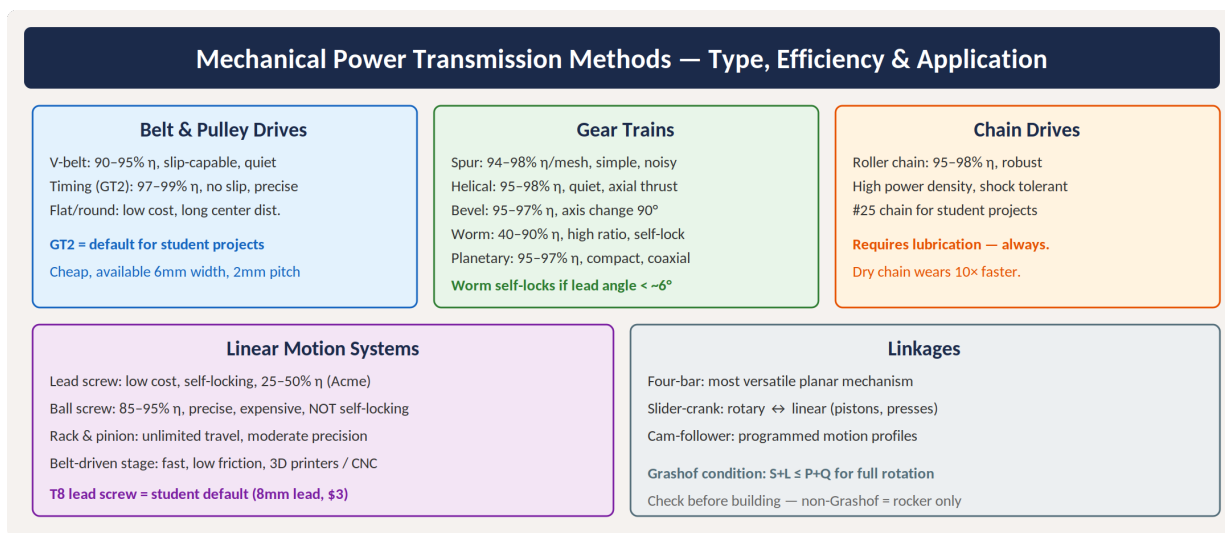


Figure 11.1: Power transmission methods with efficiency, characteristics, and recommendations.

Spur: straight teeth, 94–98%/mesh, noisy at speed. **Helical:** angled teeth, quieter, generates axial thrust. **Bevel:** intersecting shafts (90°). **Worm:** high ratio single-stage (5:1–100:1), 40–90% efficient, **self-locking** when lead angle < $\sim 6^\circ$. **Planetary:** compact, coaxial, 95–97%/stage.

Gear Selection for Student Projects

ServoCity (servocity.com) sells matched spur gear sets with standardized shafts and hubs. McMaster has individual gears in any size. For 3D-printed gears: PETG or nylon, module ≥ 1.5 , add 0.1–0.2 mm backlash.

11.4.1 Gear Train Design Process

1. Calculate required output torque and speed from the load.
2. Calculate gear ratio: $GR = \omega_{motor}/\omega_{load}$.
3. Select gear type based on noise, efficiency, and axis orientation requirements.
4. If $GR > 10$: use multi-stage (two spur stages) or single-stage worm/planetary.
5. Size gears for tooth bending stress and surface contact stress (Lewis equation for bending: $\sigma = W_t/(b \cdot m \cdot Y)$).

6. For student projects: select stock gears from ServoCity or McMaster and verify that the rated torque exceeds your load with $\geq 2\times$ margin.
7. Specify mounting: shaft diameter, key or set screw, bearing supports, and housing.

11.4.2 Backlash and Precision

Backlash is the angular play between mating gear teeth (or belt-pulley engagement). It causes positioning error and control-loop instability. Standard spur gears have 0.05–0.10 mm backlash. For precision applications: use anti-backlash gears (split gear with spring preload), ball screws (inherently low backlash), or harmonic drives (zero backlash). For most MEGN 301 projects, standard backlash is acceptable.

11.5 Shafts and Couplings

11.5.1 Shaft Design

Shafts transmit torque and support rotating components. Default: 1045 steel. Size for torque with safety factor. Keyways add $\sim 25\%$ stress concentration.

11.5.2 Shaft-Hub Connections

Keyway: rectangular key transmits torque; standard sizes from ANSI B17.1. **Set screw:** clamps hub to flat; low torque, adequate for light student loads. **D-shaft:** flat on shaft engages matching hub flat; common on stepper motors.

11.5.3 Couplings

Rigid: zero misalignment tolerance. **Flexible:** jaw coupling (elastomer spider, shock absorption), beam coupling (zero backlash), Oldham (parallel offset), universal joint (large angles).

11.6 Linear Motion

11.6.1 Lead Screws

Acme-thread, 25–50% efficient, **self-locking**. T8×8 (8 mm dia, 8 mm lead) is the student default (\$3–8 with brass nut).

11.6.2 Ball Screws

Recirculating balls, 85–95% efficient, precise, **not self-locking**. More expensive.

11.6.3 Rack and Pinion

Unlimited travel, moderate precision. Force = torque / pinion pitch radius.

11.6.4 Belt-Driven Linear Stages

Timing belt on two pulleys, carriage on linear rails. Fast, low friction. GT2 + MGN12 rail for student projects.

11.7 Linkages

Four-bar: most versatile planar mechanism. Grashof condition: $S + L \leq P + Q$ for full rotation.
Slider-crank: rotary $\leftrightarrow$ linear (pistons, presses). Stroke = $2 \times$ crank radius. **Cam-follower:** programmed motion profiles from cam shape.

11.8 Selection Framework

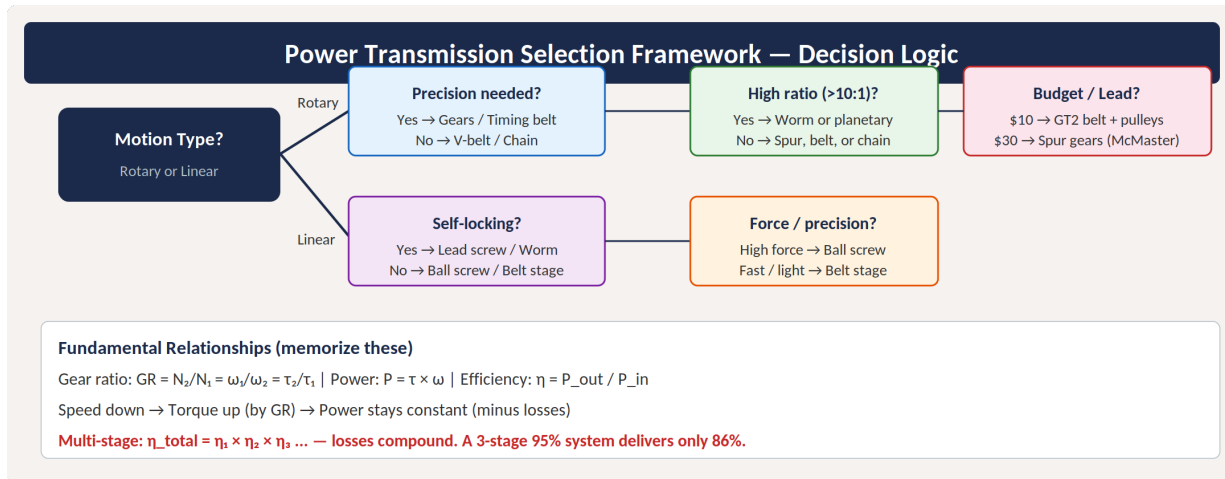


Figure 11.2: Power transmission selection: motion type, precision, self-locking, ratio, budget.

Five Key Takeaways

1. Speed down = torque up. Calculate output torque before selecting transmission.
2. GT2 belts and T8 lead screws are student defaults. Start there.
3. Efficiency compounds multiplicatively. Minimize stages.
4. Self-locking (lead screws, worms) holds position without power. Non-self-locking (ball screws, spur gears) needs a brake.
5. Every rotating joint needs a bearing. Every coupling needs defined misalignment tolerance.

Requirements Mapping: From Transmission Specs to Shall-Statements

Power transmission design generates system requirements:

- “Linear stage positions to ± 0.1 mm” $\rightarrow$ MECH-REQ: Lead screw shall have ≤ 0.05 mm backlash. Anti-backlash nut required.
- “System holds position when unpowered” $\rightarrow$ MECH-REQ: Transmission shall be self-locking (lead screw or worm gear with lead angle $< 6^\circ$).
- “Gear ratio 10:1 in minimum space” $\rightarrow$ MECH-REQ: Use planetary gearbox (≤ 40 mm diameter) with $\geq 95\%$ efficiency per stage.

Calculate the required output torque and speed *from requirements* before selecting any transmission hardware.

11.9 Practice Questions

1. A NEMA 17 stepper produces $0.4 \text{ N}\cdot\text{m}$ at 200 RPM. You need $2.0 \text{ N}\cdot\text{m}$ at 40 RPM. What gear ratio is required? What efficiency must the transmission achieve?
2. Compare GT2 timing belt vs. roller chain for a 3:1 reduction on a small conveyor. Which do you choose and why?
3. Your linear stage uses a T8 $\times$ 8 lead screw driven by a stepper at 200 RPM. Calculate the linear speed in mm/s. If the stage must move 200 mm, how long does the motion take?
4. A worm gear set has a 30:1 ratio and 60% efficiency. Input power is 50 W. What is the output power? Is this set self-locking?
5. Your project needs a mechanism that converts continuous motor rotation into a back-and-forth linear stroke of 50 mm. Propose two different mechanism solutions and compare them.
6. A 4-bar linkage has link lengths 40, 100, 70, and 80 mm. Check the Grashof condition. Which link(s) can make a full revolution?
7. Your belt-driven linear stage stalls under load. The stepper motor skips steps but does not overheat. Diagnose the root cause and propose fixes.

11.9.1 Answers

1. $GR = 2.0/0.4 = 5 : 1$ (or $200/40 = 5 : 1$). Check: $P_{in} = 0.4 \times (200 \times 2\pi/60) = 8.38 \text{ W}$. $P_{out} = 2.0 \times (40 \times 2\pi/60) = 8.38 \text{ W}$ for 100% efficiency. With real losses, the transmission must be $\geq 100\%$ to deliver full torque; meaning the motor is marginally sized. Derate: need $\geq 85\%$ efficient transmission (GT2 belt: 97%, spur gear: 94% work; worm at 60% does not).
2. GT2 belt: no lubrication, clean, quiet, 97% efficient, \$10 total. Chain: requires lube, messy, noisier, 96% efficient, \$15. For a small indoor conveyor, GT2 wins on cleanliness, cost, and simplicity. Chain is better if the conveyor sees high shock loads or outdoor contamination.
3. Linear speed = lead $\times$ RPM / 60 = $8 \text{ mm} \times 200/60 = 26.7 \text{ mm/s}$. Time = $200/26.7 = 7.5 \text{ s}$.
4. $P_{out} = 50 \times 0.60 = 30 \text{ W}$. At 30:1 ratio, the lead angle is very low ($\ll 6^\circ$), so yes; self-locking.
5. (a) Slider-crank: crank radius = 25 mm, continuous rotation $\rightarrow$ 50 mm stroke. Simple, constant stroke. (b) Belt-driven stage with limit switches: motor reverses at each end. Variable stroke and speed via firmware. Slider-crank is simpler mechanically; belt stage is more flexible.
6. $S = 40, L = 100, P = 70, Q = 80$. $S + L = 140$. $P + Q = 150$. $140 \leq 150$. Grashof satisfied. The shortest link (40 mm) can make full revolution. If grounded, it is the crank.
7. Motor torque is sufficient (no overheating) but the belt/pulley system cannot transmit the load. Causes: belt tension too low (belt slips on pulley teeth), pulley set screw loose (pulley spins on shaft), or load exceeds belt rating. Fixes: increase belt tension, verify set screws, check that belt width and tooth count match the load.

Chapter 12

Designing for Injection Molding

12.1 Injection Molding Fundamentals

12.1.1 Process Overview

Injection molding produces plastic parts by forcing molten thermoplastic into a precision steel mold under high pressure. The machine has two units: the **injection unit** (barrel, screw, heater bands, nozzle) melts pellets and injects them, and the **clamping unit** (platens, tie bars, hydraulic or toggle mechanism) holds the mold closed. Machines are rated by clamp force (tons) and shot size (oz or g).

The mold has two halves: the **cavity** (A-side, stationary, forms the outside/cosmetic surface) and the **core** (B-side, moving, forms the inside surface). The **parting line** is where the halves meet. Inside the mold, a **runner system** distributes melt from the nozzle to the gate, **cooling channels** extract heat, and **ejector pins** push the part out.

12.1.2 The Four-Phase Cycle

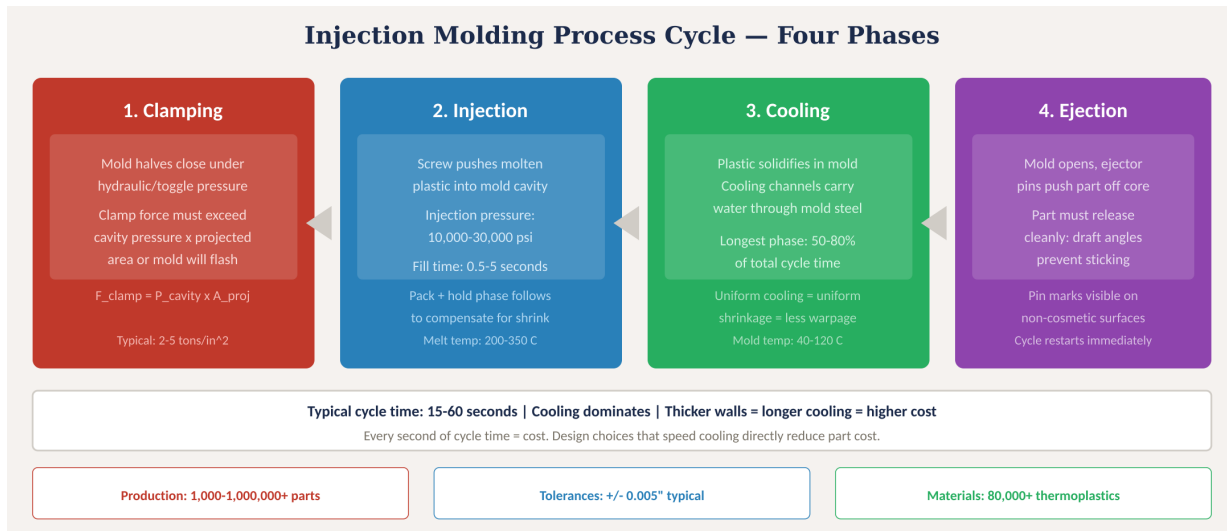


Figure 12.1: Four-phase injection molding cycle with process parameters, timing, and production capabilities.

1. **Clamping:** Mold halves close under force. Required clamp force:

$$F_{\text{clamp}} = P_{\text{cavity}} \times A_{\text{projected}} \quad (12.1)$$

Typical cavity pressures: 2–5 tons/in². A part with 10 in² projected area needs 20–50 tons of clamp force.

2. **Injection:** The screw pushes molten plastic into the cavity at 10,000–30,000 psi. Fill time: 0.5–5 seconds. A **pack-and-hold** phase follows to compensate for volumetric shrinkage as the plastic cools. Without adequate packing, parts have sink marks and are undersized. Melt temperatures range from 200 °C (PE) to 350 °C (PC, nylon).
3. **Cooling:** Plastic solidifies in the mold. This is the **longest phase**: 50–80% of total cycle time. Cooling water flowing through channels in the mold steel extracts heat. **Uniform cooling = uniform shrinkage = minimal warpage**. Mold temperatures: 40–120 °C depending on material. Cooling time scales with the *square* of wall thickness; doubling wall thickness quadruples cooling time.
4. **Ejection:** Mold opens, ejector pins push part off the core. Part must have **draft angles** so it releases cleanly. Pin marks are visible on non-cosmetic (B-side) surfaces. The cycle restarts immediately.

Cycle Time = Cost

Typical total cycle: 15–60 seconds. Every additional second of cycle time increases part cost. Wall thickness is the primary driver; every millimeter of additional wall can add 5–15 seconds of cooling per cycle. Design the thinnest wall that meets structural and flow requirements.

12.1.3 Material Selection

Material	Trade Names	Key Properties	Shrink	Common Uses
ABS	Cyclocac	Good impact, easy to mold/paint, 80 °C	0.5–0.7%	Enclosures, LEGO
PC	Lexan	High impact, optical clarity, 130 °C, UV-sensitive	0.5–0.7%	Lenses, phone cases
PP	Pro-fax	Chemical resistant, living hinge, low cost	1.0–2.5%	Containers, caps
Nylon (PA)	Zytel	High strength/stiffness, absorbs moisture	0.8–1.5%	Gears, connectors
POM	Delrin	Low friction, dimensional stability, fatigue	1.8–2.5%	Gears, zippers
PC/ABS	Bayblend	PC toughness + ABS processability	0.5–0.7%	Laptops, power tools
PBT	Valox	Good electrical, chemical resistant	1.5–2.0%	Electrical connectors
TPE/TPU	Santoprene	Flexible, rubber-like, overmoldable	0.5–2.0%	Grips, seals

Table 12.1: Common injection molding thermoplastics.

Selection criteria: Match the *cheapest* material that satisfies all of the following; mechanical loads (tensile, impact, flex), thermal environment (HDT, continuous use temperature, UL94 flammability), chemical exposure (solvents, UV, oils), aesthetics (gloss, texture, transparency, paintability), and processing (MFI, shrinkage, moisture sensitivity). Glass-fiber reinforcement increases stiffness 2–3× but reduces impact and abrades the mold.

Shrinkage Matters

When hot plastic cools it contracts. The mold cavity is cut *larger* than the finished part by the shrinkage factor. High-shrinkage materials (PP at 1–2.5%) require more generous tolerances than low-shrinkage materials (ABS at 0.5–0.7%). Crystalline polymers (PP, nylon, POM) shrink more than amorphous ones (ABS, PC, PS).

12.2 Design for Manufacturing (DFM) Rules

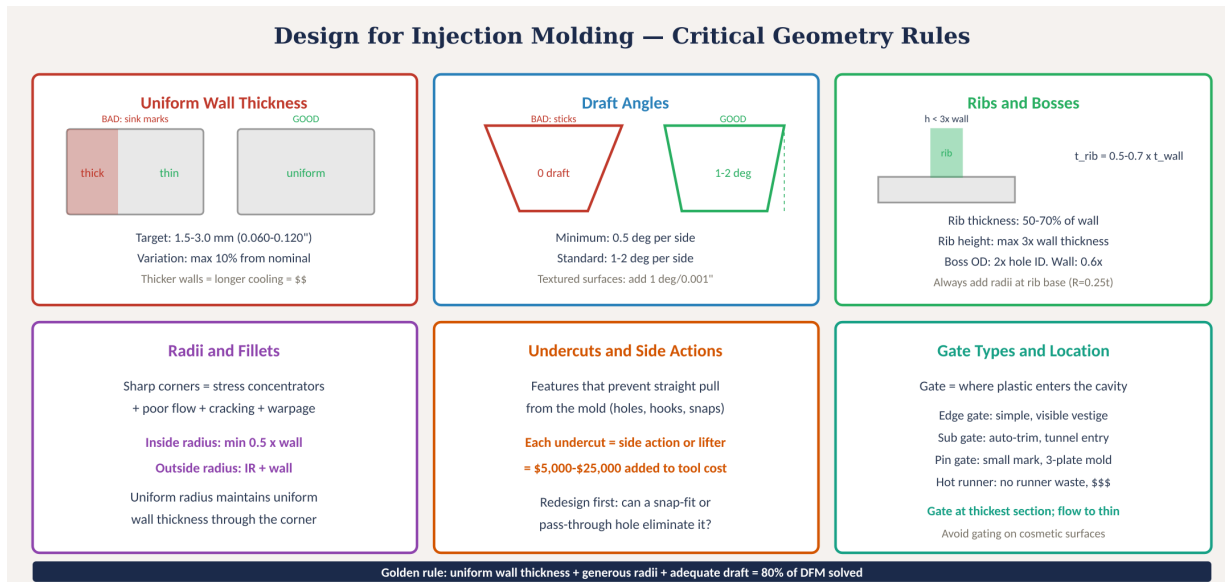


Figure 12.2: Six critical DFM geometry rules: wall thickness, draft angles, ribs/bosses, radii, undercuts, and gate location.

12.2.1 Rule 1: Uniform Wall Thickness

The single most important design parameter. Wall thickness drives cycle time ($\text{cooling} \propto t^2$), material usage, shrinkage uniformity, and defect susceptibility.

Recommended ranges: ABS: 1.5–3.5 mm. PC: 1.5–3.5 mm. PP: 0.8–3.0 mm. Nylon: 1.0–3.0 mm. POM: 1.0–3.0 mm.

Uniformity: Variation $<10\%$ of nominal. Where transitions are unavoidable, use gradual tapers at a 3:1 length-to-thickness-change ratio. Abrupt steps cause differential shrinkage, sink marks, and stress concentrations.

Coring out: Replace solid thick sections with thin uniform shells reinforced by ribs. A ribbed 2 mm wall is stiffer than a solid 4 mm wall, cools in one-quarter the time, and uses half the material.

12.2.2 Rule 2: Draft Angles

Draft is the slight taper on walls parallel to the mold opening direction that allows the part to release.

- Minimum: 0.5° per side (tight tolerance applications).

- Standard: 1° – 2° per side (recommended).
- Textured surfaces: add 1° per 0.001 in (0.025 mm) of texture depth.
- Core side often needs more draft than cavity side (part shrinks onto core).

Zero-Draft = Stuck Parts

Zero-draft vertical walls are the #1 student DFM error. Without draft, the part grips the core, ejector pins push through the wall, and the mold wears rapidly. Apply draft to *all* surfaces parallel to the pull direction; including internal ribs, bosses, and walls.

12.2.3 Rule 3: Ribs and Bosses

Ribs add stiffness without mass; bosses provide screw/insert mounting points.

Rib Proportions

- Thickness: 50–70% of adjoining wall (thicker → sink marks on opposite surface).
- Height: max $3\times$ wall thickness (taller → fill and ejection problems).
- Draft: min 0.5° per side (1° preferred).
- Spacing: min $2\times$ wall thickness between ribs.
- Base radius: 0.25 – $0.5\times$ wall thickness.
- Multiple thin ribs outperform one thick rib.

Boss Proportions

- OD: 2.0 – $2.5\times$ screw/insert hole ID.
- Wall: $0.6\times$ nominal wall thickness.
- Freestanding bosses: connect to nearest wall with a rib.
- Height: max $2.5\times$ OD.
- For heat-set brass inserts: hole ID per insert manufacturer specification.

12.2.4 Rule 4: Radii and Fillets

Sharp inside corners create stress concentrations (up to $3\times$ at zero radius), impede melt flow, and cause sink marks.

- Inside radius: min $0.5\times$ wall ($1\times$ preferred).
- Outside radius: inside radius + wall thickness.
- This maintains uniform wall thickness through the corner.
- Apply radii to *every* inside corner without exception.

12.2.5 Rule 5: Undercuts

Any feature preventing straight-pull mold separation; side holes, snap hooks, internal threads, louvers. Each undercut requires a side action (cam, lifter, or collapsing core), adding \$5,000–\$25,000 to tooling cost.

Always redesign first: Can a pass-through hole replace a blind side hole? Can a snap-fit be oriented along the pull direction? Can an internal thread become a heat-set insert installed after molding?

12.2.6 Rule 6: Gate Location

The gate is where plastic enters the cavity.

- **Place at the thickest section** so melt flows from thick to thin (ensures adequate packing).
- **Never gate on a cosmetic surface**; the gate vestige is always visible.
- Gate types: edge gate (simple, visible mark), sub gate (auto-trimmed), pin gate (small mark, 3-plate mold), hot runner (no runner waste, highest tooling cost).
- Flow simulation (Moldflow) predicts fill pattern and weld line locations.

12.3 Common Defects

Every defect has a root cause in part geometry, material, or process. Good design eliminates most defects before the mold is ever cut.

Defect	What You See	Root Cause	Design Fix
Sink Marks	Dimples/depressions on surface opposite thick sections	Thick sections cool slower; interior shrinks, pulling surface inward	Uniform wall thickness; core out thick areas; rib thickness = 0.6x wall
Warpage	Part bows, twists, or distorts after ejection	Non-uniform cooling causes differential shrinkage	Uniform walls; symmetric cooling; balanced gate location
Short Shot	Part incomplete — cavity not fully filled	Insufficient material, inadequate venting, or walls too thin to fill	Increase wall thickness in thin areas; add vents at end of fill; relocate gate
Flash	Thin film of plastic at parting line or around inserts	Clamp force too low; worn mold surfaces; excessive injection pressure	Minimize projected area; check parting line placement; mold maint.
Weld / Knit Lines	Visible line where two flow fronts meet (cosmetic + weak point)	Flow splits around holes/cores and rejoins on far side	Move gate to push weld line to non-critical area; increase wall temp
Jetting	Snake-like squiggly pattern on part surface near gate	Melt shoots across cavity before contacting a wall to form a front	Redirect gate so melt hits a wall immediately (tab gate or fan gate)

Figure 12.3: Six common injection molding defects with visual description, root cause, and design fix.

Defect	What You See	Root Cause	Design Fix
Sink marks	Dimples opposite thick sections	Thick areas cool slowly, interior shrinks	Uniform walls; core out; rib = 0.6× wall
Warpage	Part bows/twists after ejection	Non-uniform cooling → differential shrinkage	Uniform walls; symmetric cooling; balanced gate
Short shot	Incomplete part	Inadequate venting, thin walls, low pressure	Vent at end of fill; increase thin walls
Flash	Thin film at parting line	Low clamp force, worn mold, high pressure	Minimize projected area; mold maintenance
Weld lines	Visible line where fronts meet	Flow splits around holes/cores	Relocate gate; increase melt temperature
Jetting	Squiggly pattern near gate	Melt shoots across cavity before wall contact	Redirect gate so melt hits a wall immediately

Table 12.2: Common defects, root causes, and design fixes.

Weld Line Strength

Weld lines retain only 50–90% of base material strength. If a weld line falls on a load-bearing feature, relocate the gate or add ribs to reinforce. Flow simulation is the primary tool for predicting and relocating weld lines before cutting the mold.

12.3.1 Worked DFM Example: Sensor Enclosure

Requirements: outdoor IP65 enclosure for GreenShield controller, ABS material, holds PCB (60×40 mm) and two cable glands.

Design decisions:

- **Wall thickness:** 2.0 mm uniform (ABS range: 1.5–3.5 mm). Adequate for IP65 with gasket groove.
- **Draft:** 2° on all external walls. C-1 matte finish; no additional texture draft needed.
- **Ribs:** four internal ribs at 1.2 mm thick (0.6 × 2.0), 6.0 mm tall (3 × 2.0), connecting the PCB mounting bosses to the walls. Base radius 0.5 mm.
- **Bosses:** four PCB mounting bosses, OD = 6.0 mm (2.5 × 2.4 mm screw), wall = 1.2 mm, height = 10 mm. Each connected to nearest wall by a rib.
- **Radii:** all inside corners 1.0 mm (0.5 × 2.0 mm wall). Outside corners 3.0 mm (inside + wall).
- **Sealing:** 1.5 mm wide × 1.0 mm deep O-ring groove around the lid perimeter. Silicone O-ring (cord dia. 1.5 mm, compressed to 75% = 1.0 mm groove depth).
- **Gate:** edge gate on the bottom face (non-cosmetic), at the thickest section near a rib intersection.
- **Parting line:** at the lid/base junction (natural split).

12.4 Tooling and Economics

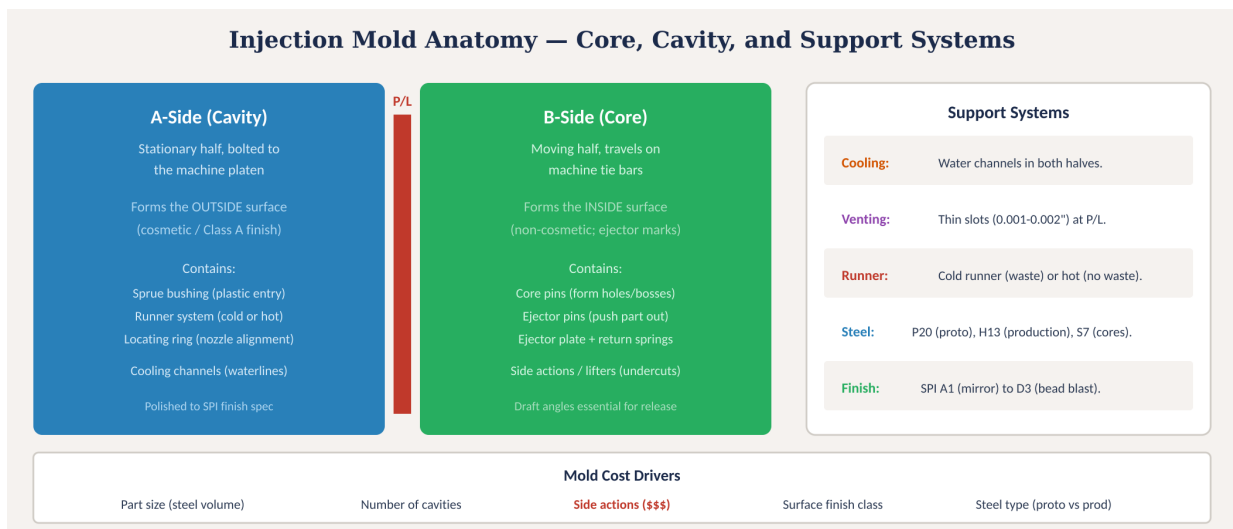


Figure 12.4: Mold anatomy: A-side (cavity), B-side (core), parting line, support systems, and cost drivers.

12.4.1 Mold Construction

The **A-side** (cavity) forms the cosmetic outside surface. It contains the sprue bushing, runner system, and locating ring. Surfaces are polished to SPI specifications (A-1 mirror to D-3 bead blast). The **B-side** (core) forms the inside surface and contains core pins, ejector pins, ejector plates, and side actions. The parting line between the halves leaves a witness mark on the part.

Support systems: cooling channels (water lines in both halves; conformal cooling can reduce cycle time 20–40%), venting (thin slots 0.001–0.002 in at parting line), runner system (cold runner produces waste; hot runner eliminates waste but costs more), and ejection system (pins, sleeves, blades, stripper plates).

12.4.2 Mold Classifications (SPI)

Class	Steel	Life (cycles)	Cost Range	Use
105 (Prototype)	Aluminum / soft steel	<500	\$5K–\$15K	Design validation, pilot runs
104 (Low Prod.)	P20 pre-hardened	<100K	\$15K–\$50K	Low volume, bridge tooling
103 (Med. Prod.)	H13 hardened	<500K	\$30K–\$80K	Medium-volume production
101 (High Prod.)	Premium hardened	1M+	\$50K–\$200K+	High-volume, long-life production

Table 12.3: SPI mold classifications.

12.4.3 Part Cost Economics

$$C_{\text{part}} = C_{\text{material}} + C_{\text{cycle}} + C_{\text{tooling, amortized}} \quad (12.2)$$

where C_{material} = weight $\times$ resin cost, C_{cycle} = cycle time $\times$ machine rate/3600, and C_{tooling} = mold cost/expected lifetime parts.

Example (small ABS enclosure, 30 g): Material: \$0.06. Cycle (25 s at \$60/hr): \$0.42. Tooling (\$25K amortized over 50K parts): \$0.50. **Total: \$0.98/part**. At 500K parts, tooling drops to \$0.05/part, total \$0.53/part.

Injection molding breaks even versus 3D printing at approximately 100–500 parts depending on part size and complexity. Below that, 3D printing is cheaper; above it, injection molding wins dramatically.

12.5 Warpage

Warpage is dimensional distortion that occurs after ejection. It is caused by differential shrinkage: when different regions of the part cool at different rates or have different molecular orientations, they shrink by different amounts, creating internal stresses that bow or twist the part.

12.5.1 Causes of Warpage

Differential cooling: one side of the wall cools faster (e.g., mold steel temperature differs between A-side and B-side). The slow-cooling side shrinks more after ejection, bowing the part toward it.

Fiber orientation: glass-filled materials shrink less in the flow direction than across it; 0.3% vs. 0.8% creates a 0.5% differential. **Gate location:** determines flow pattern and therefore orientation pattern. A poorly placed gate creates asymmetric flow that maximizes differential shrinkage.

12.5.2 Designing to Minimize Warpage

- Maintain uniform wall thickness ($\pm 10\%$).
- Equalize cooling on both sides (symmetric cooling channels).
- Use gating that produces symmetric flow.
- For glass-filled materials, gate at center to produce radial (symmetric) fiber orientation.
- Add ribs to resist bowing (ribs on the concave side of expected warp).

12.6 Surface Finish

12.6.1 SPI (PIA) Finish Standards

Surface finish is specified using SPI (Society of the Plastics Industry, now PIA) standards:

Grade	Finish	Ra (μm)	Method and Application
A-1	Mirror	0.012	Diamond buff. Optical lenses, light pipes.
A-2	High gloss	0.025	Diamond buff. Cosmetic consumer products.
B-1	Semi-gloss	0.05	600 grit paper. General cosmetic surfaces.
C-1	Matte	0.35	600 stone. Hides sink marks, fingerprints.
D-1	Textured	1.0	Dry blast. Grips, non-slip, industrial.
D-3	Rough texture	3.2	Coarse blast. Heavily textured panels.

Table 12.4: SPI/PIA mold surface finish standards.

Higher polish = higher mold cost. C-1 matte is the sweet spot for most student and industrial enclosures; hides minor defects while maintaining a professional appearance.

12.6.2 Textured Surfaces and Draft

Texture creates an effective undercut. Add 1° of additional draft per 0.001 in (0.025 mm) of texture depth. A D-1 texture (1.0 μm Ra, $\$ \sim \0.001 in depth) requires 1° extra draft beyond the baseline. Failing to add texture draft causes drag marks during ejection.

12.7 Insert Molding and Overmolding

12.7.1 Insert Molding

A pre-formed component (metal insert, PCB, bushing) is placed in the mold before injection. Plastic flows around and bonds to the insert. Use when you need: threaded metal inserts (stronger than threading into plastic), electrical contacts molded into a housing, or reinforcing metal structures embedded in plastic.

Design rules: provide ≥ 1.5 mm of plastic around the insert, add knurling or grooves for mechanical interlock, preheat metal inserts to reduce thermal shock stress, and ensure the insert is fixtured precisely.

12.7.2 Overmolding

A second material is molded over a previously molded substrate. Two-shot or two-step process. Common application: soft TPE grip over rigid ABS body (power tools, toothbrushes, medical devices). The two materials must be chemically compatible for adhesion. Common pairs: ABS/TPE, PC/TPE, nylon/TPE.

12.8 Living Hinges and Snap-Fit Clips

12.8.1 Living Hinges

A living hinge is a thin, flexible web of plastic that connects two rigid sections, allowing them to flex repeatedly without breaking. **Polypropylene (PP)** is the only commonly injection-molded material suitable for living hinges; its molecular structure allows millions of flex cycles without fatigue. Design rules: hinge thickness 0.25–0.50 mm, web length 1.0–1.5 mm, flex immediately after molding (orients molecules across the hinge, increasing life by 10×), and gate so flow is perpendicular to the hinge line.

Material Matters

ABS, PC, and nylon all crack at a living hinge within a few cycles. **Only PP** (and some PE grades) can survive millions of flexes. If your design needs a living hinge, the part must be PP; no exceptions.

12.8.2 Snap-Fit Clips

Cantilever snaps are the most common: a beam deflects during assembly, then locks behind a lip. Design for the allowable strain of your material: 2% for ABS, 4% for PP, 1.5% for PC. Maximum deflection: $y = \epsilon L^2 / (1.5t)$ where ϵ = allowable strain, L = beam length, t = beam thickness. Include a lead-in angle (30–45°) for easy assembly and a steeper retaining angle (80–90°) for secure locking. Snap-fits eliminate screw hardware; a significant DFA (Design for Assembly) advantage.

12.9 Artifact Handoff: What This Chapter Supports

Artifact	Description	Forward Reference
DFM Review	Checklist verification for each molded part	CDR deliverable
Material Selection	Resin choice with shrinkage and property data	BOM, tolerance analysis
Tooling Estimate	Mold class, cost, and timeline	Budget, schedule

12.10 DFM Checklist

Review every item before sending your CAD model to a molder.

1. Wall thickness uniform ($\pm 10\%$)? Within recommended range for material?
2. Draft angles applied to ALL surfaces parallel to pull direction? Texture draft added?

3. Ribs at 50–70% of wall? Height $\leq 3 \times$ wall? Base radii present?
4. Bosses at $2 \times$ hole ID? Connected to walls with ribs? Boss wall $\leq 0.6 \times$ nominal?
5. All inside corners radiused (min $0.5 \times$ wall)? Outside radius = inside + wall?
6. Undercuts identified? Side actions justified or redesigned out?
7. Gate location at thickest section? Not on cosmetic surface?
8. Parting line location acceptable on finished part?
9. Ejector pin locations on non-cosmetic surfaces? Adequate draft for release?
10. Shrinkage accounted for in critical dimensions? Tolerances achievable (± 0.005 in typical)?
11. Material selected? Properties verified against operating conditions?
12. Moldflow or fill analysis run? Weld lines and air traps identified?

Fix Problems in CAD, Not in Steel

Fixing a design problem in CAD takes minutes. Finding the same problem after the mold is cut costs \$5,000–\$20,000 in mold modifications and 2–4 weeks of delay. Run this checklist at your CDR before committing to tooling.

Requirements Mapping: From DFM Rules to Shall-Statements

Injection molding constraints generate system requirements:

- “Part must survive 1 m drop onto concrete” → MAT-REQ: Material shall have Izod impact ≥ 200 J/m (ABS or PC/ABS).
- “Outdoor enclosure, 5+ year UV exposure” → MAT-REQ: Material shall be UV-stabilized or painted.
- “Volume 50K units/year” → TOOL-REQ: Mold shall be SPI Class 104 (P20 steel, ≤ 100 K cycle life).
- “Assembly time < 30 s” → DFA-REQ: Enclosure shall use snap-fit closure (no fasteners). Cantilever strain $\leq 2\%$ for ABS.

12.11 Practice Questions

12.11.1 Conceptual Questions

1. Why does cooling time scale with the *square* of wall thickness rather than linearly? How does this affect part cost?
2. A part has a 4 mm thick boss connected to a 2 mm wall. Predict the defect you will see on the surface opposite the boss. What is the design fix?
3. Explain why a part shrinks onto the core side of the mold during cooling. How does this affect draft angle requirements for the core versus cavity sides?
4. Your part has a snap hook that creates an undercut on one side. The molder quotes \$18,000 for the side action. Propose two redesign alternatives that might eliminate the side action.

5. Compare ABS and polypropylene for a consumer electronics enclosure that must accept paint and withstand a 1-meter drop test. Which do you recommend and why?
6. Why must the gate be placed at the thickest section? What happens if you gate at a thin section instead?
7. A textured surface (SPI D-2, depth ≈ 0.002 in) on a 3-inch-deep wall currently has 1° of draft. Is this sufficient? If not, what is the correct draft angle?

12.11.2 Application / Scenario Questions

8. You are designing a two-piece snap-fit enclosure for an IoT sensor that will be installed outdoors. The enclosure must be IP65, withstand -20 to $+60^\circ\text{C}$, resist UV for 5+ years, and cost under \$3/part at 10,000 units. Select the material, specify wall thickness, describe the snap-fit design, and estimate whether injection molding is economically justified.
9. Your teammate's CAD model for a motor mount bracket has: 5 mm walls, zero draft on all vertical surfaces, sharp inside corners, and a boss with $\text{OD} = 1.5 \times \text{hole ID}$. List every DFM violation, predict the defects that will result from each, and specify the corrected dimensions.
10. **Cost analysis:** A part weighs 45 g in PC/ABS (\$0.003/g). Cycle time is 30 seconds on a machine at \$75/hr. The mold costs \$35,000. Calculate the total part cost at 5,000, 50,000, and 500,000 units. At what volume does injection molding become cheaper than SLA 3D printing at \$8.50/part?

12.11.3 Answers

1. Cooling is a transient heat-conduction problem governed by the Fourier equation. The characteristic time for heat to diffuse through a slab of thickness t scales as t^2/α where α is thermal diffusivity. Doubling wall thickness from 2 mm to 4 mm quadruples cooling time; potentially adding 20–40 seconds per cycle. At a machine rate of \$60/hr, each additional second costs approximately \$0.017/part. Over 100,000 parts, a 20-second cycle-time penalty costs \$17,000.
2. A sink mark (dimple/depression) on the surface opposite the boss. The 4 mm boss is twice the 2 mm wall, creating a thick section that cools more slowly. The interior shrinks and pulls the surface inward. Fix: reduce boss wall to $0.6 \times 2 = 1.2$ mm ($\text{OD} = \text{hole ID} + 2 \times 1.2$ mm). Connect the boss to the nearest wall with a thin rib.
3. As plastic cools it contracts volumetrically. The cavity half forms the outside surface, and the part shrinks *away* from it, naturally releasing. The core half forms the inside surface, and the part shrinks *onto* it, gripping tighter as it cools. This means the core side typically needs more draft (1.5° – 2°) than the cavity side (0.5° – 1°) to overcome the shrink-grip.
4. (a) Reorient the snap hook to flex along the mold-open direction; if the hook can deflect during ejection, no side action is needed (the hook “snaps” past the mold surface). (b) Replace the snap hook with a separate clip that installs post-molding via a push-fit or ultrasonic weld, eliminating the undercut entirely.
5. ABS. Both have adequate impact strength, but ABS accepts paint directly (smooth surface, good adhesion) while PP requires flame treatment or specialty primers due to its low surface energy. ABS also has a higher HDT ($\$ \sim 95^\circ\text{C}$ versus $\$ \sim 60^\circ\text{C}$ for unfilled PP), providing

more thermal margin. ABS shrinkage (0.5–0.7%) is lower and more predictable than PP (1.0–2.5%), enabling tighter dimensional control for snap-fits and mating features.

6. Gating at the thickest section ensures that the section that shrinks the most remains connected to the runner for packing pressure the longest. If you gate at a thin section, the thin area freezes first and seals off (“gates off”), cutting the thick section off from packing pressure. The result is severe sink marks and voids in the thick section because the material cannot be replenished as it shrinks.
7. No. The texture-depth rule requires adding 1° per 0.001 in of depth. At 0.002 in depth, add 2° beyond the standard 1° for a total of 3° per side. Without adequate draft, the textured surface acts like sandpaper during ejection, leaving drag marks that ruin the finish and gouge the mold steel.
8. Material: ASA (acrylic-styrene-acrylonitrile), UV-stable variant of ABS, good impact from -20 to $+80^\circ\text{C}$, IP65-rated with proper gasket design, paintable if needed. Wall thickness: 2.0 mm nominal (meets structural requirements, enables IP65 with perimeter gasket groove). Snap-fit: cantilever snaps along the mold-open axis (no side action), 2% max strain for ASA, designed with 1.5° draft. Economics: at 10,000 units with a Class 104 mold (\$25K), tooling amortization is \$2.50/part. Material ($20\text{ g} \times \0.003): \$0.06. Cycle (20 s at \$60/hr): \$0.33. Total: $\$ \sim \$2.89/\text{part}$, under the \$3 target. Injection molding is justified.
9. Violations and fixes: (a) 5 mm walls are too thick for any common plastic; excessive cooling time, sink marks, voids. Fix: core out to 2 mm uniform walls, add ribs for stiffness. (b) Zero draft on all vertical surfaces; parts will stick, ejector damage, mold wear. Fix: add minimum 1° per side to all surfaces parallel to pull. (c) Sharp inside corners; stress concentrations ($3\times$ multiplier), impeded flow. Fix: add $R = 0.5 \times 2 = 1$ mm fillet to all inside corners; outside radius $= 1 + 2 = 3$ mm. (d) Boss OD at $1.5\times$ hole ID is too thin-walled, creating a thick ring that sinks. Fix: increase OD to $2.0\times$ hole ID. Predicted defects: sink marks everywhere (thick walls + oversized boss), warpage (non-uniform cooling from variable thickness), ejection damage (no draft), cracking at corners under load (no radii).
10. Material: $45 \times 0.003 = \$0.135/\text{part}$. Cycle: $30/3600 \times 75 = \$0.625/\text{part}$. These are constant.
 At 5,000 parts: tooling $= 35000/5000 = \$7.00/\text{part}$. Total: \$7.76/part.
 At 50,000 parts: tooling $= 35000/50000 = \$0.70/\text{part}$. Total: \$1.46/part.
 At 500,000 parts: tooling $= 35000/500000 = \$0.07/\text{part}$. Total: \$0.83/part.
 Break-even vs. SLA at \$8.50/part: solve $0.135 + 0.625 + 35000/n = 8.50$. $35000/n = 7.74$.
 $n = 4,522$ parts. Above $\$ \sim \$4,500$ parts, injection molding is cheaper.

12.11.4 Application / Scenario Questions

7. You are designing an outdoor sensor enclosure in ABS. The enclosure must be IP65 (dust-tight, water-jet resistant). Specify: wall thickness, draft angles, sealing method, and surface finish. Justify each choice.
8. A client asks you to injection mold a part with a living hinge. They want to use ABS because it matches the rest of their product line. What do you tell them?
9. Your injection-molded part has sink marks opposite the boss locations. The bosses are 3.0 mm wall in a part with 2.0 mm nominal wall. What is wrong, and how do you fix it?

12.11.5 Answers to Application Questions

7. Wall: 2.0 mm nominal (meets structural, enables IP65 with perimeter gasket groove). Draft: 2° standard on all external walls; add 1° per 0.001 in of texture depth if textured. Sealing: silicone O-ring in a groove on the lid mating surface (not adhesive or gasket tape). Surface: C-1 matte (hides minor defects, professional appearance, reduces mold cost vs. polish). Internal surfaces: D-1 textured (hides ejector pin marks).
8. ABS cannot survive living hinge cycles; it will crack within 1–5 flexes. Only PP (and some PE) can form living hinges that last millions of cycles. Options: (a) redesign the part in PP; (b) use a separate PP living hinge piece bonded/overmolded to the ABS body; (c) replace the living hinge with a mechanical hinge (pin + knuckle) that can be molded in ABS.
9. Boss wall = 3.0 mm vs. 2.0 mm nominal = 150% of wall thickness. This thick section cools slowly, creating a void that pulls the surface inward (sink mark). Fix: reduce boss wall to $0.6 \times 2.0 = 1.2$ mm (60% of nominal). If the boss needs more material for screw engagement, connect it to the nearest wall with a rib (rib at 60% wall = 1.2 mm).

Chapter 13

Printed Circuit Board Design Fundamentals

13.1 PCB Structure and Materials

A printed circuit board is a layered sandwich of insulating substrate (FR-4 fiberglass-epoxy) with patterned copper conductors. Copper traces, pads, and planes are created by chemically etching unwanted copper from a continuous sheet. A **solder mask** (the green coating) protects copper from oxidation and shorts. A **silkscreen** layer prints component outlines and reference designators.

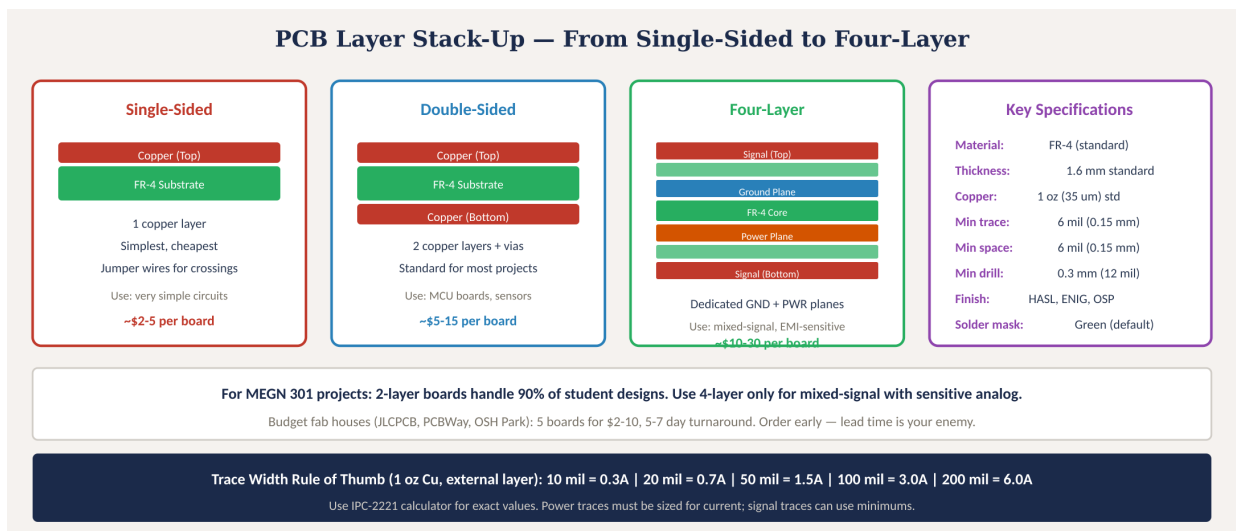


Figure 13.1: PCB layer stack-ups from single-sided to four-layer with key specifications and trace width current reference.

13.1.1 Layer Stack-Ups

Single-sided: one copper layer. Simplest, cheapest (\$2–5). Jumper wires for crossings. Very simple circuits only. **Double-sided:** copper on both sides connected by vias (plated through-holes). Standard for 90% of student projects (\$5–15). **Four-layer:** adds dedicated ground and power planes between signal layers. Use for mixed-signal or EMI-sensitive designs (\$10–30).

13.1.2 Key Specifications

FR-4 substrate, 1.6 mm standard thickness, 1 oz (35 μ m) copper, 6 mil minimum trace/space, 0.3 mm minimum drill. Surface finishes: HASL (cheapest, good for THT), ENIG (flat pads, good

for SMD), OSP (organic, short shelf life).

Trace Width for Current

A 10-mil trace on 1-oz external copper safely carries only 0.3 A. Power traces must be sized for actual current: 10 mil = 0.3 A, 20 mil = 0.7 A, 50 mil = 1.5 A, 100 mil = 3.0 A, 200 mil = 6.0 A. Use the IPC-2221 calculator for exact values. Signal traces can use minimums; power traces must be sized for current.

13.1.3 Component Packages

Component Packages — Through-Hole vs Surface Mount	
Through-Hole (THT)	Surface Mount (SMD/SMT)
Leads pass through holes in the PCB and are soldered on the opposite side.	Components soldered directly to pads on the board surface. No holes required.
DIP Dual in-line package. ICs, sockets, 0.1" pin spacing. Easy to hand-solder. TO-220 Power transistors, voltage regulators. Tab for heatsink mounting. Axial/Radial Resistors, capacitors, diodes. Axial = flat; Radial = standing. Headers Pin headers, screw terminals. 0.1" (2.54mm) pitch standard.	0805/0603 Standard resistors/caps. 0805 hand-solderable; 0603 needs skill. SOIC/TSSOP IC packages. Gull-wing leads. Hand-solderable with flux + iron. QFP/QFN Fine-pitch ICs. QFN has ground pad underneath — needs reflow/hot air. BGA Ball grid array. Cannot hand-solder. Requires reflow oven + X-ray.
Advantages: easy to solder, inspect, rework, prototype Disadvantages: large, expensive, slower assembly, fewer options	Advantages: small, cheap, fast pick-and-place, better HF performance Disadvantages: harder to hand-solder, inspect, rework; needs magnification
RECOMMENDED for MEGN 301 prototypes (unless component only available in SMD)	Use for: production, space-constrained, SMD-only parts (JLCPCB assembly can place SMD for \$8 setup + per-part)
Mixed designs are common: THT for connectors, headers, and power components; SMD for passives and ICs where THT is unavailable.	

Figure 13.2: Through-hole vs surface-mount packages with common types and soldering guidance.

Through-hole (THT): leads pass through holes. DIP, TO-220, axial/radial, pin headers. Easy to hand-solder, inspect, and rework. Recommended for MEGN 301 prototypes. **Surface-mount (SMD/SMT):** soldered to pads on the board surface. 0805/0603 passives, SOIC/TSSOP ICs, QFP/QFN fine-pitch. Smaller, cheaper, but harder to hand-solder. Use SMD only when THT is unavailable or space is critical. Mixed designs are common.

13.2 Design Workflow

Six stages: (1) **Schematic capture**; draw circuit with component symbols and net connections. (2) **Component selection**; map symbols to physical footprints, verify availability, create BOM. (3) **Board layout**; place components, route traces, pour copper fills. (4) **Design review**; peer review, check clearances, verify footprints, review Gerbers. (5) **Fabrication + assembly**; generate Gerbers, order boards, solder. (6) **Bring-up + test**; inspect, smoke test, functional test, debug.

Recommended EDA tool: **KiCad 8** (free, open-source, fully capable). Alternatives: Altium (industry standard, student license), EasyEDA (browser-based, JLCPCB integrated).

13.3 Schematic Capture Best Practices

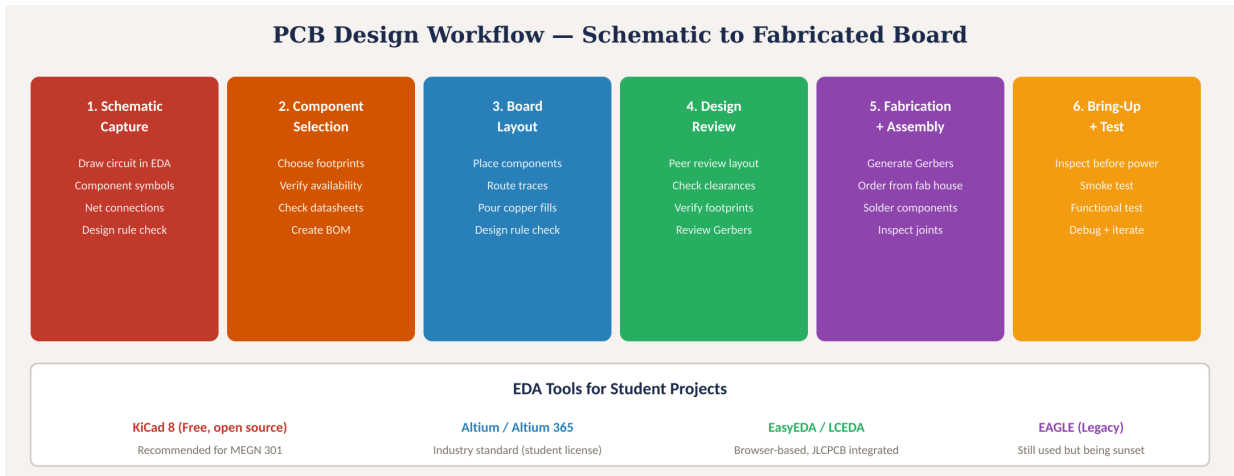


Figure 13.3: Six-stage PCB design workflow from schematic capture through board bring-up.

13.3.1 Schematic Organization

A professional schematic is a *communication document*, not just a wiring diagram. Follow these rules from IPC-2612 [20]:

1. **One functional block per sheet:** power supply on Sheet 1, MCU + peripherals on Sheet 2, motor drivers on Sheet 3, connectors on Sheet 4. Title each sheet.
2. **Signal flow left to right:** inputs on the left, outputs on the right. Power flows top (VCC) to bottom (GND).
3. **Net labels over long wires:** replace wires that cross the entire sheet with named net labels (e.g., MOTOR_PWM, SDA, VBAT). This eliminates spaghetti routing.
4. **Power flags:** explicitly show where power enters the schematic. Add decoupling caps adjacent to every IC symbol, not on a separate “capacitor page.”
5. **Consistent reference designators:** R for resistors, C for capacitors, U for ICs, J for connectors, D for diodes, Q for transistors, L for inductors.

13.3.2 Component Selection and Footprint Verification

The most expensive PCB mistake is a wrong footprint. For every component:

1. Download the **manufacturer’s datasheet** (not a distributor summary page).
2. Find the **recommended land pattern** (footprint) in the datasheet’s “Package Information” section.
3. Verify the KiCad library footprint dimensions match the datasheet **exactly**: pad size, pad pitch, courtyard.
4. If no library footprint exists, create one from the datasheet using KiCad’s footprint editor. Measure twice.
5. **Cross-check:** print the footprint at 1:1 scale and physically place the component on the printout.

The \$50 Footprint Mistake

A team ordered 5 PCBs (\$25 total) with a QFN-32 footprint that had the wrong pad pitch (0.5 mm vs. 0.65 mm). All 5 boards were scrap. The fix required a new board spin: another

\$25 + 2 weeks of lost schedule. Total cost of one unchecked footprint: \$50 + 2 weeks. The 1:1 printout would have caught this in 30 seconds.

13.3.3 Design Rule Configuration

Configure KiCad's DRC rules before starting layout:

Rule	Minimum	Recommended	Notes
Trace width (signal)	6 mil	10 mil	Wider = easier to solder rework
Trace width (power)	Per IPC calc	50–200 mil	Size for actual current
Trace spacing	6 mil	8 mil	Tight spacing increases fab cost
Via drill	0.3 mm	0.4 mm	Smaller vias cost more
Via annular ring	0.15 mm	0.25 mm	Larger = more reliable
Clearance to edge	0.3 mm	0.5 mm	Prevents copper exposure at board edge

Table 13.1: Recommended DRC rules for MEGN 301 PCBs (JLCPCB/PCBWay compatible).

13.4 Critical Layout Rules

13.4.1 Decoupling Capacitors

Place a 100 nF ceramic capacitor within 5 mm of every IC power pin, connected with the shortest possible traces. Add one 10 μ F bulk capacitor per power rail. Via directly to ground plane for the ground side. This is not optional; without decoupling, MCUs reset and ADCs produce garbage data.

13.4.2 Ground Plane

On a 2-layer board, fill the bottom layer with a ground copper pour. This provides a low-impedance return path for all signals and dramatically reduces noise. Connect the pour to GND with abundant vias. Avoid splitting the ground plane; a split forces return current to detour.

13.4.3 Component Placement and Routing

Place components before routing. Group by function. Keep high-current paths short. Route power first, then sensitive signals, then non-critical digital. Avoid 90-degree trace corners (use 45-degree). Keep analog and digital sections physically separated. Add thermal relief pads on ground-plane connections for hand soldering.

The #1 Rule

PRINT YOUR BOARD AT 1:1 SCALE AND PLACE REAL COMPONENTS ON THE PRINTOUT before ordering. This single step catches 80% of footprint errors; the most expensive PCB mistake (wrong footprint = scrap board).

13.5 Fabrication, Assembly, and Bring-Up

13.5.1 Fabrication Files

Gerber files (RS-274X format): one per layer. **Drill files** (Excellon): hole coordinates and diameters. **BOM + pick-and-place**: for assembly services. Always verify in a standalone Gerber viewer before ordering.

13.5.2 Fab Houses

JLCPCB: 5 boards for \$2–7 + shipping (5–7 day fab). PCBWay: similar. OSH Park: \$5/sq-in, US-based. Order early; total timeline is 4–6 weeks from design start to working board.

13.5.3 Board Bring-Up Protocol

Step 1 Is Mandatory: No Exceptions

Do not apply power to any board until Step 1 is complete and signed off. Skipping visual inspection is the fastest way to destroy a \$40 ESP32 module. A single solder bridge on a 0.5 mm-pitch IC creates a dead short that damages the component the instant power is applied.

1. **Visual inspection under $\geq 10\times$ magnification** (loupe, USB microscope, or stereo microscope). This is a **mandatory gate**; a teammate must co-sign the inspection before power is applied. Inspect *every* solder joint. Specific defects to look for:
 - **Solder bridges**: look between adjacent pads on fine-pitch ICs (QFP, SOIC). A bridge shorts two pins.
 - **Cold joints**: dull, grainy, or blobby solder that did not wet the pad. These create intermittent connections.
 - **Tombstoned passives**: 0603/0805 components standing on one end (one pad soldered, one lifted).
 - **Missing components**: compare the board to the BOM and the schematic. Every refdes on the silkscreen should have a component.
 - **Wrong orientation**: check polarity marks on diodes, tantalum caps, ICs (pin 1 dot), and electrolytic caps.
2. **Continuity check**: multimeter between VCC and GND on every power rail. Any reading $< 10\Omega$ indicates a short. **Do not proceed until all shorts are resolved.**
3. **Current-limited power**: set bench supply to expected voltage, current limit to 50 mA. Apply power. If the supply immediately current-limits, there is a short. Remove power and re-inspect.
4. **Voltage verification**: measure every power rail with a multimeter. Each rail should read within $\pm 5\%$ of its nominal voltage.
5. **Smoke test**: increase current limit to the expected operating current. Monitor for 60 seconds. Watch for: components getting hot (touch-test cautiously), smell of burning flux or plastic, any visible smoke.
6. **Functional test**: load firmware, verify heartbeat LED, test each subsystem individually before connecting external hardware.

13.5.4 Design Review Checklist

Before ordering PCBs, verify:

1. DRC (Design Rule Check) passes with zero errors and zero unreviewed warnings.
2. All footprints verified against component datasheets (not just library defaults).
3. Board outline correct with mounting holes matching the enclosure.
4. All connectors accessible and oriented correctly for cable routing.
5. Silkscreen readable (refdes near components, polarity marks on diodes/caps/ICs).
6. 1:1 printout with real components placed; **this catches 80% of errors**.
7. Gerber files opened in a standalone viewer and visually inspected layer by layer.
8. BOM cross-checked against schematic (every schematic component in BOM and vice versa).
9. Peer review by at least one teammate who did not do the layout.

13.5.5 Development Board Integration

When using a development board (ESP32 DevKit, Arduino Nano) instead of a custom PCB:

- Mount the dev board on a custom carrier PCB or perfboard with proper headers.
- Use screw terminals or JST connectors (not breadboard jumper wires) for all permanent connections.
- Add status LEDs and test points to the carrier board.
- Document the wiring diagram with connector pin assignments.
- Secure all wires with strain relief (zip ties, hot glue at connector exit points).

A well-wired development board that meets all requirements is superior to a custom PCB that delays the project or has errors.

13.6 EMI, Noise, and Signal Integrity

13.6.1 The Return Current Problem

Every signal current has a return current that flows through the ground plane (or ground trace) back to the source. The return current follows the path of *least impedance*; at high frequencies, this is the path directly beneath the signal trace. If the ground plane is split or missing under a signal trace, the return current detours, creating a large loop antenna that radiates EMI and picks up noise.

Rule: never split the ground plane. If you must route a trace on the ground layer, stitch the ground pour back together with vias on both sides of the crossing.

13.6.2 Analog-Digital Separation

Mixed-signal boards (analog sensors + digital MCU + motor drivers) are the norm in MEGN 301 projects. Three strategies:

1. **Physical separation:** place analog components (sensors, ADC, analog filters) on one side of the board, digital components (MCU, motor drivers) on the other.
2. **Separate power rails:** use separate voltage regulators for analog and digital sections. At minimum, use an RC filter ($10\ \Omega + 10\ \mu\text{F}$) between the digital supply and the analog supply.
3. **Single-point ground connection:** if using split ground pours (advanced), connect them at exactly one point near the ADC. For most student projects, a single continuous ground pour with careful component placement is sufficient and safer.

13.6.3 Motor Noise Mitigation on PCB

Motor PWM switching generates noise on every power and ground trace. Mitigation:

- Route motor power traces on a separate layer or board region from signal traces.
- Place ceramic capacitors (100 nF) directly across motor driver power pins.
- Add a bulk electrolytic (100–470 μ F) at the motor power input.
- Use a TVS diode on the motor power bus for transient suppression.
- Never share ground vias between motor driver and MCU sections.

13.7 Thermal Management on PCBs

Components that dissipate significant power require thermal design:

- **Thermal pads:** many voltage regulators and motor drivers have an exposed pad on the bottom of the package. This pad **must** be soldered to a copper area on the PCB and connected to internal ground planes with thermal vias (array of 0.3 mm vias on 1.0 mm pitch under the pad).
- **Copper pour as heatsink:** for SOT-223 or D-PAK regulators, extend the ground pour under and around the package. Each square centimeter of 1-oz copper on the outer layer provides approximately 60 °C/W of thermal resistance to ambient.
- **Thermal relief:** for hand-soldering, use thermal relief pads (spoke connections) on ground-connected pads. Without them, the ground plane wicks heat away faster than the soldering iron can supply it, resulting in cold joints.

Quick Thermal Check

For any linear regulator: $P = (V_{in} - V_{out}) \times I_{load}$. If $P > 0.5$ W, you need thermal analysis. If $P > 1$ W, you probably need a switching regulator instead.

Requirements Mapping: From PCB Specs to Shall-Statements

PCB design decisions directly generate system requirements:

- “Board fits in 60×40 mm enclosure” → SYS-REQ: The PCB shall have maximum dimensions of 58×38 mm with mounting holes at [coordinates].
- “Motor driver draws 3 A” → PCB-REQ: Motor power traces shall be ≥ 150 mil wide on 1-oz copper.
- “ADC reads 0–3.3 V sensors” → PCB-REQ: Analog input traces shall be routed ≥ 10 mm from motor power traces with a continuous ground plane beneath.

Every technical specification in Part II should trace back to a shall-statement in your SDRD.

13.8 Common Mistakes

Wrong footprint (scrap board), trace too narrow (burns under load), missing decoupling caps (erratic behavior), poor grounding (noise), no mounting holes (cannot secure board), connector on wrong side (cable routing impossible), silkscreen under parts (cannot read refdes). Prevention: print 1:1, run DRC, peer review, use 3D viewer.

Common PCB Layout Mistakes — What to Check Before Ordering			
Mistake	What Goes Wrong	How to Prevent	When Caught
Wrong footprint	Component doesn't fit pads; pins misaligned; board is scrap	Print 1:1 on paper; place real component on printout; verify every footprint	Board arrival
Trace too narrow	Trace burns/fuses under load current; intermittent connection	Use IPC-2221 trace width calculator for every power trace; verify in DRC	Under load testing
Missing decoupling caps	MCU resets, ADC noise, erratic digital behavior	Place 100nF cap within 5mm of every IC power pin; add bulk cap per rail	Functional testing
Poor grounding	Ground loops, noise coupling, analog drift, EMI failures	Use ground pour/plane; star-point for mixed signal; keep return paths short	Integration testing
No mounting holes	Board cannot be secured in enclosure; vibration failures	Add 4x M3 mounting holes at corners; match enclosure standoff pattern	Assembly
Connector on wrong side	Cable routing impossible; mechanical interference with enclosure	Check 3D model in EDA tool; verify against enclosure CAD before ordering	Assembly
Silkscreen under parts	Reference designators invisible after assembly; debugging impossible	Move refdes outside component outline; check silkscreen layer visibility	Assembly

Figure 13.4: Seven common PCB layout mistakes with consequences and prevention strategies.

13.8.1 Development Boards vs Custom PCBs

Use dev boards when: circuit is straightforward, form factor is not critical, timeline is tight, design is still evolving, or a suitable module already exists. Use custom PCBs when: specific form factor required, noise-sensitive analog, high reliability needed, multiple units, or clean cable routing. A well-wired dev board that meets requirements is superior to a custom PCB that delays the project.

13.9 Artifact Handoff: What This Chapter Supports

Artifact	Description	Key Deliverable For
Schematic	Complete circuit with all component values	CDR, FDR
PCB Layout	Gerber files ready for fabrication	CDR (procurement)
BOM (electronics)	Components with package, value, vendor, Plan B	CDR, FDR

13.10 Practice Questions

1. Your project has a motor driver that draws 3 A peak. What is the minimum trace width for the motor power traces on a 2-layer board with 1-oz external copper? What width would you actually use and why?
2. A teammate's PCB layout has the 100 nF decoupling capacitors placed 25 mm from the MCU power pins "because there was more room over there." Explain why this is a problem and what the correct placement is.
3. You receive your first PCB from JLCPCB. Describe the exact sequence of steps you follow before connecting the board to your system.
4. Your team is debating whether to design a custom PCB or use an Arduino Nano + sensor breakout boards. The project is a battery-powered environmental monitor that must fit inside

- a 60×40×25 mm waterproof enclosure. Argue for or against a custom PCB with specific reasoning.
5. A 2-layer board has all signal traces on the top layer and a ground pour on the bottom. A teammate routes a high-current motor trace on the bottom layer, cutting through the ground pour. What problems will this cause and how should it be fixed?
 6. Your schematic uses an LM1117-3.3 voltage regulator in an SOT-223 package to power an ESP32. The datasheet shows a thermal pad on the bottom. Your PCB layout connects the thermal pad to a small isolated copper island with no vias. Calculate the power dissipation at 5 V input, 3.3 V output, 300 mA load, and explain why the current thermal design is inadequate.
 7. You are designing a mixed-signal board with a 12-bit ADC reading a strain gauge (millivolt-level signal) and an H-bridge motor driver on the same PCB. Describe three specific layout strategies to prevent motor noise from corrupting the ADC readings.
 8. Your team's KiCad DRC reports 47 "courtyard overlap" warnings. Your teammate dismisses them: "It compiles fine, just ignore warnings." Explain why courtyard overlaps are serious and what physical problem they represent.

13.10.1 Answers

1. From the IPC-2221 reference: 3 A on 1-oz external copper requires approximately 100 mil (2.5 mm) trace width. In practice, use 150–200 mil (3.8–5 mm) to provide thermal margin, reduce voltage drop, and account for peak transients. Always add margin on power traces; the cost is board space, not money.
2. At 25 mm, the trace inductance between the cap and the IC power pin negates the decoupling effect. High-frequency noise (MHz range) sees the inductance as a high impedance and is not filtered. The cap must be within 5 mm with the shortest possible traces and a via directly to the ground plane. The loop area (VCC pin → cap → GND via → GND plane → IC GND pin) must be minimized.
3. Follow the six-step bring-up protocol: (1) Visual inspection under magnification for bridges, cold joints, and wrong-orientation parts. (2) Multimeter continuity check for VCC-to-GND shorts. (3) Current-limited bench supply at 50 mA, apply expected voltage. (4) Measure all power rails with multimeter. (5) Increase current limit to operating level, monitor 60 seconds for heat/smell. (6) Load firmware, test basic I/O.
4. Argue FOR custom PCB: the 60×40×25 mm enclosure is too small for an Arduino Nano (45×18 mm) plus breakout boards plus wiring. A custom board can integrate the MCU, sensors, and battery management into the available footprint with board-edge connectors for waterproof cable glands. The waterproof requirement also favors a single board with conformal coating over loose wiring that can vibrate and corrode. However: this requires the design to be frozen by CDR and 4–6 weeks of lead time.
5. The motor trace cuts the ground pour, forcing signal return currents to detour around the gap. This creates a large current loop that radiates EMI and couples noise into adjacent signal traces. The motor's switching noise (from PWM) will appear as voltage spikes on the ground plane near the gap, potentially corrupting ADC readings and causing MCU resets. Fix: route the motor power traces on the top layer using wide traces, keeping the bottom ground pour

continuous. If the motor trace must cross to the bottom, use a narrow bridge with ground pour restored on both sides and stitching vias.

6. $P = (5.0 - 3.3) \times 0.3 = 0.51 \text{ W}$. SOT-223 θ_{JA} without thermal vias $\approx 150^\circ\text{C/W}$, so $\Delta T = 76.5^\circ\text{C}$. At 25°C ambient: $T_J = 101.5^\circ\text{C}$, approaching the 125°C limit. At 40°C ambient (inside an enclosure): $T_J = 116.5^\circ\text{C}$, dangerous. The isolated copper island with no vias provides poor thermal coupling to inner layers. Fix: add an array of thermal vias (5–9 vias, 0.3 mm drill, 1.0 mm pitch) under the thermal pad, connecting to the ground plane on Layer 2. This reduces θ_{JA} to $\sim 60^\circ\text{C/W}$.
7. (1) Physical separation: place ADC and strain gauge conditioning on the left half of the board, motor driver on the right, with a clear “quiet zone” boundary between them. (2) Separate analog supply: use an RC filter ($10\ \Omega + 10\ \mu\text{F}$) between digital VCC and the ADC’s AVCC pin to prevent motor switching noise from reaching the analog reference. (3) Guard ring/trace: surround the strain gauge signal traces with a grounded guard trace connected to the analog ground, shielding them from electric field coupling. Additionally: route motor power traces perpendicular (not parallel) to analog signal traces when crossings are unavoidable.
8. Courtyard overlaps mean two components occupy the same physical space on the board. They will either mechanically interfere (components crash into each other during assembly), or their solder pads will short together. DRC warnings are not cosmetic; they represent physical manufacturing problems. Fix: move one component until courtyards no longer overlap. If space is genuinely tight, verify in the 3D viewer that the components physically clear each other.

Chapter 14

Microcontrollers & the ESP32

14.1 MCU Fundamentals

14.1.1 What Is a Microcontroller?

A microcontroller (MCU) integrates a CPU, memory (flash + SRAM), and peripherals (GPIO, ADC, timers, communication) on a single chip. Unlike a microprocessor (Raspberry Pi), an MCU requires no external RAM or storage; it is self-contained and runs bare-metal or RTOS code.

Key blocks: CPU core (executes instructions, 16–600 MHz), flash memory (stores program, 32 KB–16 MB), SRAM (runtime variables, 2 KB–520 KB), peripherals (GPIO, ADC, DAC, PWM, timers, watchdog, RTC), communication (UART, SPI, I2C, CAN, Wi-Fi, BLE), and power management (regulators, sleep modes, brown-out detect).

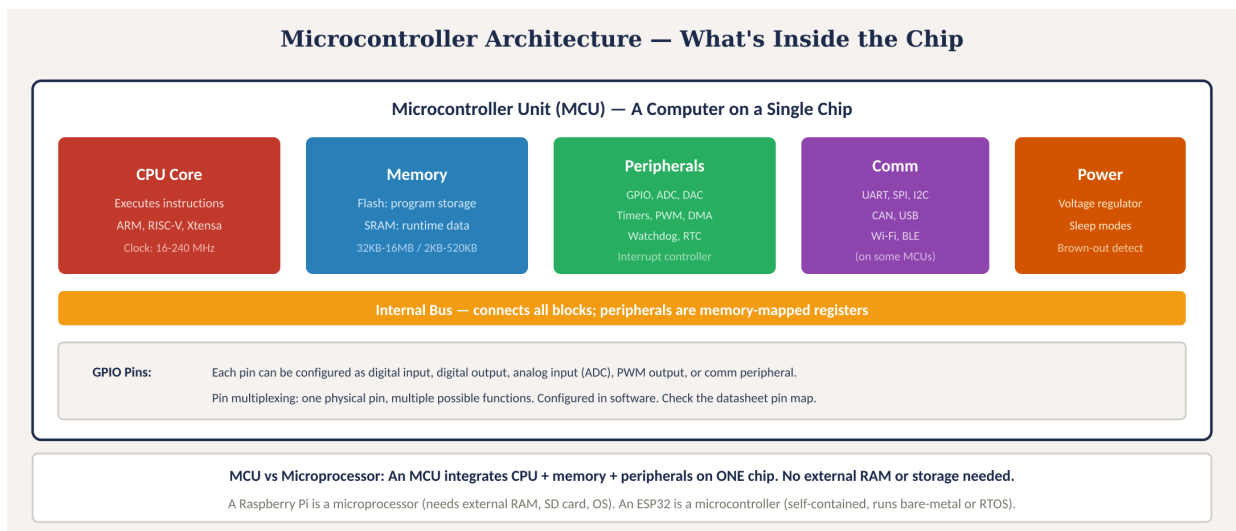


Figure 14.1: MCU internal architecture: CPU, memory, peripherals, communication, and power on one chip.

14.1.2 MCU Platform Comparison

Selection criteria: GPIO count (add 20% margin), communication protocols needed, processing power, power budget (deep sleep current for battery), ecosystem (libraries, community), and cost/availability.

MCU Platform Comparison — Selecting the Right Controller						
Feature	ATmega328P	ESP32-WROOM	STM32F4	RP2040	Teensy 4.1	Raspberry Pi
Board	Arduino Uno	ESP32 DevKit	Nucleo-F446RE	PI Pico	PIRC Teensy	PI 4/5
Core	AVR 8-bit	Xtensa dual-core	ARM Cortex-M4	ARM Cortex-M0+	ARM Cortex-M7	ARM Cortex-A
Clock	16 MHz	240 MHz	180 MHz	133 MHz	600 MHz	1.5-2.4 GHz
Flash	32 KB	4-16 MB	512 KB-1 MB	2 MB	8 MB	SD card (OS)
RAM	2 KB	520 KB	128-256 KB	264 KB	1 MB	1-8 GB
GPIO	14 digital, 6 ADC	34 GPIO, 18 ADC	50+ GPIO	26 GPIO, 4 ADC	42 GPIO	26 GPIO
Comm	UART, SPI, I2C	UART, SPI, I2C,	UART, SPI, I2C,	UART, SPI, I2C,	All + Ethernet,	All + Ethernet,
Power	~50 mA active	~80 mA (Wi-Fi off)	~100 mA active	~25 mA active	~100 mA active	~600 mA-3 A
Cost	~\$5	~\$4-8	~\$12	~\$4	~\$30	~\$35-80
Best For	Learning, simple	IoT, Wi-Fi/BLE,	Industrial, CAN,	Cost-sensitive,	High-speed DAQ,	Full OS, camera,

For MEGN 301: ESP32 is the recommended default — Wi-Fi/BLE built-in, Arduino-compatible, powerful, inexpensive, excellent ecosystem.

Figure 14.2: Six MCU platforms compared. ESP32 recommended for MEGN 301.

14.2 ESP32 Deep Dive

The ESP32-WROOM-32 is the recommended MCU for MEGN 301 projects: dual-core Xtensa LX6 at 240 MHz, 520 KB SRAM, 4 MB flash, built-in Wi-Fi (802.11 b/g/n) and Bluetooth (Classic + BLE 4.2), 34 GPIO, 18 ADC channels (12-bit), 2 DAC channels (8-bit), 16 PWM channels, 3 UART, 2 SPI, 2 I2C, 1 CAN, 10 capacitive touch pins, deep sleep at $\sim \$10 \mu\text{A}$. Cost: \$4–8 for a DevKit board.

ESP32 Deep Dive — The MEGN 301 Standard MCU	
ESP32-WROOM-32 Specifications Core: Dual-core Xtensa LX6 @ 240 MHz Memory: 520 KB SRAM, 4 MB flash (up to 16 MB) Wi-Fi: 802.11 b/g/n, 2.4 GHz, station + AP mode Bluetooth: BLE 4.2 + Classic (dual-mode) GPIO: 34 GPIO (not all exposed on DevKit) ADC: 18 channels, 12-bit (but noisy above 3.1V) DAC: 2 channels, 8-bit (GPIO 25, 26) PWM: 16 channels (LED peripheral), any GPIO Comm: 3x UART, 2x SPI, 2x I2C, 1x I2S, 1x CAN Power: 3.3V logic, 80 mA (Wi-Fi off), 240 mA (Wi-Fi) Touch: 10 capacitive touch-sense pins built-in Sleep: Deep sleep $\sim 10 \mu\text{A}$; wake on timer, touch, or GPIO Cost: \$4-8 for DevKit board (ESP32 chip $\sim \\$2$)	ESP32 Gotchas (Know These!) GPIO 6-11: connected to flash — DO NOT USE GPIO 34-39: input only — no output, no pull-up ADC2 disabled when Wi-Fi active (use ADC1 only) ADC nonlinear above $\sim 3.1\text{V}$ — calibrate or use external ADC 3.3V logic — use level shifter for 5V sensors/actuators Boot pins (GPIO 0, 2, 12, 15) — avoid for critical signals
Development Environment Arduino IDE: easiest, huge library ecosystem, recommended for 301 PlatformIO: professional, better dependency management, VS Code ESP-IDF: Espressif native SDK, FreeRTOS, maximum control MicroPython: Python on ESP32, rapid prototyping, slower execution	
ESP32 variants: ESP32-S3 (USB native, AI accelerator), ESP32-C3 (RISC-V, lower cost), ESP32-C6 (Wi-Fi 6, Thread/Zigbee).	

Figure 14.3: ESP32 specifications, gotchas, and development environment.

ESP32 Gotchas: Memorize These

- **GPIO 6–11:** connected to internal flash. DO NOT USE.
- **GPIO 34–39:** input only; no output, no internal pull-up.
- **ADC2 disabled when Wi-Fi active:** use ADC1 (GPIO 32–39) only.
- **ADC nonlinear above $\sim \$3.1\text{ V}$:** calibrate or use external ADC for precision.
- **3.3 V logic:** use level shifters for 5 V sensors/actuators.
- **Boot pins (0, 2, 12, 15):** avoid for critical signals; unexpected levels prevent boot.

14.2.1 Safe GPIO Selection

Safe for any use: GPIO 4, 13, 14, 16–19, 21–23, 25–27, 32, 33. **Input only:** GPIO 34, 35, 36 (VP), 39 (VN). **Boot-sensitive:** GPIO 0, 2, 12, 15 (usable after boot with care). **Off-limits:** GPIO 6–11 (flash), 1 (TX0), 3 (RX0).

14.2.2 Wi-Fi and Bluetooth

Wi-Fi: Station mode (connect to router), AP mode (create network), or both. ESP-NOW for peer-to-peer without router. **BLE:** GATT server/client for phone apps; lower power than Classic BT. **Classic BT:** Serial Port Profile for wireless UART.

14.2.3 Power Modes

Active ($\sim \$240\text{ mA}$ with Wi-Fi), modem sleep ($\sim \$20\text{ mA}$, radio off), light sleep ($\sim \$0.8\text{ mA}$, CPU paused), deep sleep ($\sim \$10\text{ }\mu\text{A}$, RTC only). Battery sizing example: 2000 mAh LiPo at active+Wi-Fi = 10 hours; at deep sleep with 1-min wakeups ($\sim \$0.5\text{ mA}$ average) = 166 days.

14.3 Communication Protocols

MCU Communication Protocols — When to Use Each					
Protocol	Wires	Speed	Topology	Best For	ESP32
UART	2 (TX, RX)	9600–115200 baud (up to 5 Mbps)	Point-to-point 1 device per port	Serial debug, GPS, cellular modules	3 ports (0=USB, 1,2 free)
SPI	4 (MOSI, MISO, SCK, CS)	Up to 80 MHz (ESP32 native)	Bus, 1 master, N slaves (1 CS each)	Displays, SD cards, high-speed sensors	2 buses (HSPI, VSPI)
I2C	2 (SDA, SCL)	100/400 kHz (standard/fast)	Bus, multi-master, 127 addresses	Sensors, EEPROMs, low-speed peripherals	2 buses (any GPIO pair)
CAN	2 (CANH, CANL) + transceiver	Up to 1 Mbps	Multi-node bus, message-based	Automotive, industrial, motor controllers	1 controller (needs MCP2551)
Wi-Fi	0 (wireless)	Up to 150 Mbps (802.11n)	Star (via AP) or mesh (ESP-NOW)	IoT, dashboards, cloud connectivity	Built-in 2.4 GHz
BLE	0 (wireless)	1–2 Mbps	Star, mesh (BLE 5.0)	Phone interface, low-power sensors	Built-in BLE 4.2

Decision rule: I2C for sensors (simple, 2 wires), SPI for displays/SD (fast), UART for debug/GPS, Wi-Fi/BLE for IoT. CAN for automotive.

Figure 14.4: Six communication protocols with wire count, speed, topology, and ESP32 support.

UART: 2 wires, point-to-point, 9600–115200 baud. Debug, GPS, cellular. **SPI:** 4 wires + CS, bus, up to 80 MHz. Displays, SD cards, fast sensors. **I2C:** 2 wires, bus, 100–400 kHz,

127 addresses. Most sensors (BME280, MPU6050). Needs 4.7 k Ω pull-ups. **CAN**: 2 wires + transceiver, multi-node, 1 Mbps. Automotive/industrial. **Wi-Fi/BLE**: wireless IoT.

Protocol Decision Rule

I2C for sensors (simple, 2 wires). SPI for displays and SD cards (fast). UART for debug and GPS. Wi-Fi/BLE for IoT connectivity. CAN for automotive.

14.4 Firmware and Integration

14.4.1 Firmware Best Practices

Use **state machines** (IDLE, READING, TRANSMITTING, SLEEPING). Use `millis()`-based timing instead of `delay()`. Use **interrupts** for hardware events (ISRs must be short; set a flag, process in loop). Enable the **watchdog timer** for crash recovery. Add a **heartbeat LED** as a health indicator.

14.4.2 System Integration

Power: 3.3 V regulated or 5 V via USB DevKit. Add 10 μ F + 100 nF decoupling at MCU power pins. Wi-Fi transmit draws 240 mA transient; inadequate decoupling causes brown-out resets. **Level shifting**: BSS138-based modules for I2C/SPI to 5 V devices. **I2C**: 4.7 k Ω pull-ups to 3.3 V. **Analog**: keep wires short, average readings for noise reduction.

14.5 Artifact Handoff: What This Chapter Supports

Artifact	Description	Key Deliverable For
Pin Map	GPIO assignments with function and component	CDR, FDR
Firmware Architecture	State machine, ISRs, communication protocols	CDR, FDR
Power Mode Plan	Active/sleep duty cycle for battery sizing	CDR (power budget)

14.5.1 Development Environment Setup

Arduino IDE (recommended for MEGN 301): Install Arduino IDE 2.x, add ESP32 board support via Board Manager¹, select “ESP32 Dev Module,” and verify with the Blink example. **PlatformIO**: more professional, supports version-pinned dependencies and multi-file projects. **ESP-IDF**: Espressif’s native framework; most powerful but steepest learning curve.

14.5.2 Common Integration Patterns

- **Sensor polling loop**: read all sensors in `loop()`, average 3–5 readings, apply calibration, publish to cloud/display. Use `millis()`-based timing, never `delay()`.
- **Actuator control**: state machine with IDLE, HEATING, COOLING, FAULT states. Transitions triggered by sensor thresholds with hysteresis (e.g., turn on at 3 °C, turn off at 5 °C to prevent relay chatter).

¹Board Manager URL: https://espressif.github.io/arduino-esp32/package_esp32_index.json

- **Data logging:** write sensor data to SPIFFS/LittleFS with timestamps. Upload batch to cloud on Wi-Fi schedule. Buffer locally if Wi-Fi is unavailable.
- **OTA updates:** enable ArduinoOTA for wireless firmware updates during development. Disable in production firmware for security.

14.5.3 Battery Sizing Worked Example

GreenShield off-grid: ESP32 wakes every 60 s for 2 s of active measurement (80 mA average during wake, 10 μ A during sleep). Average current:

$$I_{avg} = \frac{2 \times 80 + 58 \times 0.01}{60} = 2.68 \text{ mA} \quad (14.1)$$

With a 3000 mAh LiPo: runtime = $3000/2.68 = 1119$ hours = 46 days. Adding Wi-Fi transmit (240 mA for 1 s every 5 minutes): adds 0.80 mA average, total 3.48 mA, runtime = 36 days. Add 30% margin for battery aging: design for 25 days between charges.

14.6 Sensor Interfacing Patterns

14.6.1 Analog Sensors

Most analog sensors output a voltage proportional to the measured quantity. The ESP32's 12-bit ADC maps 0–3.3 V to 0–4095 counts. Conversion: $V = \text{count} \times 3.3/4095$. Then apply the sensor's transfer function (from datasheet) to convert voltage to engineering units.

Common problems and fixes:

- **Noise:** average 10–50 readings per sample. Use a moving average or median filter in firmware.
- **Nonlinearity:** the ESP32 ADC is nonlinear above ~ 3.1 V. Use the ESP-IDF calibration API (`esp_adc_cal`) or limit input range to 0–3.0 V with a voltage divider.
- **Offset/gain error:** calibrate against a known reference (multimeter) at two points and apply linear correction: $V_{true} = a \times V_{raw} + b$.
- **5 V sensors:** use a voltage divider or external ADC (ADS1115, 16-bit, I2C, \$3).

14.6.2 Digital Sensors (I2C)

Most modern sensors (temperature, humidity, IMU, pressure) use I2C. Basic integration:

1. Connect SDA to GPIO 21, SCL to GPIO 22 (ESP32 defaults).
2. Add 4.7 k Ω pull-up resistors from SDA and SCL to 3.3 V. **One pair per bus**, not per device.
3. Install the sensor's Arduino library via Library Manager.
4. Initialize with `Wire.begin()` and use the library's `.read()` methods.
5. If the sensor is not detected: check wiring, verify address with an I2C scanner sketch, check pull-ups with a multimeter (SDA/SCL should read 3.3 V when idle).

14.6.3 Motor and Actuator Control

DC motors: use `ledcSetup()` and `ledcWrite()` for hardware PWM at 20–25 kHz (above audible range). Never drive a motor directly from a GPIO pin. Always use a driver IC (L298N, BTS7960, DRV8871) or MOSFET with flyback diode.

Servo motors: use the ESP32Servo library. Attach to any PWM-capable GPIO. Standard range: 500–2400 μ s pulse width at 50 Hz. Power servos from a separate 5–6 V supply, not from the ESP32’s 3.3 V rail.

Relay control: drive a relay module’s signal input from a GPIO through a 1 k Ω resistor. Use optocoupled relay modules to isolate the MCU from relay coil transients.

14.7 Debugging and Troubleshooting

14.7.1 Serial Monitor Debugging

`Serial.println()` is the primary debugging tool. Best practices:

- Print sensor values with labels: `Serial.print("Temp: "); Serial.println(temp);`
- Add state machine transitions: `Serial.println("STATE: HEATING");`
- Print timestamps: `Serial.print(millis()); Serial.print(" ms: ");`
- Remove or conditionally compile debug prints before final deployment (`#define DEBUG`).

14.7.2 Common ESP32 Failure Modes

Symptom	Likely Cause	Fix
Random resets	Brown-out (insufficient decoupling)	Add 100 μ F + 100 nF at regulator output
Won’t boot	GPIO 0/2/12/15 held at wrong level	Disconnect boot-sensitive pins during upload
ADC reads garbage	ADC2 + Wi-Fi conflict, or input >3.3 V	Use ADC1 channels; add voltage divider
I2C not detected	Missing pull-ups, wrong address, SDA/SCL swapped	Check with I2C scanner; verify pull-ups
Wi-Fi disconnects	Antenna blocked by ground plane or metal	Keep antenna area clear of copper pour
Servo jitter	Shared power supply with MCU	Use separate servo power supply

Table 14.1: Common ESP32 failure modes, causes, and fixes.

Requirements Mapping: From MCU Specs to Shall-Statements

MCU capabilities generate system requirements:

- “Read 4 analog sensors” → SYS-REQ: The system shall acquire analog data from 4 channels at ≥ 10 Hz using ADC1 (GPIO 32–35).
- “Wireless data logging” → SYS-REQ: The system shall transmit sensor data to a cloud server via Wi-Fi at ≥ 1 upload per 5 minutes.
- “Battery operation for 30 days” → SYS-REQ: The system shall operate from a 3000 mAh LiPo for ≥ 30 days with a duty cycle of 2 s active per 60 s.

The battery sizing calculation in this chapter directly verifies the last requirement. Every MCU design choice must trace to a shall-statement.

14.8 Practice Questions

1. Your project requires: 4 analog sensors, 2 PWM motor outputs, I2C temperature sensor, SPI display, Wi-Fi for data logging, and battery operation. Select an MCU platform and justify your choice. Create the pin map.
2. An ESP32 project works on USB power but the MCU resets randomly when running on a 3.7 V LiPo battery through a 3.3 V regulator. What is the likely cause? Propose a fix.
3. You need to read a 0–5 V analog pressure sensor with an ESP32. What problem does this create and how do you solve it?
4. Explain why ADC2 channels should not be used in an ESP32 project that uses Wi-Fi. What happens if you do?
5. A teammate connects a motor driver's PWM input to ESP32 GPIO 8. The ESP32 fails to boot. Why? What GPIO should they use instead?
6. Your I2C temperature sensor (BME280) works perfectly on the bench, but returns NaN after you add a second I2C device (OLED display) to the same bus. Both devices use different I2C addresses. What are the three most likely causes?
7. Design a firmware state machine for the GreenShield frost protection controller. Define all states (at least 4), all transitions, and the conditions that trigger each transition. Include a FAULT state.
8. Your battery-powered sensor node needs to last 60 days on a 2000 mAh LiPo. The active current is 80 mA and the deep sleep current is 10 μ A. What is the maximum duty cycle (active seconds per minute) you can afford?

14.8.1 Answers

1. ESP32; meets all requirements: 4 ADC1 channels (GPIO 32, 33, 34, 35), 2 PWM (GPIO 25, 26 via LEDC), I2C on GPIO 21 (SDA) / 22 (SCL), SPI display on GPIO 18 (SCK) / 23 (MOSI) / 5 (CS), Wi-Fi built-in, deep sleep for battery life. No other \$4–8 MCU provides Wi-Fi + sufficient ADC + PWM + SPI + I2C simultaneously.
2. Insufficient decoupling. Wi-Fi transmit draws ~ 240 mA transient. If the regulator cannot supply this or the decoupling capacitance is too small, the 3.3 V rail droops below the brown-out threshold (~ 2.7 V) and the ESP32 resets. Fix: add 100 μ F electrolytic + 100 nF ceramic at the regulator output. Ensure the regulator can supply ≥ 500 mA.
3. ESP32 GPIO is 3.3 V maximum. Applying 5 V directly will damage the pin. Solution: use a resistive voltage divider (e.g., 10 k Ω + 6.8 k Ω) to scale 0–5 V to 0–3.0 V (staying below the 3.1 V ADC linearity limit), or use an external ADC (ADS1115) with its own voltage reference.
4. ADC2 shares hardware with the Wi-Fi radio. When Wi-Fi is active, the ADC2 peripheral is unavailable. Calling `analogRead()` on an ADC2 pin returns garbage (random values or zero). Always use ADC1 channels (GPIO 32–39) for analog readings in Wi-Fi projects.

5. GPIO 6–11 are connected to the internal SPI flash that stores the firmware. Any external connection on these pins interferes with flash communication, preventing the bootloader from loading the program. The ESP32 fails to boot or crashes immediately. Use any safe GPIO (4, 13, 14, 16–19, 21–23, 25–27, 32, 33).
6. (1) Duplicate I2C address: despite having different *default* addresses, some modules have configurable addresses that may conflict. Run an I2C scanner to verify both addresses are unique. (2) Pull-up resistor overload: each device may have its own pull-ups on the breakout board. Two sets of $4.7\text{ k}\Omega$ in parallel = $2.35\text{ k}\Omega$, which may be too strong for 3.3 V I2C. Remove the on-board pull-ups from one device. (3) Bus capacitance: the OLED's long ribbon cable adds capacitance, slowing I2C rise times. Try reducing the I2C clock speed from 400 kHz to 100 kHz.
7. States: IDLE (monitoring, heater off), HEATING (heater on, monitoring), COOLDOWN (heater off, waiting for hysteresis), FAULT (sensor failure or communication loss). Transitions: IDLE $\rightarrow$ HEATING when $\text{temp} \leq \text{setpoint}$. HEATING $\rightarrow$ COOLDOWN when $\text{temp} \geq \text{setpoint} + \text{hysteresis}$ (e.g., $+2^\circ\text{C}$). COOLDOWN $\rightarrow$ IDLE when temp stabilizes. Any state $\rightarrow$ FAULT when sensor reads out-of-range ($< -40^\circ\text{C}$ or $> 85^\circ\text{C}$) or I2C communication fails 3 consecutive times. FAULT $\rightarrow$ IDLE after manual reset via button or remote command.
8. Target runtime: 60 days = 1440 hours. Required average current: $I_{avg} = 2000/1440 = 1.39\text{ mA}$. Let t_{on} = active seconds per 60-second cycle. $I_{avg} = (t_{on} \times 80 + (60 - t_{on}) \times 0.01)/60 = 1.39$. Solving: $t_{on} \times 80 + 0.60 - 0.01t_{on} = 83.4$. $79.99t_{on} = 82.8$. $t_{on} = 1.035\text{ s}$. Maximum active time: $\sim 1.0\text{ s}$ per minute. This is tight; the team should optimize wake time or use a larger battery.

Part III

Major Project Milestones

This part provides a comprehensive guide to the three formal design reviews that gate your project’s progression. Each chapter covers one review with deliverables, evaluation criteria, and common failure patterns.

The review sequence maps to the V-Model:

- **PDR** (Sprints 1–3): left side of the V; problem definition, requirements, architecture, concept selection.
- **CDR** (Sprints 4–5): bottom of the V; detailed design, analysis, manufacturing planning.
- **FDR** (Sprint 7 + Final): right side of the V; verification, validation, documentation.

Each review has a distinct character:

- **PDR asks “Should this design work?”**; feasibility, viability, desirability.
- **CDR asks “Will this design work?”**; completeness, manufacturability, testability.
- **FDR asks “How well did it work?”**; evidence, honesty, documentation.

A strong PDR enables a strong CDR enables a strong FDR. Rushing or skipping an early review creates compounding problems at every subsequent gate.

Table 14.2: The Complete Artifact Progression

Artifact	At PDR	At CDR	At FDR
Problem Statement	Finalized	Unchanged	Unchanged
Stakeholder Map	Complete	Updated if needed	Final
ConOps	Approved	Unchanged	Unchanged
SDRD	Baselined (v1.0)	Updated (v2.0)	Final (v3.0)
Functional Decomp.	Complete	Unchanged	Unchanged
Trade Studies	Completed	Referenced	Archived
BOM	Preliminary	Finalized with orders	As-built
Schematics/Drawings	Preliminary	Finalized	As-built redlines
VCRM	All Planned	Partial results	Complete
Test Procedures	Drafted	Reviewed/updated	Executed
Integration Plan	Preliminary	Detailed	Completed
O&M Manual	Not started	Outline	Complete

Chapter 15

Preliminary Design Review (PDR)

The PDR answers one question: **“Should this design work?”** It evaluates whether the team has correctly identified the problem, written verifiable requirements, explored the solution space, and selected a concept that is technically feasible, economically viable, and schedule-achievable.

15.1 Purpose and Scope

PDR is the first formal gate. It confirms that the team has done enough front-end work to justify moving into detailed design and procurement. A team that passes PDR has *earned* the right to spend project funds. A team that fails PDR is redirected before wasting money on a flawed concept.

15.2 Deliverables

15.2.1 Problem Definition and Background Research

Complete five-element problem statement, stakeholder map with engagement strategies, ConOps with at least two operational scenarios (normal and degraded), competitive analysis (3–5 existing solutions with identified gap), and context diagram showing system boundary and external interfaces.

15.2.2 System Requirements and Traceability

SDRD (v1.0) with all system-level requirements. Each requirement has: unique ID, shall-statement, measurable acceptance criteria, rationale, verification method (T/A/I/D), and parent traceability to a stakeholder need. Forward and backward traceability demonstrated. No orphan requirements. No undecomposed parents.

15.2.3 Concept Development and Selection

Evidence of divergent thinking: morphological chart with ≥ 4 functions and ≥ 3 options each. Three or more distinct concepts developed to sketch/block-diagram level. Pugh screening matrix showing systematic elimination. Weighted decision matrix for final selection with sensitivity analysis. The selected concept must be clearly justified by the data, not by team preference.

15.2.4 Risk and Feasibility

Risk register with ≥ 5 identified risks, each rated by probability and impact. Mitigation strategy for every high/medium risk. Feasibility assessment against four lenses: technical, economic, schedule,

and operational. Any feasibility failure requires resolution before PDR.

15.2.5 Initial Test Plan

VCRM with all requirements listed and verification methods assigned. At least three test procedures drafted in full (all 10 sections). Equipment and facility needs identified. Test schedule aligned with integration timeline.

15.2.6 Concept Development and Selection: Detail

- **Reverse engineering analysis:** Documented disassembly and analysis of the COTS product, including component inventory, interface inventory, and constraints on the add-on system. Photographs of each disassembly step. Block diagram showing subsystem connections.
- **Concept generation evidence:** Morphological chart built from functional decomposition with ≥ 3 options per function. Evidence that divergent techniques (brainstorming, SCAMPER, bio-inspired) were used. Sketches of at least 3 distinct concepts.
- **Concept screening:** Pugh matrix showing qualitative evaluation against a datum.
- **Trade study:** Weighted decision matrix with criteria from requirements, documented weight rationale, data-backed ratings, sensitivity analysis, and clear recommendation.
- **Preliminary architecture:** Block diagram of selected concept showing subsystems, interfaces, and allocated functions.

15.2.7 Budget and Schedule

A preliminary budget showing: component costs (from actual vendor quotes, not estimates), fabrication costs (3D printing, PCB, machining), procurement shipping, and a 15–20% contingency. A project schedule (Gantt chart or sprint plan) with milestones, dependencies, and critical path identified. Demonstrate that the selected concept can be designed, built, and tested within the remaining project timeline.

15.2.8 Team Organization

Role assignments with clear ownership: who is responsible for each subsystem, who leads integration, who owns V&V, who manages procurement. A communication plan: meeting cadence, documentation repository, change control process.

15.3 Evaluation Criteria

Common PDR Failures

(1) Problem statement is actually a solution statement. (2) Requirements are vague or untestable (“shall be reliable”). (3) Only one concept explored; no evidence of design space exploration. (4) Trade study weights clearly biased to get the “right” answer. (5) No risk register or feasibility assessment. (6) BOM is a wish list with no prices, lead times, or vendors.

Criterion	What Reviewers Look For	Ref
Problem clarity	Solution-agnostic? All five elements present?	Ch. 1
Stakeholder completeness	All nine categories checked? Power/interest grid populated?	Ch. 1
Requirements quality	One SHALL per req? Quantitative? Testable? Implementation-free?	Ch. 2
Traceability	Every req traces to a need? Every need has a req?	Ch. 2
Concept breadth	≥ 20 concepts before screening? Morph chart complete?	Ch. 3
Selection rigor	Trade study criteria from reqs? Sensitivity analysis done?	Ch. 3
Risk identification	High-severity failures flagged? Red-zone risks have mitigation?	Ch. 3
Test plan completeness	Every req has TAID method and test description?	Ch. 4

Table 15.1: PDR evaluation criteria.

The PDR Litmus Test

Could a different team, given only your PDR documents, understand the problem, agree with your concept selection, and begin detailed design without asking you questions? If yes, you pass. If they would need to ask “what did you mean by...”; you are not ready.

15.4 Practice Questions

1. Your team’s trade study shows scores of 3.81 vs. 3.80 between two concepts. The reviewer challenges your selection. How do you respond?
2. A team presents a PDR with 15 requirements, all verified by “Test.” The reviewer asks why none are verified by Analysis, Inspection, or Demonstration. What is wrong?
3. A team passes PDR but then struggles through CDR because their requirements were actually solutions in disguise (“shall use an Arduino Mega”). How should the PDR reviewer have caught this, and what specific requirement quality test would have flagged it?
4. Your competitive analysis found no existing product that solves your problem. Does this make your project more or less risky? Explain.

15.4.1 Answers

1. A 0.01 margin is within noise. Perform sensitivity analysis: vary each weight by ± 0.05 and check if the winner changes. If it does, the selection is not robust. Consider prototyping both finalists or identifying the single criterion that differentiates them and gathering better data on that criterion.
2. Assigning “Test” to everything suggests the team did not think carefully about methods. Physical presence/labeling should be Inspection. Thermal or structural margin should be Analysis. User-observable behavior should be Demonstration. Over-reliance on Test also inflates the test schedule.
3. The PDR reviewer should apply the “implementation-free” test from the requirements quality checklist: does this requirement specify WHAT (a capability) or HOW (a specific component)? “Shall use Arduino Mega” specifies HOW. The correct requirement: “The controller shall provide ≥ 12 analog inputs, ≥ 4 serial interfaces, ≥ 256 KB flash, and support Wi-Fi connectivity.”

This is testable, implementation-free, and allows the trade study to objectively select the best platform.

4. More risky: either the problem is not real (no one else thinks it needs solving), the problem is too hard (others tried and failed), or your search was inadequate. Investigate which explanation applies. If the problem is real and unsolved, the project has high value but also high technical risk.

Chapter 16

Critical Design Review (CDR)

The CDR answers: **“Will this design work?”** It evaluates whether the detailed design is complete, manufacturable, and verifiable. CDR is the gate between design and fabrication; passing CDR authorizes the team to build.

16.1 Purpose and Scope

CDR confirms that every component is specified, every interface is defined, every manufacturing step is planned, and every requirement has a path to verification. The design is frozen after CDR; changes require formal change control.

16.2 Deliverables

16.2.1 Finalized Technical Design

Complete schematics with all component values, ratings, and part numbers. PCB layout (if applicable) reviewed and ready for fabrication. CAD models of all custom mechanical parts with engineering drawings including dimensions, tolerances, material callouts, and surface finishes. For each fabrication method: specific parameters (e.g., “FDM, PLA, 0.2 mm layer, 40% infill” or “CNC 6061-T6 per Order of Operations”).

16.2.2 Design Analysis and Verification

Analysis results for all A-method requirements: FEA stress/thermal, worst-case circuit analysis, power budget with actual component data (not estimates), link budget for wireless communications, and motor/transmission torque calculations with margin. Every analysis must show: assumption, calculation, result, margin, and verdict.

16.2.3 Manufacturing and Procurement

Finalized BOM with: actual part numbers, verified vendor availability, confirmed pricing, and realistic lead times. All long-lead items ordered. Manufacturing plan: what is being fabricated (PCBs ordered, 3D prints queued, machining scheduled), what is being purchased, and assembly sequence.

The Spring Break Procurement Trap

If CDR falls before spring break, all procurement orders must be placed at CDR, not “after break.” Parts ordered the week before break arrive during break with no one to receive them.

International orders (JLCPCB, AliExpress) need 3–4 weeks. McMaster ships next-day.

16.2.4 Updated Test Plan

VCRM updated with any new or modified requirements since PDR. All test procedures reviewed and updated based on the finalized design. Test equipment identified and availability confirmed. Test schedule with specific dates and responsible engineers. Subsystem test results from early prototyping (if available).

16.2.5 Software and Firmware

If your system includes firmware: source code under version control (Git), state machine diagram, pin mapping table, communication protocol definitions, and build/flash instructions. Code must compile and flash from the documented instructions without “tribal knowledge.” Include a hardware abstraction layer (HAL) that isolates sensor-specific code from application logic.

16.2.6 Thermal and Structural Analysis

For any component operating near its thermal limit: thermal analysis showing junction temperature under worst-case conditions with adequate margin. For structural components: FEA or hand calculations showing stress/deflection under design loads with safety factor ≥ 2.0 . Document: assumptions, model, results, margin, and verdict for each analysis.

16.2.7 Integration and Test Readiness

Integration plan: sequence of subsystem combinations with test gates at each stage. Interface verification plan: which interfaces will be tested at each stage and what constitutes pass/fail. Any subsystem demonstrations already completed (with data). Hardware/software integration strategy: how firmware will be loaded, tested, and version-controlled.

16.2.8 Risk Register Update

All PDR risks updated with current status. New risks identified during detailed design added. Top-5 risks have active mitigation plans with assigned owners and deadlines.

16.2.9 Procurement Status

For every BOM item: ordered/received/backordered status. Any substitutions from the PDR BOM documented with engineering justification. Lead time tracking for critical-path items. Budget actual vs. planned with variance explanation.

16.3 Evaluation Criteria

Common CDR Failures

Incomplete drawings: CAD exists but engineering drawings with tolerances and manufacturing notes do not. **BOM gaps:** fasteners, wire, consumables missing; budget surprises. **No Plan B:** sole-source component goes out of stock with no alternative. **DFM ignored:** poly-

Criterion	What Reviewers Look For	Ref
Design completeness	Can this be built from CDR package alone? All parts specified?	Ch. 2–3
Drawings quality	Fully dimensioned? Toleranced? Material and finish specified?	Ch. 3
DFM compliance	Polymer parts pass IM checklist? Draft, wall, ribs correct?	Ch. 12
Analysis evidence	Calculations/simulations show reqs met with margin?	Ch. 2, 4
FMEA maturity	Updated for detailed design? High-RPN resolved? Safety hazards mitigated?	Ch. 3, 6
BOM completeness	Every component listed, priced, Plan B for critical parts? Within budget?	Ch. 3
Requirements currency	Changes since PDR documented? RTM current?	Ch. 2
Test readiness	Critical procedures written? Equipment identified?	Ch. 4, 5

Table 16.1: CDR evaluation criteria.

mer parts have zero draft and non-uniform walls. **Stale FMEA:** PDR version never updated. **Late purchase requests:** ordered at CDR deadline instead of immediately after approval.

The CDR Litmus Test

Could a contract manufacturer build your system from only the CDR documents (BOM, schematics, drawings, assembly instructions) without calling you? If yes, your documentation is complete. If they would need to ask “what connector goes here?” or “what tolerance is this?”; you are not ready.

16.4 Practice Questions

1. Your CDR BOM lists “12V DC Motor” with a link to an Amazon product page. The reviewer rejects this. Why?
2. A team’s power budget shows 2.8 A total load on a 3.0 A power supply. The reviewer flags this. Why is 93% utilization a problem?
3. Your team’s schematics show a 3.3 V regulator powering the ESP32, but the thermal analysis for the regulator was never performed. The ESP32 draws 240 mA during Wi-Fi transmission. Is this a CDR pass or fail? What analysis should have been done?
4. Your PCB design uses a QFN-32 package that requires reflow soldering. Your team only has hand-soldering capability. What should you do?

16.4.1 Answers

1. Amazon listings change without notice (price, availability, even the actual product). A proper BOM entry needs: manufacturer name, manufacturer part number, key specifications (voltage, RPM, torque, current), and a stable vendor with a datasheet. Amazon is a marketplace, not a specification source.
2. 93% utilization leaves only 0.2 A margin. Motor inrush alone could draw 6–10× running current for 50–100 ms. The supply will current-limit or brown-out, causing MCU resets. Design rule: 20–25% current margin minimum. This team needs a larger supply or must reduce the load.

3. CDR fail; missing analysis evidence. Required thermal analysis: input voltage (e.g., 5 V from USB), output 3.3 V at 240 mA peak. LDO power dissipation $P = (V_{in} - V_{out}) \times I = (5.0 - 3.3) \times 0.24 = 0.41$ W. At thermal resistance $\theta_{JA} \approx 150$ °C/W for SOT-223, junction temp rise $= 0.41 \times 150 = 61.5$ °C. At 25 °C ambient: $T_J = 86.5$ °C; below the 125 °C limit but with only 38.5 °C margin. At 40 °C ambient: $T_J = 101.5$ °C; still acceptable but worth documenting. The analysis takes 5 minutes and prevents a field failure.
4. Options: (a) select a TSSOP or SOIC alternative of the same IC, (b) order a QFN breakout board/adaptor, (c) use the JLCPCB assembly service (SMT placement for \$8). Option (a) is best if available. Never plan to hand-solder QFN; it requires a hot air station at minimum and even then has high defect rates.

Chapter 17

Final Design Review (FDR)

The FDR answers: **“How well did it work?”** It evaluates the completed system against its original requirements, documents what was built versus what was designed, captures lessons learned, and delivers the operations and maintenance package.

17.1 Purpose and Scope

FDR is the capstone review. It demonstrates that the system meets (or acknowledges deviation from) every requirement, provides complete as-built documentation, and delivers everything the stakeholder needs to operate and maintain the system independently.

17.2 Deliverables

17.2.1 System Verification and Validation Results

Complete VCRM with every requirement showing PASS, FAIL (with Discrepancy Report), or WAIVED (with stakeholder-signed waiver and risk acceptance). For every test: actual measured data (not just “pass”), number of repetitions, measurement uncertainty, and test conditions. For any FAIL: root cause analysis, attempted corrective actions, and current status. Validation evidence: stakeholder demonstration that the system solves the original problem in realistic operating conditions.

17.2.2 Final Design Documentation

As-built BOM (what was actually used, not what was planned). As-built schematics reflecting any changes made during integration. As-built CAD models and drawings reflecting any modifications. Firmware source code with version tag and build instructions. All design files archived in a team repository with README.

17.2.3 As-Designed vs. As-Built Comparison

A table comparing the CDR design intent with the actual implementation. Every deviation must be documented with: what changed, why it changed, what impact the change had on requirements, and whether the change was approved under change control. This comparison is the audit trail that demonstrates design discipline.

17.2.4 Design Process Documentation

Decision log: key decisions made during the project with rationale and alternatives considered. Change log: all changes after CDR baseline with approval records. Risk register: final status of all identified risks (mitigated, accepted, or realized). Lessons learned: what worked, what did not, and what the team would do differently.

17.2.5 Operations and Maintenance Materials

Quick-Start Guide (1 page, 10-minute setup), Operating Manual (complete reference), Maintenance Manual (procedures with photos), Troubleshooting Guide (symptom-based flowcharts), spare parts list with vendors and lead times, and training materials.

17.2.6 External Team Testing Results

Results from having another team execute at least one of your test procedures. Document: which procedure was tested, what ambiguities the external team identified, what procedure updates resulted, and whether the external team achieved the same results. This is one of the most valuable FDR deliverables; it proves your documentation is truly reproducible.

17.2.7 Demonstration Video

A 3–5 minute video showing the system operating in its intended environment (or a representative simulation). The video should demonstrate: power-on sequence, normal operation, at least one sensor reading verified against a reference, actuator response to a trigger condition, and communication/alerting function. This video serves as permanent evidence of system functionality.

17.2.8 As-Designed vs. As-Built Comparison: Template

Item	As-Designed (CDR)	As-Built (FDR)	Approved?
MCU	ESP32-WROOM-32	ESP32-WROOM-32	N/A (same)
Temp sensor	DS18B20	BME280	CR-003, approved
Enclosure material	ABS (injection mold)	PETG (3D print)	CR-007, approved
Relay	SRD-05VDC	SRD-12VDC	CR-009, not approved

Table 17.1: As-designed vs. as-built comparison template.

Every deviation must have a Change Request (CR) number, engineering justification, impact assessment on requirements, and approval status. The unapproved relay change (12 V coil on a 5 V rail) is a design error that must be resolved.

17.2.9 Lessons Learned: Beyond “Start Earlier”

Good lessons learned are specific, actionable, and traceable to evidence:

- **Bad:** “We should have communicated better.” (Vague, no action.)
- **Good:** “I2C pull-up ownership was undefined in the ICD, causing 8 hours of debugging. Action: every ICD must explicitly state which side provides pull-ups, termination, and bias voltages.” (Specific problem, specific fix.)
- **Bad:** “We should have started procurement earlier.” (Generic.)

- **Good:** “We ordered the GSM module from AliExpress 3 weeks before CDR. It arrived 2 weeks late due to customs delay. Action: use McMaster-Carr (next-day) or DigiKey (3-day) for critical-path components; international vendors only for non-critical items with 4-week buffer.” (Specific, actionable.)

17.2.10 Reflection and Future Work

Honest assessment: did the system solve the original problem? What requirements were not met and why? What would a v2.0 address? What skills did the team develop? This section demonstrates engineering maturity; the ability to critically evaluate your own work.

17.3 Evaluation Criteria

Common FDR Failures

(1) VCRM shows “PASS” for requirements that were never actually tested. (2) No measured data; just assertions. (3) As-built documentation does not match the physical system. (4) O&M manual is missing or is just a copy of the design report. (5) No lessons learned or reflection.

The FDR Litmus Test

Could a new team, given only your FDR package, operate, maintain, troubleshoot, and eventually improve your system without ever talking to you? If yes, your knowledge transfer is complete. If they would need to call you for “that one trick to get it to boot”; you have not finished.

17.3.1 Budget Final Accounting

Final cost accounting: planned vs. actual for every BOM category. Total project cost including labor hours (if tracked). Cost drivers identified: what was more expensive than expected and why. This is valuable data for future project planning.

17.3.2 Schedule Performance

Planned vs. actual milestone dates. Critical path analysis: was the critical path what you expected? Where did schedule slip occur and why? Sprint velocity analysis: did productivity improve over the project? This retrospective drives continuous improvement.

17.4 Evaluation Criteria

Common FDR Failures

VCRM with gaps: requirements never tested, no explanation. **PASS without evidence:** “all requirements met” but no measured values or data. **Hidden failures:** known defects not reported. **Stale documentation:** as-built system has changes not reflected in drawings. **No O&M materials:** system cannot be operated by anyone except the build team. **Generic lessons learned:** “we learned to start earlier” without specific, actionable insights.

Criterion	What Reviewers Look For	Ref
VCRM completeness	Every req has final status with evidence? No unaddressed items?	Ch. 4
Test rigor	Measured values reported (not just PASS/FAIL)? Repeated trials? Boundary tests?	Ch. 4
Failure honesty	Failures documented with root cause? Waivers justified? No hidden defects?	Ch. 4
As-built accuracy	Documents match what was actually built? BOM, drawings, wiring current?	Ch. 5
O&M readiness	Quick-start usable? Spare parts identified? Maintenance plan actionable?	Ch. 6
Design evolution	Journey from PDR → CDR → FDR documented with rationale?	All
Lessons learned	Specific and actionable? Demonstrate growth in engineering judgment?	;
Budget accounting	As-designed vs. as-built cost? Variances explained?	Ch. 3

Table 17.2: FDR evaluation criteria.

17.5 The Complete Review Progression

The three reviews form a narrative: PDR asks “Should it work?” (problem + concept), CDR asks “Will it work?” (design completeness), FDR asks “How well did it work?” (evidence + documentation). Each review builds on the previous. A strong PDR enables a strong CDR enables a strong FDR. Skipping or rushing an early review creates compounding problems downstream.

17.6 Practice Questions

1. Your VCRM at FDR shows 18 PASS, 2 FAIL, and 1 PARTIAL PASS out of 21 requirements. How do you present this to the stakeholder?
2. A team’s FDR as-built BOM has 8 components that differ from the CDR BOM, but the change log is empty. What does this indicate?
3. Your stakeholder says “the system works great but I wish the display were bigger.” Is this a verification failure or a validation failure? Explain.

17.6.1 Answers

1. Present the complete VCRM transparently. For the 18 PASS: evidence summaries. For the 2 FAIL: root cause analysis, corrective actions attempted, and recommendation (fix, waive, or defer). For the PARTIAL PASS: which criteria passed, which failed, risk assessment, and recommendation. The stakeholder decides on open items.
2. No change control was exercised. The team made ad-hoc substitutions during build without documenting why, assessing impact, or seeking approval. This is a process failure that undermines confidence in the entire build. Any undocumented change could have invalidated a test result.
3. Validation failure. The system met all written requirements (verification passed). But the stakeholder’s actual need for readability was not captured as a requirement (e.g., “display shall be readable at 2m distance”). The requirements were incomplete; a Chapter 1 failure that propagated to FDR.

Industry SE Parallels

In industry, the three-review gate structure maps to the standard SE review cycle: **PDR** corresponds to the System Requirements Review (SRR) and Preliminary Design Review. **CDR** corresponds to the Critical Design Review and Test Readiness Review (TRR). **FDR** corresponds to the System Verification Review (SVR) and Operational Readiness Review (ORR). The artifact maturity progression is consistent with INCOSE SEH Sec. 4.9 (Technical Reviews) and NASA SP-2016-6105 Chapter 6. The “should work / will work / how well did it work” framing is widely used in defense acquisition (DoD 5000 series) and commercial product development.

Glossary of Acronyms and Key Terms

Term	Definition
ABET	Accreditation Board for Engineering and Technology
ADC	Analog-to-Digital Converter
BLE	Bluetooth Low Energy
BLDC	Brushless Direct Current (motor type)
BOM	Bill of Materials; complete list of every component
BSPP	British Standard Pipe Parallel (thread standard)
BSPT	British Standard Pipe Taper (thread standard)
CAN	Controller Area Network (communication bus)
CDR	Critical Design Review: “Will this design work?”
ConOps	Concept of Operations; stakeholder-facing system description
COTS	Commercial Off-The-Shelf (product)
DFA	Design for Assembly
DFM	Design for Manufacturability
DR	Discrepancy Report
EMC/EMI	Electromagnetic Compatibility / Interference
ESC	Electronic Speed Controller (for BLDC motors)
FDR	Final Design Review: “How well did the design work?”
FEA	Finite Element Analysis
FMEA	Failure Modes and Effects Analysis
FMECA	Failure Modes, Effects, and Criticality Analysis
FDM	Fused Deposition Modeling (3D printing)
GPIO	General Purpose Input/Output
I2C	Inter-Integrated Circuit (2-wire serial bus)
ICD	Interface Control Document
INCOSE	International Council on Systems Engineering
IP (rating)	Ingress Protection (e.g., IP67 = dust-tight, 1 m submersion)
LDO	Low Drop-Out (voltage regulator)
MTBF	Mean Time Between Failures
MTTR	Mean Time To Repair
MCU	Microcontroller Unit
MoSCoW	Must / Should / Could / Won’t (prioritization)
NEMA	National Electrical Manufacturers Association (motor frame sizes)
NPT	National Pipe Thread (tapered, U.S. standard)
NPS	Nominal Pipe Size
PDR	Preliminary Design Review: “Should this design work?”
PWM	Pulse Width Modulation
RPN	Risk Priority Number (Severity $\times$ Occurrence $\times$ Detection)
RTM	Requirements Traceability Matrix

Term	Definition
SDRD	System Design Requirements Document
SPI (comms)	Serial Peripheral Interface (4-wire bus)
SPI (mold)	Society of the Plastics Industry (mold finish standards)
TAID	Test, Analysis, Inspection, Demonstration (verification methods)
THT	Through-Hole Technology (component mounting)
SMD/SMT	Surface-Mount Device / Technology
TVS	Transient Voltage Suppressor (protection diode)
UART	Universal Asynchronous Receiver/Transmitter
UNC	Unified National Coarse (thread standard)
UNF	Unified National Fine (thread standard)
V&V	Verification and Validation
VCRM	Verification Cross-Reference Matrix

Exam Quick-Reference Cheat Sheet

Requirements

- SHALL = mandatory; SHOULD = goal
- One shall per requirement
- Quantitative, testable, implementation-free
- TAID: Test, Analysis, Inspection, Demonstration

V&V

- Verification: built right (meets specs)
- Validation: right system (meets needs)
- VCRM tracks every req to its test status
- PASS/FAIL/PARTIAL. Always with data

Motors

- $\tau = K_t \times I$; $V_{back} = K_e \times \omega$
- Stall current = $V/R_{winding}$ (6–10× rated)
- Design at 50–75% of continuous torque
- Every motor needs a matched driver

Power

- LDO: low noise, $V_{out} < V_{in}$, η = ratio
- Buck: efficient step-down, EMI risk
- Boost: step-up from low voltage
- 20% current margin minimum

Fasteners

- $T = K \cdot F \cdot d$ (torque-preload)
- Thread engagement: $1.5d$ steel, $2d$ aluminum

- Always specify grade in BOM
- Nyloc or Loctite for vibration; not split washers

Bearings

- Default: deep groove ball (6000 series)
- $L_{10} = (C/P)^3 \times 10^6 / (60n)$
- 2RS = sealed. Never hammer.
- Press fit inner ring; sliding fit outer ring

Transmission

- $GR = N_2/N_1 = \omega_1/\omega_2 = \tau_2/\tau_1$
- $P = \tau \times \omega$; $\eta_{total} = \eta_1 \times \eta_2 \times \dots$
- Self-locking: lead screws, worm gears
- Student defaults: GT2 belt, T8 lead screw

Injection Molding DFM

- Uniform wall $\pm 10\%$; draft $\geq 1^\circ$
- Rib = 60% wall; boss OD = $2 \times$ hole ID
- Inside radius $\geq 0.5 \times$ wall
- Living hinge: PP only, 0.25–0.5 mm

PCB Layout

- 100 nF decoupling within 5 mm of every IC
- Ground pour on bottom layer
- Print 1:1 and place real components
- 10 mil trace = 0.3 A; 100 mil = 3.0 A

Why Projects Fail. The Top 20

1. Problem statement is actually a solution statement.
2. Missing stakeholders discovered after PDR.
3. Requirements are vague, untestable, or compound.
4. Only one concept explored; no design space coverage.
5. Trade study weights manipulated to get the “right” answer.
6. No competitive analysis; reinventing an existing product.
7. BOM has no part numbers, no vendors, no prices.
8. Procurement delayed until after spring break.
9. Motor sized at stall torque instead of continuous.
10. Power supply sized for average current, not inrush peak.
11. I2C pull-up ownership undefined in ICD.
12. Integration done “big-bang” (everything at once).
13. Testing only at room temperature.
14. Test procedures that only the author can execute.
15. VCRM updated only at FDR (should evolve from PDR).
16. No configuration management; “which firmware version?”
17. O&M manual not written until the night before FDR.
18. As-built documentation does not match the physical system.
19. Change log empty despite 8 BOM substitutions.
20. No lessons learned or honest reflection on what went wrong.

Each of these failures has been observed in actual MEGN 301 projects. They are listed in approximate chronological order; early failures (1–6) cause more damage because they compound through the project lifecycle.

The cost of late discovery: A problem discovered at PDR costs 1 hour to fix (rewrite a requirement). The same problem discovered at CDR costs 1 week (redesign a subsystem). At FDR, it costs the entire grade (the system does not work and there is no time to fix it). This is the “1-10-100 rule” of systems engineering: fix problems at the earliest possible stage.

#	Failure	Caught At	Fix Cost
1	Solution masquerading as problem	PDR	1 hour
5	Trade study weights manipulated	PDR	2 hours
8	Procurement delayed past spring break	CDR	2–3 weeks
11	I2C pull-ups undefined	Integration	8 hours
12	Big-bang integration	Integration	1–2 weeks
17	O&M manual not written	FDR	2 days (rushed)

Table 17.4: Representative fix costs for common project failures.

ABET Student Outcome Mapping

MEGN 301 addresses the following ABET Student Outcomes for mechanical engineering:

SO	Description	MEGN 301 Coverage
1	Identify, formulate, and solve complex engineering problems	Ch. 1–3 (problem definition through concept selection)
2	Apply engineering design to produce solutions meeting specified needs	Ch. 2–6 (requirements through O&M), Part III (design reviews)
3	Communicate effectively with a range of audiences	Part III (PDR/CDR/FDR presentations and documentation)
5	Function on a team, establish goals, plan tasks, meet deadlines	Sprint structure, design review gates
6	Develop and conduct appropriate experimentation, analyze data	Ch. 4 (V&V and test plans)
7	Acquire and apply new knowledge as needed	Ch. 7–12 (technical reference applied to design projects)

Table 17.5: ABET Student Outcome mapping for MEGN 301.

GreenShield: Complete Worked Example

This appendix traces the GreenShield Automated Greenhouse Frost Protection System through every phase of the design-build process, consolidating the chapter-by-chapter GS boxes into a single narrative.

Phase 1: Problem Definition (Chapter 1)

Problem: Small-farm greenhouse operators [WHO] cannot maintain above-freezing temperatures in unheated hoop houses during Colorado Front Range winters [WHERE], resulting in crop losses averaging \$12,000/season [WHY], within a \$500 budget using existing 120 VAC infrastructure [CONSTRAINTS].

Stakeholders: Primary user (operator), beneficiary (crops/farm revenue), sponsor (operator/farm), maintainer (operator or farm hand), regulator (NEC electrical code, county building code), integrator (existing greenhouse structure), trainer (operator's spouse or employee).

ConOps: System monitors air temperature continuously. When temperature drops below a user-adjustable setpoint (default 3 °C), it activates a supplemental heater and sends an SMS alert. In degraded mode (communication failure), it continues autonomous operation. The operator can adjust setpoint and check status remotely.

Phase 2: Requirements & Architecture (Chapter 2)

Key Requirements:

1. SYS-REQ-001: Monitor temp at ≥ 1 Hz, accuracy ± 0.5 °C, range -20 to $+60$ °C. [T]
2. SYS-REQ-002: Activate heater when temp $\leq$ user setpoint. Hysteresis: 2 °C. [T]
3. SYS-REQ-003: SMS alert within 60 s of heater activation. [T]
4. SYS-REQ-004: Autonomous mode on comm failure. [D]
5. SYS-REQ-005: Operate on 120 VAC, < 50 W standby. [T]
6. SYS-REQ-006: Enclosure IP44 minimum (splash-resistant). [I]
7. SYS-REQ-007: All field connections tool-free (connectors, not solder). [I]

Architecture: Five subsystems. Sensor (DS18B20), Controller (ESP32), Actuator (relay + ceramic heater), Communication (SIM800L GSM), Power (12 V adapter + buck converters).

Phase 3: Concept Generation & Selection (Chapter 3)

Three concepts evaluated via weighted decision matrix. Concept 1 (Grid-Connected, Wi-Fi) selected: lowest cost (\$85), simplest integration, meets all requirements. Off-Grid Solar concept deferred to v2.0.

Risk register: top risk = MCU brown-out during relay switching (RPN 12). Mitigation: 470 μ F decoupling, separate relay power trace.

FMEA top item: relay stuck closed (RPN 108). Mitigation: thermal fuse on heater as independent safety cutoff.

Phase 4: V&V Planning (Chapter 4)

VCRM established at PDR with 7 requirements, all methods assigned. Three test procedures drafted: TP-001 (temperature accuracy), TP-002 (heater activation threshold), TP-003 (SMS alert timing).

Test environment: ME department environmental chamber for boundary testing (0 °C and 50 °C). Ice bath as backup for 0 °C test point.

Phase 5: Integration & Debug (Chapter 5)

Integration sequence: (1) Power subsystem alone (verify 5 V and 3.3 V rails). (2) Add sensor (verify I2C communication). (3) Add relay (verify GPIO control without brown-out). (4) Add GSM module (verify SMS send). (5) System-level 48-hour burn-in.

Integration failure encountered: I2C communication intermittent. Root cause: missing 4.7 k Ω pull-up resistors; neither the sensor breakout nor the ESP32 DevKit provided them. ICD updated to specify pull-up ownership on the controller side.

Phase 6: Operations & Maintenance (Chapter 6)

Wear items: relay contacts (rated 100,000 cycles; at 10 cycles/night $\times$ 150 nights/year = 67 years), DS18B20 sensor (no wear items but verify annually against reference thermometer), SIM800L antenna connector (inspect annually for corrosion).

O&M manual: Quick-start (1 page), operating manual (4 pages), maintenance schedule (1 page), troubleshooting flowchart (1 page), spare parts list with McMaster/DigiKey part numbers.

Final Results (FDR)

VCRM: 6 PASS, 1 PARTIAL PASS (SYS-REQ-001 accuracy was ± 0.6 °C at 0 °C boundary; missed by 0.1 °C). Root cause: DS18B20 factory calibration drift at low temperatures. Corrective action: polynomial correction applied in firmware, re-test showed ± 0.4 °C across full range. Updated status: PASS.

Total cost: \$92 (\$8 under budget). Build time: 6 weeks. Lessons learned: (1) always define pull-up ownership in ICDs; (2) test at temperature boundaries early, not at FDR; (3) the 470 μ F capacitor fix for brown-out should have been in the original design, not discovered during integration.

Quick Reference Tables

These tables consolidate the most commonly needed design values. For full context, see the referenced chapters.

Fastener Grades and Proof Loads

Grade/Class	Tensile (MPa)	Proof (MPa)	Marking	Typical Use
SAE Gr. 2 / ISO 4.6	400	225	None / “4.6”	General purpose
SAE Gr. 5 / ISO 8.8	830 / 800	585 / 640	3 lines / “8.8”	Structural
SAE Gr. 8 / ISO 10.9	1035 / 1040	895 / 940	6 lines / “10.9”	High strength
ISO 12.9	1220	1100	“12.9”	Socket head caps

Thread Engagement by Material

Steel: $1.5d$ min. Aluminum: $2.0d$. Cast iron: $2.0d$. Plastic: $2.5d$ (consider threaded insert).

PCB Trace Width vs. Current (1-oz External Copper, IPC-2221)

Trace (mil)	10	20	50	100	200
Safe Current (A)	0.3	0.7	1.5	3.0	6.0

Wire Gauge Reference (Chassis Wiring, Single Conductor)

AWG	Dia. (mm)	Max (A)	Common Use
30	0.25	0.5	Wire-wrap, breadboard jumpers
26	0.40	1.3	Ribbon cable, signal wiring
22	0.64	3.0	General hookup, sensor cables
18	1.02	7.0	Motor power, relay wiring
14	1.63	15	High-current bus, battery cables

Injection Molding DFM Rules

IP Rating Quick Reference

IP20: finger-safe, no water. IP54: dust-protected, splash-resistant. IP65: dust-tight, water jets. IP67: dust-tight, 1 m submersion 30 min. IP68: continuous submersion per spec.

Parameter	Rule
Wall thickness (ABS/PC)	1.5–3.5 mm, uniform $\pm 10\%$
Wall thickness (PP)	0.8–3.0 mm
Draft angle	1° – 2° per side; add 1° per 0.001 in texture depth
Ribs	50–70% of wall thickness, height $\leq 3 \times$ wall
Bosses	OD = 2.0–2.5 $\times$ hole ID; connect to walls with ribs
Inside radii	$\geq 0.5 \times$ wall; outside = inside + wall

Key Formulas

Bearing life: $L_{10} = (C/P)^p \times 10^6 / (60n)$ hours [$p = 3$ ball, $10/3$ roller]

Bolt torque: $T = K \cdot F \cdot d$ [$K=0.20$ dry, 0.15 lube, 0.12 anti-seize]

Motor torque: $\tau = K_t \times I$ Back-EMF: $V_e = K_e \times \omega$

Gear ratio: $GR = N_2/N_1 = \omega_1/\omega_2 = \tau_2/\tau_1$

Clamp force: $F_{clamp} = P_{cavity} \times A_{projected}$

Availability: $A = MTBF / (MTBF + MTTR)$

References

- [1] INCOSE, *Systems Engineering Handbook*, 4th ed., Wiley, 2015.
- [2] NASA, “Systems Engineering Handbook,” SP-2016-6105 Rev 2, 2016.
- [3] B. S. Blanchard and W. J. Fabrycky, *Systems Engineering and Analysis*, 5th ed., Pearson, 2011.
- [4] A. Kossiakoff et al., *Systems Engineering Principles and Practice*, 3rd ed., Wiley, 2020.
- [5] K. Ulrich and S. Eppinger, *Product Design and Development*, 7th ed., McGraw-Hill, 2020.
- [6] G. Pahl and W. Beitz, *Engineering Design*, 3rd ed., Springer, 2007.
- [7] G. Kotonya and I. Sommerville, *Requirements Engineering*, Wiley, 1998.
- [8] S. Robertson and J. Robertson, *Mastering the Requirements Process*, 3rd ed., Addison-Wesley, 2012.
- [9] IEEE, “IEEE 1012: Standard for System, Software, and Hardware Verification and Validation,” 2016.
- [10] R. A. Malloy, *Plastic Part Design for Injection Molding*, 2nd ed., Hanser, 2010.
- [11] D. V. Rosato and M. G. Rosato, *Injection Molding Handbook*, 3rd ed., Springer, 2000.
- [12] B. S. Blanchard, *Logistics Engineering and Management*, 6th ed., Pearson, 2004.
- [13] R. S. Pressman, *Software Engineering: A Practitioner’s Approach*, 9th ed., McGraw-Hill, 2019.
- [14] IEC 60300, “Dependability Management,” International Electrotechnical Commission.
- [15] Protolabs, “Design Guide: Injection Molding,” [protolabs.com](https://www.protolabs.com/design-guide/injection-molding/), accessed 2025.
- [16] IPC, “IPC-2221B: Generic Standard on Printed Board Design,” 2012.
- [17] KiCad Project, “KiCad Documentation,” [kicad.org](https://www.kicad.org/), accessed 2025.
- [18] H. W. Ott, *Electromagnetic Compatibility Engineering*, Wiley, 2009.
- [19] M. I. Montrose, *PCB Design for Real-World EMI Control*, Wiley, 2000.
- [20] IPC, “IPC-2612: Sectional Requirements for Electronic Diagramming Documentation,” 2010.
- [21] R. Khandpur, *Printed Circuit Boards: Design, Fabrication, Assembly and Testing*, McGraw-Hill, 2005.
- [22] Espressif Systems, “ESP32 Technical Reference Manual,” V5.1, 2023.

- [23] SKF Group, “Rolling Bearings Catalogue,” PUB BU/P1 10000/2 EN, 2018.
- [24] R. G. Budynas and J. K. Nisbett, *Shigley’s Mechanical Engineering Design*, 11th ed., McGraw-Hill, 2020.